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Theoretical Learning

Theoretical and novel analysis, second draft fixture

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Introduction

This thesis-like document, outlines specific contributions, theoretical developments, and experimental investigations on the **theoretical encoding theory**, automata theory and computer-based systems, simulation logics, **statistical learning theory** and **general learning theory**, within the standard outcome also concluded of statistical physics emulations, and explicit specialized movements. Overall, the document posits itself in the *synthesis package*, in which old ideas are reframed, old formalism being look from different angles of perspective, clarification of concepts, restructuralization of different frameworks and their axioms, presumptions, conjectures, hidden dilemma, and structural failure modes, and proposal of newer insights, outlooks, and theories. In specialities, we focus on the *reformulation of automata systems*, of which applies to current practical reduction to computer encodings, *presumption bias* in changing encoding system and the task of encode or decode, the recursive natures of such computing systems, and the justification of models and AI system living on such automata frame itself. On statistical learning, we focus on the proxy of the problem of *double descent* to highlight theoretical-empirical mismatch, theoretical holes and potential failure modes apparent of the old learning theory, reformulation and synthesis of existing frameworks and usages, and novel contribution to a new theory itself, with experimental toy models for investigations. Additionally, we also emphasize the practical notions and selective simulation or emulations encoding on machine systems, plus the general notion of the learning concepts. In conjunction, collaborations and tangent line of thoughts also consider the statistical learning interpretation, of such learning system.

The core authoring contribution comes from **Bui Gia Khanh**, from Department of Physics, Hanoi University of Science, Vietnam National University, with experimental results in majority coming from **Luong Van Tam**, **Dang Tri Trung** and **Daud Shahbaz**. Specifically, Luong Van Tam also participated as the main contributor in the chapter about statistical physical interpretation of the learning dynamics. We would like to thank any particular contributors participating in this document and research progress itself. Until further funding is acquirred, we would like to clarify of open-source and self-funded research approach on this document.

Background and motivation

The new advancements of Artificial Intelligence (AI), and thus techniques in conjunction to it, for example, the **learning theory**, or the technique of knowledge encoding, the latent space specifications, modelling constructions, agent-theoretic formalism, **actionable system feedback** techniques, and so on. Within such, resurfaced the utilization of (super)computers and so on, within the baseline of computational theory of construction, and subsequently, computational learning theory / statistical learning theory (CoLT/SLT). Such system relies on the algorithmic approaches and procedural, computable representation of the automata systems to create models with capacities capable of being called as AI, and in a narrower sense, capable of adjusting or self-regulating response upon which it is provided of observations and situation-aware information. Such is where the word *machine learning* comes, and situated itself as the main anecdotal reason or capability of AI, for an artificial intelligence system to be intelligent, is posited to have the learning process. Of a long history in development, ever since the few first systems and classical notions of AI, McCulloch and Pitts [1943], Rosenblatt [1958a], Minsky and Papert [1969], Hebb [1949], Robbins and Monro [1951], Widrow and Hoff [1960b], Hestenes and Stiefel [1952] of the old paradigm in symbolic and neuronal approaches of discrete formulaic learning, to the looser *statistical, data-based learning*, particularly as of Valiant [1984], Vapnik [1999], Kearns and Vazirani [1994], Hopfield [1982], Fukushima [1980], Cybenko [1989], Bishop [1995], Haykin [1994], Kohonen [1982], Geman et al. [1992], Nemirovski and Yudin [1983], Rattray et al. [1998], Mohri et al. [2012], and up to modern development of such materials, such as, Jacot et al. [2018], Khan and Rue [2024], Buschjäger et al. [2020], Manin and Marcolli [2024], Devlin et al. [2018], Goodfellow et al. [2016], Kaplan et al. [2020], Bronstein et al. [2017, 2021], Luo and Tseng [1992], Tseng [2001], Friedman et al. [2007], Hu et al. [2021b], Wright [2015], Janik and Witaszczyk [2021], and so on, of the topics about the modern consideration of learning theories, optimization techniques and interpretation landscape, formal languages of the learning processes, as well as architectures on which modern systems are based on; more advanced techniques are developed, for example, as *reinforcement learning*, as seen of Sutton and Barto [2018], Bertsekas and Tsitsiklis [1996], Bertsekas [2017], Szepesvári [2010], Lattimore and Szepesvári [2020], Kaelbling et al. [1996], Watkins and Dayan [1992], Bellman [1957], Williams [1992], Konda and Tsitsiklis [2000], Mnih et al. [2015, 2016], Silver et al. [2016], Schulman et al. [2015, 2017], Lillicrap et al. [2015], Haarnoja et al. [2018], Arulkumaran et al. [2017], Ernst et al. [2005], Fu et al. [2020], Kostrikov et al. [2021], or *online learning* of Shalev-Shwartz [2012], Cesa-Bianchi and Lugosi [2006], Zinkevich [2003], Hazan [2016] and Littlestone and Warmuth [1994] as authoritative examples. Even further, is toward Large Language Model, or LLM (Vaswani et al. [2017a], Devlin et al. [2019], Brown et al. [2020a], Zhao et al. [2023, 2024], Radford et al. [2019, 2018], Raffel et al. [2020], Touvron et al. [2023a,b], Chowdhery et al. [2022], Ouyang et al. [2022], Wei et al. [2022c], ?], Hoffmann et al. [2022], Bai et al. [2022]), where cutting edge efforts and so on are to build agentic systems capable of replacing certain labours kind of human in real systems, and so on improving quality of life. The scope is truly large, the field of applications and its limit overwhelmingly extended, and a much richer landscape was formed in those fields.

However, even with such advancements, certain pushback started to emerge. Those are *structural pushbacks* against previously held assumptions, foundational problems regarding concepts taken for granted of hand-waived, or simply is not considered, structural behaviours out of scope of the dynamical trajectory that classical theory tracks (i.e. a theoretical sensory problem), and inadequacy of theory against empiricism, practiced in the modern AI landscapes. Such mismatch highlights substantial loopholes and opaque understanding and patchworks workaround that was present within the field, in which making a large portion of important problems simply too hard to solve analytically or conclusively, as for example, grokking, seen of Power et al. [2022a], or *double descent*, as of Belkin et al. [2019a], Nakkiran et al. [2019], in which exposes the loose nature of the definition on model selections and its landscape, model complexity, and evaluation on model theoretic approaches. The push for information theoretic learning, statistical interpretation, discourses among different interpretation of the learning setting and so is also for artificial intelligence in general, plus the concerns on explainability of artificial intelligence, of which the modern paradigm poorly offer such, as seen of such pushes on Doshi-Velez and Kim [2017], Lipton [2018], Ribeiro et al. [2016], Lundberg and Lee [2017], Molnar [2020] . Foundational problems still retain its strength unquestioned or unresolved, such as Dreyfus [1965, 1972], Dreyfus and Dreyfus [1986], Suchman [1987], Brooks [1991], Searle [1980], McCarthy and Hayes [1969], Harnad [1990]. On the more modern side, of contemporary critiques, Pearl [2009], Marcus [2018], Sutton [2019], and the modernized Chinese Room Arguments. The *frame problem*, while problematic and is often ignored by modern practitioners, often resurfaced when discussions about explainability and operability are in range, as well as the epistemological notions of artificial operatives, or proofs that such artificial constructs truly 'understand', or 'learn', and its particular computational nature. More than all, is the apparent *theoretical entropy* observed from the landscape of development itself. While empiricism pushes forward, such encouraged particular fields, or rather somewhat all of them, to introduce different analogous interpretation, framing, concepts, or treatments in and of their own to the mechanical details of artificial intelligence. While not solving the inherent foundational problems or assumptive framework that it was based on, various different theories are used to tackle, in a heuristically employed manner, to patch different deviations or unexplainable phenomena together.

Chapter 1. Preliminaries

There are several preliminaries to this particular document, many of which are basic or rather facts regarding different notations, different theoretical knowledges, and so on, covering from mathematics of algebra to functional analysis, mathematical simulation and mathematical modelling, optimization theory and convergence bounds, concentration inequalities, coursework about computational theories, and so on. This will be condensed and precise, and does not aim to replace traditional knowledge or textbooks on such matter. For more information, please refer to authoritative sources as ideal.

1.1 Concentration inequalities

A lot of results in statistical learning is proved using bounds. For example, if $X_1, \dots, X_n$ are independent Bernoulli (θ) random variables representing the outcomes of a sequence of n tosses of a coin with bias (probability of head), then for any $\epsilon \in (0,1)$:

$$\mathbb{P}\Big(|\hat{\theta}_n - \theta \geq \epsilon\Big) \leq 2e^{-2\epsilon^2}, \quad \text{for } \hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{1.1}$$

For the fraction of heads in $X^n = (X_1, \dots, X_n)$. Since $\theta = \mathbb{E}\hat{\theta}_n$, 1.1 says that the sample (or empirical) average of X_i concentrates sharply around the statistical average. Bound like these are fundamental in statistical learning theory (well, mostly for proof, not going to lie). For now, let us learn the technique used to derive such bounds for settings much more complicated than coin tossing.

1.1.1 What are concentration inequalities?

In a probabilistic setting, we are sometimes interested of the random fluctuations of functions of independent random variables. *Concentration inequalities* quantify such statements, typically by bounding the probability that such a function differs from its expected value (or from its median) by more than a certain amount.

The search for concentration inequalities has been a topic of intensive research in the last decades in a variety of areas because of their importance in numerous applications. Typically, this falls into the range of designing non-deterministic (by the definition of such words), dynamic randomized algorithms or procedure of interest, while able to bound them with concentration inequalities to a certain given degree of high probabilistic 'accuracy', for example. Among the areas of applications, without trying to be exhaustive, we mention statistics, learning theory, discrete mathematics, statistical mechanics, random matrix theory, information theory, and high-dimensional geometry (while not actually sure why geometry is in here).

Informally, the *concentration phenomenon* asserts that if $X_1, \dots, X_n$ are independent random variables, then $f(X_1, \dots, X_n)$ does not deviate much from its mean provided that $f(x_1, \dots, x_n)$ is not too sensitive in any of its coordinate x_i. While concentration properties for sums of independent random variables were thoroughly studied and fairly well understood in classical probability theory, powerful tools to handle more general functions of independent random variables were not introduced until the appearance of *martingale methods* in the 1970s, by Yurinskii (1976), Maurey (1979), Milman and Schechtman (1986), Shamir and Spencer (1987) and McDiarmid (1989).

Note 1.1.1. *While the topic itself is much more complex than it is guaranteed to be, a martingale can be defined as a sequence of random variables (i.e. for a stochastic process) for which, at a particular time, the conditional expectation of the next value in the sequence is equal to the present value, regardless of all prior values. A basic definition can be achieved. For a process* $(M_n)_{n\geq 0}$*, it is called martingale if:*

- *For every $n \geq 0$, the expectation $\mathbb{E}[M_n]$ is finite, and hence equivalently, $\mathbb{E}[|M_n|] < \infty$.*
- *For every $n \geq 0$ and all $m_n, m_{n-1}, \dots, m_0$ we have:*

$$\mathbb{E}(M_{n+1} \mid M_n = m_n, \dots, M_0 = m_0) = m_n \tag{1.2}$$

1.1.2 Relevant toolset

Of our book chapter, by a *concentration inequality* we most often mean an upper bound for the probability that a real-valued random variable Z differs from its expected value by more than a certain amount. In other word, we seek upper bounds for tail probabilities of the form

$$\mathbb{P}[Z - \mathbb{E}(Z) \geq t], \quad \mathbb{P}[Z - \mathbb{E}(Z) \leq -t] \tag{1.3}$$

where $t > 0$. Of course, here we assume implicitly that the expected value exists, as for the probability to be included of the measurable space.

To derive a lot of our concentration bounds that will solve such problem, however, we have to develop a few basic tools in the analysis. This will include Markov's inequality, Chebyshev's inequality, the Chernoff bound (often called the Chernoff trick), and Hoeffding's lemma.

1.1.3 Markov's inequality

Markov's inequality

Let $Y \in \mathbb{R}$ be a nonnegative random variable. Then for any $t > 0$, we have the *Markov's inequality* :

$$\mathbb{P}(Y \geq t) \leq \frac{\mathbb{E}[Y]}{t} \tag{1.4}$$

Before jumping straight into a proof, let us see what are we looking at from the perspective of the inequality.

Proof. The proof is straightforward:

$$\begin{aligned} \mathbb{P}(Y \geq t) &= \mathbb{E}[\mathbb{1}_{(Y \geq t)}] \\ &\leq \frac{\mathbb{E}[Y \mathbb{1}_{(Y \geq t)}]}{t} \\ &\leq \frac{\mathbb{E}[Y]}{t} \end{aligned} \tag{1.5}$$

We use the fact that

$$\mathbb{P}(Y \in A) = \int_A P_Y(dy) = \int_Y \mathbb{1}_{(x \in A)} P_Y(dy) = \mathbb{E}[\mathbb{1}_{Y \in A}] \tag{1.6}$$

Additionally, we also have $Y \geq t > 0$ implies $Y/t \geq 1$, and $Y \geq 0 \implies Y\mathbb{1}_{(Y \geq t)} \leq Y$ which implies their expected value. □

If ϕ denotes a nondecreasing and nonnegative function defined on a possibly infinite interval $I \subset \mathbb{R}$, and if Y denotes a random variable taking values in I, then Markov's inequality implies that for every $t \in I$ with $\phi(t) \neq 0$,

$$\mathbb{P}(Y \geq t) \leq \mathbb{P}[\phi(Y) \geq \phi(t)] \leq \frac{\mathbb{E}[\phi(Y)]}{\phi(t)} \tag{1.7}$$

This can be used to derive the Chebyshev's inequality later.

1.1.4 Chebyshev's inequality

The common application of Markov's probability is Chebyshev's probability. Let X be an arbitrary real random variable. Then for any $t > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\mathrm{Var}[X]}{t^2} \tag{1.8}$$

Where $\mathrm{Var}[X] := \mathbb{E}[|X - \mathbb{E}X|^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ is the variance of X. The prominent role of Chebyshev's inequality is not only explained by historical reasons, but also among all the absolute moments of the form $\mathbb{E}[|Z - \mathbb{E}(Z)|^q]$, the variance is typically the easiest to handle.

Proof. Let X be a random variable, so $(X - \mathbb{E}[X])^2$ is a non-negative random variable. Hence, we can apply Markov's inequality:

$$\begin{aligned} \mathbb{P}(X - \mathbb{E}[X] \geq t) &= \mathbb{P}\left((X - \mathbb{E}[X])^2 \geq t^2\right) && (1.9) \\ &\leq \frac{1}{t^2}\mathbb{E}\left[(X - \mathbb{E}[X])^2\right] && (1.10) \\ &= \frac{1}{t^2}\mathrm{Var}[X] && (1.11) \end{aligned}$$

□

The property of Chebyshev's inequality contains several constraints, though arguably better than Markov's. Chebyshev's inequality does not require that the random variable is non-negative. However, it also requires that we know the variance in addition to the mean. Hence, the goal of Chebyshev's inequality is to bound the probability that the RV is far from its mean in either direction. While in principle Chebyshev's inequality asks about distance from the mean in either direction, it can still be used to give a bound on how often a random variable can take large values, and will usually give much better bound than Markov's inequality - since we have more knowledge of the distribution.

We can use Chebyshev's inequality to prove a soft variation of the law of large number, or the *weak law of large number*.

Theorem 1.1.1 (Weak Law of Large Number). *Let $X_1, \ldots, X_n$ be a sequence of i.i.d. random variables with mean μ. Let $\bar{X}_n = (1/n)\sum_{i=1}^n X_i$ be the sample mean. Then $\bar{X}_n$ converges in probability to μ. That is, for any $\epsilon > 0$,*

$$\lim_{n\to\infty} \mathbb{P}\left(|\bar{X}_n - \mu| > \epsilon\right) = 0 \tag{1.12}$$

Proof. By the property of the expectation and variance of sample mean consisting of n i.i.d. variables, for $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. By Chebyshev's inequality,

$$\lim_{n\to\infty} \mathbb{P}\left(|\bar{X}_n - \mu| > \epsilon\right) \leq \lim_{n\to\infty} \frac{\mathrm{Var}(\bar{X}_n)}{\epsilon^2} = \lim_{n\to\infty} \frac{\sigma}{n\epsilon^2} = 0 \tag{1.13}$$

thus recover our desired equation. □

1.1.5 McDiarmid's inequality

A generalization of Hoeffding's inequality is the McDiarmid's, or Bounded Difference inequality, where the quantity of interest is some function of the data, i.e. $S_n = \phi(X_1, \ldots, X_n)$. Some restrictions on ϕ are required to get exponential bounds.

Theorem 1.1.2 (McDiarmid's Inequality). *Let $X_1, \ldots, X_n$ be independent random variables, where X_i has range $\mathcal{X}_i$. Let $f : \mathcal{X}_1 \times \cdots \times \mathcal{X}_n \to \mathbb{R}$ be any function with the $(c_1, \ldots, c_n)$-bounded difference property: for every $i = 1, \ldots, n$ and every $(x_1, \ldots, x_n), (x_1', \ldots, x_n') \in \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ that differ only in the i-th coordinate, we have:*

$$|f(x_1, \ldots, x_n) - f(x_1', \ldots, x_n')| \leq c_i \tag{1.14}$$

For any $t > 0$:

$$\mathbb{P}\Big([f(X_1, \ldots, X_n)] - \mathbb{E}[f(X_1, \ldots, X_n)] \geq t\Big) \geq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right) \tag{1.15}$$

Proof. Write $\mathbb{E}_i[\cdot]$ to denote expectation conditioned on $X_{1:i} := (X_1, \dots, X_i)$. Therefore, $g_i(X_{1:i}) := \mathbb{E}_i[f(X_{1:n})]$ is, as the notation suggests, a function of $X_{1:i}$. Now define the following random variables:

$$\begin{aligned} Y_i &:= g_i(X_{1:i}) - g_{i-1}(X_{1:i-1}), \\ A_i &:= \inf_{x_i \in \mathcal{X}_i} g_i(X_{1:i-1}, x_i) - g_{i-1}(X_{1:i-1}), \\ B_i &:= \sup_{x_i \in \mathcal{X}_i} g_i(X_{1:i-1}, x_i) - g_{i-1}(X_{1:i-1}). \end{aligned}$$

These random variables satisfy

$$\begin{gathered} Y_i \in [A_i, B_i], \quad \mathbb{E}_{i-1}[Y_i] = 0, \\ \sum_{i=1}^{n} Y_i = f(X_{1:n}) - \mathbb{E}[f(X_{1:n})]. \end{gathered}$$

Furthermore, we claim that $[A_i, B_i]$ is always an interval of length at most c_i. We defer the proof of this claim to later.

To bound the probability that the sum of the Y_i's is at least t, we use the Chernoff bounding method. Let $S_i := Y_1 + \cdots + Y_i$. Then

$$\begin{aligned} \mathbb{E}[\exp(\lambda S_n)] &= \mathbb{E}[\exp(\lambda(Y_n + S_{n-1}))] && (1.16) \\ &= \mathbb{E}[\mathbb{E}_{n-1}[\exp(\lambda Y_n)] \exp(\lambda S_{n-1})] && (1.17) \\ &\le \exp(\lambda^2 c_n^2/8)\, \mathbb{E}[\exp(\lambda S_{n-1})] && (1.18) \\ &= \exp(\lambda^2 c_n^2/8)\, \mathbb{E}[\exp(\lambda(Y_{n-1} + S_{n-2}))] && (1.19) \\ &= \exp(\lambda^2 c_n^2/8)\, \mathbb{E}[\mathbb{E}_{n-2}[\exp(\lambda Y_{n-1})] \exp(\lambda S_{n-2})] && (1.20) \\ &\le \exp(\lambda^2 c_n^2/8) \exp(\lambda^2 c_{n-1}^2/8)\, \mathbb{E}[\exp(\lambda S_{n-2})] && (1.21) \\ &\vdots && (1.22) \\ &\le \exp(\lambda^2 c_n^2/8) \cdots \exp(\lambda^2 c_1^2/8) && (1.23) \\ &= \exp\left(\frac{\sum_{i=1}^n c_i^2}{4} \cdot \frac{\lambda^2}{2} \right). && (1.24) \end{aligned}$$

Above, the inequalities all follow from Hoeffding's inequality. Therefore, S_n is $\sum_{i=1}^n c_i^2/4$ sub-Gaussian. The probability bound now follows from the Cramer-Chernoff inequality for subgaussian random variables.

It remains to prove that $[A_i, B_i]$ is an interval of length at most c_i. We show that $B_i - A_i \le c_i$. By independence of $X_1, \dots, X_n$, we have

$$\begin{aligned} g_i(X_{1:i-1}, x_i) &= \mathbb{E}[f(X_{1:i-1}, x_i, X_{i+1:n}) \mid X_{1:i-1}, X_i = x_i] && (1.25) \\ &= \mathbb{E}[f(X_{1:i-1}, x_i, X'_{i+1:n}) \mid X_{1:i-1}], && (1.26) \end{aligned}$$

where $X'_{i+1:n}$ is an independent copy of $X_{i+1:n}$. Then

$$\begin{aligned} B_i - A_i &= \sup_{b \in \mathcal{X}_i} g_i(X_{1:i-1}, b) - \inf_{a \in \mathcal{X}_i} g_i(X_{1:i-1}, a) && (1.27) \\ &= \sup_{a,b \in \mathcal{X}_i} g_i(X_{1:i-1}, b) - g_i(X_{1:i-1}, a) && (1.28) \\ &= \sup_{a,b \in \mathcal{X}_i} \mathbb{E}[f(X_{1:i-1}, b, X'_{i+1:n}) - f(X_{1:i-1}, a, X'_{i+1:n}) \mid X_{1:i-1}] && (1.29) \\ &\le \mathbb{E}\Big[\sup_{a,b \in \mathcal{X}_i} |f(X_{1:i-1}, b, X'_{i+1:n}) - f(X_{1:i-1}, a, X'_{i+1:n})| \Big| X_{1:i-1} \Big] && (1.30) \end{aligned}$$

$$\leq c_i. \tag{1.31}$$

The last step above uses the bounded differences property of f. □

1.1.6 The Cramer-Chernoff method

One more important bounding method that can be discussed in this section is the Cramer-Chernoff bound. This method determines the best possible bound for a tail probability that one can possibly obtain by using Markov's inequality with an exponential function $\phi(t) = \exp(\lambda t)$. This simple technique leads to surprisingly sharp bounds in many cases. We work out some simple examples, and then will proceed with some of its analytical properties.

Let Z be a real-valued random variable. For $\lambda \geq 0$, Markov's inequality implies

$$\mathbb{P}[Z \geq t] \leq \exp(-\lambda t)\mathbb{E}[\exp(\lambda Z)] \tag{1.32}$$

Since this inequality holds for all values of $\lambda \geq 0$, one may choose λ to minimize the upper bound. Defining the logarithm of the moment generating function as

$$\phi_Z(\lambda) = \log(\mathbb{E}[\exp(\lambda Z)]), \quad \forall \lambda \geq 0 \tag{1.33}$$

and introducing the supremum of such generating function being

$$\phi_Z^*(t) = \sup_{\lambda \geq 0}(\lambda t - \phi_Z(\lambda)) \tag{1.34}$$

We obtain the Chernoff's inequality: Chernoff's inequality

$$\mathbb{P}[Z \geq t] \leq \exp\left[-\left(\sup_{\lambda \geq 0}(\lambda t - \phi_Z(\lambda))\right)\right] = \exp[-\phi_Z^*(t)] \tag{1.35}$$

The function $\phi_Z^*(t)$ is called the *Cramer transform* of Z. Usually, this is also called the Legendre transform (for example, in ?) or Legendre-Fenchel transform. The Cramer transform is characterized of its shape by satisfying the *Cramer's condition*, Cramer condition

Definition 1.1.1 (Cramer's condition). *Let M_X denote the moment-generating function of a given random variable X, that is, $M_X(t) = \mathbb{E}[\exp(tX)]$. Then the random variable X satisfies the Cramer condition if there exists $c > 0$ such that $\mathbb{E}[\exp(|cX|)] < \infty$. If X satisfies the Cramer condition, then we call M_X to be well-defined (taking finite values) on a connected neighbourhood, containing the interval $[-c, c]$, of zero and moreover possess the expansion*

$$M_X(t) = \sum_{n=0}^{\infty} \frac{\mathbb{E}[X^n]}{n!} t^n \tag{1.36}$$

where $|t| < t_0$ and $t_0 > c$.

The function itself is a nonnegative function, since $\phi_Z(0) = 0$. If $\mathbb{E}[Z]$ exists, then the convexity of the exponential function and Jensen's inequality imply that $\phi_Z(\lambda) \geq \lambda\mathbb{E}[Z]$ and therefore, for all negative values of λ, $\lambda t - \phi_Z(\lambda) \leq 0$ whenever $t \leq \mathbb{E}[Z]$. This means that we may formally extend the supremum over all $\lambda \in \mathbb{R}$ of the Cramer transform,

$$\phi_Z^*(t) = \sup_{\lambda \in \mathbb{R}}(\lambda t - \phi_Z(\lambda)) \tag{1.37}$$

from this, formally, we treat the right-hand expression of the often known, as what we call the *Fenchel-Legendre dual function* of ϕ_Z. Thus, at every $t \geq \mathbb{E}[Z]$, the Cramer transform coincides with the Fenchel-Legendre dual. That is not to say they are the same, however the range of λ is relevant.

Chapter 2. Classical learning theory

Classical learning theory, as most popular of Valiant [1984], Vapnik [1999], Vapnik and Chervonenkis [1971], is the cornerstone of modern learning landscape. It was largely adopted, along the side of computational empiricism, as the main interpretation framework of machine learning since the 2000s. To understand the limitations in application, and different out-of-bound failure modes (as already mentioned of unresolvable phenomena, and structural requirement failures), it is suggested that a thorough coverage of the topic is required.

What we have done, what we have been investigated, including mathematical modelling, can be said to be the *static construction*. What this means is that, albeit intrinsic of collecting and transforming resources, information, or any given kind of input - given the black box, in-out system treatment. This comes as a cost - the model requires human touches to even conform to certain structures, knowledge, tasks, and utilities. Mitigate this issue requires we to come up with a notion that automates the acquisition and relative understanding (butchering the word understanding, since our model currently does not understand *anything*, strictly speaking), and it is called (and found) as the *learning theory* of machines.

2.1 Precursors to learning theoretic

Before the introduction of learning theory as a practical and theoretical process, functional models, usually mathematical models, mostly relies on acute, specific solution, correct formulation of given form, or rather being done solely in a static sense. What this means must be followed through by historic development to understand why it is called static, and subsequently what is then considered dynamic.

Mainly mathematics (but not restricted to such, sometimes usual construction is alright) has been used to formalize, simplify, express, and encapsulate the fraction of reality through the language of quantification, to make **models** that befit the knowledge and the respective fraction observed. Simply said, considering mathematics as the foremost language of abstraction (removing physical realization), quantification (give us the notion of quantity), and the build up of structures within said language, it is fundamentally useful in decoding and expressing reality in the way that makes it understandable, interpretable, logical (the word logical brings around the roundabout), easy to quantify components, actions, factors, objects, and perhaps many further benefits of using such language. Thus, one of the most fundamental example of using the language of mathematics to formalize and quantize reality to be a functional construct, is physics. And its development throughout history also says plenty bit on what is static.

A theory is often considered a static construct, by virtue of how it is formed. It begins by setting up axioms, assumptions, simplification, the target of the theory, and development on which it is built on upon. Further development will follow through the main foundation upward, without many changes to the theory. If one is to modify the foundation, it would mean usually to create new theory on said basis. This process of transition is typically thought in philosophy as the *negation of the negation*, or in propositional logic, double negation. Fortunately, the transition is perhaps, if not totally very natural of natural science and overall interpretation of the world that we often use. We can attribute this to the fact that we do not understand the world in its entirety, Furthermore, mathematical modelling on itself is not perfect, and cannot capture intrinsically every given thing available, for specific methods - basically, our hands are tied, and is not perfect either way. That is why there must be, and would be this type of changes throughout the theory itself. Building a structure is the same, and so much for a system.

Most of the current, widely perceived general model construction and the way that artificial intelligence was created (under the umbrella term of automated thinking machine, artificial is one, and machine learner later on was subjected to be the equivalent, more *advanced* development) belongs to either the specific type of inference parameterized model, or symbolic model (AI) - which utilizes the propositional logic ϕ of the truth functionality $\mathcal{V}^n \to \mathcal{V}$. Under said notion, precursor AI, for even specialized task, was conducted in the way that it is designed by experts, for particular problem of knowledge implementation, and then use logic to deduce from said logic pool reasonable course of action, or rationalization; historically, Minsky and Papert [1969] and the symbolic AI practitioners encode hard logics and baseline logical learning to itself. This is partially because of the computational limit of the time - at the time of 1950s to 1970s, and the lack of networks of information and data. Unfortunately, such design choice led to a limitation of the *frame problem* widely present in predicate-based artificial intelligence.

One major theme of the disadvantage of the phenomenological approach on statistically-assumed space, is the fact that it is sensitive to changes. That is, given the holding parameters k for the Boole's coefficient of a spring under stress, for certain deviating interference, by for example, $2k$, then the effective phenomenological model usefulness is reduced by the same factor. Trivially, one can then simply expand the system to accommodate the various setting of k, but such task requires human-inferred choices and modifications, and loss of generalization, in comparison to the mechanistic modelling approach. It is well-understood that without certain dynamic process, and objectively the primitive notion of learning, a phenomenological sometimes might be significantly difficult to handle, based on the immense dimensionality requirement of the parameters used in relation to the system.

Afterward, it is perceived that expert inputs and traditional knowledge acquisition and their form as logical proposition is not adequate - in fact, it is unable to be scaled up. Yes, it is good - even by today's standard - but given a larger, more comprehensive and more complex problem, the system break down to be unsustainable and unattainable. Thereby, we need something else. Something of the quality that makes the model *learn for itself*, rather than waiting for developer's implementation and expert input. That comes of as, eventually not so surprising, an ability that human possesses: *learning*.

In another sense, learning can also be conjured to be the action of *computational reasoning with uncertainty*, as opposed to the general notion of artificial agent, a type of model which makes use of, as previously mentioned, logics and relational discrete spaces. We would look into this aspect later, however, it is also suggested to note that, learning can also still happen, within the logical region of a model. By that, we think of the internal space and external space - the classical definition for learning theory and its distinction in such case is typically the classification between *external-dependent reasoning* and *internal-dependent reasoning* (which depends only on the propositional space).

2.2 Transparency shift

The learning theory started from the early 1970s. By the sense of classical, we do not mean by, as currently called, of non-deep learning theory, but rather the old treatment of theoretical boundaries and rigorous formalization of the concept itself. The nature of the learning theory can also be brought back from certain proposition or argument, most probably, the fact that it is mostly from the black-box interpretation - hence making almost all model being phenomenological by default.

In historical term, such learning problem is derived out from the problem of **adaptation**. Supposedly, given a system M, in an environment E, can it derive itself toward a goal p? Or rather, can an *agentic machine* capable of existing own their own, in the environment produced by the perceived world? Such requires implicitly the objective of survival p for the machine in such space, and the desired pattern for the respective survival. Typically, this is reduced to finding the right answer from information, or deduction logics toward accurate functionals, as seen in symbolic-based chain logical rules. Without constructing and considering the internal mechanism of the model, we ask the following basic questions:

1. What is the environment? What mode of existence the model is enabled in it? For example, if the world itself is considered symbolically, then it means that the environment it sees happens in the form of causality, as every observation is connected to each other of a symbolic decision graph, thus make use of *first-order predicate calculus* on causal chain, as seen in mostly Clocksin and Mellish [2003], Sterling and Shapiro [1994], Norvig [1992], Brachman and Levesque [2004], Philip C. Jackson [1985], Lucas [1989], Giarratano and Riley [1998], with looser version scaled to *expert systems*, where objects are encoded into the environment via human expert's consideration. The downside on this approach is that observational spaces are limited to what the environment truly encodes - for example, different discrete type of features and decision domains. Thus, the environment itself is *not universal*, in the sense that the connection from reality or larger domain is simplified to the decision network of the environment itself. On the other hand, statistical models encapsulate *observational encoding*, potentially on infinite feature's value range, and use the observed statistics as truth instead. Such removes the rigid structural consideration, and enforce phenomenological interpretation of the data-centric view as

production of actual operations and situations in real life. While still limited to sensors or measurement scope, it provides more freedom, patterns, and so on, in exchange for the lack of mechanistic extraction.

2. What is the constraint in which the environment forces on the existence of the model? What is the interaction between the model and its world? What are its allowed dynamics and actions? What is the specified state of the world, static, dynamic, mutable, or objectively considered as facts? Those are questions from the internal construct of the environment, as well as actionable domain of the model of interest, which lives in it. For example, a model in symbolic sense can only be a *predicate processor* with weighted path, such that certain predicative decisions are enforced instead of others, for morality (wrong or right on ethics), and so on.

3. What are the failure modes of the environment that *discourage* the model from doing certain parts? In typical setting, we assume that in the operation of such model inside the environment, it will encounter *problems*; however, the question is to encode it, and toward such, the question elaborates to what kind, how, and the role of such inhibitions.

Those questions are then answered by giving adaptation in a form of **trial and errors**. Effectively, this means learning a scenario or target concept by treating them in a form that can be given of *causal relationship*. The simplest and most straightforward of such is the construct to be function, i.e. $f : x \to y$. This permits partial realization of the system at worst of such framework, and total predictability of such system at best. Suppose a system contains $(x, y, z, q_1, \ldots, q_k)$ for relative set of $q_i \leq k$ elements of quantification. Then, the system can have the class of all *directed function* mapping from $x \to y$, thus in a directed graph can be represented using directed edge, with the arrow pointing to dependent value in respective frame. In general, however, such relationships are often two-way, thus guarantee the existence of the inverse function. Then, a system can have binary multivariate relationship to be encoded of such causal form, for example, $v_1 : x \to z$, $v_2 : (x, y) \to q_3$, and so on. The set of all of such function in the system can be considered as the edge class (in a sense, a graph structure without fixed numerical representation is really effective for this type of assumption). The conjecture is this: the model can, estimate certain thread of this space, and in ideal case, find a causal chain good enough to infer different information about it. We call this specific node the **universal node**. Whether this node exists or not in the structure is unknown, and much like other form of assumptions and typical structural view, it is not totally guaranteed but of the bias in constructing such structure by itself. Nevertheless, much of learning theory exists, or rather, use this for approximation of systems.

2.2.1 Ladder of functionals

2.3 Classical setup

In simplicity, in classical learning theory in the modern paradigm (data-centric), our model is called the **hypothesis** $h \in \mathcal{H}$ of certain **hypothesis class** $\mathcal{H}$. This basically means that h is configured to hypothesize of the one-one relationship of the observational space. The other ingredient of the learning theory includes:

1. The concept class versus function class can be fully incorporated into one as $\mathcal{C}$, if one is to assume the form of any given concept as $c : X \to Y$ for another set Y which can lies in the sigma-algebra $\mathcal{S}$, or the measurable mapping as f.

2. The task for the hypothesis, and the designer (us) is to figure out how to best approximate c, assuming arbitrary notion of knowledge, prior information, such that h correctly *mimic* the behaviour and phenomenon observed by c. We call this loosely as the **learning problem**. Naturally, learning problem is a statistical problem, since their setting is fully observational.

We are given our information of the system by the form of a dataset $\mathcal{S}$, called *observation*. We first define the **observations**. The set of observation $\mathcal{S}$ is presentable in the space of 2-tuple $(\mathcal{X}, \mathcal{Y}) \subset (\mathbb{R}^d, \mathbb{R}^p)$. Usually, hence, our data will be realized in the Borel σ-algebra $\mathcal{B}$. The full dataset is *discrete*, contain of n instances of observations, and is denoted by

$$\mathcal{S} = \{(x_1, y_1), \ldots (x_n, y_n)\} \subset (\mathcal{X} \times \mathcal{Y})^n \tag{2.1}$$

Here, $\mathcal{X}$ is called the *ground* or input space, and $\mathcal{Y}$ is called the *target* or output space. By default, in general, the input space $\mathcal{X}$ is supposed to follow some special distribution $\mathcal{D}$. This treatment branches from the ambiguity of the origin of $x \in \mathcal{X}$ for elements of the input space, that is, $x \sim \mathcal{D}$ by a random sampling process. This assumption also ensures relative population representability. By this, we mean that the observation captured fully describe what was happening with the entire concept - even including the pathological and outlier. We also assume that this sample of observation is **identically and independently sampled** of a given sampling process.

The **learning problem** divides the act, with respect to configuration the chain of component into two parts. The first one is for any measurable space of the model, after defining the system, the dependency, and the question, we have to gain the *accuracy measure*, or *phenomenological aberration* of a given model, and for the respective problem. Depends on the specific requirement of the problem, the measure can change, but, they have the same landscape. The second one requires the development of a *scheme* to dynamically move the given hypothesis, under fixed description, into a supposed optimal point in which under specific threshold, our model is 'perfect' for the concept. That is, *mimic c by h in a perfect way*. It is perceived, though, that the objective of learning the concept perfectly is impossible. Hence, we will often opt for the latter half of approximating it to a given threshold θ.

The learning problem - which is most importantly described by the phase of *phenomenological aberration* (or data-based replication) - is then formulated as followed. Given the machine learning model expressed a hypothesis h of the hypothesis class $\mathcal{H}$, the learning theory aims for creating a procedure to learn either elements of the concept class $\mathcal{C}$ of all concepts $c : \mathcal{X} \to \mathcal{X}$, or the function class $\mathcal{F}$ of all functions $f : \mathcal{X} \to \mathcal{Y}$, which is usually set to $\{0, 1\}$ or $[0, 1]$. This distinction is trivial, hence, if it is clear, we will talk about the concept class $\mathcal{C}$ only as representative form. The learner $\mathcal{L}(h)$ consider the set of possible hypothesis $\mathcal{H}$, in which might not coincide with $\mathcal{C}$. It receives a partial image of sample $S = (x_1, \dots, x_2, \dots, x_n)$ drawn i.i.d. according to $\mathcal{D}$ as well as the label $(c(x_1), \dots, c(x_n))$. This constitutes the dataset $\mathcal{S}$, which are based on specific concept $c \in \mathcal{C}$ for the hypothesis to learn. The task is then to use (or *extract*) meaningful information to select a hypothesis $h_S \in \mathcal{H}$ that accurately mimic c, with marginal error Θ. The notation h_S stands for all hypothesis that can be inferred from the range of the dataset (the first argument, $\mathcal{X}$).

The marginal error Θ is considered of two parameters, the empirical error $\hat{R}(h)$ and the generalization error $R(h)$. This can be justified as following.

To compare between h and c, we use the metric provided by $\mathcal{Y}$. Then, define certain type of function called *loss function*, also called objective, error function, denoted by $\ell\{h, c\}$ such that it measures the difference between h and c. This measure is arbitrary, and we assume no form of it, aside from the fact that it also acts and return values on certain real subset of values. The following definitions provide us with the notion of *empirical error* and *generalization error*. Notice that one is empirical - indeed, since we only can work on certain fixture of c, i.e. the dataset provided, and the other one is generalized - meaning the true error toward the actual target concept $c \in \mathcal{C}$.

empirical risk

Definition 2.3.1 (Empirical risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $S = (x_1, \dots, x_m)$. For some particular $\epsilon > 0$, the* ***empirical error*** *or* *empirical risk* *of h is defined by*

$$\hat{R}_S(h) = \mathop{\mathbb{P}}_{x \in S \sim \mathcal{D}}[\ell\{h(x), c(x)\} \leq \epsilon] = \frac{1}{m}\sum_{i=1}^{m} \ell\{h(x_i), c(x_i)\} \tag{2.2}$$

generalization risk

Similarly, the generalization risk defined in the same fashion, but extending beyond the dataset $\mathcal{S}$.

Definition 2.3.2 (Generalization risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$ on $\mathcal{X}$. For some particular $\epsilon > 0$, the generalization error or risk of h is defined by*

$$R(h) = \mathop{\mathbb{P}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\} \leq \epsilon] = \mathop{\mathbb{E}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\}] = \int_{x \in \mathcal{D}} \ell\{h(x), c(x)\}\, dP(x) \tag{2.3}$$

The notion of empirical and generalization error clearly indicates the issue of two concepts - one observable 'component' concept c' presented by the dataset, and the actual concept c itself. Hence, we assume, of the universal $\mathcal{X}$ set belongs to c, there exists $\mathcal{X}_c$ and $\mathcal{X}_{c'}$, hence we have two distributions, for example, denoted by $\mathcal{D}_c$ and $\mathcal{D}_{c'}$.

There exists a problem in the underlying theory of data-based statistical learning. The question is as simple. In the statistical learning setting, one gain the dataset $(\mathcal{X}, \mathcal{Y})$. Because we are using the *black-box interpretation*, the concept that is supposed to be in $c : \mathcal{X} \to \mathcal{Y}$ is said to be governed by a probability distribution $\mathcal{P}$ on said data. Now, depends on the type that you want, it can either be **dense**, that is, there are infinitely many $x \in \mathcal{X}$ as the phase space, and the probability is on y, that is, $\mathcal{P}(Y)$. Or, you can remove the strictly dense assumption and get the probability $P(\mathcal{X}, \mathcal{Y})$ of joint probability between the two instead. We know from the start that the dataset is not always representative, neither captures the entirety of the concept c. Hence, there must exist a particular $\mathcal{D}$ of true probability distribution that describes c. In a perfect world, we would not need to see c as probability distribution - since in said perfect world, everything might as well be deterministic, but the best one can do is to be able to get close to the presumptions probability distribution of the event. Then, the question is, what is the *inherent correlation* between $\mathcal{P}$ and $\mathcal{D}$? To answer this question, we might as well need some divergence metric to see their diverging property. This will eventually be the KL-divergence measure, which we will discuss later. A disadvantage of KL-divergence, however, is its inability to take into account the structural problem of the system, as well as its model. We might as well address this in our later section.

Using the above notion of generalization, we can separate the learning problem into two goals - one to minimize $\hat{R}(h)$, and one to minimize $R(h)$. As we have been saying, and for Theorem 6.1.3, one of our main assumption, as well as a guarantee under uniform convergence, is the fact that we must have $\hat{R}(h) \approx R(h)$ for larger and larger dataset size, or the empirical space. Generally, this branches off to be two problems.

Definition 2.3.3 (Empirical learning problem). *We present the formal form of the empirical learning. Suppose we have a target, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. The problem is to use certain algorithm $\mathcal{A}$ using $\mathcal{S}$, to obtain a hypothesis h^* for a fixed $\mathcal{H}$ such that:*

$$R(h^*) = \min_{h\in\mathcal{H}} \hat{R}(h) = \min_{h\in\mathcal{H}} \mathop{\mathbb{E}}_{x\sim\mathcal{D}, x\in\mathcal{S}} \ell\{h(x), c(x)\} \tag{2.4}$$

The hypothesis h^* is often called the *empirical best*, for it being the minimal, finite hypothesis of the lowest loss evaluation on the entire observation space $\mathcal{S}$. There exists no certified assumption regarding whether h^* aligns with the minimal generalization error.

Definition 2.3.4 (Generalization learning problem). *We present the formal form of the generalization learning problem. Suppose we have a target, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. Supposed we have an algorithm $\mathcal{A}$ that for fixed hypothesis space $\mathcal{H}$, 6.5 holds true. The problem is to use certain algorithm $\mathcal{A}'$ such that, under limited availability, to obtain $\boldsymbol{h}$, satisfies:*

$$R(\boldsymbol{h}) = \min_{h\in\mathcal{H}} R(\boldsymbol{h}) \leq \{\epsilon\}, \quad \epsilon > 0 \tag{2.5}$$

For a set of risk bounds ϵ. If the setting is deterministic, then there exists $\epsilon = 0$.

We can show that normally, the problem of empirical learning and generalization learning is not the same. That is, if one tries to solve the empirical learning problem to certain measure, then they will fail the generalization problem. This can be illustrated by figure 2.1. This problem between the disparity of the empirical and generalization target is what we often called as the *Allegory of the cave*. This is the phenomena where we interpolate too much of the empirical space, when turning to the generalization space, the model instead is marginally significantly inaccurate. Here, we risk being dependent ourselves on false hypothesis obtained by empirical data acquisition, but also have to aim for the general form of the target by itself, which is often assumed unobtainable or impossible to know of prior belief. Hence, either that we take a skewed representation of reality through empirical data, or we diverge from such in a manner that gets closer to the ground truth's concept. At least in this setting, it is then observed as above that empirical learning problem is a subproblem toward generalization problem - one have to solve the empirical learning problem before attempting the generalization one. We also want to introduce the term **data shape** and **concept shape** for data $\mathcal{S} \in c$ and the shape of c by itself, to reflect this difference.

For the generalization problem's best model in a fixed $\mathcal{H}$, we have the notion of a **Bayes model**.

Definition 2.3.5 (Bayes risk). *Given a distribution $\mathcal{D}$, the Bayes error R^* is defined as the infimum of the errors achieved by measurable functions $h : \mathcal{X} \to \mathcal{Y}$:*

$$R^* = \inf_{h\in\mathcal{H}} R(h) \tag{2.6}$$

Definition 2.3.6 (Bayes model). *A hypothesis h with $R(h) = R^*$ is called a Bayes hypothesis or Bayes classifier, and is taken by*

$$\forall x \in \mathcal{X}, \quad h_{\mathrm{B}} = \operatorname*{argmax}_{y\in\mathcal{Y}} \mathbb{P}[y \mid x] \tag{2.7}$$

The Bayes model is understood to be the minimal of $R(h)$. And, by our observations, $\hat{R}(h) \neq R(h)$ in circumstances.

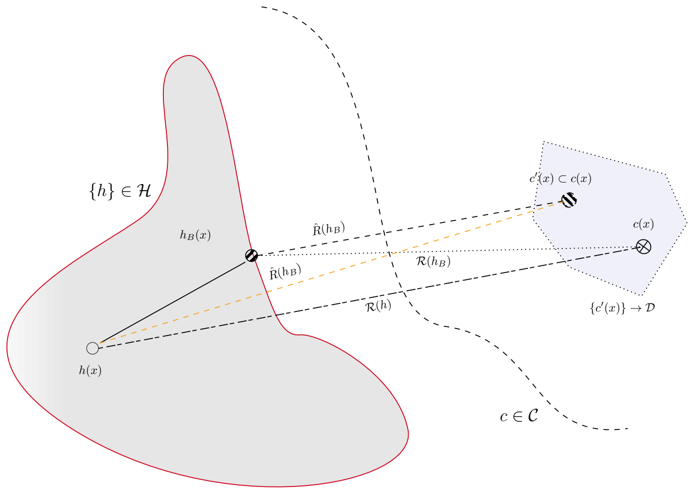


Figure 2.1: An illustration of statistical learning theory on the evaluation of the risks and errors, during learning process. c' is presented in the 'orbital' vicinity around c, with its distance of certain metric define how 'accurate' the reconstruction from distribution can be. Of the hypothesis set $\mathcal{H}$, there exists the Bayes hypothesis h_B and an arbitrary 'random' hypothesis h, and their respective measure.

Overall, in the classical learning setting, we can dissect the problem into what constitute the following cleaner version of the setting, includes both the generalization learning and empirical learning problem. Again, under ideal condition, and ideal setup, the learning setting will have empirical learning and generalization learning to be on the same track. Later on, we would formalize what exactly on the same track means.

Setting 2.3.1. *Given an op-space $\mathcal{X} \times \mathcal{Y}$, define the concept class $\mathcal{C}$ of all $c : \mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{Y})$ of all power set for both, where $\mathcal{X} \sim \mathcal{D}$ for some distribution $\mathcal{D}$ of the probabilistic assumption. The learning problem consists of defining a hypothesis class $\mathcal{H}$, for h being an arbitrary hypothesis, either with specific architecture or not, and a learning algorithm $\mathcal{A}$, of a* **learner** *such that the following is acquired:*

- *Set $\epsilon, \delta > 0$ be the error bound and the confidence measure (that is - the probability of failure) of h, for ℓ the evaluative measure of $\{h(x), c(x)\}$, define $\hat{R}(h)$ being the empirical risk of h on the observation set $\mathcal{S}$ available. Then, the empirical learning problem is tasked to find the hypothesis such that $\hat{R}(h) = \min_{h_0 \in \mathcal{H}} \hat{R}(h_0) \leq \epsilon$ for some arbitrary h_0, with confidence $1 - \delta$. This h is called the empirical best.*
- *Similarly, define $R(h)$ the generalization error, assumed outside $\mathcal{S}$. Then, the generalization learning problem is tasked to find the hypothesis such that $\hat{R}(h) = \min_{h_0 \in \mathcal{H}} R(h_0) \leq \epsilon$ for some arbitrary h_0, with confidence $1 - \delta$. This h is called the Bayes model.*
- *The generalization setting requires choosing the optimal point for both h^* of the empirical best, and h_B for Bayes hypothesis.*

To choose and to compare between the empirical best, and the Bayes hypothesis (generalization best), we need certain notions to compare the two, algorithm to choose and evaluate, and metric to determine how comparison should be performed, with both measurable. At best, we need to find the correlation between the subconcept and the actual target concept itself.

2.4 Optimal learning - PAC-learning

The PAC-learning process (Shalev-Shwartz and Ben-David [2014], Mohri et al. [2012], Hajek and Raginsky [2021]), or rather, *Probably Approximately Correct* learning, connects three complexity measures - the sample space complexity, representation complexity, and optimization complexity, to the evaluation of the statistical learning process. It refers to the categorization of various learning procedure that satisfies the Probably Approximately Correct criteria in its name, as such being followed. The theory was proposed first by Valiant [1984], of whom then won the Turing Award, but his contribution stayed in the shadow in much of the end of the century, until the formal foundation of machine learning theory, of which then it becomes the stay ground on the standard discussion on the story of machine learner and algorithmic guarantees. As such, it is the de-facto foundation of statistical-based learning. The materials here based heavily on Mohri et al. [2012] and Kearns and Vazirani [1994], aside from interpretations.

For now, we assume no structure of h and c. They can be functions, partial functions, relations, complex algorithms, or others. In a typically learning setting, we also have the argument of *preliminary knowledge*, presented in literatures of the term *inductive bias*. For example, if the concept class $\mathcal{C}$ is assumed, then we say the setting is **model-specific**. If there exists no hard assumption on $\mathcal{C}$, then the learning setting is said to be **model-free**. For clarification, we also use in the examples the 0-1 loss function, $\ell = \delta(h, c)$ for the Kronecker delta, such that

$$\hat{R}(h) = \frac{1}{n}\sum_{i=1}^{n} \delta_{h(\mathbf{x}_i)\neq y_i}, \text{ where } \delta_{h(\mathbf{x}_i)\neq y_i} = \begin{cases} 1, & \text{if } h(\mathbf{x}_i) \neq y_i \\ 0, & \text{o.w.} \end{cases} \tag{2.8}$$

We can later on replace δ and the binary setting with others, albeit the complexity will increase on the formal derivation for others loss functions. One of the main reason why we often choose $0-1$ loss error in our analysis, comes from a variety of reasons. But, from what can be seen, it reduces the complexity of the problem, as well as raises the absoluteness of the loss measure. Indeed, naturally, from a soft point of view, the mean square loss measure is more natural of $\mathcal{L}_2$ norm on a representation space. As we have been saying, the loss function is chosen can influence the problem setting that one might expect the model to act on. $1-0$ measure effectively reduces the problem to the absolute *binary classification problem*, where it is either correct or not.

The *learning algorithm* $\mathcal{A}$ can be interpreted to belong to a specific set of algorithm A such that it gives a sequence $\{A_i\}$. If there exists such that $|\{A_i\}| = n$ for a fixed n, and can be explicitly defined and expressed, then the algorithm is *deterministic*. Otherwise for arbitrary $m > 1$ varying in magnitude, and with added uncertainty - that is, no sequence $\{A_i\}$ is the same, the algorithm is said to be *probabilistic*, and by extension, a *stochastic process*. We will give the definition of the stochastic process later on. Then, the algorithm is denoted by

$$\mathcal{A} = [h]\{A_1, \dots, A_n\} = \{A_i\}_{i=1}^{n}$$

Under this line, we separate the classification of c into two types: model-specific means that c is deterministic - it is supposed to belong to certain class $\mathcal{C}$, or rather, its description can be contained into certain concept class (which ultimately is some priori). Model-free means that c is not apparent to be able to separated to a concept class, which adds uncertainty, though all of them are still supposed to be at least a transformation $c : V \to U$ of arbitrary space V, U, by the setting of mathematical functional relationship.

Definition 2.4.1 (Model-specific learning). *A learning scenario is called model-specific learning if, for $R(h)$ being a measure-theoretic metric on the hypothesis h, we are given the dataset $D = \{(\mathcal{X}, \mathcal{Y})\}$ with distribution $x \sim \mathcal{D}$, problem is to find a hypothesis $h \in \mathcal{H}$ that satisfies:*

$$R(h) = \mathop{\mathbb{P}}_{x\sim\mathcal{D}}[h(x) \neq c(x)] = \mathop{\mathbb{E}}_{x\sim\mathcal{D}}[1_{h(x)\neq c(x)}] \tag{2.9}$$

In the same way, we define the model-free learning, where the distribution is extended to $\mathcal{Y}$. This is to facilitate the uncertainty of the concept class. If the concept class $\mathcal{C}$ is deterministic, then there would be no ambiguity of the concept $c \in \mathcal{C}$ - we would only have to learn $\mathbb{P}(y \mid x)$, to predict or reconstruct accordingly. In such case, the only factor in consideration is the ground space $\mathcal{X}$, or rather, the relationship between x and y itself. model-free learning

Definition 2.4.2 (Model-free learning). *A learning scenario is called model-free learning if, for $R(h)$ being a measure-theoretic metric on the hypothesis h, we are given the dataset $D = \{(\mathcal{X}, \mathcal{Y})\}$ with distribution $(x, y) \sim \mathcal{D}$. Problem is to find a hypothesis $h \in \mathcal{H}$ that satisfies*

$$R(h) = \mathop{\mathbb{P}}_{(x,y)\sim D}[h(x) \neq y] = \mathop{\mathbb{E}}_{(x,y)\sim D}[1_{h(x)\neq y}] \tag{2.10}$$

In both setting, we take the stance of absoluteness, using measure $\neq$, similar to a type of discrete decision process called *classification*, for a finite class counts of the target class. This notion can be aggravated to be generalized to other classes, simply by replacing $h(x) \neq y$ by a valued-function, or by considering specific measure (for example, *Bregman measure* for strictly convex functions). In the context of n-dimensional Euclidean space, and with the isomorphism $\mathbf{E} \simeq \mathbb{R}^n$ of suitable basis, the measure is simply the finite-accuracy Euclidean distance $d(h(x), c(x))$.

Before analysing this problem in a PAC-like fashion, let us recall the setting in which we are operating upon.

- The goal of the learning game is to learn an unknown target set, but the target set is not arbitrary. Instead, there is a known and rather strong constraint on the target set — it is a rectangle in the plane whose sides are parallel to the axes. Under certain analysis, we can call this as having indeed a strong *priori* to the concept $c \in \mathcal{C}$ possible.
- Learning occurs in a probabilistic setting. Examples of the target rectangle are drawn randomly in the plane according to a fixed probability distribution which is unknown and unconstrained.
- The hypothesis of the learner is evaluated relative to the *same* probabilistic setting in which the training takes place, and we allow hypotheses that are only approximations to the target. The tightest-fit strategy might not find the target rectangle exactly, but will find one with only a small probability of disagreement with the target.
- We are interested in a solution that is efficient: not many examples are required to obtain small error with high confidence, and we can process those examples rapidly - that is, for time efficiency t. For it to be controllable, it is in our interest to know if we can run the learning process in polynomial time, that is, with P-complexity class.

Ideally, we would want our model to be able to do the following:

- The number of calls to the concept c, or by extension, the oracle process $EX(c, \mathcal{D})$, is small, in the sense that it is bounded polynomially by some parameters.
- The amount of computation performed is small.
- The algorithm outputs a hypothesis h such that Θ is small. Recall that $\Theta_h = (\hat{R}(h), R(h))$, but generally, the ideal goal would be to reduce to the infimum of $R(h)$.

We are now ready to specify the definition of Probably Approximately Correct learning. For now, we designate this as a preliminary definition, because we will add more features into it.

Definition 2.4.3 (PAC Learning, Preliminary Definition). *Let $\mathcal{C}$ be a concept class over X. We say that $\mathcal{C}$ is **PAC-learnable** if there exists an algorithm $\mathcal{L}$ that for every concept $c \in \mathcal{C}$, for every distribution $\mathcal{D}$ on X, and for all $0 < \epsilon < 1/2$ and $0 < \delta < 1/2$, if $\mathcal{L}$ is given access to $EX(c, \mathcal{D})$ for its output, and inputs ϵ, δ, then with probability at least $1 - \delta$, outputs a hypothesis $h \in \mathcal{H}$ satisfying $R(h) \leq \epsilon$. This probability is taken over the random examples drawn by calls to $EX(c, \mathcal{D})$, and any internal randomization of $\mathcal{L}$.*

*If $\mathcal{L}$ further run in time polynomial in $1/\epsilon$ and $1/\delta$, we say that $\mathcal{C}$ is efficiently PAC learnable. We will sometimes refer to the input ϵ as the **error parameter**, and the input δ as the **confidence parameter**.*

The hypothesis $h \in \mathcal{H}$ of the PAC-learning algorithm is thus approximately correct, with high probability, hence we got the name. Usually, $\mathcal{H}$ can not coincide with $\mathcal{C}$. However, a stronger form of the preliminary PAC-learning can set $h \in \mathcal{C}$, which guarantee the existence of a near-perfect hypothesis class.

For the preliminary PAC learning model, there are two important problems with it, or rather, remarks on how it is constructed.

First, the parameter pair ϵ, δ control two types of failure to which a learning algorithm in the PAC model is inevitably susceptible. The error parameter ϵ is necessary since there may be only a negligible probability that a small random sample will distinguish between two competing hypotheses that differ on only one improbable point in the instance space. The confidence on only one improbable point in the instance (ground) space. The confidence parameter δ is necessary since the learning algorithm may occasionally be extremely unlucky, and draw a terribly "unrepresentative" sample of the target concept - for example, a sample consisting only of repeated draws of the same instances. The best we can hope for is that the probability of both types of failure can be made arbitrarily small at a modest cost.

Second, we notice that we demand a PAC learning algorithm perform well with respect any distribution $\mathcal{D}$. This is a very strong requirement, which is moderated by the fact that we only evaluate the hypothesis of the learning algorithm with respect to the same distribution $\mathcal{D}$.

2.4.1 Representation size and PAC-configuration

An important issue that has not been mentioned in our previous formation of PAC learning, is the fundamental distinction between a concept and its representation.

Consider a class of concepts defined by the satisfying assignments of certain formulae. A concept from this class that satisfies such formulae, can be represented by a formula f, a truth table, or any given formulae that is tautologically equivalent formulae f' to f. An example can be shown in high-dimensional Euclidean space $\mathbb{R}^n$, we may choose to represent a convex polytope (the hell is this?), we can specify the problem by either specifying its vertices, or specifying linear equations for its faces, and these two representation scheme, while being of the same problem, can differ in size.

In each of these examples, we are fixing some representation scheme - that is, a precise method for encoding concepts - and then examining the size of the encoding for various concepts. Other natural representation schemes that we are familiar it could be the vanilla neural network, or decision trees, though we cannot say exactly that the representation of neural network is indeed of the 'vanilla' representation. Taken example of the boolean formulae, for example, the boolean parity function $f(x_1, \dots, x_n) = x_1 \oplus \cdots \oplus x_n$ which can be computed by a circuit of $\wedge, \vee$ and $\neg$ gates whose size is bounded by a fixed polynomial in n, but to represent this same function as a disjunctive normal form (DNF), requires size exponential in n. In these representation schemes, there is an obvious mapping from the representation (can be decision tree, or neural network) to the set or boolean function that is being represented. There is also a natural measure of the size of given representation in the scheme (for example, the number of weights, or neuron in a neural network).

representation scheme

Since our PAC-learning algorithm for its model, follows the phenomenological interpretation, hence it is experimental data-only, it has absolutely no information about which, if any, of teh many possible representations is actually being used to represent the target concept in reality. However, it matters greatly which representation the algorithm chooses for its hypothesis, since the time to write this representation down is obviously a lower bound on the running time of the algorithm. At the end though, note that this is purely a computational concern at a glance, yet, further analysis into the problem might stem out certain consideration that previously unseen, for example, that fact that the representation scheme actually matters more in sense of a system analysis and its dynamic.

Formally, a *representation scheme* for a concept class $\mathcal{C}$ is defined as followed to capture this notion of representation size, and can be defined as below.

Definition 2.4.4 (Representation scheme). *For a given concept class $\mathcal{C}$, a **representation scheme** for such concept class is a function $\boldsymbol{\mathcal{R}} : \Sigma^* \to \mathcal{C}$, where Σ is the finite alphabet of symbols. We call any configuration $\sigma \in \Sigma^*$ such that $\boldsymbol{\mathcal{R}}(\sigma) = c$ a **representation** of c, under $\boldsymbol{\mathcal{R}}$. For any given c, the representation scheme might not be unique.* [1]

To capture the notion of representation size, we assume that associated with $\boldsymbol{\mathcal{R}}$, there is a mapping size: $\Sigma^* \to N$ that assigns a natural number $size(h)$ to each representation $h \in \Sigma^*$. Note that we allow $size(\cdot)$ to be any such mapping results obtained under a particular definition for $size(\cdot)$ will be meaningful only if this definition is natural. In a more simple case, and arguably realistic setting, we can take $\Sigma = \{0, 1\}$, thus, we have a binary encoding of concepts, and define $size(h)$ to be the length of h in bits. Although we will use other definitions of size when binary representations are inconvenient, our definition of $size(\cdot)$ will always be within a polynomial factor of the binary string length definition.

So far, this definition is applicable only to representations, that is, to strings $h \in \Sigma^*$. We would like to extend this to $c \in \mathcal{C}$. Since the learning algorithm has access only to the input-output behaviour of c, in the worst case, it must assume that the simplest possible mechanism is generating this behaviour. Thus, we define $size(c)$ to be the infimum, that is:

representation size

$$size(c) = \min_{\boldsymbol{\mathcal{R}}(\boldsymbol{\sigma})=c} \{size(\sigma)\}$$

In other words, $size(c)$ is the size of the smallest representation of the concept c in the underlying representation scheme $\boldsymbol{\mathcal{R}}$.

Definition 2.4.5 (Representation complexity). *Given a concept $c \in \mathcal{C}$, the **representation complexity** of c is defined to be the size of the smallest representation of the concept c in the underlying representation scheme $\boldsymbol{\mathcal{R}}$, that is,*

$$size(c) = \min_{\boldsymbol{\mathcal{R}}(\boldsymbol{\sigma})=c} \{size(\sigma)\}$$

for any string σ of the representation scheme $\boldsymbol{\mathcal{R}}$ that still represents c.

[1] Perhaps representation scheme can be further realized by applying a Borel σ-algebra on top, and put on it a perspective measure to handle the notion of representation size. For example, in the traditional example of axis-aligned rectangle, one such representation scheme could be the real number schema $\boldsymbol{R} : (\Sigma \cup \mathbb{R})^* \to \mathcal{C}$, which then can utilize the extended notion of a Lebesgue measure over $\mathbb{R}$ and its spaces.

Under such definition, the more "complex" the concept c is with the respective chosen representation scheme, the larger $size(c)$ is. This is also where the concept of the concept class $\mathcal{C}$ comes from - it comes from the *representation class indiction* that we have in mind some fixed concept classes we study by their representation scheme.

From this consideration of the representation size for the language in which the problem setting is expressed in, we gain the complete definition of PAC-learning for a fixed representation priori of the concept class.

Definition 2.4.6 (PAC-learning). *A concept class $\mathcal{C}$ is said to be* ***PAC-learnable*** *if there exists an algorithm $\mathcal{A}$ and a polynomial function $poly(\cdot,\cdot,\cdot,\cdot)$ of 4-argument such that for any $\epsilon > 0$, $\delta > 0$, for all distribution $\mathcal{D}$ on $\mathcal{X}$ and for any target concept $c \in \mathcal{C}$, the following holds for any sample size $m \geq poly(1/\epsilon, 1/\delta, n, size(c))$:*

$$\Pr_{S\sim\mathcal{D}^m}[R(h(S)) \leq \epsilon] \geq 1-\delta$$

for a given error measure $R(h_S)$. If $\mathcal{A}$ further runs in $poly(1/\epsilon, 1/\delta, n, size(c))$, then $\mathcal{C}$ is said to be efficiently PAC-learnable. When such algorithm exists, we call $\mathcal{A}$ a PAC-learning algorithm.

2.4.2 Generalization bounds

The theory based its goal around the target of **generalization learning problem**, hence, most of the bounds can be focus on achieving generalization target. Superficially, we assume the following is true of specific h when trying to resolve the generalization problem.

Assumption 2.4.1 (Structural assumptions). *We assume that $h \in \mathcal{H}$ is expressive enough on the hypothesis class, and algorithm $\mathcal{A}(h)$ is capable enough to extrapolate h^* of the empirical best on the empirical space $\mathcal{S}$.*

Under such assumption, to introduce the explicit objective of generalization bounds, we first need to introduce the categorization of the hypothesis based of one criterion: consistency. We say that an algorithm is a consistent learner for a concept class C using hypothesis class $\mathcal{H}$, if for all n, for all $c \in C_n$ and for all m, given $\mathcal{S}$, as input, $x_i \in X_n$ outputs, then the algorithm L outputs a hypothesis $h \in \mathcal{H}_n$ such that $h(x_i) = c(x_i), i = 1, \ldots, m$. Then h is called the **consistent learner**. In other word, a consistent hypothesis is the hypothesis that is consistent with all the training data provided to it from the concept c. The following section contains the debate between consistent hypothesis (i.e., for example, admitting no error on training data, almost perfect - maybe even so), and inconsistent hypothesis, of which have general defects from the actual concept, or even different by a margin.

Famous results contain two types of bound for classical PAC: the finite $\mathcal{H}$, and consistent hypothesis, and finite $\mathcal{H}$, but inconsistent. Infinite $\mathcal{H}$ assumes different stochasticity, and in consistent bound, generalization depends on the variances between empirical and generalization space, which is not covered in the theory, though in general setting the metric-induced distance can be small and controllable, which makes PAC applicable on said domain. For consistent case, we will consider the general sample complexity bound, or equivalently, a generalization bound for consistent hypothesis in the case where the cardinality $|\mathcal{H}|$ is finite. We will assume, as such, that the target concept c is in $\mathcal{H}$.

Theorem 2.4.2 (Learning bound - finite $\mathcal{H}$, consistent case). *Let H be a finite set of functions mapping from $\mathcal{X} \to \mathcal{Y}$. Let $\mathcal{A}$ be an algorithm that for any target concept $c \in H$ and i.i.d. samples S returns a consistent hypothesis H_S, such that $\hat{R}(h_S) = 0$. Then for any $\epsilon, \delta > 0$, the inequality $\Pr_{S\sim\mathcal{D}^m}[R(h_S) \leq \epsilon] \geq 1-\delta$ holds if*

$$m \geq \frac{1}{\epsilon}\left(\log|H| + \log\frac{1}{\delta}\right)$$

This sample complexity result admits the following equivalent statement as a generation bound: for any $\epsilon, \delta > 0$, with probability at least $1-\delta$,

$$R(h_S) \leq \frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right) \tag{2.11}$$

Theorem 2.4.3 (Learning bound - finite $\mathcal{H}$, inconsistent case). *Let $\mathcal{H}$ be a finite hypothesis set. Then, for any $\delta > 0$, with probability at least $1-\delta$, the following inequality holds:*

$$\forall h \in \mathcal{H}, \quad R(h) \leq \hat{R}_S(h) + \sqrt{\frac{\log|\mathcal{H}| + \log 2/\delta}{2m}} \tag{2.12}$$

The proofs are left in the Appendix section. In general, theorem 2.4.2 shows that when the hypothesis set $\mathcal{H}$ is finite, a consistent algorithm $\mathcal{A}$ is a PAC-learning algorithm, since the sample complexity is dominated by a polynomial in $1/\epsilon$ and $1/\delta$. The generalization error of consistent hypothesis is upper bounded by a term that decrease w.r.t m. The decreases rate of $O(1/m)$ is guaranteed by this theorem. For the inconsistent theorem 2.4.3, it indicates that for a finite hypothesis set $\mathcal{H}$, then

$$R(h) \leq \hat{R}_S(h) + O\left(\sqrt{\frac{\log_2 |\mathcal{H}|}{m}}\right) \tag{2.13}$$

The term $\log |\mathcal{H}|$ can be interpreted as the number of bit needed to represent $\mathcal{H}$. A larger sample size m guarantees better generalization, and the bound increases with $|\mathcal{H}|$, but only logarithmically. Note that the bound suggests seeking a trade-off between reducing the empirical error versus controlling the size of the hypothesis set: a larger hypothesis set is penalized by the second term but could help reduce the empirical error, that is the first term. But for a similar empirical error, it suggests using a smaller hypothesis set. This can be viewed as an instance of the so-called *Occam's Razor*, of which can be said as: *Plurality should not be posited without necessity*, or, the simplest explanation is best.

2.4.3 Agnostic-general PAC-learning

For the definition of PAC-learning in the above setting, for model-specific notion, there are some issues with it, more specifically, with the type of *strong assumptions* that it holds.

1. The *representation size* and boundary only works for priori given of the concept class $\mathcal{C}$. In case that we cannot determine this, and there are a large potential for that to fail in capturing the minimal representation scope of the concept, the bound simply fails to a certain extent - given that our definition includes the polynomial bound of the concept class itself.
2. The assumption that the target concept c belongs to $\mathcal{C}$ means that we are trying to fit a hypothesis to data, which are a priori known to have been generated by some member of the model class defined by $\mathcal{C}$. However, in general, we may not want to assume much about the data generation process, and instead would like to find the best fit to the data at hand using an element of some model class of our choice.
3. The assumption that the training features are labelled noiselessly rules out the possibility of noisy measurements or observations.
4. Even if the above assumptions were somehow true, we would not necessarily have a priori knowledge of the concept class $\mathcal{C}$, containing the priori knowledge of the concept class $\mathcal{C}$ containing the target concept or function. In that case, the best we could hope for is to pick our own model class and seek the best approximation to the unknown target concept among the elements of that class.

Agnostic case is considered the most general scenario of supervised learning, where the distribution $\mathcal{D}$ is defined over $\mathcal{X} \times \mathcal{Y}$, and the training data is a labelled sample S drawn i.i.d. according to $\mathcal{D}$:

$$S = \{(x_1, y_1), \ldots, (x_m, y_m)\}$$

The learning problem is to find a hypothesis $h \in \mathcal{H}$ with small generalization error

$$R(h) = \mathop{\mathbb{P}}_{(x,y)\sim\mathcal{D}}[h(x) \neq y] = \mathop{\mathbb{E}}_{(x,y)\sim\mathcal{D}}[1_{h(x)\neq y}] \tag{2.14}$$

This more general scenario is referred to as the stochastic scenario. Within this setting, the output target is a probabilistic function of the input. The stochastic scenario captures many real-world problems where the label of an input point is not unique[2]. That said, the case of model-free is can be constructed in a simple manner, where we forgo any

[2]Considering two set X and Y. A probabilistic function from X to Y assign to each $x \in X$ a relevant sub-distribution of elements of Y, rather than a single value $y \in Y$. Such a sub-distribution is a function $\delta : Y \to [0, 1]$ such that

$$\sum_{y\in Y} \delta(y) \leq 1$$

In other words, a probabilistic function $X \to Y$ is a function $f : X \to (Y \to [0, 1])$, interpreted as $f : X \times Y \to [0, 1]$ such that for all $x \in X$,

$$\sum_{y\in Y} f(x, y) \leq 1$$

assumptions on the behaviours or structures of $\mathcal{C}$ and its associated c-model, and thus the observable pairs is simply $(x, y) \sim \mathcal{D}_{mf}$. Here, we denote $\mathcal{D}_{mf}$ to separate it from $\mathcal{D}$ of which is a singular x-distribution; however, we can also write $\mathcal{D}_{ms}$ and $\mathcal{D}_{mf}$ in certain specific cases. Per Hajek and Raginsky [2021], the treatment of model-agnostic (no model c assumption), targets the ingredient sets of,

1. Sets $\mathcal{X}, \mathcal{Y}$ and $\mathcal{U}$. Here, $\mathcal{U}$ is the correlated h-functional result, i.e. the hypothesis model, covering the case of **probabilistic models.**
2. A class $\mathcal{P}$ of probability distribution on $Z := \mathcal{X} \times \mathcal{Y}$. Here, the distribution covers both.
3. A class $\mathcal{H}$ of functions $h : \mathcal{X} \to \mathcal{U}$.
4. A loss function as objective, $\ell : \mathcal{Y} \times \mathcal{U} \to [a, b]$, preferably $[0, 1]$. Additional overhead measures are also required.

The *agnostic PAC-learning* format is then modified by considering $\mathcal{Y}$ as also a random variability, and thus can be stated as followed.

Definition 2.4.7 (Agnostic PAC-learning). *A concept class $\mathcal{C}$ is said to be **PAC-learnable** if there exists an algorithm $\mathcal{A}$ and a polynomial function $poly(\cdot, \cdot, \cdot, \cdot)$ of 4-argument such that for any $\epsilon > 0$, $\delta > 0$, for all distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$ and for any target concept $c \in \mathcal{C}$, the following holds for any sample size $m \geq poly(1/\epsilon, 1/\delta, n)$:*

$$\Pr_{S \sim \mathcal{D}^m} \left[R(h(S)) - \min_{h \in \mathcal{H}} R(h) \leq \epsilon \right] \geq 1 - \delta \tag{2.15}$$

If $\mathcal{A}$ further runs in $poly(1/\epsilon, 1/\delta, n)$, then $\mathcal{C}$ is said to be efficiently agnostic PAC-learnable. When such algorithm exists, we call $\mathcal{A}$ an agnostic PAC-learning algorithm.

Under such lens, PAC-on-agnostic can be realized also using a more enumerative bound-on-definition instead:

Definition 2.4.8 (agnostic-PAC learning, Hajek and Raginsky [2021]). *We say that a learning algorithm for a problem $(\mathcal{X}, \mathcal{Y}, \mathcal{U})$ and interpretations $(\mathcal{P}, \mathcal{H}, \ell)$ is PAC-learnable to accuracy $\epsilon > 0$, if, we have*

$$\lim_{n \to \infty} \sup_{P \in \mathcal{P}} P^n \left(Z^n : \hat{R}_P(h_n) > \inf_{h \in \mathcal{H}} R_P(h) + \epsilon \right) = 0 \tag{2.16}$$

*An algorithm that is PAC-learnable to accuracy ϵ for every $\epsilon > 0$ is said to be a PAC-learning algorithm. A learning problem is said to be model-free learnable if there exists such algorithm, for every $A \in \mathcal{A}$ of such class to be **PAC**-learning algorithm class.*

PAC-learning bound argues in the sense of *computational complexities*, and *space complexities*. Specifically, the term $poly(1/\epsilon, 1/\delta, n, size(c))$ hopes to bound the required learning process to polynomial time, specified by 4 parameters. Here, n is the *input dimension*, $size(c)$ is the encoding complexity of the target concept, $\epsilon > 0$ is the *accuracy bound*, and δ is the confidence bound - the probability that h fails.

2.4.4 Empirical and structural algorithms

Suppose that for a concept class $\mathcal{H}$, we can then partition it to $\mathcal{H} = \bigcup_{k \in \mathbb{N}} \mathcal{H}_k$ for finite k hypotheses. Assume that they satisfy the uniform convergence (Shalev-Shwartz et al. [2010]), which means in the arbitrary path landscape, one can guarantee ℓ-reduction, of m samples on $\mathcal{S}$ to:

$$\lim_{m \to \infty} \sup_{\mathcal{D}} \mathop{\mathbb{E}}_{S \sim \mathcal{D}^m} \left[\sup_{h \in \mathcal{H}} [R(h) - R_S(h)] \right] = 0 \tag{2.17}$$

or, can also be expressed of equal to the following expression, for functions ϵ_k satisfies that for all k and $\delta \in (0, 1)$, $\lim_{m \to \infty} \epsilon_k = 0$.

$$\forall k \in \mathbb{N}, \quad \mathop{\mathbb{P}}_{S \sim \mathcal{D}^m} \left(\sup_{h \in \mathcal{H}_k} R(h) - \hat{R}_S(h) \leq \epsilon_k(m, \delta) \right) \geq 1 - \delta \tag{2.18}$$

Then such is said to be possible as a ruling dynamic when for a learning problem, the empirical risks of hypotheses in the hypothesis class converges to their population risk uniformly, with a distribution-independent rate. The result of this is that a problem then can be considered learnable with the ERM rule, for any given characterization. We state the following definition.[3]

Definition 2.4.9. *A learning problem is learnable if there exist a learning rule $\mathcal{A}$ and a monotonically decreasing sequence $\epsilon_{cons}(m)$ such that $\epsilon_{cons}(m) \overset{m\to\infty}{\longrightarrow} 0$ and*

$$\forall \mathcal{D}, \quad \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[F(\mathcal{A}(S)) - F^*\right] \leq \epsilon_{cons}(m), \tag{2.19}$$

with $F^* = \inf_{h\in\mathcal{H}} R(h)$. A learning rule $\mathcal{A}$ for which this hold is denoted as a universally consistent learning rule.

There are certainly a fundamental class of rule or algorithm $\mathcal{A}$ to resolve this particular uniform convergence of classes. One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, one strategy is the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to 'generalize' this heuristically is apprehended on the hypothesis.

Definition 2.4.10. *A rule $\mathcal{A}$ is an ERM (Empirical Risk Minimizer), denoted $\mathcal{A}_{ERM}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m)\sum_{c\in C}\ell(h,c)$, over the observational space S,*

$$\hat{R}_S(\mathcal{A}_{ERM}(S)) = \hat{R}_S(h_S) = \inf_{h\in\mathcal{H}} \hat{R}_S(h) \tag{2.20}$$

Hence, the ERM tries to obtain $\mathrm{ERM}(h') = \arg\min_{h\in\mathcal{H}} \hat{R}(h)$.

The problem of overfitting still somewhat resides in the setting of the method. Hence, usually, we have an implicit term in addition, called the *regularizer*, of which is as followed.

Definition 2.4.11. *A rule $\mathcal{A}$ is an SRM (Structural Risk Minimizer), denoted $\mathcal{A}_{ERM}$ if it minimizes the empirical risk $\hat{R}(h) = (1/m)\sum_{c\in C}\ell(h,c)$, over the observational space S,*

$$\hat{R}_S(\mathcal{A}_{SRM}(S)) = \hat{R}_S(h_S) + \lambda r(h) = \inf_{h\in\mathcal{H}} \hat{R}_S(h) + \lambda r(h) \tag{2.21}$$

of the regularizing term $r(\cdot)$. Hence, the SRM tries to obtain $\mathrm{ERM}(h') = \arg\min_{h\in\mathcal{H}} \hat{R}(h)$, while taking into consideration a forcing term r, usually defined on the model complexity.

Hence, the SRM is inherently stronger than ERM. The regularizer $\lambda r(h)$, in specific case of convex loss landscape and reasonably well-behaved irregularities between domains, and of the structural regularization term $r_s(h) = \sum_{i,j} w_{ij}^2$ for weights of ij index, then we have the following theorem.

Conjecture 2.4.1 (ERM-SRM growth). *Consider the regulizer $\lambda r(h)$ for $h \in \mathcal{H}$. Assuming structural regularization $r_{\mathcal{S}}(h) = \sum_{i,j} w_{ij}^2$ for linear-layering of ij index, for ERM and SRM optimizer classes,*

$$\begin{aligned} &\mathrm{ERM}\left(\inf_{h\in\mathcal{H}} \hat{R}_S(h), \ell\right) \\ &\qquad \approx o\left(\mathrm{SRM}\left(\inf_{h\in\mathcal{H}} \hat{R}_S(h) + \lambda r(h), [\ell, \lambda r(h)]\right)\right) \end{aligned} \tag{2.22}$$

[3] The reason why $r(\cdot)$ is not characterized in this preliminary section, is because of their uncertainty of effect. Hypothetically, $r(\cdot)$ is an additional structure (bias) of which apply a new gradient field in conjunction to the loss gradient field itself. This addition is uncertain of its content and analysis, since it depends on various stochastic dynamics of the model. We reserve from using regularization for now because of such reason.

Where the little-o notation is used for saying of the optimization strength per arbitrary time-similar notion of algorithmic evolution. The reason for this change is because of the restriction domain that SRM indicted on the model optimization landscape. Effectively, we introduce the structural bias $\lambda r(h)$ that forces the morph of the total loss landscape. In well-behaved case, this is usually effective to a point, however in realistic system and pathological cases, such can lead to the c' approximation in c-space, just as we introduced in the reformulation of learning system. In this paper, we would like to presume the ERM method in question as the famous *gradient descent*. We refer to Achlioptas, Ruder [2017] for general view of the algorithm, and Zhang [2019] for a source on particularly gradient descent algorithm on deep learning models.

One particularly troublesome and should take serious consideration into interpreting it, is the fact of the insufficiency on ERM. The following proposition is rather questionable.

Proposition 2.4.4. *For any sample S, the following inequality holds for the hypothesis returned by ERM:*

$$\mathbb{P}\Big[R(h'_S) - \inf_{h\in\mathcal{H}} R(h) > \epsilon\Big] \leq \mathbb{P}\Big[\sup_{h\in\mathcal{H}} |R(h) - \hat{R}_S(h)| > \frac{\epsilon}{2}\Big] \tag{2.23}$$

We will prove this proposition in later section. For now, this proposition bounds the error of ERM, with respect to the probability of the generalization distance $|R(h) - \hat{R}_S(h)|$ between the two errors. As we might have suspected, the performance of ERM is typically very poor. This is because the algorithm disregards the complexity of the hypothesis set $\mathcal{H}$: in practice, either it is not complex enough, in which case the error can be large, or $\mathcal{H}$ is so rich, that the complexity makes the model shoot out of the complexity range. Additionally, in many cases, determining the ERM solution is computationally intractable.

2.4.5 The PAC-ERM notion

Without restricting ourselves to specific ERM construction, we can still apply PAC on the set of ERM algorithms in its generality of itself. Within such understanding, the empirical risk minimization strategy is often motivated by the hope that *$R(h)$ and $\hat{R}(h)$ are not too far-distant apart*. Particularly, it also implicitly agree with statements conjectured with law of large number, for $n \to \infty$, $|R(h) - \hat{R}(h)| \approx \epsilon$ of irreducible noise. To what extent is this true, we have to take certain boundary on which might have been previously shown. We refer to authoritative introductory texts like Alquier [2024], for both PAC-ERM and later sections on PAC-Bayes variation.

Proposition 2.4.5 (Alquier [2024]). *For any $h \in \mathcal{H}$, for any $\delta \in (0, 1)$,*

$$\mathbb{P}_S\left(R(h) > \hat{R}(h) + C\sqrt{\frac{\log(1/\delta)}{2n}}\right) \leq \delta \tag{2.24}$$

Proof. Applying Hoeffding's inequality onto $U = \mathbb{E}[\ell_i(h)] - \ell_i(h)$,

$$\mathbb{E}_S\left[\exp tn[R(h) - \hat{R}(h)]\right] \leq \exp\left[(nt^2C^2)/8\right] \tag{2.25}$$

For any $s > 0$, we have by Markov's inequality,

$$\begin{aligned}\mathbb{P}_S(R(h) - \hat{R}(h) > s) &= \mathbb{P}_S\left(\exp nt[R(h) - R] > \exp(nts)\right)\\ &\leq \frac{1}{\exp(nts)}\mathbb{E}_S\left[e^{nt[R(h)-\hat{R}(h)]}\right]\\ &\leq \exp\left(\frac{nt^2C^2}{8} - nts\right)\end{aligned} \tag{2.26}$$

We can make this bound as tight as possible, by optimizing the choice of t. For $t = 4s/C^2$, we have

$$\mathbb{P}(R(h) - \hat{R}(h) > s) \leq \exp\left(\frac{-2ns^2}{C^2}\right) \tag{2.27}$$

This means that, for a given h, the empirical risk $\hat{R}(h)$ cannot be much smaller than the risk $R(h)$. The order of this "much smaller" can be better understood by introducing $\delta = \exp\left(-2ns^2/C^2\right)$, of which substituting δ to s yields the required proposition. □

Proposition 2.4.5 bounds the asymptotic term such that $R(h)$ will exceed $\hat{R}(h)$ by certain margin, at worst $1/\sqrt{n}$. However, this is not enough to justify the use of ERM (which indeed use the near-space approximation from empirical to general risk), since the proof only applies for h that was fixed above in terms of complexity. For those that is implicit of the data itself, we cannot apply them[4]. The usual approach though, is to use the inequality

$$R(h_{ERM}) - \hat{R}(h_{ERM}) \leq \sup_{h\in\mathcal{H}} \left[R(h) - \hat{R}(h)\right] \tag{2.28}$$

and to prove a version of Proposition 2.4.5 that would hold uniformly on $\mathcal{H}$.

Theorem 2.4.6. *Assume that* $\mathrm{card}(\mathcal{H}) = M < +\infty$*, for any* $\delta \in (0,1)$*,*

$$\mathbb{P}_S\left(R(h_{ERM}) \leq \inf_{h\in\mathcal{H}} \hat{R}(h) + C\sqrt{\frac{\log\left(M/\delta\right)}{2n}}\right) \geq 1-\delta \tag{2.29}$$

Proof. We upper bound the supremum in the inequality by

$$\begin{aligned}\mathbb{P}_S\left(\sup_{h\in\mathcal{H}}[R(h) - \hat{R}(h) > s]\right) &= \mathbb{P}_S\left(\bigcup_{h\in\mathcal{H}}\left\{[R(h) - \hat{R}(h)] > s\right\}\right)\\ &\leq \sum_{h\in\mathcal{H}} \mathbb{P}_S(R(h) - \hat{R}(h) > s)\\ &\leq M\exp\left(\frac{-2ns^2}{C^2}\right)\end{aligned} \tag{2.30}$$

This time, put $\delta = M\exp\left(-2ns^2/C^2\right)$ we gain

$$\mathbb{P}_S\left(\sup_{h\in\mathcal{H}}[R(h) - \hat{R}(h)] \leq C\sqrt{\frac{\log\left(M/\delta\right)}{2n}}\right) \leq \delta \tag{2.31}$$

Thus it satisfies

$$\mathbb{P}_S\left(\sup_{h\in\mathcal{H}}[R(h) - \hat{R}(h)] \leq C\sqrt{\frac{\log\left(M/\delta\right)}{2n}}\right) \geq 1-\delta \tag{2.32}$$

Using the inequality

$$\mathbb{P}_S\left(R(h_{ERM}) \leq \hat{R}(h_{ERM})C\sqrt{\frac{\log\left(M/\delta\right)}{2n}}\right) \geq 1-\delta \tag{2.33}$$

As $\mathcal{H}$ is finite, $\hat{R}(h_{ERM}) = \inf_{h\in\mathcal{H}} \hat{R}(h)$. □

This bound coincides for the *Probably Approximately Correct* PAC-bounds, specifically for algorithm of ERM principle, since it approximates $R(h_{ERM})$ within $C\sqrt{\log\left(M/\delta\right)/2n}$ with probability $1-\delta$, and once again, under very strong arguments. Certain version of this theorem's result can be derived to 'predict' double descent, but not giving its the diagnostic view.

[4]This means models such ass *support vector machine* (SVM) or *Gaussian process regression* of which the empirical optimization function and the hypothesis itself, relies on the number of data points available.

2.4.6 Noise, Mohri et al. [2012]

noise

Using the notion of the Bayes classifier and Bayes error, one can define the *noise* of a given learning setting under PAC-learning consideration, as followed.

Definition 2.4.12 (Noise). *Given a distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, the noise at point $x \in \mathcal{X}$ is defined by*

$$noise(x) = \min_{x \in \mathcal{X}} \{\mathbb{P}[1 \mid x], \mathbb{P}[0 \mid x]\}$$

The average noise or the noise associated to $\mathcal{D}$ is then the minimum possible error aside from intrinsic model error, and is calculated by $\mathbb{E}[noise(x)]$

Thus, the average noise is precisely the Bayes error, that is, $noise = \mathbb{E}[noise(x)] = R^*$. The noise is a characteristic of the learning task indicative of its level of difficulty. A point $x \in \mathcal{X}$, for which noise is close to $1/2$, is sometimes referred to as *noisy* and is of course a challenge for accurate prediction. Such is understandable, and is often enough, however, we note that structurally, it is missing particular components or decomposition components that would eventually make up the builks of its claims.

2.5 Simplistic learning - Occam's Razor

The PAC model that we introduced defined learning directly in terms of the predictive power of the hypothesis output by the learning algorithm. That is, given a hypothesis $h \in \mathcal{H}$ and the learning algorithm $\mathcal{L}$, the learning involves switching this dynamic of the predictive power of the hypothesis, over certain setting and problem. We can, as it is possible to apply this measure, to a learning algorithm because we made the assumption that the instances are drawn, from a probabilistic perspective, from a fixed distribution $\mathcal{D}$ then measured the predictive power with respect to the same distribution.

Our new notion, dubbed as *Occam learning* (our source is mainly taken from the exposition of Kearns and Vazirani [1994]), would let us consider a different definition of learning that makes no assumptions about how the instances in a labelled sample are chosen, though we will assume that the labels are generated by a target concept chosen from a known class. Instead of measuring the predictive power of a hypothesis, the new definition judges the hypothesis by how succinctly it explains the observed data (or a labelled example). The crucial difference between PAC learning and Occam learning, is then that in PAC learning, the random sample drawn by the learning algorithm is intended only as an aid for reaching an accurate model of some external process, while in the new definition, we are concerned only with the fixed sample before us, and not any external process. By that, we simply reduce it to somewhat similar of the statistical estimation problem setting, though not exactly the same.

This also contains the substance of which we often call the *Occam's razor*, telling that overly complex scientific theories should be subjected to a simplifying knife. If we equate "simplicity" with representational succinctness, then another way to interpret Occam's razor would be to say learning is the act of finding a pattern in the observed data that facilitates a compact representation or compression of this data. In our simple concept learning setting, succinctness is measured by the size of the representation of the hypothesis concept. Equivalently, we can measure succinctness by the cardinality of the hypothesis class used by the algorithm, for if this class is small then a typical hypothesis from the class can be represented by a short binary string, and if this class is large then a typical hypothesis must be represented by a long string. Thus, an algorithm is an *Occam algorithm* if it finds a short hypothesis consistent with the observed data.

Additionally so, it would be plenty interesting to see that Occam's razor somehow falls into the range of what would be expected of the *bias-variance tradeoff*. This perhaps, will prompt us to push further into the inquiry of double descent, which empirically, breaks the principle of the Lord William Occam.

2.5.1 Occam Learning and Succinctness

Let $X = \cup_{n \geq 1} X_n$ be the instance space, let $\mathcal{C} = \cup_{n \geq 1} \mathcal{C}_n$ be the target concept class, and let $\mathcal{H} = \cup_{n \geq 1} \mathcal{H}_n$ be the class of hypothesis representation. The notation is preferably clarified: X_n is under the binary representation typically concerned $\{0, 1\}^n$, or a boolean string of length n. Hence, $\mathcal{H}_n$ and $\mathcal{C}_n$ are subsequently concept and hypothesis class specified for such string space. In this part, we will assume, unless explicitly stated otherwise, that the hypothesis representation scheme of $\mathcal{H}$ uses a binary alphabet, and we define $size(h)$ to be the length of the bit string h. Also, recall that for a concept $c \in \mathcal{C}$, $size(c)$ denotes the size of the smallest representation of c in $\mathcal{H}$. Let $c \in \mathcal{C}_n$ denote the target concept. A labelled sample S of cardinality m is a set of pairs: [5]

$$S = \{(x_1, c(x_1)), \dots, (x_m, c(x_m))\}$$

An **Occam algorithm** L takes as input a labelled sample S, and outputs a short hypothesis h, "relatively so", that is consistent with S. By consistent, we mean as one definition above, $h(x_i) = c(x_i)$ for each i, and for short we mean that $size(h)$ is a sufficiently slowly growing function of n, $size(c)$ and m. This is formalised in the following definition.

Definition 2.5.1 (Occam algorithm). *Let $\alpha \geq 0$ and $0 \leq \beta < 1$ be constant. L is an (α, β)-**Occam algorithm** for $\mathcal{C}$ using $\mathcal{H}$ if on input a sample S of cardinality m labelled according to $c \in \mathcal{C}_n$, L outputs a hypothesis $h \in \mathcal{H}$ such that:*

- *h is consistent with S, or $h(x_i) = c(x_i)$ for all $x_i \in S_x$.*
- *$size(h) \leq (n \cdot size(c))^\alpha m^\beta$*

*We say that L is an **efficient** (α, β)-Occam algorithm if its running time is bounded by a polynomial in n, m and $size(c)$.*

The growth complexity $\mathcal{O}(n \cdot size(c))^\alpha m^\beta$ is somewhat difficult to scale down. However, we can somewhat try to prove if it is convergence or not.

Theorem 2.5.1. *Fix n. Then $\mathcal{O}(n \cdot size(c))^\alpha m^\beta$ is convergence or divergence depends on the ratio of α and β.*

Proof. Fixed n, then we have the class $\mathcal{C}_n$ to be finitely restricted of n representation class. Then, $size(c)$ for $c \in \mathcal{C}$ is also constant. We then reduce the first complexity measure to be $\mathcal{O}(\lambda^\alpha m^\beta)$. Since $0 \geq \beta < 1$, it follows that as β increase, $size(h)$ monotonically decrease. This is also the case for α in range $[0, 1)$. Hence, it converges to at least n, and diverges accordingly for arbitrary $\alpha > 1$, though the trend will not as strong if $m > n$, or $m \gg n$. □

Now, in what sense can we say that the output h of an Occam algorithm succinct? First, let us assume that $m \gg n$, so that the above bound can be effectively simplified to $size(h) < m^\beta$, for some $\beta < 1$. Since the hypothesis h is consistent with the sample S, H allows us to reconstruct the m labels $c(x_1) = h(x_1), \dots, c(x_m) = h(x_m)$, and is given only the unlabelled sequence of instance $x_1, \dots, x_m$. Thus the m bits $c(x_1), \dots, c(x_m)$ have been effectively *compressed* into a much shorter string h of length at most m^β. Note that the requirement $\beta < 1$ is quite weak, since a consistent hypothesis of length $O(mn)$ can always be achieved by simply storing the sample S in a table (at a cost of $n + 1$ bits per labelled example) and giving an arbitrary answer for instances that are not in the table. We would be certainly not expecting such a hypothesis to have any predictive power.

By sheer observation of the definition we have been constructing, the $(1, 0)$-Occam algorithm is the largest deviation, with the bound $size(h) \leq n \cdot size(c)$. This means it can at most, have the representation size equal to *all* representation partition of $c \in \mathcal{C}$. It is also good to note the misconception, or rather the misinterpretation: $size(c)$ is different from n, even though they are particularly the same from first glance. It is because n is actually the representation size of the instance string, and not the concept representation string, which utilizes another alphabet entirely. Though, suppose they are of the same alphabet, then $X = \{\Sigma\}_n$, then again, there exists a number q such that $q \geq n$. In the case where $q = n$, it is most likely that the representation is constrained of the linear transformation order, of which does not change the string length.

Let us observe that even in the case $m \ll n$, the shortest consistent hypothesis in $\mathcal{H}$ may in fact be the target concept, and so we must allow $size(h)$ to depend at least linearly on $size(c)$. We will see cases where this makes it easier to efficiently find a consistent hypothesis — by contrast, computing the shortest hypothesis consistent with the data is often a computationally hard problem.

Theorem 2.5.2 (Occam's Razor). *Let L be an efficient (α, β)-Occam algorithm for $\mathcal{C}$ using $\mathcal{H}$. Let $\mathcal{D}$ be the target distribution over the instance space X, let $c \in \mathcal{C}_n$ be the target concept, and $0 < \epsilon, \delta \leq 1$. Then there is a constant $a > 0$ such that if L is given as input a random sample S of m examples drawn from $EX(c, \mathcal{D})$, where m satisfies:*

$$m \geq a \left(\frac{1}{\epsilon} \log \frac{1}{\delta} + \left(\frac{(n \cdot \text{size}(c)^\alpha)}{\epsilon} \right)^{1/1-\beta} \right) \tag{2.34}$$

[5]This section is taken, and studied, from a pretty old book. It would be then observed that a lot of the concept, including the representation, is taken in a rather computer science fashion such as for the hypothesis $\mathcal{H}$ to be binary representation.

then with probability at least $1-\delta$ *the output* h *of* L *satisfies* $error(h) \leq \epsilon$*. Moreover,* L *runs in time polynomial in* n*,* $size(c)$*,* $1/\epsilon$ *and* $1/\delta$*.*

Notice that as β tends to 1, the exponent in the bound for m tends to infinity. This corresponds with the assumption of intuition that the length of the hypothesis approaches that of the data itself, then the predictive power of the hypothesis is diminishing.

To apply this theorem into variational setting, we turn to the arguably more general definition, measuring representational succinctness by the cardinality of the hypothesis class rather than the representationally supposed bit length $size(h)$. Then, theorem 2.5.2 will be the special case of such theorem. To make this precise, let $\mathcal{H}_n = \cup_{m\geq 1}\mathcal{H}_{n,m}$.

Theorem 2.5.3 (Occam's Razor, Cardinality Version). *Let* $\mathcal{C}$ *be a concept class and* $\mathcal{H}$ *a representation space. Let* L *be an algorithm such that for any* n *and any* $c \in \mathcal{C}_n$*, if* L *is given as input a sample* S *of* m *labelled examples of* c*, then* L *runs in time polynomial in* n*,* m*, and* $size(c)$*, and output an* $h \in \mathcal{H}_{n,m}$ *that is consistent with* S*. Then there is a constant* $b > 0$ *such that for any* n*, any distribution* $\mathcal{D}$ *over* X_n*, and any target concept* $c \in \mathcal{C}_n$*, if* L *is given as input a random sample from* $EX(c, \mathcal{D})$ *of* m *examples, where* $|\mathcal{H}_{n,m}|$ *satisfies:*

$$\log|\mathcal{H}_{n,m}| \leq b\epsilon m - \log\frac{1}{\delta} \tag{2.35}$$

or equivalently, where m *satisfies* $m \geq (1/b\epsilon)(\log|\mathcal{H}_{n,m|} + \log(1/\delta))$ *then* L *is guaranteed to find a hypothesis* $h \in \mathcal{H}_n$ *that with probability at least* $1-\delta$ *obeys* $error(h) \leq \epsilon$*.*

We do not mention, or necessarily claim that L is an efficient PAC learning algorithm. In order for the theorem to apply, we must pick m large enough so that $b\epsilon m$ dominates $\log|\mathcal{H}_{n,m}|$.

Proof of Theorem 2.5.3. We first give a proof of theorem 2.5.2, by proving the more general form in theorem 2.5.3, of which then is considered a special case to prove. Firstly, we say that a hypothesis $h \in \mathcal{H}_{n,m}$ is bad if $error(h) > \epsilon$, where the error is of course measured with respect to the target concept c and the target distribution $\mathcal{D}$. Then, by the independence of the random examples, the probability that a fixed bad hypothesis h is consistent with a randomly drawn sample of m examples from $EX(c, \mathcal{D})$ is at most $(1-\epsilon)^m$. Using the union bound, this implies that if $\mathcal{H}' \subseteq \mathcal{H}_{n,m}$ is the set of all bad hypotheses in $\mathcal{H}_{n,m}$, then the probability that some hypothesis in $\mathcal{H}'$ is consistent with a random sample of size m is at most $|\mathcal{H}|(1-\epsilon)^m$. We want this to be at most δ, since $|\mathcal{H}'| \leq |\mathcal{H}_{n,m}|$ we get a stronger condition if we solve for $|\mathcal{H}_{n,m}|(1-\epsilon)^m \leq \delta$. From this, we obtain

$$\log|\mathcal{H}_{n,m}| \leq m\log(1/1-\epsilon) - \log(1/\delta) \tag{2.36}$$

Using the fact that $\log(1/1-\epsilon) = \Theta(\epsilon)$, we get the statement of the theorem. □

Proof of Theorem 2.5.2. Let $\mathcal{H}_{n,m}$be the set of all possible hypothesis representation that the (α, β)-Occam algorithm L might output when given as input a labelled sample S of cardinality m.

Since L is an (α, β)-Occam algorithm, every such hypothesis has bit length at most $(n \cdot size(c))^\alpha m^\beta$, thus implying that $|\mathcal{H}_{n,m}| \leq 2^{(n\cdot size(c))^\alpha m^\beta}$. By Theorem 2.5.3, the output of L has error at most ϵ with confidence at least $1-\delta$, provided

$$\log|\mathcal{H}_{n,m}| \leq b\epsilon m - log\frac{1}{\delta} \tag{2.37}$$

Transposing, we want m such that

$$m \geq \frac{1}{b\epsilon}\log|\mathcal{H}_{n,m}| + \frac{1}{b\epsilon}\log\frac{1}{\delta} \tag{2.38}$$

The above condition can be satisfied by picking m such that both

$$m \geq \left(\frac{2}{b\epsilon}\right)\log|\mathcal{H}_{n,m}|, \quad m \geq \left(\frac{2}{b\epsilon}\right)\log\frac{1}{\delta} \tag{2.39}$$

will effectively hold. Choosing $a = 2/b$ yields the statement in the theorem. [6] □

[6] Why is the pairing of (a, b) value is as such, we are not so sure.

2.6 Discrete space empiricism - Rademacher complexity

The jump toward discrete space is natural, as information are often discrete on infinite and continuous phase space (i.e. the domain, for example, $\mathbb{R}$-values). The hypothesis sets typically used in machine learning are infinite, even in toy model, as the rectangular learning problem itself can be expressed by infinitely many rectangles. But the sample complexity bounds are uninformative when dealing with infinite hypothesis sets – for arbitrary construction of $\mathcal{H}$. One could ask whether efficient learning from a finite sample is even possible when the hypothesis set $\mathcal{H}$ is infinite. The basic goal with the Rademacher complexity is this exact question, for the first complexity treatment. The general idea for doing so consists of reducing the infinite case to the analysis of finite sets of hypotheses, and then proceed as previous sections. Authoritative sources provided mainly the insight and theories in this section, as of Mohri et al. [2012] and Bartlett and Mendelson [2002].

2.6.1 Rademacher descriptions

We will continue to use $\mathcal{H}$ to denote a hypothesis set as in the previous chapters. Many of the result of this section are general and hold for arbitrary loss function, or $L : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$. In what follows, $\mathcal{G}$ will generally be interpreted as the *family of loss functions associated to* $\mathcal{H}$, mapping from $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ to $\mathbb{R}$. That is,

$$\mathcal{G} = \{g : (x, y) \mapsto L(h(x), y) : h \in \mathcal{H}\}$$

However, the definitions are given in the general case of a family of arbitrary functions $\mathcal{G}$, mapping from an arbitrary input space $\mathbb{Z}$ to $\mathbb{R}$. Thus, we can say that we endowed upon the landscape a measure set $\mathcal{G}$ of all singular mapping.

The **Rademacher complexity** captures the richness of a family of functions by measuring the degree to which a hypothesis set can fit random noise. This is perhaps one of the very *empirical complexity notion* that would be seen, as for the random noise idea, that is. ERC

Let G be a family of functions mapping from $Z \to [a, b]$, $S = (z_1, \ldots, z_m)$ a fixed sample of size m with elements in Z. Then, the **empirical Rademacher complexity** of G with respect to the sample S is defined as:

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}\left[\sup_{g \in G} \frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i)\right]$$

where $\boldsymbol{\sigma} = \{\sigma_1, \ldots, \sigma_m\}^\top$ with σ_is independent uniform random variables taking values in $\{-1, +1\}$. The random variable σ_i are called **Rademacher variables.** Rademacher variables

The empirical Rademacher complexity can be rewritten as:

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \mathop{\mathbb{E}}_{\sigma}\left[\sup_{g \in G} \frac{\boldsymbol{\sigma} \cdot \mathbf{g}_S}{m}\right]$$

for $\mathbf{g}_S$ denoting the vector of values taken by g over the sample S: $\mathbf{g}_S = (g(z_1), \ldots, g(z_m))^\top$. The inner product measures the correlation of $\mathbf{g}_S$ with the vector of random noise $\boldsymbol{\sigma}$. The supremum $\sup_{g \in \mathcal{G}}(\boldsymbol{\sigma} \cdot \mathbf{g}_S/m)$ is a measure of how well the function class $\mathcal{G}$ correlates with $\boldsymbol{\sigma}$ over the sample S. Thus, the empirical Rademacher complexity measures on average how well the function class correlates with random noise on the dataset. What will the random noise be like? Well, not much. For a fixed sample size, it captures how **random** the loss function distribution is. A random noise system is inherently chaotic by default, even if we restrict its value in $\{-1, +1\}$ [7]. Hence, if the model is good enough on certain dataset, that the noisy, useless data configuration performs "somewhat pretty well", then from that, we can know the complexity of the model by its flexibility in such noisy fitting. That is, in the topic of fitting, at least. So the richer or more complex families $\mathcal{G}$ can generate more vectors $\mathbf{g}_S$ and thus better correlate with random noise on average.

We come to the definition of *the* Rademacher complexity. Rademacher complexity

Definition 2.6.1 (Rademacher complexity). *Let $\mathcal{D}$ denote the distribution according to which samples are drawn. For any integer $m \geq 1$, the **Rademacher complexity** of $\mathcal{G}$ is the expectation of the empirical Rademacher complexity over all samples of size m drawn according to $\mathcal{D}$.*

$$\mathfrak{R}_m(\mathcal{G}) = \mathop{\mathbb{E}}_{S \sim \mathcal{D}^m}[\hat{\mathfrak{R}}_S(\mathcal{G})] \tag{2.40}$$

[7] Actually, this is indeed a problem. What will happen if the value is totally synchronous for a specific configuration that is, though statistically impossible, yet is the solution to the finite space?

We are now ready to present our first generalization bound based on Rademacher complexity.

Theorem 2.6.1. *Let $\mathcal{G}$ be a family of functions mapping from $\mathcal{Z}$ to $[0,1]$. Then, for any $\delta > 0$, with probability at least $1-\delta$ over the draw of an i.i.d. sample S of size m, each of the following holds for all $g \in \mathcal{G}$:*

$$\mathbb{E}[g(z)] \leq \frac{1}{m}\sum_{i=1}^{m} g(z_i) + 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log(1/\delta)}{2m}} \tag{2.41}$$

and

$$\mathbb{E}[g(z)] \leq \frac{1}{m}\sum_{i=1}^{m} g(z_i) + 2\mathfrak{R}_m(\mathcal{G}) + 3\sqrt{\frac{\log(2/\delta)}{2m}} \tag{2.42}$$

The final term in both equations is typically much smaller than the Rademacher complexity. Note that they are one-sided uniform deviation bounds, and that they are also *data-dependent* bound, just like how Gaussian process regression is also functionally data-dependent.

Proof. For any sample $S = (z_1, \ldots, z_m)$, and any $g \in \mathcal{G}$, we denote $\hat{\mathbb{E}}_S[g]$ by the empirical average of g over S. The proof then consists of applying McDiarmid's inequality to function Φ defined for any sample S by:

$$\Phi(S) = \sup_{g\in\mathcal{G}}\Big(\mathbb{E}[g] - \hat{E}_S[g]\Big) \tag{2.43}$$

This is the supremum of the different between the average and the empirical average for all $g \in \mathcal{G}$. If there is anything, this is similar to the following rational choice: just simply change the inequality to:

$$\begin{aligned} \mathbb{E}[g(z)] - \frac{1}{m}\sum_{i=1}^{m} g(z_i) &\leq 2\mathfrak{R}_m(\mathcal{G}) + 3\sqrt{\frac{\log(2/\delta)}{2m}} \\ \mathbb{E}[g(z)] - \hat{\mathbb{E}}_{S'}[g(z)] &\leq 3\mathfrak{R}_m(\mathcal{G}) + a\sqrt{\frac{\log(2/\delta)}{2m}} \end{aligned} \tag{2.44}$$

in which we then take the supremum. The supremum is for the pretty trivial choice: for this equality to hold true then the largest element of the LHS $\mathbb{E}[g(z)] - \hat{\mathbb{E}}_{S'}[g(z)]$ to be less than the RHS, for any δ (for equation 2.41) and $\delta/2$ (for equation 2.42). Note that the same thing is applicable.

Let S and S' be two samples differing by exactly one point, say z_m in S and z'_m in S'. Then, since the difference of suprema does not exceed the supremum of the difference, we have:

$$\Phi(S') - \Phi(S) \leq \sup_{g\in\mathcal{G}}\Big(\hat{\mathbb{E}}_S[g] - \hat{\mathbb{E}}_{S'}[g]\Big) = \sup_{g\in\mathcal{G}} \frac{g(z_m) - g(z'_m)}{m} \leq \frac{1}{m} \tag{2.45}$$

Similarly, we can obtain

$$\Phi(S') - \Phi(S) \leq \frac{1}{m} \implies \big|\Phi(S) - \Phi(S')\big| \leq \frac{1}{m} \tag{2.46}$$

By McDiarmid's inequality, for any $\delta > 0$, with probability at least $1 - \delta/2$, the following holds.

$$\Phi(S) \leq \mathop{\mathbb{E}}_{S}[\Phi(S)] + \sqrt{\frac{log2/\delta}{2m}} \tag{2.47}$$

We next bound the expectation of the right-hand side as follows:

$$\mathbb{E}_S[\Phi(S)] = \mathbb{E}_S\left[\sup_{g\in\mathcal{G}}\Big(\mathbb{E}[g] - \hat{\mathbb{E}}_S(g)\Big)\right] \tag{2.48}$$

$$= \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \left(\mathbb{E}_{S'}[\hat{\mathbb{E}}_{S'}(g)] - \hat{\mathbb{E}}_S(g) \right) \right] \text{ from 6.1.3} \tag{2.49}$$

$$= \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \mathbb{E}_{S'} \left[\hat{\mathbb{E}}_{S'}(g) - \hat{\mathbb{E}}_S(g) \right] \right] \tag{2.50}$$

$$\leq \mathbb{E}_{S,S'} \left[\sup_{g \in \mathcal{G}} \left(\hat{\mathbb{E}}_{S'}(g) - \hat{\mathbb{E}}_S(g) \right) \right] \tag{2.51}$$

$$= \mathbb{E}_{S,S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \left(g(z_i') - g(z_i) \right) \right] \tag{2.52}$$

$$= \mathbb{E}_{\sigma,S,S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \left(g(z_i') - g(z_i) \right) \right] \tag{2.53}$$

$$\leq \mathbb{E}_{\sigma,S,S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i') \right] + \mathbb{E}_{\sigma,S} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} -\sigma_i g(z_i) \right] \tag{2.54}$$

$$= 2\, \mathbb{E}_{\sigma,S} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i) \right] = 2\mathfrak{R}_m(\mathcal{S}). \tag{2.55}$$

We note that there are several implicit choices, especially for the properties of the subadditivity of the supremum, $\sup A + B = \sup A + \sup B$.

The reduction to $\mathfrak{R}_m(\mathcal{G})$ in equation 2.55 yields the bound in the previous equation (the first one), using δ instead of $\delta/2$. To derive a bound in terms of $\hat{\mathfrak{R}}_S(\mathcal{G})$, we observe that, by definition, changing one point in S changes $\hat{\mathfrak{R}}_S(\mathcal{G})$ by at most $1/m$. Then, using again McDiarmid inequality, with probability $1 - \delta/2$, the following holds:

$$\mathfrak{R}_m(\mathcal{G}) \leq \hat{\mathfrak{R}}_S(\mathcal{G}) + \sqrt{\frac{\log 2/\delta}{2m}} \tag{2.56}$$

Finally, we use the union bound to combine inequalities together, which yields

$$\Phi(S) \leq 2\hat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log 2/\delta}{2m}} \tag{2.57}$$

which matches the inequality. □

The above inequality can be somewhat extended to be formulated, using the empirical Rademacher complexity rather than the expected Rademacher complexity.

Corollary 2.6.1. *Suppose that a sample S of size m is drawn according to distribution $\mathcal{D}$. Then for any $\delta > 0$, with probability at least $1 - \delta$, the following holds for all $g \in \mathcal{G}$:*

$$\mathbb{E}[g(z)] = L(g) \leq \underbrace{L_S(g)}_{\hat{\mathbb{E}}_S[g]} + 2\mathfrak{R}_S(\mathcal{G}) + O\left(\sqrt{\frac{\log 1/\delta}{m}} \right) \tag{2.58}$$

Proof. We may consider the empirical Rademacher complexity $R_S(G)$ as a function of the points $z_1, \ldots, z_m$ that comprise the sample S. Changing one of the z_i to a new value z_i' changes $R_S(G)$ by at most $1/m$. Applying McDiarmid's inequality with $c = 1/m$, and $\epsilon = \sqrt{\log 2/\delta/2m}$ we have that with probability at least $1 - \delta/2$,

$$R_S(G) \leq R_m(\mathcal{G}) + \sqrt{\frac{\log 2/\delta}{2m}} \tag{2.59}$$

By union bound, with probability at least $1 - e\delta$, the inequality both holds. This implies that our corollary holds for all $g \in \mathcal{G}$ with probability at least $1 - 2\delta$. Replace δ by $\delta/2$ yields the required result. □

The following result relates the empirical Rademacher complexities of a hypothesis set $\mathcal{H}$ and to the family of loss functions $\mathcal{G}$ associated to $\mathcal{H}$ in the case of binary loss (zero-one loss).

Lemma 2.6.2. *Let $\mathcal{H}$ be a family of functions taking values in $\{-1,+1\}$, and let $\mathcal{G}$ be the family of loss functions associated to $\mathcal{H}$ for the zero-one loss:*

$$\mathcal{G} = \{(x,y) \mapsto 1_{h(x) \leq y} : h \in \mathcal{H}\}$$

For any sample $S = \{(x_i, y_i)\}_m$ of elements in $\mathcal{X} \times \{-1,1\}$, let $S_\mathcal{X}$ be the projection *over $\mathcal{X}$: $S_\mathcal{X} = (x_1, \dots, x_m)$. Then, the following relation holds:*

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \frac{1}{2}\hat{\mathfrak{R}}_{S_\mathcal{X}}(\mathcal{H}) \tag{2.60}$$

Proof. For any sample $S = ((x_1,y_1),\dots,(x_m,y_m))$ of elements in $\mathcal{X} \times \{-1,+1\}$, by definition, the empirical Rademacher complexity of $\mathcal{G}$ can be written as:

$$\begin{aligned}
\widehat{\mathfrak{R}}_S(\mathcal{G}) &= \mathbb{E}_\sigma\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i \mathbb{1}_{h(x_i)\neq y_i}\right] \\
&= \mathbb{E}_\sigma\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i \frac{1-y_i h(x_i)}{2}\right] \\
&= \frac{1}{2}\mathbb{E}_\sigma\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m -\sigma_i y_i h(x_i)\right] \\
&= \frac{1}{2}\mathbb{E}_\sigma\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i h(x_i)\right] \\
&= \frac{1}{2}\widehat{\mathfrak{R}}_S(\mathcal{H}),
\end{aligned}$$

where we used the fact that $\mathbb{1}_{h(x_i)\neq y_i} = (1 - y_i h(x_i))/2$ and the fact that for a fixed $y_i \in \{-1,+1\}$, σ_i and $-\sigma_i y_i$ are distributed in the same way.

□

The lemma then implies by taking expectations, that for any $m \geq 1$,

$$\mathfrak{R}_m(\mathcal{G}) = \frac{1}{2}\mathfrak{R}_m(\mathcal{H}) \tag{2.61}$$

These connections between the empirical and average Rademacher complexities can be used to derive generalization bounds for binary classification in terms of the Rademacher complexity of the hypothesis set $\mathcal{H}$. This leads to the following result on the Rademacher complexity bounds on binary classification.

Theorem 2.6.3 (Rademacher complexity bounds - binary classification). *Let $\mathcal{H}$ be a family of functions taking values in $\{-1,+1\}$ and let $\mathcal{D}$ be the distribution over the input space $\mathcal{X}$. Then, for any $\delta > 0$, with probability at least $1-\delta$ over a sample S of size m drawn according to $\mathcal{D}$, each of the following holds for any $h \in \mathcal{H}$.*

$$R(h) \leq \hat{R}_S(h) + \mathfrak{R}_m(\mathcal{H}) + \sqrt{\frac{\log 1/\delta}{2m}} \tag{2.62}$$

and

$$R(h) \leq \hat{R}_S(h) + \mathfrak{R}_m(\mathcal{H}) + 3\sqrt{\frac{\log 2/\delta}{2m}} \tag{2.63}$$

Proof. This follows immediately from the above theorem and lemma. □

2.6.2 Growth function

Here, we will show how the Rademacher complexity can be bounded in terms of the *growth function*. . The idea encoded of growth function simply asks for *how many* truly different hypotheses are there when we only observe m input from infinitely discrete possible inputs of c.

growth function

Definition 2.6.2 (Growth function). *Given a binary class of functions $\mathcal{H}$, we define the growth function $\Pi_{\mathcal{H}} : \mathbb{N} \to \mathbb{N}$ as:*

$$\Pi_{\mathcal{H}}(m) = \max_{\{x_1,\dots,x_m\}\subseteq\mathcal{X}} |\{h(x_1),\dots,h(x_m) : h \in \mathcal{H}\} \tag{2.64}$$

In other words, $\Pi_{\mathcal{H}}(m)$ is the maximum number of distinct ways in which m points can be classified using hypotheses in $\mathcal{H}$. This growth function is then also a measurement of the richness of a class of function, and each one of these distinct classification is called a *dichotomy*. The growth function then counts the number of dichotomies that are realized by the hypothesis. This differs from the Rademacher complexity, in the way that it is independent of the distribution of points sampled, hence is purely combinatorial. Also, the growth function is conducted for all $h \in \mathcal{H}$ of the finite hypothesis space. If it is the binary case, then there is at most 2^m categorization configuration. To relate these two together, theoretical consideration uses the Massart's lemma.

Lemma 2.6.4 (Massart's lemma). *Let A be a finite subset of $\mathbb{R}^m$, and let*

$$\max_{\vec{y}\in A} ||\vec{y}||_2 \leq r$$

Then

$$\mathop{\mathbb{E}}_{\sigma_1,\dots,\sigma_m}\left[\sup_{\vec{y}\in A}\sum_{i=1}^{m}\sigma_i y_i\right] \leq r\sqrt{2\log|A|} \tag{2.65}$$

This lemma offers not much aside from an interesting bound. Using this result, we can now bound the Rademacher complexity in terms of the growth function.

Corollary 2.6.2. *Let $\mathcal{G}$ be a family of functions taking values in $\{-1,+1\}$. Then the following holds:*

$$\mathfrak{R}_m(\mathcal{H}) \leq \sqrt{\frac{2\log\Pi_{\mathcal{H}}(m)}{m}} \tag{2.66}$$

Proof. For a fixed sample S, we denote by $\mathcal{G}_{|S}$ the set of vectors of function values $(g(x_1),\dots,g(x_m))^\top$ for $g \in \mathcal{G}$. Since $g \in \mathcal{G}$, $|g' \in \mathcal{G}_{|S}| \leq \sqrt{m}$. We can then apply Massart's lemma as:

$$\begin{aligned}\mathfrak{R}_m(\mathcal{G}) =& \mathop{\mathbb{E}}_{S}\left[\mathop{\mathbb{E}}_{\boldsymbol{\sigma}}\left[\sup_{u\in\mathcal{G}_{|S}}\frac{1}{m}\sum_{i=1}^{m}\sigma_i u_i\right]\right] \\ \leq& \mathop{\mathbb{E}}_{S}\left[\frac{\sqrt{m}\sqrt{2\log|\mathcal{G}_{|S}|}}{m}\right]\end{aligned} \tag{2.67}$$

By definition, $|\mathcal{G}_{|S}|$ is bounded by the growth function. Thus,

$$\begin{aligned}\mathfrak{R}_m(\mathcal{G}) \leq& \mathop{\mathbb{E}}_{S}\left[\frac{\sqrt{m}\sqrt{2\log\Pi_{\mathcal{G}}(m)}}{m}\right] \\ =& \sqrt{\frac{2\log\Pi_{\mathcal{G}}(m)}{m}}\end{aligned} \tag{2.68}$$

which concludes the proof. □

From this and the generalization bound, we will also yield the following generalization bound in terms of the growth function.

Corollary 2.6.3 (Growth function generalization bound). *Let $\mathcal{H}$ be a family of function taking values in $\{-1,+1\}$. Then, for any $\delta > 0$, with probability at least $1-\delta$, for any $h \in \mathcal{H}$:*

$$R(h) \leq \hat{R}_S(h) + \sqrt{\frac{2\log \Pi_{\mathcal{H}}(m)}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}} \tag{2.69}$$

This growth function bound can also be derived directly. The computation of the growth function is also, pretty weird, and is often not always convenient, especially because of its combinatorial nature.

2.7 Equivalent spatial separation - VC-theory

VC, or Vapnik-Chervonenkis theory (Vapnik and Chervonenkis [1971], Vapnik [1999], Sauer [1972], Shelah [1972]), base a lot of its results and theories on both the growth function, and the combinatorial nature of which it takes form. The problem of the formation of VC theory is simple. We have seen that a consistent learner can be used to design a PAC-learning algorithm, provided the output hypothesis comes from a class that is not too large, in particular, of any polynomial class $poly(\cdot,\cdot,\cdot,\cdot)$. However, one certain assumption and condition on the PAC-learning algorithm measure is that it only works on (the result, that is) finite hypothesis class or concept class. Concept classes that are uncountably infinite are often used, for example, linear halfspaces, or more familiar, the hypothesis on such space is the **linear classifier** or **perceptron**. In this section, we would particularly see how we can use a new capacity measure called VC dimension of a concept class to make a consistent learner to be PAC-learnable. All theories and practices using the VC-dimension as one of its main ingredient is then called the VC theory, from authoritative sources. We also note that from this point on, there exists more fundamental extensions of the VC-theory, or alternatives. This includes algorithm-dependent bounds, of Sachs et al. [2023], treatments of learning bounds based on stochastic algorithmic stability as in Bousquet and Elisseeff [2002], similar discrete and combinatorial view as of Natarajan's dimension (Natarajan [1989]), pseudo-dimension (Pollard [1984]), information theoretic approach (Xu and Raginsky [2017b]), or representational complexity form into *approximate description length*, as seen in Daniely and Katzhendler [2022].

The main concept of VC-theory lies in the apparatus of capacity as the **VC-dimension**. The VC-dimension is often regarded as a purely combinatorial notion, but it is often easier to compute than the growth function, or the Rademacher Complexity, nevertheless holding equivalence to both and statements of PAC-learning. As we shall see, the VC-dimension is a key quantity in learning, and is directly related to the growth function.

shattering

To define the VC-dimension, we will have to define the notion of *shattering* . As we recalled of the previous sections on the growth function, on the hypothesis set $\mathcal{H}$, a dichotomy on a set S is one of many possible ways of labelling the points in S using a hypothesis $\mathcal{H}$. A set S of $m \geq 1$ points is said to be shattered by a hypothesis set $\mathcal{H}$ when $\mathcal{H}$ realizes all possible dichotomies of S, that is when $\Pi_{\mathcal{H}}(m) = 2^m$.

Definition 2.7.1 (Shattering). *We say that a finite set $S \subset \mathcal{X}$ is* **shattered** *by C, if $|\Pi_C(S)| = 2^{|S|}$. S is shattered by C if all possible dichotomies over S can be realized by C.*

VC-dimension

We now can define the notion of the *dimension* for a concept class $\mathcal{C}$.

Definition 2.7.2 (VC-dimension). *The VC-dimension of a hypothesis set $\mathcal{H}$ is the size of the largest set that can be shattered by $\mathcal{H}$,*

$$\mathrm{VCdim}(\mathcal{H}) = \max_{m\in S}\{m : \Pi_{\mathcal{H}}(m) = 2^m\} \tag{2.70}$$

Note that, by definition, if $\mathrm{VCdim}(\mathcal{H}) = d$, then there exists a set of size d that can be shattered. However, it does not mean that all sets of size d or less are shattered, because this only get you the existential criteria. Similar to the growth notion, if we are to look at the definition of VC-dimension, we see the similar problem: it is very computationally heavy for experimental purposes. However, its treatment of the learning theory through discretized notion, and separation conjunctive space is powerful, making it effective to express many hypothesis classes.

halfspace

A famous example illustrating this concept is the problem of halfspace partition. The **halfspace** hypothesis space is the set of hypotheses that consist of a hyperplane in a d-dimensional coordinate space that classifies a feature vector

$\phi(x) \in \mathbb{R}^{d+1}$ as either $-1, 1$ based on which side the of the hyperplane it lies. Here, d represents the number of features on item x.

Mathematically, to define the halfspace hypothesis space, consider the domain $\mathcal{X} = \mathbb{R}$ and concept set $\mathcal{Y} = \{-1, 1\}$. The class $\mathcal{H}_{gl}$ of general linear classifiers, or linear halfspaces, is defined as

$$\mathcal{H}_{gl} = \{h_{\boldsymbol{w},b} \mid \boldsymbol{w} \in \mathbb{R}^d, b \in \mathbb{R}\} \tag{2.71}$$

where

$$h_{\mathrm{W},b}(\boldsymbol{x}) = \operatorname{sgn}(\langle \boldsymbol{x} \cdot \boldsymbol{w} \rangle + b) = \operatorname{sgn}\left(\left(\sum_{i=1}^{d} x_i w_i\right) + b\right) \tag{2.72}$$

General linear halfspaces in $\mathbb{R}^d$ can be viewed as *homogenous linear halfspaces* in $\mathbb{R}^{d+1}$ via a simple transformation. We define the class of homogenous linear halfspaces $\mathcal{H}_l$ as

$$\mathcal{H}_l = \{h_{\boldsymbol{w}} \mid \boldsymbol{w} \in \mathbb{R}^d, b \in \mathbb{R}\}$$

where

$$h_{\mathrm{W}}(\boldsymbol{x}) = \operatorname{sgn}(\langle \boldsymbol{x} \cdot \boldsymbol{w} \rangle) = \operatorname{sgn}\left(\sum_{i=1}^{d} x_i w_i\right) \tag{2.73}$$

Let $\boldsymbol{x} = h_{\boldsymbol{w},b}(\boldsymbol{x}) \in \mathbb{R}^d$, for $\boldsymbol{w} = (w_1, w_2, \ldots, w_d) \in \mathbb{R}^d$ and $b \in \mathbb{R}$ be given, we then define

$$\boldsymbol{x}' = (1, x_1, \ldots, x_d) \in \mathbb{R}^{d+1} \tag{2.74}$$

$$\boldsymbol{w}' = (1, w_1, \ldots, w_d) \in \mathbb{R}^{d+1} \tag{2.75}$$

Then we get $\langle \boldsymbol{x} \cdot \boldsymbol{w} \rangle + b = \langle \boldsymbol{x}' \cdot \boldsymbol{w}' \rangle$ for all $x \in \mathbb{R}$, and thus

$$h_{\boldsymbol{w},b}(\boldsymbol{x}) = h_{\boldsymbol{w}'}(\boldsymbol{x}')$$

Let $\mathcal{X}$ be a set of $d+2$ points. By Radon's theorem, it can be partitioned into two sets $\mathcal{X}_1$ and $\mathcal{X}_2$ such that their convex hulls intersect. Observe that when two sets of points $\mathcal{X}_1, \mathcal{X}_2$ are separated by a hyperplane, their convex hulls are also separated by that hyperplane. Thus, $\mathcal{X}_1$ and $\mathcal{X}_2$ cannot be separated by a hyperplane and $\mathcal{X}$ is not shattered. Combining our lower and upper bounds, we have proven that $\mathrm{VCdim}(\mathcal{H} \in \mathbb{R}^d)$ is $d+1$.

The VC-dimension of many other hypothesis sets can be determined or upper-bounded in a similar way. In particular, the VC-dimension of any vector space of dimension $r < \infty$ can be shown to be at most r. The next result, known as *Sauer's lemma*, clarifies the connection between the notions of growth function and VC-dimension.

Sauer's lemma

Theorem 2.7.1 (Sauer's lemma). *Let $\mathcal{H}$ be a hypothesis set with $\mathrm{VCdim}(\mathcal{H}) = d$. Then, for all $m \in \mathbb{N}$, the following inequality holds:*

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^{d} \binom{m}{i} \tag{2.76}$$

2.8 Summary

In the above section, we have introduced the notion of the learning problem, in its simplistic form and mathematical formalism. With such, we developed a setting of which is reasonable to guarantee learning and the existence of a hypothesis, PAC-learning, complexity analysis (Rademacher complexity, growth function, VC-dimension) and application of them on gaining learning bounds, and some of the examples on typical learning problems. Further than such, we have some remarks and problems to discuss.

The learning theory result itself is limited to particular class of result. Most of the results from classical learning setting either are optimization bound, "learning bound", or convergence criteria bounds. Some of them are concerned of the complexity of the model and vice versa, and a lot of them then are concerned with connecting the complexity notion to the learning bound itself - more like a dual relationship. Those results are good, for example, PAC-learning is *asymptotic learning bound* of time complexity, which makes sense of the time and the bounding method is restricted to (ϵ, δ) pair, or by some versions of VC-learning which is the *combinatorial discrete complexity bounding* method for those model itself. However, aside from the setting is very much limited, the notion of generalization is more or less generally just being empirical and generalization error (or risk) without sufficient practical heuristic special case, the results from which does not give us much idea about what to do and how to achieve the capable result in such bound. They *guarantee* results as probable, or measurable, but give us no effective ruler to measure it. Some of the measure is improbable, for example, the growth function complexity in various case, and generality of such. Furthermore, complexity is very not so well-defined, especially in the numerical space encoding. Some of the topic, while very interesting, is not discussed, or rather cannot be incorporated into the existing consideration.

The learning setting itself, from this setting, misses certain requirement to define it in a concrete sense. For example, its inclusion set contains the set X of instances in observational space, σ-algebra $\mathcal{S}$ on X, the family $\mathcal{P}$ of probability measure on $(X, \mathcal{S})$ (which is quite optional, give it or take), and subset $\mathcal{C} \subseteq \mathcal{S}$ called the *concept class*, or else a family $\mathcal{F}$ of measurable mapping $X \to [a, b]$ of closed set on the Borel σ-algebra, called the *function class*. These formalisms are often implicitly defined, yet uncertain of their actual property and interpretation. For example, in some testing cases, we would want to know the deterministic degree of the probabilistic measure space $\mathcal{P}$, for whether p_X will collapse into singular behaviour - which indeed probabilistically can be done using elementary probability, however, for different system and different concept, different setting, this proves to be rather difficult. Usually, $[a, b] = [0, 1]$. Furthermore, there are also the definition for the *encoding space* $\mathcal{E}(\cdot, \cdots)$ of all fundamental components, of the ambiance space $\mathbb{A}$ and the representation space $\mathcal{R}$ together, with a distinct encoding rule that is not included. In utmost importance, encoding space needs to be one of the main focus of the theory, however, reality often proves to be unpleasant. Some of the definitions and further clarification is not also included in the rigours, leaving questions that is unattainable of the setting provided. Granted, the theory is minimized, but for example, such is to introduce the distribution $\mathcal{D}$ into the preliminary setting, one must clarify and indeed, concretely provide the fundamental setting of the probabilistic nature of the system - in which sense and when will the system becomes probabilistic, and under what circumstances the probabilistic nature of it varies. Couples that with another theoretical idea of control/process theory, in which the learning theory is often settled to be discrete process, leaves much clarification to be found. Worst, philosophically speaking, and developmental speaking, the notion of *learning* is still very premature of interest - which lead directly to the case where everyone has frameworks and all, but not a lot actually works, and many overlaps up to almost identical.

Aside from all of those, we have an inconsistent framework to deal with. Usually, as we see, there exists no concrete structure to the model that we are asking of. Most of the time, the landscape of practical application changes so fast that we also do not have the ability to structuralize it. As of an interdisciplinary field, bring ideas of a maturing subject with uncertainty also come with difficulty of raising a good hallmark to the problem. While it is true that we have two domains - the *classical structure* of model with design in mind closer to inference, and the *modern structure* which almost identical now to *deep neural structure*, focus on something akin to prediction rather than not, most of the time, this classification is a poor classification in the grand scheme of all, especially if we take into account others model and structures that lie outside of it (Markov chain, generative evolutional structure, etc., programming synthesis).

Thereby, a new construction is recommended to exist, to formalise the theory itself, and to further down increase the development speed since we are pretty much lagging behind, both in formalisation and practical implementation. Should the new theory arise in this book, it must fix almost all the classical theory's pitfall, and further on of the *concurrent theory* being discussed.

Chapter 3. Introduction to Double Descent

The analysis of learning action, machine learning and related practices in the field theoretically has been made by utilizing Computational Learning Theory (CoLT) Valiant [1984] and Statistical Learning Theory (SLT) Vapnik [1999]. These two theories, while overlapped, provides a general framework in analysing and justifying learning actions and learning model constructions, aiding in the formation of modern practices. One of the famous insight using such framework is the *bias-variance tradeoff* Geman et al. [1992], which states that model complexity and generality inversely affect each other, thus guarantee the need for a safe bracket between them. However, recent literatures, Belkin et al. [2019a] has indicated the fallout of such dilemma by the new phenomenon called *double descent*, where there exists an interpolation threshold that renders the current statistical justification for bias-variance inconsequential to a given region. Various 'anomaly' has also been detected similarly, with varying degree of sophistication and potential.In this paper, we would like to review through progress made in this particular field of research, and the phenomena potential explanations up until now.

3.1 Introduction

The modern machine learning landscape is dominated and guaranteed by the foundation of theoretical machine learning, specifically computational learning theory (CoLT) and statistical learning theory (SLT). The theory of machine learning and its modern practice has been developed and researched, of a substantial portion by empirical and heuristic approach, either by advancing new practices, architectures or by try-and-test modification. From its early onset of a regression estimator $\theta(x, y)$ on the linear regression problem, machine learning has developed substantially. Certain model concept with great successes includes regression-classification model, Bayesian modelling, generative learning model, support vector machine, Gaussian processes, and more. Of all such, the more formal and complex model architecture created, is the concept of a *neural network*. With increasingly sophisticated architecture, heuristic approach become popular, the fast-paced advancement of the field comes with new method, new results, new observations, and its far-reaching application which led to even bigger and larger scale deployment, there has been questions about the formation and status of a theoretical ground, a rigorous matter on the side of **theoretical machine learning**.

While rigorous and well-formulated in a sense, classical and theoretical machine learning was dwarfed by the modern advancement of machine learning as a whole, leading to several anecdotal problems regarding the interpretation of phenomena, the re-evaluation of the theory to fit the more updated analysis, and explanation to more sophisticatedly designed system. This and many more, plus as present, many of such advancements and improvements are heuristic, and the general theory and conceptual understanding remain limited, led to the choice to often opt for analogies and empirical workaround. Ultimately, this resulted in multiple 'failures' in explaining, interpreting, and predicting behaviours of new phenomena appeared in the modern landscape of machine learning. While there exists novelty of analysis, and different theoretical schemes to analyse such, particular problems remain in the form of irregularity and conflict of interpretations between frameworks.

Our main focus is to review on the topic, analyse existing analysis and observations, and understanding of the umbrella term classical learning theory, for the recent phenomena observed named 'double descent'. Statistical learning theory (SLT) and computational learning theory (CLT) Vapnik [1999], Mohri et al. [2012], Shalev-Shwartz and Ben-David [2014], Hajek and Raginsky [2021], Bousquet et al. [2020] has been prominent in constructing a well-rounded formal theory surrounding learning problems, models, and machine learners analysis. Valuable insights have been dissected from treatment of statistical theory and mathematical modelling on models, including the *bias-variance tradeoff* Geman et al. [1992], Domingos [2000c], which serves as a bound for efficient learning and model configuration (or complexity). However, recently there has been observations of *double descent* Belkin et al. [2019a], Schaeffer et al. [2023], Nakkiran et al. [2019], Lafon and Thomas [2024] which refute the famous tradeoff assumption, and hence brings question to the establishment of the theory, as well as several assumptions and insight in the framework. Further events and phenomena observed also includes grokking and triple, to n-descent Davies et al. [2023], d' Ascoli et al. [2020]. Furthermore, many problems of defining and formalizing notions used in designing and implementing machine learning models are inconclusive, such as, for example, *model complexity* and others.

3.1.1 Outline

The main contributions of the paper lie in the heart of the problem of *machine learning theoretical*, and specifically *double descent*. As such, we focus on the following:

1. Theoretical analysis of the machine learning framework, and reinterpretation of the empirical-generalization framework. We explicitly redetermine perspectives and formulation, about the problem of generalization learning, and model selection principles. Additionally, we also formalize the algorithmic objective of the objective system.

2. Observations and insight reviews and critical analysis. We analyse current knowledge, different analysis of the phenomena, and provide critiques, critical analysis on limitation, constraints, different assumptions and fallacy observed in those results, and verification or extension of experiments. This means analysing existing papers specific of details, and intrinsic of certain functional class.

3. Providing our own insight to the phenomena of double descent, and hence the need for new analysis or different theory scheme to be utilized in solving the problem.

Even though we already have the **theoretical attacks and analysis** of the learning theory in much greater details above, such details are simply too large to be totally relevant to our formal investigation on double descent. Largely speaking, it is much more relevant in our further sections targeting the reformulation of the descent theory rather than not. Thus, we will *reintroduce* some of those notions in a much denser form, as followed in the following critique.

While this is non-conclusive of the problem of double descent, we want to emphasize the importance of such issue to be further than simply the dilemma between model complexity and risk error. It hints at the mismatch of knowledge, fuzzy nature of certain concepts, failure modes being apparent that indicates the intricacy of the sensitivity of the theory when encountering general structures, and the immaturity in which the theoretical landscape contains many noises.

3.1.2 Related works

On itself, the phenomena *double descent* has been investigated somewhat comprehensively, firstly introduced by Belkin et al. [2019a], which gives it distinctive saddle escape illustration. Further analysis was made by several literatures, particularly attributed the existence of double descent to the concept of model complexity and inductive bias, and potential explanations of such. Nakkiran et al. [2019] expanded the phenomena into deep neural network models. Their conclusion is reached by considering the perturbation of a learning procedure $\mathcal{T}$ on the effective model complexity $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ defined in the paper, separating the eventual phenomena into regions of observations, thereby in one way or another, predicting the tendency of double descent. This is also the first, arguably, "concise" definition of double descent, even though its nature is an empirical definition, including the notion of model complexity. Recently, there is also the Neural Tangent Kernel (Jacot et al. [2018]) which can be used to explain double descent, however further researches are still required. Preliminary works are done by Lafon and Thomas [2024], Schaeffer et al. [2023], and Liu and Flanigan [2023] on the role of optimization in double descent. Davies et al. [2023] attempted to unifying grokking with double descent, theorized a possibility of similarity during the generalization phase transition of the inference period, and Olmin and Lindsten [2024] attempted to explain epoch-wise double descent within the model of two-layer linear network. However, it is mentioned that it can also be expanded into deep nonlinear networks.

Further literatures on double descent varies, but fairly new, ever since their discovery articles in Belkin et al. [2019a] and Nakkiran et al. [2019]. Since then, we have Shi et al. [2024], Schaeffer et al. [2023], Quétu and Tartaglione [2023a], Lafon and Thomas [2024], Neal [2019], Davies et al. [2023], Quétu and Tartaglione [2023b], Liu and Flanigan [2023], Mei and Montanari [2020, 2019a], Adlam and Pennington [2020], Transtrum et al. [2025], Liu et al. [2021], Allerbo [2025], Pezeshki et al. [2021], Brellmann et al. [2024], Nakkiran [2019b], Mei and Montanari [2019b], Heckel and Yilmaz [2020], Nakkiran et al. [2020], Belkin et al. [2020], Yang and Suzuki [2024], Spiess et al. [2023], Nakkiran et al. [2021], Zhang [2024], Cherkassky and Lee [2024], Nakkiran [2019a]. An even more exotic version of double descent, dubbed *triple descent* (or more), is discovered in d' Ascoli et al. [2020].

3.2 Foundational investigation

Several foundational notions must be clarified before it is able to take in the major works about double descent. Specifically, we need to clarify the structure of classical learning, relevant structures, notions of underfitting, overfitting, overparameterization and underparameterization, double descent and what it is counteracting, and so on. Such will also be followed by a section strictly the particular solutions and empirical corrections of theories.

For the need of analysing the phenomenon of double descent, we begin by listing and reviewing different foundational notions that describe the typical theory on the learning dynamics, and the learnability problem. Such ranges from statistical learning theory, as seen of Vapnik [1999], Hajek and Raginsky [2021], Mohri et al. [2012], and the more practical learning theory, as seen in James et al. [2009].

The setting of typical machine learning system comes in two-fold, either from the outer objective of the *observation space* $\{x_i, y_i\}_{i\leq n}^{n}$, denoted $\mathcal{S}$ for any 2-tuple of that form, or the general, oracle-induced view in which there exists the main form, or the oracle $c \in \mathcal{C}$, of which form the functional spaces. We would like to review these two outlooks.

Because we are to introduce certain formalism, or at least interpretation of the learning process, as well as the different dynamical components associated, the preliminary would be first reserved to clarify such notions, alongside fixing usual notations that would be used accordingly. Historically speaking, the learning theory started from the early 1970s. By the sense of classical, we do not mean by, as currently called, of non-deep learning theory, but rather the old treatment of theoretical boundaries and rigorous formalization of the concept itself. In such development, lies the formal setup of learning, and then, the task of which developing it, until we can even reach the *Probably Approximately Correct* learning (PAC) and Vapnik-Chervonenkis theory of maximal dimension, Neural Tangent Kernels (Jacot et al. [2018]), and so on.

The nature of the learning theory can also be brought back from certain proposition or argument, most probably, the fact that it is mostly from the black-box interpretation - hence making almost all model being **phenomenological** by default. An interesting aspect of such, though, is the learning process itself, or we call it so.

3.2.1 Classical notion of learning

Classically, we can see that the problem of double descent is not inherently a problem that can be explained only by considering model complexity and the error rate. As such, it requires a framework that consider multiple factors together, and interpreting different hypothesis. Nevertheless, classical theory does indeed point to certain factors or aspect that should be of concern. For example, we have seen plenty of analysis that indicates that *sample complexity* might be the biggest factor, considering the behaviour is intrinsically operable and perhaps observable in almost any model, except for certain model that is specialized of the data structure itself. Even by then, the error still can be obtained given large enough model disparity to the concept class. There are attempts to, indeed, analyze double descent from a VC perspective (Lee and Cherkassky [2022]), however, the paper falls into the same pitfall as others in their interpretation and analysis.

Nevertheless, even though we have said plenty of the disadvantage of the classical theory in terms of statistical learning, their insight is valuable in determining the possibility of dynamic in question. There is a reason why they are versatile and structurally valuable enough that our majority of understanding are formed from such framework itself. In simplicity, within the principle of working, our model is called the **hypothesis** $h \in \mathcal{H}$ of certain **hypothesis class** $\mathcal{H}$. The target objective is typically concerned of, denoted by $c \in \mathcal{C}$, as to be of a structural class $\mathcal{C}$, in which observations creates information in the form of the observable set $\mathcal{S}$. The set of observation $\mathcal{S}$ is presentable in the space of 2-tuple $(\mathcal{X}, \mathcal{Y}) \subset (\mathbb{R}^d, \mathbb{R}^p)$. Here, $\mathcal{X}$ is called the *ground* or input space, and $\mathcal{Y}$ is called the *target* or output space. By default, in general, the input space $\mathcal{X}$ is supposed to follow some special distribution $\mathcal{D}$. This treatment branches from the ambiguity of the origin of $x \in \mathcal{X}$ for elements of the input space, that is, $x \sim \mathcal{D}$ by a random sampling process. This assumption also ensures relative population representability[1]. By this, we mean that the observation captured fully describe what was happening with the entire concept - even including the pathological and outlier. We also assume that this sample of observation is **identically and independently sampled** of a given sampling process.

The task for the hypothesis, and the designer (us) is to figure out how to best approximate c, assuming arbitrary

[1]By population representability, we mean that the density is well-illustrated, such that the data fully captures the entire operational space that is relevant and meaningful of the underlying concept.

notion of knowledge, prior information, such that h correctly *mimic* the behaviour and phenomenon observed by c. We call this loosely as the **learning problem**. Naturally, learning problem is a statistical problem, since their setting is fully observational.

The **learning problem** divides the act, with respect to configuration the chain of component into two parts. The first one is for any measurable space of the model, after defining the system, the dependency, and the question, we have to gain the *accuracy measure*, or *phenomenological aberration* of a given model, and for the respective problem. Depends on the specific requirement of the problem, the measure can change, but, they have the same landscape. The second one requires the development of a *scheme* to dynamically move the given hypothesis, under fixed description, into a supposed optimal point in which under specific threshold, our model is 'perfect' for the concept. That is, *mimic c by h in a perfect way*. It is perceived, though, that the objective of learning the concept perfectly is impossible. Hence, we will often opt for the latter half of approximating it to a given threshold θ

Setting in the second part, most of our problem is concerned with the issue of predicting the unseen states of being, or rather, the *generalization problem*. This then perhaps, classify the problem of statistical learning theory to *mostly* predictive, or rather, we are concerned of prediction model. This is subsidized by the inner *estimation problem* for an estimation model. What is the difference between the two? First, the distinction comes from the trivial on-sight fact that estimation problems and their models, comes of as a closed-form working space. That is, they work on a closed space of what is observed, and extract properties, determine statistically the behaviours, characteristic of that system alone and only. In other word, an estimator *uses data* to guess and learn about the true state, and properties of the setting (usually also said to be the *true state of nature*). However, we notice a discrepancy - not always will we get the entirety of the problem that we are interested at. For some reason, for example, our data is only quite a portion, lacking a bit valley, a bit of patterns here and there by the blank spaces where observations left behind. Then, what if we can generalize pass that? What if we also want to somewhat estimate correctly for unseen portion of our concept class? This is called the predicting problem we have said earlier, in which we use the estimation already established, and add some flavours into what can possibly constitute a larger view - a more entire view of the learning concept. From a statistical perspective, the model itself in statistical learning theory, would have to both estimate the intrinsic empirical distribution and concept $(c_{\mathcal{S}}, \mathcal{D}_{\mathcal{S}})$, and then somehow, leaves room for generalization to the actual distribution and concept $(c, \mathcal{D})$, which is particularly of more interest than not in practical settings. Here, the notation simply means two thing: the facade of which c is the true concept, and $\mathcal{D}$ is the uncertain, probabilistic presumption of the structure in which c seems to exhibit, thus there then also exists $c_{\mathcal{C}} \equiv c'$ of the proxy concept c' that 'seems right' in comparison to the true concept, and also its observation-based distribution.

With just that much narrow scope of the claims, we can then, at least formalising ourselves of particularly this layer of understanding, toward which we will inevitably have to procure from mathematical formulation, of the learning process, as followed.

Definition 3.2.1 (Learning process, provisional). *A procedure $\mathcal{P}$ is called a **learning process**, given two **constructs** or **model** $h \in \mathcal{H}$, $c \in \mathcal{C}$ if it follows an **error measure** or operator $\ell(\cdot,\cdot)$ such that, for $\mathcal{P}_i \in \mathcal{P}$ indicating the ith iteration of the procedure $\mathcal{P}$, h_i the ith hypothesis iteration of $\mathcal{P}$, minimize the operator that it chooses, that is,*

$$\exists n, i \in \mathbb{N}^*, n \geq i, \quad \ell(h_n, c) = \min_{h \in \mathcal{H}} \ell(h_i, c), \tag{3.1}$$

In most results of the statistical learning theory, the majority would, and will use the setting of *supervised learning framework*. The reason why can be justified in simple terms. Consider unsupervised learning. Under the scope of unsupervised learning, the model essentially goes into a free-mode, where there exists no metric that is non-context enough for evaluating the theoretical form of the model. For a given density estimation model ψ, there can be many ways to 'think of evaluating it' on how well it groups objects together. This means there are not so much meaningful information can be extracted from examining the behaviour of the model, including its performance. Furthermore, since it is random as much as a random process, no conclusive bounds or theoretical theorem can be made on the maximal-minimal problem of a given model. Hence, most of the learning theory is concerned of supervised learning setting, because it is a controlled, closed, well-defined system of interest.

3.2.2 Preliminary - precursor double descent estimation

In a glance, we can fairly setup a condition on a type of specific learning criterion specification, to predict the existence of the pattern for double descent. The PAC-learning process (Shalev-Shwartz and Ben-David [2014], Mohri et al. [2012], Hajek and Raginsky [2021]), or rather, *Probably Approximately Correct* learning, connects three complexity measures - the sample space complexity, representation complexity, and optimization complexity, to the evaluation of the statistical learning process. It refers to the categorization of various learning procedure that satisfies the Probably Approximately Correct criteria in its name, as such being followed. The theory was proposed first by Valiant [1984], of whom then won the Turing Award, but his contribution stayed in the shadow in much of the end of the century, until the formal foundation of machine learning theory, of which then it becomes the stay ground on the standard discussion on the story of machine learner and algorithmic guarantees. As such, the de-facto foundation is such.

PAC-learning procedure gives, by assumption of a **consistent learner**, two different generalization bounds for two different cases. We first define the notion of a consistent learning algorithm, or consistent learner, for a concept class C and hypothesis h.

Definition 3.2.2 (Consistent Learner). *We say that an algorithm is a consistent learner for a concept class C using hypothesis class $\mathcal{H}$, if for all n, for all $c \in C_n$ and for all m, given*

$$S = \{(x_1, c(x_1)), (x_2, c(x_2)), \dots, (x_m, c(x_m))\}$$

as input, $x_i \in X_n$ outputs, then the algorithm $\mathcal{A}$ outputs a hypothesis $h \in \mathcal{H}_n$ such that

$$h(x_i) = c(x_i), i = 1, \dots, m$$

A consistent hypothesis is the hypothesis that is consistent with all the training data provided to it from the concept c. The two bounds then contain the debate between consistent hypothesis (i.e., for example, admitting no error on training data, almost perfect - maybe even so), and inconsistent hypothesis, of which have general defects from the actual concept, or even different by a margin. This equates to observational noise σ, typically used in simulation of noisy feature and spread, and also for randomization. Inconsistent then means the further this particular noise is increase, the higher the lower bound of the error can be for any particular measure, in comparison to the true concept c in consistent case. We need to note that this inherently pushes the consistency as for both the algorithm L, and the hypothesis $\mathcal{H}$. Decomposing such definition, we can clearly see two assumptions:

Assumption 3.2.1 (Consistent learner, condition 1). *The algorithm is capable of extrapolate the entire coverage structure of class $\mathcal{H}$. Thus, $\mathcal{A}(R, \ell)$ of the objective condition is guaranteed upon all class $\mathcal{H}$.*

Assumption 3.2.2 (Consistent learner, condition 2). *There exists $h \in \mathcal{H}$ that absolutely equals to c. That is, for the concept c of interest, on the arbitrary expression space of which contains $\mathcal{H}, \mathcal{C}$, then for a subset $\mathcal{C} \supset \mathcal{C}' \supset c$ as the size of $|\mathcal{C}'| \to 0$, $\mathcal{C}' \subset \mathcal{H}$.*

On such guarantee of the formalization and assumption of coverage, PAC-learning famously leaves two results on the learning bound of the learning setup. The first one concerns of the form as to be consistent, and the second case is for where the second assumption fails. However, we have no correct formalization of this failure mode. Specifically, the learning bound intricacy states that of the **global correctness**, that is, absolutely fitting. Usually this is not true, and the inconsistent case is much more possible, but yet again loose as to only examine the case where consistency fails, not *how much* or how many.

Theorem 3.2.3 (Learning bound - finite $\mathcal{H}$, consistent case). *Let H be a finite set of functions mapping from $\mathcal{X} \to \mathcal{Y}$. Let $\mathcal{A}$ be an algorithm that for any target concept $c \in H$ and i.i.d. samples S returns a consistent hypothesis H_S, such that $\hat{R}(h_S) = 0$. Then for any $\epsilon, \delta > 0$, the inequality $\Pr_{S \sim D^m}[R(h_S) \leq \epsilon] \geq 1 - \delta$ holds if*

$$m \geq \frac{1}{\epsilon}\left(\log |H| + \log \frac{1}{\delta}\right)$$

This sample complexity result admits the following equivalent statement as a generation bound: for any $\epsilon, \delta > 0$, with probability at least $1 - \delta$,

$$R(h_S) \leq \frac{1}{m}\left(\log |\mathcal{H}| + \log \frac{1}{\delta}\right) \tag{3.2}$$

Proof. See Mohri et al. [2012] for the original proof. □

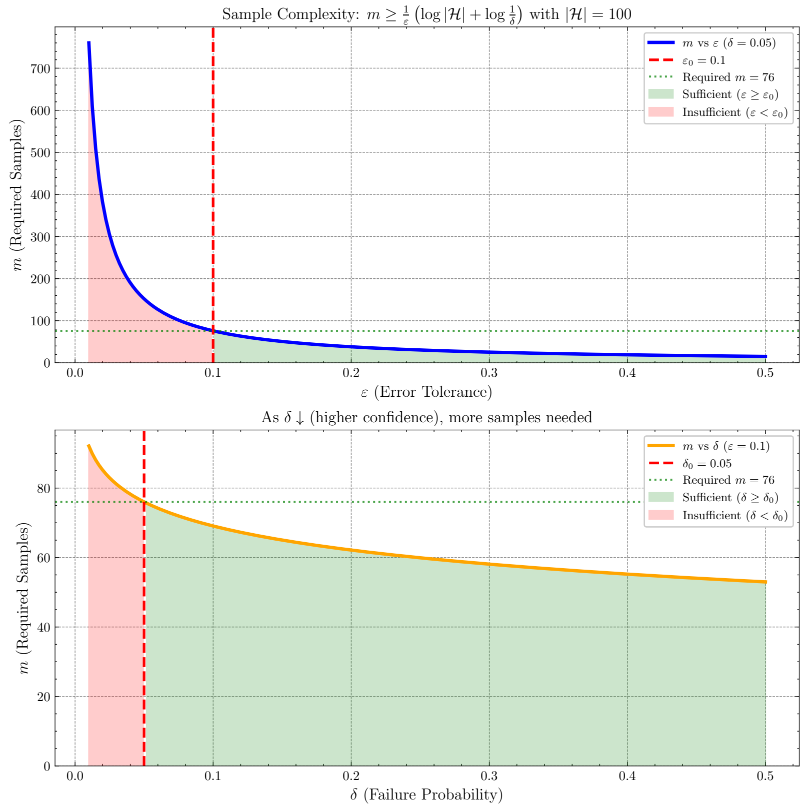


Figure 3.1: Illustration of of sample complexity in terms of cardinality $m = |S|$, with respect to ϵ or δ measure.

The theorem is illustrated as followed in Figure 3.1 and Figure 3.2:

As for inconsistent case, the margin of error exists, thus $\mathcal{H}$ is bounded like the following.

Theorem 3.2.4 (Learning bound - finite $\mathcal{H}$, inconsistent case). *Let $\mathcal{H}$ be a finite hypothesis set. Then, for any $\delta > 0$, with probability at least $1 - \delta$, the following inequality holds:*

$$\forall h \in \mathcal{H}, \quad R(h) \leq \hat{R}_S(h) + \sqrt{\frac{\log|\mathcal{H}| + \log 2/\delta}{2m}} \tag{3.3}$$

Proof. This proof is based on parts of Mohri et al. [2012] and the section about PAC theorems. In the most general case, there may be no hypothesis in $\mathcal{H}$ consistent with the labelled training sample. This, in fact is the typical case in practice, where the learning problems may be somewhat difficult or the concept classes more complex than the hypothesis set used by the learning algorithm.

To derive the learning guarantees in the more general setting, we would use the following corollary, of which relates the generalization error and empirical error of a single hypothesis.

Corollary 3.2.1. *Fix $\epsilon > 0$. Then, for any hypothesis $h : X \to \{0, 1\}$, the following inequalities hold:*

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h) - R(h) \geq \epsilon\right] \leq \exp\left(-2m\epsilon^2\right) \tag{3.4}$$

$$\underset{S\sim\mathcal{D}^m}{\mathbb{P}}\left[\hat{R}_S(h) - R(h) \leq -\epsilon\right] \leq -\exp\left(-2m\epsilon^2\right) \tag{3.5}$$

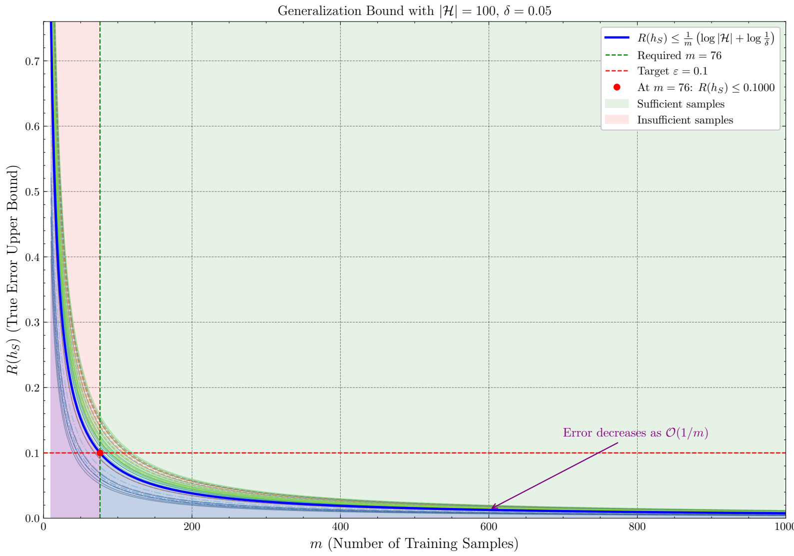


Figure 3.2: Illustration of of generalizational error/risk in terms of cardinality $m = |S|$, with respect to changing $\mathcal{H}$ cardinality and δ-value measure. Blurred line indicates different cases or values, the deviation is rather small, thus the bound is very tight.

By the union bound, this implies the following two-sided inequality:

$$\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}\left[|\hat{R}_S(h) - R(h)| \leq \epsilon\right] \leq 2\exp\left(-2m\epsilon^2\right) \tag{3.6}$$

Proof. This can be proved using Hoeffding inequality. Recall that the inequality is stated for $Z_1, \ldots, Z_n$ independent random variable, for range $z_i \in [a_i, b_i]$ with probability one, $S_n = \sum_{i=1}^n Z_i$ then for all $t > 0$:

$$\mathbb{P}(S_n - \mathbb{E}(S_n) \leq -t) \leq \exp\left(\frac{-2t^2}{\sum(b_i - a_i)^2}\right)$$

and

$$\mathbb{P}(S_n - \mathbb{E}(S_n) \geq t) \leq \exp\left(\frac{-2t^2}{\sum(b_i - a_i)^2}\right)$$

We can derive this for our case of the two empirical and generalization risk. Remember from 6.1.3 that we have $R(h) = \mathbb{E}[\hat{R}_S(h)]$, we can transform the phrase to be:

$$\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}\left[\hat{R}_S(h) - R(h) \geq \epsilon\right] = \mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}\left[\hat{R}_S(h) - \mathbb{E}[\hat{R}_S(h)] \geq \epsilon\right]$$

with respect to $\mathcal{D}^m$. The form of $\hat{R}_S(h)$ has an additional $1/m$, hence, for simplification, we will take the form

$$\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}\left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m\right]$$

By Hoeffding inequality, we have:

$$\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}\left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m\right] \leq \exp\left(\frac{-2m^2\epsilon^2}{\sum(b_i - a_i)^2}\right) \tag{3.7}$$

which is

$$\mathop{\mathbb{P}}_{S\sim\mathcal{D}^m}\left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m\right] \leq \exp\left(\frac{-2m\epsilon^2}{(b-a)^2}\right)$$

Here, our $\sum(b_i - a_i)^2$ was replaced by a more general bound which contains only the sufficient amount of m samples. This is because the sum follows from m samples, which are taken as random variable of choice, then, for $a \leq Z_i \leq b$

$$\frac{m^2}{\sum(b_i - a_i)^2} = \frac{m^2}{m(b-a)^2} = \frac{m}{(b-a)^2}$$

Since our hypothesis is mapping to $\{0, 1\}$, hence our result space is discretely bounded in the normal range, the inequality reduces to

$$\underset{S \sim \mathcal{D}^m}{\mathbb{P}} \left[\hat{R}_S(h)/m - \mathbb{E}[\hat{R}_S(h)]/m \geq \epsilon m \right] \leq \exp\left(-2m\epsilon^2\right) \tag{3.8}$$

Rearranging the left-hand-side to the original yields the inequality. Doing the same for the second case, and using the union bound, we get the two-sided inequality for $|\hat{R}_S(h) - R(h)|$. □

Such bound can be naturally extended toward any arbitrary range $[a, b]$, as for Hoeffding inequality being defined on the arbitrary closed interval instead. Then, we have the general bound

$$\underset{S \sim \mathcal{D}^m}{\mathbb{P}} \left[|\hat{R}_S(h) - R(h)| \leq \epsilon \right] \leq 2 \exp - \frac{2m\epsilon^2}{(b-a)^2} \tag{3.9}$$

Setting the right-hand side of the final expression to be equal to δ and solving this for ϵ yields immediately the following bound for a single hypothesis.

Corollary 3.2.2 (Generalization bound - single hypothesis). *Fix a hypothesis $h : \mathcal{X} \to \{0, 1\}$ or more generally onto $h : \mathcal{X} \to [a, b] \subset \mathbb{F}$. Then, for any $\delta > 0$, the following inequality holds with probability at least $1 - \delta$:*

$$R(h) \leq \hat{R}_S(h) + \sqrt{\frac{\log 2/\delta}{2m}} \tag{3.10}$$

The desired bound can then be realized as extending the proved corollary bound, by noticing that,

$$\hat{R}_S(h) + \sqrt{\frac{\log 2/\delta}{2m}} < \hat{R}_S(h) + \sqrt{\frac{\log |\mathcal{H}| + \log 2/\delta}{2m}} \tag{3.11}$$

for all $|\mathcal{H}|$. □

For a looser bound, we can ultimately use an ϵ-consistent learner criteria in such case. PAC-learning setting raises some peculiar insight toward the problem itself. First, is the learning bound itself of which raises the importance of several factors — the dataset size, m, of which is used as the **total resources required** for approximating such function. Furthermore, the set complexity $\mathcal{H}$ is also used, albeit questionably inconsistent in the term $\log |\mathcal{H}|$, which implies at least consideration on the cardinality of the hypothesis class. In a finite domain, we notice that PAC-learning of consistent case is exactly the case of interpolation, assuming non-smooth and noisy data, and the inconsistent case describe the wider range of approximation-based case, but does not consider an approximation-aware bound. The design of $|\mathcal{H}|$ also does not allow for certain exact hypothesis class size measurement, except for combinatorial case where $\mathcal{H}$ is indeed finite. Nevertheless, the bound indeed indicates that for m increase, the hypothesis class often do much better in general. One particularly troublesome notion of which we will meet, in the inconsistent case, is the reversal of guarantee compared to bias-variance form. Specifically, the inconsistent, model-agnostic version of Theorem 3.2.4 is as followed:

Theorem 3.2.5 (Inconsistent case, model-agnostic, finite $\mathcal{H}$). *For finite $\mathcal{H}$, then for any $\delta > 0$, up to ϵ-accuracy, with probability at least $1 - \delta$, the following inequality holds:*

$$R(h_S) \leq \min_{h_S \in \mathcal{H}} \hat{R}_S(h_S) + \sqrt{\frac{\log |\mathcal{H}| + \log 2/\delta}{2m}} \tag{3.12}$$

such that

$$\sqrt{\frac{\log|\mathcal{H}| + \log 2/\delta}{2m}} \leq \epsilon \tag{3.13}$$

for $m > 0$ samples. We also say that the error varies by $\mathcal{O}(\sqrt{1/m})$), such that

$$R(h_{\mathcal{S}}) \leq \min_{h_{\mathcal{S}}\in\mathcal{H}} \hat{R}_{\mathcal{S}}(h_{\mathcal{S}}) + \mathcal{O}\left(\sqrt{\frac{\log|\mathcal{H}|}{m}}\right) \tag{3.14}$$

We denote, again, $\hat{R}_{\mathcal{S}}$ the empirical risk given on set $\mathcal{S}$ of observations, and $h_{\mathcal{S}}$ the hypothesis obtained from dynamics regarding such observational set.

This theorem directly indicates that, for bigger $|\mathcal{H}|$, the rate of which m increases can also mitigate bias-variance, thus bigger dataset can mitigate such issues of overfitting. Indeed, we notice that $\mathcal{O}(\log|\mathcal{H}|) \leq \mathcal{O}(m)$ linear terms, thus if the algorithm $\mathcal{A}$ exhibits such dynamics in which dataset varies, bigger hypothesis space should not increase significantly of the error measure. However, this contradicts, in a sense, with the usual term of notion to bias-variance, of which suggest otherwise, hence overfitting. Or, rather insidiously speaking, the interpretation is wrong, such that bias-variance concern fixed m situation, while such bound indicates, or at least suggest, of variational m.

While this does not cover the case of which $|\mathcal{H}|$ is argued to be infinite, such is then also be bounded, as for any measure, temporary called $\mu : \mathcal{H} \to \mathcal{F}$ for field $\mathcal{F}$ of which can be evaluated to $\mathbb{R}$ or else, then such is bounded in the contribution terms, and thus of the framework in which we use to express the learning dynamics. It is also then to be concerned of different classes of $\{\mu\}$ itself, as a limiting factor. This, perhaps, will lead to a potential solution of double descent.

Indeed, a loose analogy, or rather speculation can follow. While criticisms of classical uniform-convergence PAC learning correctly observe that one-dimensional capacity sweep indeed, fail to predict or explain the modern double descent curve, such cannot be said in certain cases of the inconsistent setting. Suppose we introduce the parameter ρ of the growth regime on $\mathcal{M}(\mathcal{H}) : \mathcal{H} \to \mathbb{F}^k$ of the complexity measure or the size of the hypothesis space, in which usually $k = 1$ and the value is on the real field, for $\gamma = |\mathcal{S}|$ as the sample size, such that we have τ correlates to a particular trajectory in which hence the escape trajectory is then controlled subsequently. We can then interpret the τ parameter as encoding the moving **interpolation threshold**. The parameter τ then can be computed, accordingly for the optimization term as

$$\mathcal{O}\left(\sqrt{\frac{\log|\mathcal{H}|}{m}}\right) \equiv \sqrt{\frac{\log\tau + \log 2/\delta}{2\tau}} \tag{3.15}$$

This particular trajectory, can indeed, create certain cases where the right-half of the curve is similar to the trajectory of double descent. In general, investigation would see for

$$\mathcal{O}\left(\sqrt{\frac{\log|\mathcal{H}|}{m}}\right) \equiv \sqrt{\frac{\log q(\tau) + \log 2/\delta}{2q(\tau)}} \tag{3.16}$$

for arbitrary function $q(\tau), p(\tau)$ where it should exhibit double descent terminal terms. However, this task is really hard aside from simply guessing. A simple guess on structural reformulation would be that

$$\min_{h_{\mathcal{S}}\in\mathcal{H}} \hat{R}_{\mathcal{S}}(h_{\mathcal{S}}) \approx \frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right), \tag{3.17}$$

$$\mathcal{O} \approx I\sqrt{\frac{log(\tau - p) + \log(2/\delta)}{2(\tau - p)^{\epsilon} + 1}} \tag{3.18}$$

Thus the sum would be defined in a more orthodox way, as

$$\frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right) + I\sqrt{\frac{log(\tau - p) + \log(2/\delta)}{2(\tau - p)^{\epsilon} + 1}} \tag{3.19}$$

for arbitrary constant I, p, of which now the addition operator would be slightly adjust, such that the sum will contain only the LHS of the operator if the RHS is in its undefined domain, and otherwise they add on together. The factor $(\tau - p)$ appears in both previously the sample size m and the hypothesis space size $|\mathcal{H}|$. This result is highly speculative, as it is an educated guess on the error landscape, however, still produces the approximated term on double descent. Such is then of the following hypothesis.

Hypothesis 3.2.1 (The flawed PAC-descent). *Let $\mathcal{H}$ be the hypothesis class, for moving parameter τ called the* ***interpolation threshold****. We define the interpolation offset region, $\Delta(\tau) \equiv \Delta := \tau - p$, and the regularized deviation term $\eta, \kappa > 0$. Then, for specific $q(\tau)$, where we set $\tau = |\mathcal{H}|$ of the hypothesis class size, and $\tau = m$ case of the number of samples on $\mathcal{S}$, then $R(h_{\mathcal{S}})$ is bounded by, for at least probability of $1 - \delta$:*

$$R(h_{\mathcal{S}}) \leq \begin{cases} \frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right) & \Delta \leq -\gamma \\ \frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right) + \lambda(\Delta) I\sqrt{\frac{log(\tau-p)+\log(2/\delta)}{2(\tau-p)^{\epsilon}+1}} & |\Delta < \gamma| \\ \frac{1}{m}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right) + I\sqrt{\frac{log(\tau-p)+\log(2/\delta)}{2(\tau-p)^{\epsilon}+1}} & |\Delta < \gamma| \end{cases} \tag{3.20}$$

for $I > 0$, $\lambda(\Delta) \in [0, 1]$ the smooth terms on transition region, where taking $\gamma = 0$, $\lambda(0) = 1/2$ returns the discrete separation case.

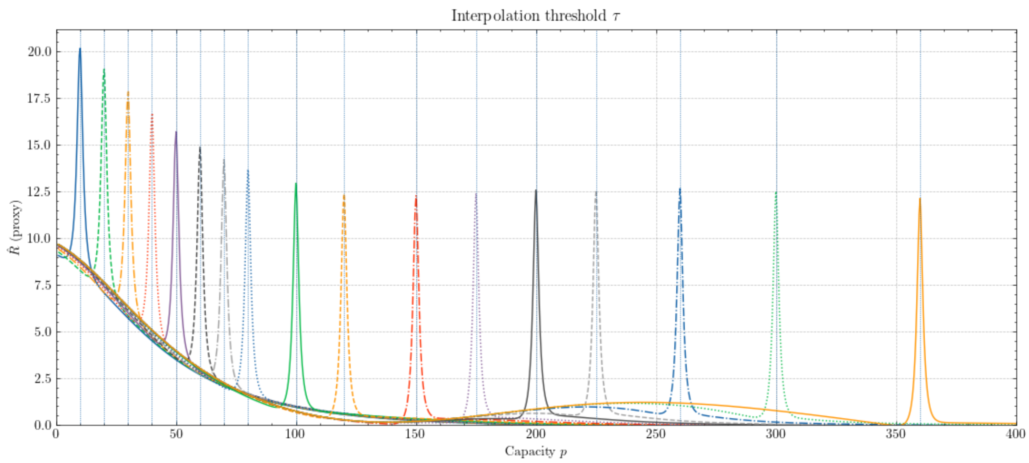


Figure 3.3: Prediction made by certain configurations of Hypothesis 3.2.1.

While imperfect, this encodes in the simplest form of PAC-system, why such bound can somewhat say of the double descent phenomena, in terms of the singular trajectory error instead. But it also reveals the problematic trouble within doing this, essentially move the theory into the untested and unprepared zone in which orthordox choices has to be made, but no interpretation is available. The ineffective side of such derivation is that:

1. Ill-defined terms, $\log(\tau - p)$ is extremely brittle and thus makes for an interpolation threshold, but theoretically, this is inexplainable, also for $(\tau - p)^{\epsilon}$ for the error ϵ bounded. Even with the regularized term, such is forced on because of necessity, not structural reason.

2. Circularity of the concept is that we are treating on the assumption of the escape trajectory, i.e. $(\tau - p)$, will fit both the hypothesis space size and the sample size. Indeed, such procured the double descent curve, but does not explain how and why.

3. Continuity and unit mismatch is indeed troublesome, as there exists no way to tell specific $q(\tau)$ that preserves the parameterization to τ. Continuity means having to form piece-wise terms, hence we result in an abrupt bounding instead.

Inherently, τ is also empirical, and thus $q(\tau)$. Such expression in general appears in KL-convergence of PAC-Bayes, and linear models isotropic design gives $\sigma^2 p/(m-p-1)$, which is fairly similar to what we have. Nevertheless, one should realize that current theories are not equipped to express this in particularly detailed insight. In certain cases, it only serves as heuristics or empirical presumptions, or rather theoretical speculation in a non-constructive way. Imitating double descent aside, this prompts us to seriously tackle the double descent problem, or otherwise the messiness of analysis will continue.

3.3 The bias-variance - double descent dilemma

Before continuing further, we must review the problem statement that led to the problem of double descent. We will have a look at the theory, the original conflict, and the illustrative phenomenon in action. In general, the learning setting in consideration of statistical learning theory is often supervised learning, for guarantee of a plausible upper bound result.

3.3.1 Bias-variance tradeoff

In classical sources, James et al. [2009], Goodfellow et al. [2016], Hajek and Raginsky [2021], Mohri et al. [2012], Shalev-Shwartz and Ben-David [2014], the principle of bias-variance tradeoff is a classical principle used in the categorization of technique called *model selection*. Specifically, this refers to the progress of choosing the best predictor h^* that minimize the generalization error that is the main goal of a particular machine learning model, during the process of training and inference. Originally, the theory that bias and variance often come in to conjunction is **Estimation theory**, of which there exists an estimator $\hat{\theta}$ that use a set of observations, to estimate the concept, or the underlying mechanics of certain system that outputted the observation set, by the **probabilistic perspective** - that is, it is governed by appearance by an independently and identically distributed process of parameterized probability $\theta \in \Theta$. Here, parameterized means being expressed by a set, often finite, of parameters, by E. L. Lehmann [1998], Paninski [2005]. It is only when Geman et al. [1992] introduced it to machine learning that we have an equivalent structure that is similar to bias-variance in machine learning.

Precursor (Geman et al. [1992])

The original definition of bias-variance tradeoff by Geman et al. [1992] is first constructed using the means-square error, which is regarded as a normal measure in the real encoding space. Their approach is to justify bias-variance via decomposition of the loss function ℓ, for such to find an alternative reasonable form of such loss landscape. Suppose of a regression problem to construct a hypothesis function $f(x)$ from $(x_1, y_1, \dots, x_N, y_N)$ for the purpose of generalization - that is, predicting unseen variational values for different pair $(x_j, ?)$ such that $? = y_j + \epsilon$ for a conceivable implicit error. To be explicit about the relation of this problem, or f on the given data $\mathcal{D} = \{(x_i, y_i) \mid i \le N\}$, denote $f(x; \mathcal{D})$ instead of f, the natural mean-square measure as a predictor is:

$$\mathcal{M}(f, y) = \mathbb{E}\left[((y - f(x; \mathcal{D})))^2 \mid x, \mathcal{D}\right] \tag{3.21}$$

for $\mathbb{E}[\cdot]$ the expectation wrt to a distribution P. Decomposing the right-hand side, we have:

$$\begin{aligned}\mathcal{M}(f, y) &= \mathbb{E}\left[((y - f(x; \mathcal{D})))^2 \mid x, \mathcal{D}\right] \\ &= \mathbb{E}\left[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D}\right] + (f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2\end{aligned} \tag{3.22}$$

Here, $\mathbb{E}\left[(y - \mathbb{E}[y \mid x])^2 \mid x, \mathcal{D}\right]$ does not depend on $\mathcal{D}$, but simply the statistical variance of y given x. The term $(f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2$ is considered a natural measure of effectiveness on $\mathbb{R}^n$ as a singular predictor of y. Now, for $\mathbb{E}_{\mathcal{D}}\left[(f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2\right]$ which depends on the training set $\mathcal{D}$ in its computation, is decomposed into the form of *bias-variance decomposition* terms, by derivation:

$$\begin{aligned}&\mathbb{E}_{\mathcal{D}}\left[(f(x; \mathcal{D}) - \mathbb{E}[y \mid x])^2\right] \\ &\quad = \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})] - \mathbb{E}[y \mid x]\}^2}_{\text{bias term}} \\ &\quad + \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})])^2\right\}}_{\text{variance term}}\end{aligned} \tag{3.23}$$

We summarize this in the following statement.

Theorem 3.3.1 (Bias-variance decomposition). *Suppose the model $f(x;\mathcal{D})$ for the data $\mathcal{D} = (x_i, y_i)$ and its parameter x is defined. For y_i of the target concept's responses y, and consider a regression problem with the loss measure $\mathcal{M}(f, y)$ of mean squared risk, the following statement is true:*

$$\begin{aligned}\mathbb{E}\big[\mathcal{M}(f,y)\big] &= \mathcal{B}(f,y) + \mathcal{V}(f,y)\\ &\quad + \mathbb{E}\Big[\mathbb{E}\left[((y - f(x;\mathcal{D})))^2 \mid x, \mathcal{D}\right]\Big]\end{aligned} \tag{3.24}$$

for $\mathbb{E}[\,\cdot \mid x, \mathcal{D}]$ any expression with dependencies on x and $\mathcal{D}$. The bias and variance term is subsequently expressed by

$$\mathcal{B}(f,y) = \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})] - \mathbb{E}[y \mid x]\}^2}_{bias}, \tag{3.25}$$

$$\mathcal{V}(f,y) = \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x;\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})])^2\right\}}_{variance} \tag{3.26}$$

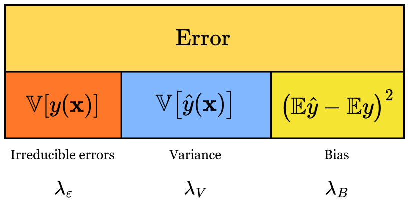


Figure 3.4: **Decomposition of the error term into 3 parts.** Respectively, the irreducible error $\mathbb{V}[y(\mathbf{x})]$, the variance (blue) and the bias (dark yellow). All of them are assumed to take up 100% of the error observed in such composition. The proportion is included using the three coefficient $\lambda_\epsilon, \lambda_V, \lambda_B$ where $\lambda_\epsilon + \lambda_V + \lambda_B = 1$.

The above decomposition principle is often expressed into a form where there exists the intrinsic noise Brown and Ali [2024]:

$$\begin{aligned}\mathbb{E}_D\left[\mathbb{E}_{xy}\left(y - \hat{f}(x)\right)^2\right] &= \mathbb{E}_x\left[\left(y^* - \mathbb{E}_D[\hat{f}(x)]\right)^2\right]\\ &\quad + \mathbb{E}_x\left[\mathbb{E}_D\left(\hat{f}(x) - \mathbb{E}_D[\hat{f}(x)]\right)^2\right]\\ &\quad + \mathbb{E}_{xy}\left[(y - y^*)^2\right]\end{aligned} \tag{3.27}$$

In all, almost all representation of bias-variance decomposition are the same. For simplicity of notation, we adopt the similar form in the standard case of the paper Adlam and Pennington [2020]:

$$\mathbb{E}\left[\hat{y}(\mathbf{x}) - y(\mathbf{x})\right]^2 = (\mathbb{E}\hat{y}(\mathbf{x}) - \mathbb{E}y(\mathbf{x}))^2 + \mathbb{V}\left[\hat{y}(\mathbf{x})\right] + \mathbb{V}[y(\mathbf{x})] \tag{3.28}$$

A main common theme of criticism toward bias-variance tradeoff is the fact that the decomposition is much more general, and intrinsic for the class of *mean squared loss*. However, when considering the naturalness of an error measure and then its direct applicant, mean square error on real space of sufficient support and measure comes of as a very natural choice of an error measure, at least in consideration of the regression setting; for that, we then can also define classification as another top-layer above a regression's similar real continuous space, i.e. a decision layer. Furthermore, it can also be shown Brown and Ali [2024], Pfau [2013] that it also holds for the class of Bregman divergence measure.

What does the decomposition say about model selection principle in general? In general, bias-variance is typically presented of its decomposition only, and the asymptotic prediction of its behaviours. In fact, one of the reason that it became the rule-of-thumb for ML practitioner, as well as generally statistical learning (Lafon and Thomas [2024] provides a quite rigorous treatment of bias-variance tradeoff in the section on statistical learning theory) solidify the trade-off as a particular model selection principle. Generally, this tradeoff can be summarized as followed:

Theorem 3.3.2 (Bias-variance tradeoff). *For the expected loss of any given hypothesis h, the bias $\mathcal{B}(f,y)$ and variance $\mathcal{V}(f,y)$ is inversely proportional, that is, $\mathcal{B}(f,y) \propto \lambda^{-1}\mathcal{V}(f,y)$ for some proportionality λ that may or may not be constant. In the most general case possible, $\lambda = -1$ on the entire error range.*

The theorem can be typically seen as a conjecture instead. Regardless, the tradeoff is then stated to be of inverse proportionality. Indeed, statistically, we have such tradeoff on a statistical framework in a more concrete sense. For the bias to increase, variance will increase, of which the criterion is inverse - we would like to have more bias but lower variance, and vice versa, according to such theory. Such drives the theory to choose on its model basing on the two interpretations - of increasing capacity but lead to accumulating too much noise in typical statistically noisy systems, or reducing them too much that leads to incapacity to realize higher-order patterns. Thus, variance aims to reduce the *volatility* of the model performance across admitted region of operation, and bias aims to reduce the inherent errors for under-configured models. That said, if we do not want to simply let them be related of strict equality, we can establish an inequality instead, that is, for

$$\mathbb{E}_D\left[\mathbb{E}_{xy}\left(y - \hat{f}(x)\right)^2\right] \geq \mathbb{E}_x\left[\left(y^* - \mathbb{E}_D[\hat{f}(x)]\right)^2\right] + \mathbb{E}_x\left[\mathbb{E}_D\left(\hat{f}(x) - \mathbb{E}_D[\hat{f}(x)]\right)^2\right] \tag{3.29}$$

We do not inherently claim that noise will add to the fix of the equality as a case, intentionally to capture this gap between them. We do note that the equality often only holds in cases of mean-square error on Euclidean metric space. Per usual, we would like to use the abstract notion between $\mathcal{B}$ and $\mathcal{V}$ instead of explicit metric.

Expansion and variations

The bias-variance decomposition and tradeoff is not general. Indeed, works, by Geman et al. [1992], Sharma and Aiken [2014], Domingos [2000d], Adlam and Pennington [2020], Yang et al. [2020] and more all attempted to reconstruct and reformulate bias-variance, and it did leave out a picture of uncertainty regarding the concept. Specifically, there are certainly different way to do the bias-variance decomposition, as classified in Adlam and Pennington [2020] paper. Some of this are particularly specialized in *neural network* training, basing on the main structure of modern machine learning in neural network formalism.

1. **Semiclassical approach**: The semiclassical approach is fairly simple. We can rewrite to another variation of the classical approach to be

$$\begin{aligned}\mathop{\mathbb{E}}_{h\in\mathcal{H}}[\ell(h)] &= \mathbb{E}_x\mathbb{E}_\epsilon\left[\hat{y}(x) - y(x)\right]^2 \\ &= \mathbb{E}_x\left(\mathbb{E}_\epsilon[\hat{y}(x)] - y(x)\right)^2 + \mathbb{E}_x\mathbb{V}_\epsilon\left[\hat{y}(x)\right]\end{aligned} \tag{3.30}$$

   and instead, consider the *structural conformation* of the decomposition, or rather, taking the expectation within $\theta \in \Theta$ of the structural description of $h \in \mathcal{H}$. Then, we shall have

$$\begin{aligned}\mathop{\mathbb{E}_{\mathrm{SC}}}_{h\in\mathcal{H}}[\ell(h)] &= \mathbb{E}_x\mathbb{E}_\theta\mathbb{E}_\epsilon\left[\hat{y}(x) - y(x)\right]^2 \\ &= \mathbb{E}_x\mathbb{E}_\theta\left(\mathbb{E}_\epsilon[\hat{y}(x)] - y(x)\right)^2 + \mathbb{E}_x\mathbb{E}_\theta\mathbb{V}_\epsilon\left[\hat{y}(x)\right]\end{aligned} \tag{3.31}$$

   Do note that the order matters, since the expectation can be constant of a given ordering. And expected value over constant is just the constant.

2. **Symmetric decomposition**: This approach is fairly difficult, we refer to the paper itself of its own justification on the decomposition. The symmetric decomposition relies on the shape of the input, $\mathcal{X} = \{P, D, \epsilon\}$, of which you can remove the ϵ term and retain some decomposition on P and D per distribution.

Generalized decomposition

According to Domingos [2000b,a], the bias-variance decomposition can usually be treated as being controlled by three factor error, with their respective coefficients $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ for such:

$$
\begin{aligned}
&\mathbb{E}_{\mathcal{D}}\left[d((f, \mathbb{E}[y \mid x])\right] \\
&\quad = \lambda_1 \mathrm{Bias}(f, y) + \lambda_2 \mathrm{Var}(f, y) + \lambda_3 \epsilon(\mathcal{D}) \\
&\quad = \lambda_1 \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})], \mathbb{E}[y \mid x]\}}_{\text{bias term}} \\
&\qquad + \lambda_2 \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x;\mathcal{D}), \mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})])\right\}}_{\text{variance term}} \\
&\qquad + \underbrace{\lambda_3 \epsilon}_{\text{irreducible error}}
\end{aligned}
$$

where $\epsilon(\mathcal{D})$ is the irreducible error of the system, depends (theoretically) on the intrinsic imperfection of the dataset $\mathcal{D}$. This method of which the bias-variance-irreducible error are 'fit' into the total error by three coefficients $\lambda_1, \lambda_2, \lambda_3$ instead. This does not only scale up the B-V-IE triplet, but also leaves out the inexplainable gap between the scaling, and the actual normal measure.

Approximation-estimation tradeoff

Another closely related notion to bias-variance is the concept of approximation-estimation tradeoff. We refer to Mohri et al. [2012], Lafon and Thomas [2024] for some analysis and mentions.

Let $\mathcal{H}$ be a family of functions mapping $\mathcal{X} \to \{1, -1\}$. This is the particular case of **binary classification**, in which can be straightforwardly extended to different tasks and loss functions. The *excess error* of a hypothesis $h \in \mathcal{H}$, is the difference between its error $R(h)$ and the Bayes error R^*. This can be decomposed to be the following:

$$R(h) - R^* = \left(R(h) - \inf_{h \in \mathcal{H}} R(h)\right) + \left(\inf_{h \in \mathcal{H}} R(h) - R^*\right) \tag{3.32}$$

The first bracket contains the **estimation error**, and the second bracket contains what is called the **approximation error**. The estimation error depends on the hypothesis h selected. It measures the error of h with respect to the infimum of the error achieved by hypotheses in $\mathcal{H}$, or that of the best-in-class hypothesis h^* when that infimum is reached. The approximation error measures how well the Bayes error can be approximated using $\mathcal{H}$. It is a property of the hypothesis set $\mathcal{H}$, a measure of its richness.

Model selection consists of choosing $\mathcal{H}$ with a favourable trade-off between the approximation and the estimation error. However, in the most general case, this will be done, but not in practice, as it requires the underlying distribution $\mathcal{D}$ to be known to determine R^*, which is not possible. In contrast, the estimation error can be bounded, or can be analysed, using particular metric and analysis. It is however, worth to note that bias-variance and approximation-estimation have a very complicated relationship, for example, in Brown and Ali [2024] shown that they are indeed not the same, and is in fact two different decomposition, where one is the others' component. To see this, let us compare the two of them in details.

3.3.2 Double descent

Nevertheless, of bias-variance and the analogous statistical learning theory concept, the target is the same. It is the dilemma of which is presented in **Occam's razor**, for choosing the sufficient model of good complexity, or bias, for tradeoff of its generalization ability, or variance. Then there must exist a sweet spot between the axis of bias and variance, since they are as exhibited above inversely proportional to each other. However, double descent seemingly broke the status quo, and insists on an interesting phenomenon - under the same setting, if we 'crank' the complexity high enough, we will then reach a point then called the **interpolation threshold**, such that the trend reverse and the error rate, instead of being theorized to go up, goes down to a certain line of lower bound.

The first identification of the double descent phenomena dated back to the paper of Belkin - Belkin et al. [2019a], in which the title is literally "reconciling" modern machine learning practice and the bias-variance tradeoff. In modern machine learning practice, or state-of-the-art developments, models are now bigger than ever. If to notice,

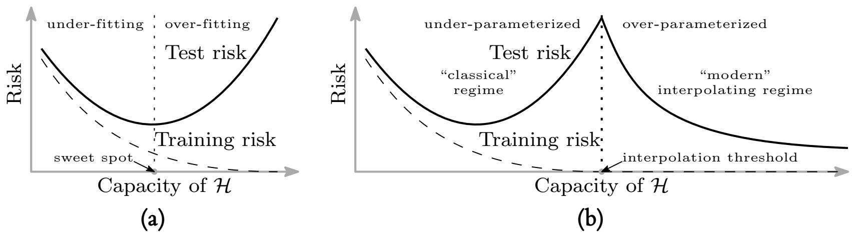


Figure 3.5: **Curves for training risk (dashed line) and test risk (solid line).** (a) The classical *U-shaped risk curve* arising from the bias-variance trade-off. (b) The *double descent risk curve*, which incorporates the U-shaped risk curve (i.e., the "classical" regime) together with the observed behaviour from using high capacity function classes (i.e., the "modern" interpolating regime), separated by the interpolation threshold. The predictors to the right of the interpolation threshold have zero training risk. Reproduced from Belkin et al. [2019a].

we will see that currently models are inherently large, for example, a normal large language model will have from 900 millions (900M) to a few billions, for example 10 billions (10B) parameters. That is not taking into account the overall dynamics and structure of the model, which dictates the operating range and efficiency of the model itself. These model, based on the neural network architecture are somewhat trained to exactly fit (or interpolate) the data, almost certainly so that it turn from a prediction setting to an estimation setting. By statistical learning theory, this would be considered overfitting, and yet, they often obtain very high accuracy on test data.

The main finding that Belkin found is a pattern for how the apparent performance on unseen data depends on model capacity and the mechanism underlying the emergence of double descent. When function class capacity is below the "interpolation threshold", learned predictors exhibit the classical U-shaped curve from Figure 3.5. The 'modern' interpolating regime marks the opposite trend to the right, where the risk starts to decrease up to a lower bound, which then can be called the *optimal descent bound*.

> *The bottom of the U is achieved at the sweet spot which balances the fit to the training data and the susceptibility to over-fitting: to the left of the sweet spot, predictors are under-fit, and immediately to the right, predictors are over-fit. When we increase the function class capacity high enough (e.g., by increasing the number of features or the size of the neural network architecture), the learned predictors achieve (near) perfect fits to the training data—i.e., interpolation. Although the learned predictors obtained at the interpolation threshold typically have high risk, we show that increasing the function class capacity beyond this point leads to decreasing risk, typically going below the risk achieved at the sweet spot in the "classical" regime.* (Belkin et al. [2019a])

At this point, we would like to also clarify the term interpolation threshold in this case. Usually, under the pretext of overfitting and underfitting, interpolation threshold means the following:

Definition 3.3.1 (Interpolation threshold). *Assuming a typical supervised learning setting, for a hypothesis h within class structure $\mathcal{H}$, and a supposed concept c of the class $\mathcal{C}$. Suppose $[X, Y] \subset c \in \mathcal{C}$ being the dataset, or observable space of c on which h is tested on. Under the* ***empirical generalization learning*** *procedure, partitioning $[X, Y]$ into $[X, Y]_1$ and $[X, Y]_2$, for error evaluated on the first set, $\hat{R}_1(h)$ called training error, and the second set, $\hat{R}_2(h)$ called test error, with optimization applied on the first set, then evaluated statically on the second, the* ***interpolation threshold*** *with respect to a particular complexity measure of h, $\mathcal{M}(h)$, is the point $\mathcal{M}_0(h)$ where it is possible for $\hat{R}_2(h)/\hat{R}_1(h) \approx 1$, for $\hat{R}_1(h) \ll \epsilon$, $\hat{R}_2(h) \ll \epsilon$, with $\epsilon \geq 0$.*

This basically means that what is learnt, with respect to the model complexity that gauge *how much* and *what* the model can learn, perfectly fits the concept c itself, from the observation space on its own. We can say that, that specific sense, we gain the assumption that the observation space is *fully representative* of the concept class, and the concept class is then fully realized by the model. Such point where that is possible, or there then such behaviour is possible, is called the interpolation point. The definition relies on the statement 'where it is possible', hence it still, does not give us a particularly sound notion on to how and what constitute such possibility. The term empirical generalization learning comes from the fact that the procedure in practice of estimating and optimizing train-test-validation set in itself, is an imperfect empirical approximation of generalization testing. While the fact that we cannot gauge generalization error, in a purely blind black-box setting is settled (Mohri et al. [2012], Sugiyama [2015], Shalev-Shwartz and Ben-David [2014]), the method is then to try to mimic such by using generalization-via-hidden observations. To facilitate this, deliberately, the empirical learning procedure is only dynamic during training session, and static throughout the whole testing and validation testing section, and further on.

By default, in general theory, such interpolating point is dangerous. Not only because it is inherently difficult to know exactly what the dataset constitute, whether it is actually representative of the concept that gives rise to it, or the structural information, noises, and interferences in between. It is hence impossible, to a certain point, to truly know the effectiveness of the model beside interpolating the current dataset which forms the observable details.

Another prominent discovery to look at is Nakkiran et al. [2019], on the double descent of deep learning models. The paper serves as the first observation upon which double descent is observed on modern, deep learning theoretic models, especially complex one like Transformer or ResNet networks.

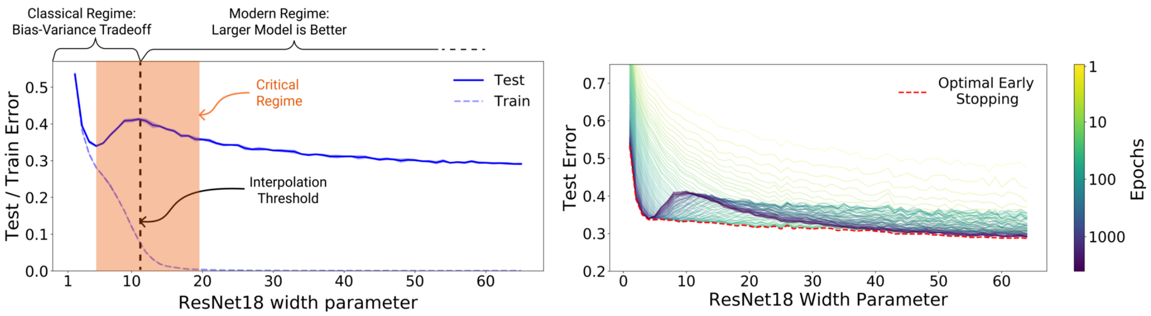


Figure 3.6: **Left**: Train and test error as a function of model size, for ResNet18s of varying width on CIFAR-10 with 15% label noise. **Right**: Test error, shown for varying train epochs. All models trained using Adam for 4K epochs. The largest model (width 64) corresponds to standard ResNet18. Resued from Nakkiran et al. [2019].

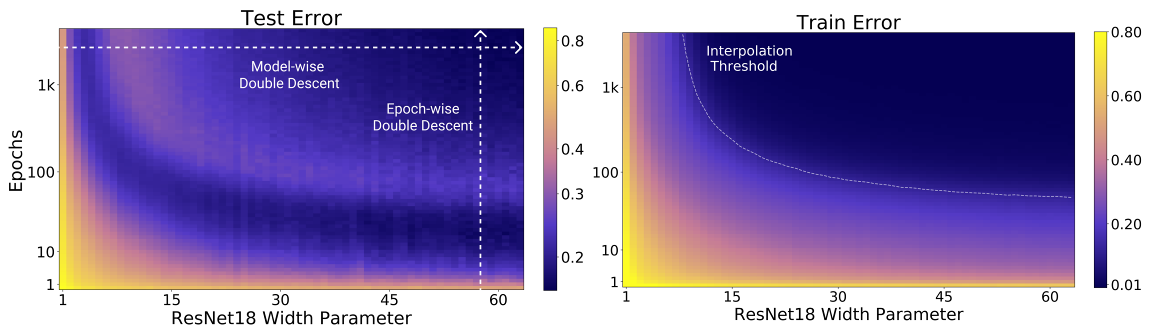


Figure 3.7: **Left**: Test error as a function of model size and train epochs. The horizontal line corresponds to model-wise double descent–varying model size while training for as long as possible. The vertical line corresponds to epoch-wise double descent, with test error undergoing double-descent as train time increases. **Right** Train error of the corresponding models. All models are Resnet18s trained on CIFAR-10 with 15% label noise, data-augmentation, and Adam for up to 4K epochs. Reused from Nakkiran et al. [2019]

They define *effective model complexity* of $\mathcal{T}$ (w.r.t. distribution $\mathcal{D}$) to be the maximum number of samples n on which $\mathcal{T}$ achieves on average ≈ 0 *training error*. This is an entirely empirical definition, similarly per definition as VC-dimension.

Definition 3.3.2 (Effective Model Complexity). *The* Effective Model Complexity *(EMC) of a training procedure $\mathcal{T}$, with respect to distribution $\mathcal{D}$ and parameter $\epsilon > 0$, is defined as:*

$$\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) := \max\left\{n \mid \mathbb{E}_{S\sim\mathcal{D}^n}[\mathrm{Error}_S(\mathcal{T}(S))] \leq \epsilon\right\}$$

where $\mathrm{Error}_S(M)$ *is the mean error of model M on train samples S.*

Using this definition, their main hypothesis can then be stated as the following three-fold regions:

Hypothesis 3.3.1 (Generalized Double Descent hypothesis, informal). *For any natural data distribution $\mathcal{D}$, neural-network-based training procedure $\mathcal{T}$, and small $\epsilon > 0$, if we consider the task of predicting labels based on n samples from $\mathcal{D}$ then:*

Under-parameterized regime. *If* $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ *is sufficiently smaller than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*

Over-parameterized regime. *If* $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T})$ *is sufficiently larger than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.*

Critically parameterized regime. *If* $\mathrm{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) \approx n$*, then a perturbation of $\mathcal{T}$ that increases its effective complexity might decrease* **or increase** *the test error.*

It is to notice that this definition is also only observational. By that, it only outlines the specific region of interest where the paradigm shift is identified through experimental results. It has no effective predictive power, or rather descriptive power more than setting up hypothesis with respect to the effective model complexity, and some arbitrary perturbation. Nakkiran et al. [2019] also noticed this difficulty in providing a theoretical definition and theorem regarding such hypothesis, as said in their manuscript. The existence of the critical regime is also not defined.

The behaviour itself has been particularly investigated, in certain settings. For example, before Belkin et al. [2019a], Advani and Saxe [2017b] investigated the generalization error in neural networks of high-dimensional measure. Of such, various behaviours that bear similarity to double descent can be observed. Belkin et al. [2018] expanded on his previous research on the particular subset of kernel learning models, providing a theoretical analysis of specifically the Laplacian kernel on standard neural network. Mei and Montanari [2020] investigated random feature regression in such regard, and also found similar results.

The fact that double descent is not well-defined itself is a problem on its own. Up to the author's knowledge and of myself, there has been no effective definition or given description that outlines specifically how double descent can be structured. Instability in reproducing experiments and the effective range of the phenomena remains a question on its own.

3.4 What is double descent, then?

Ever since Belkin et al. [2019a], despite the general chaotic view of double descent that seems to hold up pretty well, many researches have been conducted throughout, trying to observe or explain it, as we briefly mentioned Nakkiran et al. [2019] as one of such main paper. Before attempting to do anything, and to further contribute with hypotheses forward, we will have to resolve to review previous coursework on such matter. Most of those works include some on theoretical analysis, some on dissection of optimization method then speculation of possible hindsight, some on experiments and observable effects, and so on. Furthermore, sections specifically designed to host experiments and observations will also offer a clearer insight into how this can be done.

3.5 Current researches

The main goal of attacking double descent stemmed from the observation in which there exists irregularity in the model expression behaviours. As such, the first thing to do is to contrive the consensus on 'what' constitutes double descent, at least per observable results, by delivering a conjecture. We deliver the phenomenological conjecture as below.

Conjecture 3.5.1. *On the landscape of double descent, model complexity versus generalization error, there exists a fundamental point $\mathcal{M}_0$ of model complexity axis, where $dR(h)/d\mathcal{M} < 0$ for $\mathcal{M} > \mathcal{M}_0$, which is the interpolation point.*

While this is not perfect, it provides us with a general idea on where to look for double descent, as the *resultant action* of the model complexity observation on learning error. The assumption here, with respect to previous bias-variance disposition, is also that such error correlates strictly to the general risk. Typically, question remains as to see:

1. What is exactly double descent? As far as we can say, there exists the interpolation threshold in which the dynamic changes, but don't know why. Such exists and occur within the complexity of deep neural networks, and so is also analogous form in classical models. It is suspected, reasonably so, that it is related to the terms of bias-variance, however, it is not fully developed and no further inquiry is made beside the above conjecture as empirical observations.

2. Where does it lead? What is the setup? In truth, the setup problem in machine learning has been quite troubling. As mentioned, statistical theory lives in the idealized system, and thus many of its bounds hold in such domain. However, there exists the fundamental mismatch between how models are in practice, or rather in a semi-theoretical regime, compared to the rigidity of classical systems. The setup itself is then impractical, for example of the assumption in which the distribution drew out of x is fully Gaussian, or isotropically Gaussian, which is not helpful in practice.

3. What counts as the measure, and what we even measure? This is more of the concern toward what counts as model complexity, and the structural constraints that comes with it. Usually this is directed toward neural network, but even then, parameter-based complexity is often a poor choice of representation, thus begs the question what is even it, and why suddenly it appears there, within the dynamic of such?

4. Why would such an optimization space induce double descent? Surpassing what we have, the bounds that we are having indicates the strong trajectory in which downward trend is the only way, as for the generalization capacity is increased by increased transparency, i.e. $|\mathcal{S}|$ of the observation set, and by increased capacity $|\mathcal{H}|$ on certain induced or 'natural system' (convex system topology). However, this seems to not be the case.

With this, we are ready to follow up the literatures in provision of said questions. Let us also recite what happens within the setting of many double descent results.

3.5.1 Discovery of double descent

While double descent is discovered by Belkin et al. [2019a] and further on modern deep learning landscape of Nakkiran et al. [2019], it remains that the conclusion is not reached if there exists some fundamental models that would not have double descent, the consistency of double descent through different models entirely and so on. This is mainly because of a particular 'shock effect', since before double descent, bias-variance holds the candle as the main principle of choosing and optimizing models to its optimal saddle point. Suddenly by then we observe double descent, which introduces the interpolating threshold $\mathcal{M}_0$, that breaks this principle. There is then the open question on how far-reached is double descent in consideration of different architectures, and this is reflected in several papers above. As of now, the main structures that have been verified includes

- **Classical models** Yang and Suzuki [2024], Nakkiran [2019b], Schaeffer et al. [2023], Lafon and Thomas [2024] (linear regression and classical polynomial), Li and Sonthalia [2024] (linear regression upon least square), Liu et al. [2021], Zhang [2024], Allerbo [2025] (kernel methods/regressions), Brellmann et al. [2024] (Classical MSBE-LSTD[2] optimization), Harzli et al. [2023] (Gaussian process). Some papers also have experiments on Random Fourier Feature Networks, and so on, but not substantial (including Belkin et al. [2019a] paper).
- **Deep neural networks**: Luo and Dong [2023], Yang and Suzuki [2024], Pezeshki et al. [2021] (CNN, VGG, ResNet, and DenseNet), Nakkiran et al. [2019] (Transformer), Shi et al. [2024] (GNN).

Most of the experiments are concerned of seeing and analysing double descent, basic analysis and experimental observations, test case scenarios for different models and setting and so on. All of the above models are then considered to be, concretely, reported of observing double descent, through various testing scenario on them. Review papers and literatures can be found as Schaeffer et al. [2023], Hu [2021], and Shi et al. [2024] on specifically, graph neural network. However, GNN would requires its separated discussion, so we will only briefly mention it as such.

3.5.2 Theoretical attacks

Because of its intrinsic nature, where double descent breaks the long-standing principle of learning selection, theoretical resolutions has been employed to tackle such problem. Indeed, for example, Cherkassky and Lee [2024] and Lee and Cherkassky [2022] aim toward such direction by understanding the behaviours of double descent using

[2]Mean-Squared Bellman Error and Least-Squared Temporal Difference

the classical VC-theory approach. However, such papers only introduce, or relate double descent to VC-theory via phenomenological approach - for example, reducing the VC-dimension while fixing or reducing training error, or by examining a potential proxy between the VC-bounds and potential factors encapsulated in the quantities involved. They however, do raise voices of certain inherent problems lies within the theory of learning and in general, learning models.

Beyond such, certain discovery and inquiries help in having a clearer picture on the behaviour of double descent through various structures. For example, Hastie et al. [2019], under pretext of studying high-dimensional ridgeless (ℓ_2 norm) interpolation, analyse the case of *isotropic features*, *correlated features*, misspecified model (feature observation inhibited), on ridgeless interpolation of linear model, and soon as nonlinear model and latent space model. While strong, their results are restricted to a specific class of assumptions and structure, and is doubtful to be extended further. Allerbo [2025] investigated and find a particular double descent trace when changing kernel during training, specified in the term of the minimum allowed bandwidth. The kernel bandwidth $\sigma \geq 0$ of a specific kernel filter determines the information smoothing, hence implies the complexity abides by the kernel structure, as well as determine the smoothness of information indicted by the term.

The sentiments that are shared in between researchers on the topic of double descent vary, but have some common roots, most of them coming from the difficulty to fully assess double descent and its factor components. While double descent concerns the evaluation of *generalization (test) error*, and the *model complexity* notions, the error in question is often implicitly defined or loosely considered in one way or another, and model complexity has no consensus on its definition. In fact, this points out a very potential, if not already mismatch in the theoretical development. While optimizations and phenomenological aberration of models, as well as building and constructing models at will are well-developed, the definitions on which they are defined on is not fully understood, if not undefined. As of now, literatures on model complexities like et al., Hu et al. [2021b], Janik and Witaszczyk [2021], Truong [2025], Luo et al. [2024] introduce their variations or a copula of different definitions altogether, where some are concerned with its effective complexity, some are concerned with its expressive or structural complexity with different ways to define the model underneath, from parameters to units, to random standard variables (Neal et al. [2018]). There is also the rift in consideration between classical theory, of which mainly is for classical models, and modern deep neural network structures of which do not conform to such analysis.

3.5.3 High-dimensional ridgeless interpolation - Hastie et al. [2019]

Aligning with our previous experiments, Hastie et al. [2019] conducted investigation into the class of ridgeless approximation criterion on linear-nonlinear classes disparity. The investigation itself is thorough in parameter-term. Specifically, they investigated the case of which we have **isotropic features**, where asymptotic risk depends on the weight w only through its norm $||w||_2$; **anisotropic features**, where the model risk depends on the geometry of the pair (Σ, w) for $\Sigma = \mathrm{Cov}(x_i)$ on the dataset $(y_i, x_i)_{i\leq n}^n \in \mathcal{S}$; and the case of nonlinear model, where the model itself is tested with added structure - x_i is obtained by passing $z_i \sim \mathcal{N}(0, I_d)$ through a one-layer neural network with random first layer weights.

Overall, the research itself is rather insightful, however, one problem with this research is that, it is fully theoretical with only deterministic prediction and numerical computation on heavily specified, synthetically generated data. Such is to say that the potential for such assumptions to hold in large system is uncertain, and whether if it will apply or transfer to higher structure, for example, similar construction but of neural network formalism. More specifically, the setting of this ridgeless case is constrained. Assuming the **dataset** as $(\mathbf{X}, \mathbf{Y})$ in form of $(\mathbb{R}^{n\times p}, \mathbb{R}^n)$ for the design/input space matrix $\mathbb{X}$ and response compression vector $\mathbf{Y}$, and parameter vector $\boldsymbol{\beta} \in \mathbb{R}^p$. The fundamental model is of the form

$$y_i = x_i^\top \beta^* + \epsilon_i, \quad y_i' = x_i^\top \boldsymbol{\beta} \tag{3.33}$$

From such, a **ridgeless regression estimator** is defined by the following optimization scheme, where

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}\in\mathbb{R}^p\,:\,\mathbf{X}\boldsymbol{\beta}=\mathbf{Y}} \left\{ \|\boldsymbol{\beta}\|_2^2 \ :\ \exists\,\{\lambda_k\}_{k=1}^\infty,\ \lambda_k \downarrow 0 \text{ such that } \boldsymbol{\beta} = \arg\min_{\tilde{\boldsymbol{\beta}}\in\mathbb{R}^p} \left(\|\mathbf{X}\tilde{\boldsymbol{\beta}} - \mathbf{Y}\|_2^2 + \lambda_k \|\tilde{\boldsymbol{\beta}}\|_2^2 \right) \right\}. \tag{3.34}$$

Of the penalized complexity on L_2 norm. Note that we should not confuse the ordering of the interpolant to treat λ ridge term as equal to fit; that is, the requirement of the minimization problem is to say, among all interpolants, choose the one that would be preferred by an infinitesimal amount of ridge regularization. Thus, removing λ-ridge bears no side effect to the minimization structure, but only removes a particular selection criterion; within the ridge-based analytical framework, this criterion serves to single out a canonical representative among interpolants. In other word, we say that ridgeless impose the minimal Occam's Razor case of ridge-specification singular model canonical optimizer, and only then hidden structure by ridge is realized. By construction equivalently, this suggests that a ridge regression model requires such condition on any differential-based optimization algorithm. Differentiate with respect to β of the trainable parameters (i.e. actual moving average),

$$\nabla_\beta \left(||\mathbf{X}\boldsymbol{\beta} - \mathbf{Y}||_2^2 + \lambda ||\boldsymbol{\beta}||_2^2 \right) = 2\mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} - \mathbf{Y}) + 2\lambda\boldsymbol{\beta} \tag{3.35}$$

On this differential landscape, the stability points that would be present means that any points admissible on such will result in

$$\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} + \lambda\boldsymbol{\beta} = \mathbf{X}^\top \mathbf{Y} \tag{3.36}$$

$$(\mathbf{X}^\top \mathbf{X} + \lambda)\boldsymbol{\beta} = \mathbf{X}^\top \mathbf{Y} \tag{3.37}$$

and thus leading to the invertability observation for $\lambda > 0$ of the matrix encoding, thus

$$\hat{\beta}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda)^{-1} \mathbf{X}^\top \mathbf{Y} \tag{3.38}$$

is an admissible form that should be satisfied at such landscape on itself. Ridgeless appears when you enforce the following limit of the solution, where we get to the minimum norm least squares estimator of

$$\hat{\boldsymbol{\beta}} = \lim_{\lambda \to 0^+} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}, \quad \hat{\beta} = \lim_{\lambda \to 0^+} \hat{\beta}_\lambda$$

on the matrix design space. Ridgeless here means that *regularization is taken zero from the positive side*, or rather that we take the regularization effect to be minimal yet present in the optimization scheme. A problem can be identified. Without ridge or small negligible ridge, this assumes the most hardwired-free form of the optimization problem, where optimization can be considered without structural penalization, but **only in this sense**. The reason why direction 0^+ matters can also be easily explained as there exists no meaningful way of encoding, in a strict simple sense, negative structural complexity measure. However, we note that the optimization here assumes that the loss-landscape is admissible of the L_2 form on Euclidean metric. In reality, what we have constructed of the learning template suggests a different kind of metric, a pseudometric. Such is to say, we are assuming these two metrics to be equivalent in the same form or shape as to be admissible of both. This is a very strong assumption, and overall, the system itself contains mismatch and extremely high structural assumption. If in ideal environment, such algorithm and formation process might turn out to be sufficient and correct. However, if this is not always the case, and the pseudometric equivalent condition implicitly assumed here can break, then pathological behaviours occur naturally of the unseen structure outside the framework consideration. Those pathologies are not emergent; moreover, they are precluded by definition. If we are to observe more, then the problem of multiple critical points might also is only an artefact of the Euclidean mismatch, outside of linear-enforced, convex setting. Thus, we can further observe another layer of the strong assumption, this time on the properties of the space. Extending this ideal setting, especially to more complex models, where the system are free and chaotic enough even in constrained operation (layer-frame and equivalent neuron units) is ill-posed, and can be reasonably considered a potential pressure point toward why there exists the mismatch between realistic models and theoretical presumptions.

Furthermore, from the model itself, and further on of the linear regression setting, is the fact that we have lossless information availability (true fit data) with sufficiently small $\epsilon_i > 0$. Assuming uniform ϵ further, we can see that under such view, any model can be reconstructed sufficiently, as for there exists only $\mathcal{O}(1)$ growth on parameter space. With small mismatch between $\boldsymbol{\beta}^*$ and $\boldsymbol{\beta}$ of the learner, such error can be high during optimization interval due to compression of parameters into a smaller set of parameters, while on high-dimensional, over-parameter case, dilution of parameters contains admissible dynamic of diluting functional form of a smaller set of parameters to a larger one, thus also explains in spirit the formation of double descent. However, this is not rather reflected in the framework. If ϵ_i are high, we still only consider a portion of it into variance control of the model, and nowhere there exists generalization. In fact, linearized model is very poor in gauging generalization behaviours, as the model itself, with the dataset synthetically generated in a supposed adversarial case (two linear models against each other), assumes quietly the structural relational space of f and f^* to be monotonic, and often with a narrow deviation corridor. Meaningful noise also is restricted, and thus can be considered into variance control. Adding this with the full-transparent optimization problem where there exists no statement about meaningful divergent or unseen data, generalization is hardly discovered here, but only ad-hoc parameter stabilization, non-permutating in order measure.

That is, model can reconstruct parameters, but they are almost never the same, both in order and magnitude if they are of same expression class.

The effectiveness of ridge observations and so on are basically phenomenal, and contains no sufficient theoretical insight on partition dynamics. In spirit, the theoretical treatment is not structural-free, nor the definition of generalization or unseen structure is sound. This is also troublesome as to rely on this to take on the problem of double descent. If we are to notice, the problem of double descent bears no resemblance to this setting in any conceivable way, at least when it was discovered. This pulls back to the dilemma of the definition of double descent, where on one research it is about descent on sample space, and on one hand it is on fully-admissible, population representative, full-access data as the paper we are discussing.

3.5.4 Sample-wise double descent - Nakkiran [2019b]

Regarding double descent, Nakkiran [2019b] explored the problem, in which the increment of sample actually hurts linear regression process. As always, the setting considers isotropic $x \sim \mathcal{N}(0, I_d)$. Here, the observation and claim is that the number of samples can indeed, hurt the end result for test MSE on the model. This results intwo two claims, mainly of the decomposing bias-variance analysis. First is the overparameterized risk claim, in which is defined of:

Claim 3.5.0.1 (Overparameterized Risk). *Let $\gamma := \frac{n}{d} < 1$ be the underparameterization ratio. The bias and variance are:*

$$B_n = (1-\gamma)^2||\beta||^2 \tag{3.39}$$

$$V_n \approx \gamma(1-\gamma)||\beta||^2 + \sigma^2\frac{\gamma}{1-\gamma} \tag{3.40}$$

And thus the expected excess risk for $\gamma < 1$ is:

$$\mathbb{E}[\bar{R}(\hat{\beta})] \approx (1-\gamma)||\beta||^2 + \sigma^2\frac{\gamma}{1-\gamma} \tag{3.41}$$

$$= (1-\frac{n}{d})||\beta||^2 + \sigma^2\frac{n}{d-n} \tag{3.42}$$

and the second one is about the underparamterized case, which is as the second claim.

Claim 3.5.0.2 (Underparameterized Risk, Hastie et al. [2019]). *Let $\gamma := \frac{n}{d} > 1$ be the underparameterization ratio. The bias and variance are:*

$$B_n = 0 \quad , \quad V_n \approx \frac{\sigma^2}{\gamma - 1}$$

Those claims are troublesome, in the sense that they indict heavily on the assumptions of which, for example, suddenly the bias collapse to zero without any indication or further justification. The variance is specifically designed for a replica-style curve fitting instead, of which the approximating relation suggest the not-so-useful notion, in which such deviation from the claim can be wide and far yet still right in terms of descriptive behaviours (converges to 0). Vanishing terms transition from the first claim to the second claim inherently accept bias of the system itself, of which is then trouble some to proceed against. And the list of assumptions on fine structures are too many in spirit, especially the linear algebra mapping in which derivations of computation are used in the paper itself. To truly justify this, we need something beyond what is described here.

3.5.5 Multiscale feature learning dynamics - Pezeshki et al. [2021]

3.5.6 Reinforcement learning and double descent - Brellmann et al. [2024]

3.5.7 Out-of-Distribution Detection - Ammar et al. [2025]

3.5.8 Gaussian processes and neural network - Harzli et al. [2023]

3.5.9 Double descent on GNN - Shi et al. [2024]

3.5.10 Weak feature double descent - Belkin et al. [2020]

3.5.11 Grokking, Emergence, and Double Descent - Huang et al. [2025]

3.5.12 Variations on analysis

Many sub-variations and analysis alongside major substantiated discoveries listed above can help in analysis. Some of them are toward linking different existing dynamics to large neural network in non-linear fashion, or generalizing approximation of linear network toward nonlinear systems.

Dropout and double descent - Yang and Suzuki [2024]

Dropout is a relatively well-established regularization technique for training deep neural networks, usually by exclusion of different neurons during processes, as seen i Srivastava et al. [2014a] to control overfitting. Accordingly, Yang and Suzuki [2024] predicted or stated that regularization techniques can nullify double descent, and thus also dropout. The model considered is linear regression model, for p covariates $x \in \mathbb{R}^p$ and response or output $y \in \mathbb{R}$ are related by

$$y = x^\top \beta_0 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{3.43}$$

The situated dropout is as followed. Given training data $z^n = \{(x_i, y_i)\}_{i=1}^n$, for $X = [x_1, \dots, x_n]^\top$ and $y = [y_1, \dots, y_n]^\top$, the estimation of weight β by $\hat{\beta}(z^n)$ (model's weight), to minimizing the training error

$$\mathop{\mathbb{E}}_{r \sim \mathcal{B}er} ||y - (r \odot X)\beta||_2^2 = ||y - \gamma X\beta||_2^2 + (1-\gamma)\gamma||\Gamma\beta||_2^2 \tag{3.44}$$

for $\odot$ denoting the element-wise product on L^2 norm space, each element of $R \in \mathbb{R}^{n \times p}$ takes one and zero with probabilities γ and $1-\gamma$, for the distribution $\mathcal{B}er(\gamma)$ corresponding to such, and $\Gamma = \operatorname{diag}(X^\top X)^{1/2}$. The final description gives the form of the effective dropout objective. Originally from the paper, they consider it exactly of a Tikhonov regularization method. Set $\beta' = \gamma\beta$, then

$$\begin{aligned} &||y - X\beta'||_2^2 + \frac{1-\gamma}{\gamma}||\Gamma\beta'||^2, \\ &\hat{\beta}_{n\gamma} = \left(X^\top X + \frac{1-\gamma}{\gamma}\Gamma^\top\Gamma\right)^{-1} X^\top y \end{aligned} \tag{3.45}$$

minimization. While experiments regarding such regularization works in such domain and was illustrated as to be reasonably effective, fallacies in claims and insight remains, especially in the nullification of double descent. The problem here is that, any given amount of dropout is, accordingly to the author, similar to Tikhonov or ridge regularization on structural complexity. As explained in the c' optimization of c-space in preliminary sections, such induces a different, much smaller system in which the additional structural attractor points toward the region of low regularizing complexity throughout the domain. Such superposition between the $\{\lambda r(h)\}(\mathbb{R})$ landscape and the classical loss $\hat{R}$ only morph the dynamic setting to a different region, and thereby bypass double descent in such case for absolute non-overfitting convergence to $\min(\hat{R}_S(h))$. This works well on well-behaved and loosely stable case, but in a lot of scenarios, such cannot be guaranteed, especially for pathological systems. Furthermore, Speaking in terms of the *inductive bias*, mismatch between model and effective regularization becomes a problem, as there exists no effective complexity measure for the regularization term $r(h)$. Some are ill-conditioned, for example, $r(h) = \sum_j w_{ij}^2$ for square terms ij-indexed weights only take into account the reasonable weight system, without the semantic itself, or meaningful weights. A forced dropout might potentially in certain less ideal cases, be more destructive, and introduces double descent back again, until the stochastic process shifts meaningful weight to a more resource constrained system. This means we have to induce, or at least formalize if stochastic process can recover meaningful semantic after dropout randomly. We can test this somewhat, by introducing a small toy test on the proposed dropout estimator:

Theorem 3.5.1 (Expected test error, Yang and Suzuki [2024]). *Let $\alpha = C < \frac{1}{(1+\sqrt{p/n})^2}$, the expected test error is*

$$f(\alpha) = \left\{1 - a\alpha + 2\alpha^2 \frac{p}{n}\right\} ||\beta_0||^2 + \sigma^2\alpha^2(\alpha+1)\frac{p}{n} + O\left(\frac{1}{n^2}\right) \tag{3.46}$$

with $\alpha = \gamma/(1-\gamma)$.

While the theorem is proved, we then must know if such theorem correctly correlates to even small dynamic systems. However, as it stands, we should not read the paper as to say that dropout resolves double descent, but rather that *double descent* region of solution space is effectively ignored by reduced solution space on dropout condition, especially within ideal setting, and linear functional class, not even multilinear models. Alignment outside of this ideal setting introduces the problem of brittle prediction, and might as well nullify this result.

Chapter 4. Interpretation

There are a lot of problems regarding how we treat the problem, how we see it, and how we actively analyse it. Most of them lies in the space of statistical learning theory, since bias-variance on itself has variations and justification that is seemingly very natural in statistical theory, like the other version name *approximation-estimation error tradeoff*. All of them contribute somewhat to the current outlook, including operational factors and hidden assumptions underneath the framework of statistical learning, in which might only hold in special regions and not of a wider landscape, and non-applicable to richer constructions of semantic. Thus, we are prompted to look into both the **structural** problems, i.e. specifically on the reliance of bias-variance connection toward risk, for example, the fact that it is not general of a decomposition; the oblique definition of model structures and gaps between different discipline of model building; different operational definition of risks and hypothesis space, and the *operational problems*, of different modelling setting; different treatment in consideration of the learning scenarios; different conditions and mathematical structures enabled while also overclaim patterns seen within such space, and so on so forth. While we do not claim to solve or identify all of them, the baseline of this section is to **note down** such disparity. Following up such section would be the interpretation of double descent in its form, and inconclusively of what is the nature of it.

4.1 Structural and operational problems

There are several structural problems that require attention in this section. That said, it can be observed rather easily by our incapacity to express or explain recent structural failures or phenomena that appear of large system. Despite its prevalence in particular research interest, double descent itself has no definite statement to describe its particular behaviour. This is also true for a list of other phenomena, such as *grokking* (Power et al. [2022a]), *emergence behaviours* (Wei et al. [2022b], typically discussed in large language models, which is understandable of large-scale specific ruling system), *implicit bias* (Soudry et al. [2024]), and *neural scaling law* (Kaplan et al. [2020], Wei et al. [2022b]). There is also *catastrophic forgetting* (van de Ven et al. [2024]) which is still fairly native of large systems; nevertheless, those phenomena will be of target later on (or discussed in later chapters). The fact remains that we only *partially understand* these particular results with conjectures and theories to support the facilitation of new insights, new analytical lens, and nothing much. Without a clear definition of what is even double descent, sometimes, we cannot proceed, lest to say even solving it. This is a problematic issue, since the current theory is so messy and not rigid that many even turns to exotic choices, like topological informatics learning theory, *Homotopy Theoretic and Categorical Models* (Manin and Marcolli [2024]), and several more. Without clear definitions, it remains ambiguous whether double descent constitutes a single phenomenon, a family of related behaviours, or an artefact of particular modelling and evaluation choices.

Thus, prompt several distinction classes that are needed to take into consideration. One is toward the class between **small and large** models and what do they mean structurally or semantically; one is about **model description and richness** between them; another one is between **different metrics and evaluating spaces**, between different interface and *operational procedure*, i.e. between algorithm-agnostic and algorithm-dependent results or observations. They would be detrimental of the analysis, since the concept of bias-variance itself hedges on such gap between theoretical treatment and empirical observations. Operational problems are rather easier to spot aside from operational procedure. For example, the concept of bias and variance can be computed differently in different algorithmic settings, the risk profile can also be different, inherent design might introduce structural reduction to finite cases of the evaluation landscape, and different state of deformations of uncontrolled modules in randomized setting might deviate results or damp the observation down in noise. Such will have to be taken into consideration before moving into further sections focusing on experiments.

4.1.1 Bias-variance and the definitive form

Bias-variance tradeoff is not new in concept, their definitions on computation has been existing before the formal foundation of modern learning theory. However, it is not trivial on how it became the statement of overreach, much to the disparity observed between classical and deep neural network model. An analysis might be sufficed to analyse such disparity in treatment of the general working space. Mainly, models in classical settings are considered to be of **parameterized models**, and also the non-parameterized version also. With such in mind of the classical setting for which probing object is a structurally unspecified relation $x \mapsto y$, we can define the following definition as parameterization using $\Psi_\Gamma : \Theta \times X \to Y$ such that for functions $f \in \mathcal{F}_\Gamma$, $f(x) \approx \Psi_\Gamma(\theta, x)$.

Definition 4.1.1 (Parameterization). *On arbitrary function space $\mathcal{F} : \{X\} \to \{Y\}$, of which a system admits its operation in such form, let $\mathcal{F}_\Gamma \subseteq Y^X$ denote a class of functions admitting representation under a fixed mathematical semantic Γ (e.g. algebraic, geometric, compositional, or dynamical semantics). Then such system is transformed to a similar* ***parameterized system*** *if and only if it admits the factorization form, or if there exist a parameter space Θ and a family of operators $\Psi_\Gamma : \Theta \times X \to Y$, where Γ is the mathematical semantic used, θ is the above-level replacement of direct relational connection into variable control of θ, and variations in y being controlled by θ, such that for functions $f \in \mathcal{F}_\Gamma$, $f(x) \approx \Psi_\Gamma(\theta, x)$.*

Thus, it is in-spirit, a definition of parameterization of model that we can use. Most definition of parameterization differs from this form, not inherently a disadvantage, to presume independent choice of selecting a family of parameterization; undermine the structural form that are explicit in Γ; using parameter as controllers that can be tuned of optimization, and often merge such description together with loss and training. The loss of Γ is rather troublesome, but they are often mediated in a weak form using similarity of elementary compositions, i.e. most structural forms in ML models are often elementarily composable, though there exists transcendental expressions of which can be formed during process and not inherent of the syntax itself. Concretely, Γ is the hypothesis about *how relations are connected of each other*, i.e. the grammar of admissible operators. When Γ is absent or mismatched, optimization is free to exploit the structural gaps by directly specify θ-only optimization on non-structured parameters, and can indeed imperfectly gauge the underlying hypothesis model structure itself. Two things can happen in this case: either optimization reduces empirical loss by fitting idiosyncrasies of the training set (overfitting to structure outside the true operator family or admissible approximation of the family to another external family); or optimization selects arbitrary parameter instantiations along many uninformative directions, producing unstable solutions with high variances. Those are potentially two failure mode on wrong operators and worse variance, and thus also correlate to the arbitrary notion of generalization, where this is particularly troublesome since some form of optimization setting assumes Γ_c of the concept $c \in \mathcal{C}$. From such, we state the frame in which bias-variance is calculated.

One of the main question that should be asked of the modern landscape is that, would modern models even **admit bias-variance** in any reasonable way?

4.1.2 The definitions of double descent

Double descent is often considered the direct opposition to the viewpoint of bias-variance control, and often not more. More fundamentally, the term **double descent** does not currently admit a formal definition beyond an empirical characterization of observed risk behaviour. While several works have suggested interpreting double descent as an instance of emergent behaviour, such interpretations necessarily remain conjectural, as emergence itself lacks a precise and operational definition within contemporary learning theory. As a result, appeals to emergence do not resolve the underlying indeterminacy of the phenomenon, but rather shift it to a different conceptual level.

Empirically, double descent has been reported in a variety of settings, while being absent in many others, even under closely related modelling and data conditions. This variability complicates attempts to treat double descent as a unified phenomenon and raises whether it constitutes a single mechanism, a collection of regime-dependent effects, or an artefact of specific modelling and evaluation choices. In the absence of necessary and sufficient conditions for its occurrence, theoretical accounts are forced to accommodate mutually incompatible behaviours, thereby undermining their explanatory specificity. Consequently, existing frameworks face a dual difficulty: they neither reconcile naturally with classical bias-variance analyses, nor align with the assumptions underlying earlier generalization results, such as uniform convergence over well-behaved hypothesis classes. Without a precise characterization of the object under study, it remains unclear how such theories can be systematically compared, falsified, or extended.

	Bias-variance
Structures:	We use the setup of a model M structured into $(\theta, I, \{f\})$, where $\{f\}$ is the set of descriptions using θ of the internal model's components, and I as the external receiver of connector. In effect, a model is structurally the set of all $\{I, (\theta, \{f\})\}$ with respect to the component form θ, and variation of the external connector I, often can be considered input, and if θ is reduced to variable, we call it a **parameterized model** that creates implicit linkage between I and its intended transformation from I, using θ. Furthermore: 1. Input space I and output space O, typically referred to as X, Y, measurable space taking values from the real field $\mathbb{R}$. Typical set of all feasible values of input and output forms can be specified of $(\mathbb{R}^d, \mathbb{R}^k)$ or for output to be a small embedded subset $Y = \{y_1, \ldots, y_n\}$ being finite and discrete on $\mathbb{R}$. 2. Euclidean metric $\|\cdot\|$ is used for all real-valued components. 3. System relation between X, Y is the **function space** Y^X or a subset $\mathcal{F}_\Gamma \subset Y^X$, where Γ is the chosen semantic or representation class. Such system uses also the **parameter space** $\Theta \subseteq \mathbb{R}^p$ for finite p. 4. Hypothesis model, i.e. the learnable model admits the **parameterized form** if able, as Definition 4.1.1, or $\Psi_\Gamma : \Theta \times X \to Y$. A model instance is $\hat{f}_\theta(x) = \Psi_\Gamma(\theta, x)$. We denote all models as $h \in \mathcal{H}$ for $\mathcal{H}$ the class of all specific Γ and intended value space on $\theta_{\mathcal{H}}$. 5. We assume there exists an **objective oracle**, denoted either c or EX, of which belongs to a class $\mathcal{C}$ of all oracles fitting certain description (θ_c, Γ_c). We assume this objective oracle is a part of the general oracle class $\mathcal{O} \supset \mathcal{C}$. Furthermore, we assume **knowledge** of this oracle is presented only by fixed instantiation of it, giving non-factorable description of linkage in the form of *dataset*, $D = \{(x_{,i}, y_i)\}_{n=1}^{n}$ being finite and drawn from observations or probing of c. One more assumptions that can be removed: c exists and knowable of existence; in case this does not happen, we say the setting is *oracle-black boxed*. In case partial information are possible, we say that the setting is semi-transparent. The dataset cutoff implicitly creates $c' \in \mathcal{C}'$ of a proxy concept class, such that lowest condition remains $\mathcal{C} \cap_{\Gamma,\theta} \mathcal{C}' \neq \varnothing$, and so on of the inclusion structure. 6. We apply a **statistical interpretation** on the dataset of observables D to capture the observations' occurrence patterns replacing the internal Γ exposition, into probabilistic distribution P on $X \times Y$ of variations. Simplest assumption can be determined of being i.i.d. or identically and independently sampled, and thus observation are further assumed to come from instantiation of the **sampling** action. 7. Loss function $\ell : Y \times Y_c \to \mathbb{R}_{\geq 0}$ is defined on comparison space between h and c. On Euclidean space, use L_2 norm or square loss as measure. The semantic can change, thus we say the loss system contains $\{\ell_i\}$ of variational applied form, such that $\{\ell_i\}_{i=1}^{3}$ being $\ell_3(\ell_2(\ell_1(\cdot,\cdot)))$. Define **population (expected)** error named as risk, taking expected value over the entire space. 8. Operations making use of ℓ defines the **objective** for finding suitable h such that to reduce $\ell(y, y')$ of their representative observables.
Operations:	Nothing for now.

4.1.3 Gradient descent and metric mismatch

Gradient descent is an effective approach to realize the case of **empirical risk minimization** into the formation of practical encodable algorithm, within certain structural constraints, or rather the reverse is true. For ERM to be applied, we require certain spaces of admissible structure:

Chapter 5. Experimental investigations

For understanding and analysing double descent and bias-variance trade-off, and furthermore in later section on identifying double descent, we would like to use test models specifically for exhibiting bias-variance, as well as testing hypothesis and forming theoretical conjectures. This is because of the troublesome nature of the phenomena itself, which proves difficult to analyse without sufficient experimental observations. Furthermore, specific analysis on these types of model might prove useful for generalization toward more difficult and complex structure.

Our major focus is on three types of model. First two are classical models and system, linear models on the real space $\mathbb{R}^n \to \mathbb{R}$. Second, is the type of **support vector machine** (SVM) models to highlight the analysis toward data-based fundamental analysis. The third one is the class and collection of different *neural network structures* (MLP, or variations). Our goal is then to expansively analyse, formalize, making conjectures and hypothesis and test them, to find, force double descent out, or identify it on the learning landscape. We note that structures of those models are not well-generalized, hence analysis from one might not apply or apply only in small conjunction. This approach is rather quite helpful in its own merit, as its choice and preference will gauge our parameters than we gauge the phenomena of the parameter it induces.

5.1 Experimental planning

This section of the manuscript will explicitly contain instructions, setup, methodology, and so on, with regard to the experimental details of the experiments itself.

5.1.1 Setting: gradient-based optimization

In our usual experiment, we would most specifically, use a closely-regarded family of gradient-based algorithms, such as gradient descent (Cauchy [1847], Robbins and Monro [1951], Widrow and Hoff [1960a], Hestenes and Stiefel [1952], Fletcher and Reeves [1964]), mirror descent, natural descent, and so on (Nemirovski and Yudin [1983], Amari [1996, 1998], Yang and Amari [1998], Rattray et al. [1998], Gunasekar et al. [2021], Luong et al. [2016], Xiao and Zhang [2014], Sohl-Dickstein [2012], Raskutti and Mukherjee [2013]). While implementation and details of the specific algorithm can be different, they all shares some of the same concepts of which making them possible for optimization in the specified iterative domain. The most general formulation possible is the following algorithm on convexity:

Algorithm 1: Gradient descent on general convex function

Data: $(x_i, y_i) \in \mathcal{S}$, η, $\nabla(\cdot)$ operator.
Result: Expected optimization to the minimal error on $\ell(\cdot,\cdot)$.
begin
  $x_1 \leftarrow$ randomly initialized points;
  for $t \leftarrow 1$ *to* T **do**
    $x_{t+1} \leftarrow x_t - \eta \nabla f(x_t)$
  end
end
return $\hat{x} = 1/T \sum_{t=1}^{T} x_i$

Nevertheless, one should be cautioned of the intrinsic assumption on which gradient descent of this type assumes. It assumes *convexity*, which can only be possible, if the landscape itself is supposedly, assumed to contain such global minima. Said global minima in a richer language can be described as the epistemic collapse of the learning structure, under the arbitrary optimization space. This is a *very serious assumption*, and should it fail, as seen in a lot of non-populated or general observation space, the entire procedure is invalid. It is also troublesome as to rely on the singular objective (numerically encoded) space in which it is trivial to realise that there are indeed, local minima. This scaling, with respect to both locality and global consideration, must be considered carefully, since *every single contribution* might distort the experiment in a very unexpected way. In a more sophisticated explanation, we would be more aware of the structural conformation that gradient descent make of the encoding space.

5.1.2 Experiments list

The following describes and outline specific requirements updated of experiments, aside from done experiments. Currently, there remains the *neural network* in plain form functional approximator experiment, *polynomial neural network (customized)* experiments, validation of *support vector machine (SVM)* on generalized task space experiments, different large-scale neural networks (VGG, ResNET, LLM) experiments set, and toy model for investigations. Specifically:

1. **Linear models on abstract $\mathbb{R}$-conjunction space**[1]: Different kind of components-based on linear behaviours, in different setting and so on, of different testing scenario (regression, classification).
2. **Polynomial-type models on abstract conjunction space**: Similar to the above linear case, this one however allow for polynomial-style components, i.e. degree k terms up to nth degree.
3. **Support Vector Machine (SVM) on abstract spaces and sample space**: The SVM model is rather interesting (within which it has regression and classification as main functionality), because it goes reverse from perspective - instead of creating models to fit data, it uses data to create fitting model.
4. **Neural network on-layer, abstract functional spaces**: This is the set of experiments for layer-based neural networks, on abstract functional spaces of analysis. The list includes vanilla (l, n)-mass neural network with mixed activation patterns, CNN, VGG, ResNet, spatial neural networks, RNN, LSTM, and if able, certain precursor of LLM systems.

All of those models base their purposes in *component-based interpretation*, such that the scope in which their components are allowed, change the dynamics and so on of the model. The theoretical problem of which those experiments are targeted on is to mark the singular hindsight - *simply how* model behaves under controlled structural changes and setting changes. Because of the lacking in such deference of approaches, it also depends heavily on how the adversarial set (dataset, concept) itself is targeted of, or coverage of such. Such is to say, we also need the *precursor theory* of component treatments. Which is explained above, but such is also particularly extensible to explain also grokking (Power et al. [2022b], Liu et al. [2023]), benign overfitting (Bartlett et al. [2020], Muthukumar et al. [2021]), *neural collapse* (Papyan et al. [2020], Han et al. [2022]), optimization-based anomaly or phase-transition similar form (Cohen et al. [2021], Fort et al. [2019]), and the fuzzy, diverged front of explanations in deep learning phenomena, as criticized by Razeghi et al. [2025].

Polynomial neural networks

Because of the sensitive context of double descent, polynomial suffers the same fate of design, as so is many of the experiments. Its design implementation details must be explicit. That is, of which can be seen in the polynomial transition model. That said, the setting itself is rather more difficult to assess.

Originally, it is asked of the experimental team to set up **adversarial testing**, i.e. testing two mismatched polynomial models together, in a singular testing system. So, we will have (p_k, p_c) for $k \in [a, b]$ of complexity specification, in this case, the number of initialized degree of interest. As such, the experimental system is between two polynomials. But this turns out to have certain problem: *domain range*. The domain value, when admitted of the input $x \in \mathbb{R}$ or $\mathbf{x} \in \mathbb{R}^n$ of n-dimensional input sequence, is rather difficult to control, especially for $[a, b]$ that is nonstandard in $[-1, 1]$, and not of normalized unit. Usually, we understand the polynomial transition space as the modular $\mathbb{R} \to \mathbb{R}^h \to \mathbb{R}$ mapping for singular n-dimensional hidden field, the polynomial basis terms. However, if we are to upgrade it to multivariate polynomial, such changes dramatically, and so is the structure itself. Thereby, so is the

[1] Conjunction space simply means that the full numerical encoding space of all activities involved

behaviours on ranges. The justification for nonstandard, free monomial is that for increased value range, the pattern is not chaotic, but rather smoothed out by dominant terms, for example, the n-degree terms contribution in univariate case. Similar case happens when taken into consideration n-dimensional polynomial mapping on $\mathbb{R}^n \rightarrow \mathbb{R}^m$, but only this time the dominance is local to individual dimension itself. In general,

Conjecture 5.1.1. *The probability of dataset representing wrong n-univariate polynomial p_n, on finite arbitrary space $\mathcal{D} = [a, b]$, for $|\mathcal{D}| \rightarrow \infty$, increases, in presence of $\epsilon \sim \mathcal{N}(a, \sigma)$ noise.*

The main fundamental reason is because of the ill-conditioned Vandermonde basis, in which is typically used in polynomial expression. Elsewhere, on finite space $[a, b]_m$ on m samples, it is then conjectured that the chance of interpolation is much greater, and much more deviated from the actual concept shape if stray over large enough $[a, b]$, thus also influence the error rate and gradient-based learning form of different weights. The goal is then, to construct rich enough polynomial concept class, based in this new discrete, finite range and initialization goals, with the main goal is to find sufficient polynomial construction on arbitrary $[a, b] \in \mathbb{R}$ that y foes not explode. This is fundamentally hard, and if unable to procure, we would have to choose to go for the normalized space on $[-1, 1]$. This is also for computational purpose, since `float128`, of which is the main data point form, is numerically unstable with large operations and the density of operations as such without sufficient safety control, in which it is very hard to reconcile.

One more point to note. What we have seen is based on the perception of typical machine learning setting that **monomial Vandermonde basis** is the polynomial representation space. In fact, this is not true, as there are many as Lagrange polynomial, Newtonian basis, Legendre polynomials, and different classes of orthogonal polynomials. However, without structural operational analysis, such addition and generalization is hard to tract of the structural droplet they introduce.

The following will list the amount or configuration that test should be taken into consideration, first with the guideline.

1. **Carefulness on numerical processing**: Introducing polynomial models meaning that for each component, it is reducibly equivalent to n linear components, for each component, with multiplicative combinator. So for example, a system of full n-degree model sequentially, would have $n! + 1$ linear equivalent mass size, including the bias on numerical space. Therefore, the value in practical calculation can be very large, even in $[-1, 1]$ as the pairwise normalized. Cares then must be taken to draw that out.
2. **Minimal implementation**: Use the least, but consistent code possible. Each code must be recommended to be designed modularly, such that you can snap it in others with only relevant knowledge as to *what it does* and *what is the typing allowed* of the unit itself, aside from utility systems.
3. **Loop collapse**: The order of loop, or changing test case is very necessary to be taken into account of. First one, is the **system test loop**, where the setting of the environment for operation, i.e. noise, concept allowed (dimension or description or class of types), number of samples, and so on. The second one is the **hypothesis-action loop**, of the empirical action of *train-test-validate-cross* partition of the observational space $\mathcal{S}_n$ to mimic generalization in oracle setting, the hypothesis spaces, and so on. The final one is the **operational level**, where you specify the loss-landscape, learning metric, aggregated measures, loss function measure, actions on learning algorithms, the learning algorithm configurations (for example, learning rate, etc.), and so on.

Aside from such, each layer in the final point *procure measures and statistics*. For example, at system test loop, you have properties of the dataset itself. Distribution and shape or form of the model as the second loop, and so is the jump path, approaches, pathology of the reconstructed loss space. Such should be taken into consideration. The list of model is then:

1. Polynomials, function concept set: This is where we do not require the setting of two-polynomial model configuration together and to study them using a class of different elementary function, only fit the input-output space. For example, singular $sin(\cdot)$ class that fits $\mathbb{R} \rightarrow \mathbb{R}$ is ideal, or else, or different pathological systems.
2. Polynomial, adversarial $[-1, 1]$ range: Construct both, only in $[-1, 1]$ range for evaluation, since higher range collapses into smoother values.
3. Polynomial, changing basis construction, $[-1, 1]$ range: Use modules, like `numpy.polynomial.legendre`, or `scipy.special.legendre`, and so on.

4. Polynomial, changing basis (plus monomial), $[-1, 1]$ range, input-dropout (filter): Filter out, in the form of hidden information from the hypothesis polynomial, and not the concept polynomial, to simulate ordered information mismatch.

One of the main examples would be to pitch different kind of models together, for example, targeting Legendre polynomial class on monomial going against Newtonian polynomials, or the Vandermonde basis going against other structural models as well.

5.1.3 Preliminaries on observations - Schaeffer et al. [2023]

Conveniently, one of the few analytical sufficient experimental analysis that we can get, is from Schaeffer et al. [2023] paper on demystifying double descent. While the experimental setup is rather well-defined and interesting, it is also rather limited in scope, of which we want to dive deeper into it and further the analysis that was provided.

5.1.4 Linear function class $h(u)$ - Lafon and Thomas [2024]

Linear-linear scenario learning is one particularly well-studied case of representing double descent or not, usually through simplicity in analysis (of the similar ease of proof compared to binary classification). Models in this setting can be said to share the same *parameter space* with no additional structure or scaling mask (like in monomial basis). The model that is representative of this preliminary observation is from Lafon and Thomas [2024], specifically, linear regression (function approximation) with Gaussian noise, and of univariate case. Consider the family class $(\mathcal{H}_p)_{p\in[\![1,d]\!]}$ of linear functions $h : \mathbb{R}^d \mapsto \mathbb{R}$ where exactly p components are non-zero ($1 \leq p \leq d$), we have the following model.

Definition 5.1.1 (Lafon and Thomas [2024]). *For $p \in [\![1, d]\!]$, $\mathcal{H}_p$ is the set of functions $h : \mathbb{R}^d \mapsto \mathbb{R}$ of the form:*

$$h(u) = u^T w, \quad \text{for } u \in \mathbb{R}^d$$

With $w \in \mathbb{R}^d$ having exactly p non-zero elements. The complexity here itself can be interpreted as the effective random initialization, trimming down data of hidden details instead - this is indicated by the amount of information it observes given d dimension, and p-hypothesis dimension. [2]

Using such model, they consider the setting of minimizing the customized mean square loss,

$$\min_{w\in\mathbb{R}^d} \frac{1}{2}\|\boldsymbol{X}w - \boldsymbol{Y}\|^2 \to \min_{w\in\mathbb{R}^p} \frac{1}{2}\|\boldsymbol{X_{\sim p}}w - y\|^2 \tag{5.1}$$

of the sub-problem on p cardinality, where again p is the dimensionality or independent degree of freedom. This results in the following theorem that 'mimic' double descent on this linear regression model.

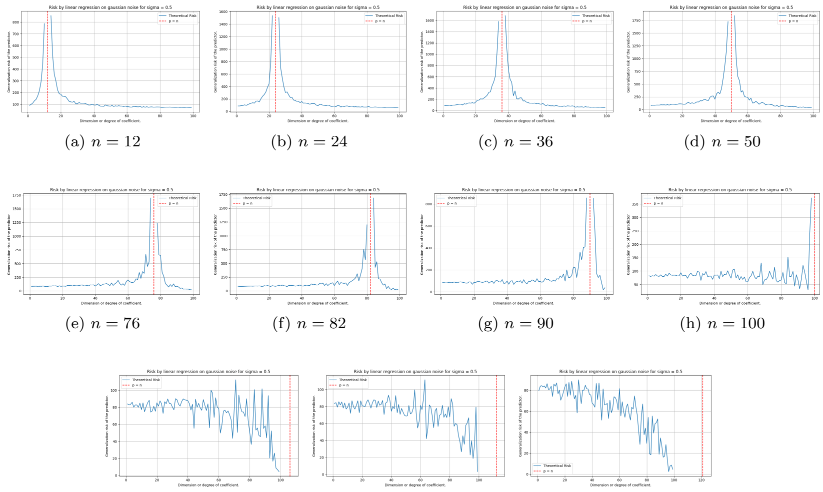
Theorem 5.1.1 (Double descent on linear regression - Lafon and Thomas [2024]). *Let $(x, \epsilon) \in \mathbb{R}^d \times \mathbb{R}$ independent random variables with $x \sim \mathcal{N}(0, I)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2)$, and $w \in \mathbb{R}^d$. we assume that the response variable y is defined as $y = x^T w + \sigma\epsilon$. Let $(p, q) \in [\![1, d]\!]^2$ such that $p + q = d$, $\boldsymbol{X_{\sim p}}$ the randomly selected p columns sub-matrix of X. Defining $\hat{w} := \phi_p((\hat{w}), [\hat{w}])$ with $(\hat{w}) = \boldsymbol{X}^+_{\boldsymbol{\sim p}}y$ and $[\hat{w}] = 0$.*
The risk of the predictor associated to $\hat{w}$ is:

$$\begin{aligned}
&\mathbb{E}[(y - x^T\hat{w})^2] \\
&= \begin{cases} (\|w_{\sim q}\|^2 + \sigma^2)(1 + \frac{p}{n-p-1}) & p \leq n-2 \\ +\infty & n-1 \leq p \leq n+1 \\ \|w_{\sim p}\|^2(1 - \dfrac{n}{p}) & \\ \quad + (\|w_{\sim q}\|^2 + \sigma^2)(1 + \dfrac{n}{p-n-1}) & p \geq n+2 \end{cases}
\end{aligned} \tag{5.2}$$

The point $p = n$ is representative of the interpolation threshold from underparameterization to overparameterization.

[2] According to the *manifold hypothesis*, there are chances where the actual concept can be interpreted and mimicked by seeing that it lies on a lower-dimension manifold inside its actual expressive dimensional space.

We present two parts - first are those fundamentals predicted by the theorem, and second are verifications of the theorem with our own version of testing. The first experimental setting is then conducted with fixed d, varying p for each varying n samples. Since this is a least square result, preliminary run is conducted using one-shot error-risk measure on least square closed-form approximation. Note that $p \leq d$, hence we are regarding the domain of the term underparameterized and equal parameterization setting. The result is presented in Figure 5.1



(a) $n = 12$ (b) $n = 24$ (c) $n = 36$ (d) $n = 50$

(e) $n = 76$ (f) $n = 82$ (g) $n = 90$ (h) $n = 100$

(i) $n = 106$ (j) $n = 112$ (k) $n = 121$

Figure 5.1: Theorem 5.1.1 behaviours on randomized setting. For this model, we have $d = 100$, $\sigma = 0.5$, and variational n. Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

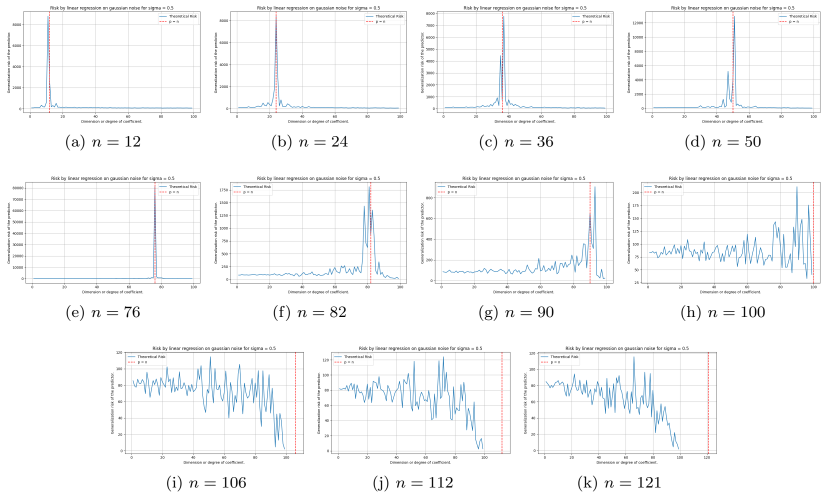
As we see from this particular experimental result, the theorem predicted a loose, left-peak absent, double descent curve estimation for a variety of test case with more and more samples in increment. There exist some fundamental conclusion that we have about the shift of the interpolation threshold upon to $p = n$, which is reflected in the piecewise prediction. However, this pattern breaks down for n higher than specific constant, for this case, starting from $n \approx 100$ to beyond. From here, at least from testing scenario, the test indicates that predictions suggest a breakdown of the double descent estimation. Beyond this, a simple semi-chaotic convergences is observed, from $n = 106, 112, 121$.

Interestingly, verifying this notion of easy enough to simplify it into what the statement holds under particular notion. Namely, as we find out, the above double-descent replicating formula can actually be realized by the parameter-wised error, comparing the parameter between the least-square solution and the actual concept. The formula for this is expressed by the exposed MSE on parameter function, for $\beta_{p,k}$ the coefficient value estimated for the k-th selected feature when the model is restricted for randomization of p features,

$$\mathrm{MSE}_{\mathrm{param}} = |\hat{w} - w|_2^2 = \sum_{j=1}^{d} (\hat{w}_j - w_j)^2 \tag{5.3}$$

where $\hat{w}_j$ is chosen such that it is equal 0 for $j \notin S$, otherwise $\beta_{p,k}$ for $S = \{i_1, \ldots, i_p\}$ is the set of selected feature indices. Nevertheless, it is then found dubious why this is true, as for the original statement of Lafon and Thomas [2024] indicates correlation to the prediction error of the result, $x^\top \hat{w}$ instead. The result of evaluating $\mathrm{MSE}_{\mathrm{param}}$ is presented in Figure 5.2.

If, however, we turn our attention to the statement itself, of which refers to the prediction error $\mathrm{MSE}_{\mathrm{pred}}$, defined



(a) $n = 12$ (b) $n = 24$ (c) $n = 36$ (d) $n = 50$

(e) $n = 76$ (f) $n = 82$ (g) $n = 90$ (h) $n = 100$

(i) $n = 106$ (j) $n = 112$ (k) $n = 121$

Figure 5.2: Contrary to Theorem 5.1.1, this is the behaviours on randomized setting for standard parameter-wised mean square error measure (MSE-on-parameter). For this model, we have $d = 100$, $\sigma = 0.5$, and variational n. Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

by the equation

$$\mathrm{MSE}_{\mathrm{pred}} = \frac{1}{n}\left\| y - X_p\,\hat{\beta} \right\|_2^2 = \frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - x_p^{(i)\,T}\,\hat{\beta} \right)^2 \tag{5.4}$$

on h and c accordingly, a strong case of this happens to be not following the pattern indicted by the theorem. Rather than being the interpolation point of return, $p = n$ is now the convergence point of the prediction error instead. Test result is shown in Figure 5.3.

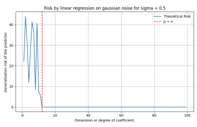
Thence, the theorem gives us two things - the interpolation point, in this setting, refers to the parameter error for their double descent behaviour. However, on the prediction results' error measure, the interpolation point correlates to the point where the defined generalization risk in such case equals zero. These patterns hold up to certain point. However, the test case is only up to $d = p$. An unexplored test case for $p > d$ indicate the surprising fact according still, to the theorem: the generalization risk takes double descent as a region with disturbance in the middle, but bias-variance appears afterward. This is illustrated in Figure 5.4, where the same theorem is applied but in the domain which is usually called as *overparameterized region*.

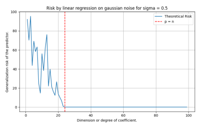
Such can then be seen as a prediction that double descent in certain case, can be extrapolated to be the transition between the two curvature of bias variance, or the abnormal point on the error curve measure. Such must then be verified, both in the experiment's correctness, and so on that lead to such result. Nevertheless, if true, it would mean that double descent indeed has another implication of itself.

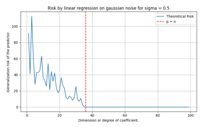
5.1.5 Extension

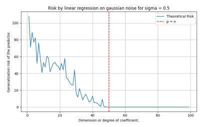
The current theory prompted for a more general treatment of the linear function class, as in-line with exploring reasonable existing alternatives in interpretation (Nakkiran [2019a]). We have much to desire of, for example, extension of prediction for stochastic or iterative algorithms, consideration on statistical formations, uncertainty metrics. Or furthermore, different kind of linear models, including multilinear models and so on, with variational setting meaningful to the problem space.

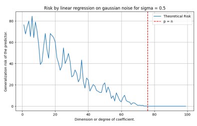
Let us states some hypothesis about the previous experimental cases. First, is the lack of the begin-tail of the prediction on double descent of linear random cutoff features model. This can be explained informally as to be resulted from the uniformity of the operating space. Specifically, all parameters of the $\mathbb{R}^n$ parameter space, with respect to the

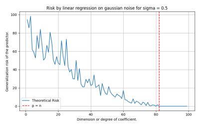

(a) $n = 12$

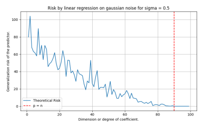

(b) $n = 24$

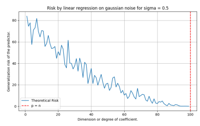

(c) $n = 36$

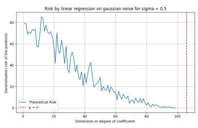

(d) $n = 50$


(e) $n = 76$


(f) $n = 82$


(g) $n = 90$


(h) $n = 100$


(i) $n = 106$


(j) $n = 112$

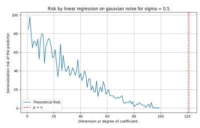

(k) $n = 121$

Figure 5.3: Contrary to Theorem 5.1.1 and both figure 5.2, this is the behaviours on randomized setting for standard parameter-wised mean square error measure (MSE-on-prediction). Especially, we do not see double descent pattern in this setting, as it is. For this model, we have $d = 100$, $\sigma = 0.5$, and variational n. Full test cases are for $p = [1, 100]$, $n = \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121\}$ accordingly. The test function of the concept itself is the same linear model, but with different input only.

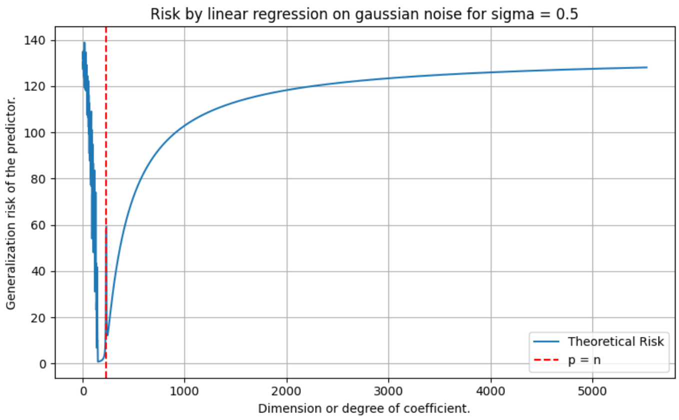


Figure 5.4: Similar experiment according to setting and result of Theorem 5.1, where p is in the range $[1, 5523]$. Here, we can see small double descent modelled exactly at the interpolation threshold, while the later region exhibit similar phenomenon as bias-variance tradeoff.

dual space on such $f : \mathbb{R}^n \to \mathbb{R}$ projection is defined, are of the same order of magnitude. That is, for $f_{i,k}(x), f_{j,k}(x)$ of arbitrary parameter indexing on the set k of shared parameter choice, and $i \neq j$ being the distinct choice, then we say $f_{i,k}(x) = \mathcal{O}(f_{j,k}(x))$ of the growth of both function on value ranges. Approximately speaking, the choice of parameter does not dramatically growth to a point of no return, like in later examples of polynomial approximation model. Such special model of multi-variable linear projection of scalar output also reinforces the numeric growth of the numerical encoding expression on the hypothesis model. The behaviour of the MSE-on-parameter, the internal risk, is also explainable using the same notion, such is that for a black-box case approximation, such linear model does not allow for average parameter deviation per approximation to be oversaturated and skewed (for example, the degree of polynomial directly counteract this, by separating expressive complexity by growth degree). How such hypothesis can be evaluated, we will have to make it for the general extension experiment, which we will have to conduct.

Let us define the more general class of linear model, denoted $\mathcal{H}_L$, in the following structure of the linear model. We also note that the previous linear class $\mathcal{H}_p$ only solves for the problem of underparameterized, hidden parameters p'. Let $k \in \mathbb{N}$ denote the number of input vectors, and let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ represent the input vectors where each $\mathbf{u}_i \in \mathbb{R}^q$ for $i \in \{1, 2, \dots, k\}$. Here, $q \in \mathbb{N}$ denotes the dimensionality of each input vector, with the convention that $q = 1$ corresponds to scalar inputs (i.e., $u_i \in \mathbb{R}$).

For a given dimension parameter $d \in \mathbb{N}$, we define $\mathcal{H}_L$ as the set of all functions $h : (\mathbb{R}^q)^k \to \mathbb{R}$ that can be expressed in the unified form:

$$h(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) = \sum_{j=1}^{d} w_j \cdot \phi_j(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) + b \tag{5.5}$$

where $\mathbf{w} = [w_1, w_2, \dots, w_d]^\top \in \mathbb{R}^d$ is the weight vector, $b \in \mathbb{R}$ is the bias term, $\phi_j : (\mathbb{R}^q)^k \to \mathbb{R}$ is a feature map for $j \in \{1, 2, \dots, d\}$ that transforms the input vectors into a scalar feature, we consider the following important special cases, distinguished by their choice of feature map ϕ_j. If we do not have such feature mapping of which preserves relative linearity, then we said it is the basic linear class dynamic.

Definition 5.1.2 (Linear class, $q = 1$). *Consider class $h : (\mathbb{R}^{d+1})^q \to \mathbb{R} \in \mathcal{H}_{\mathsf{Lin}}$ of all functionals of the form as Equation 5.5. Then, when $q = 1$ of scalar observables, i.e., $u_i \in \mathbb{R}$ for $i \in \{1, \dots, k\}$, then the function is simply as:*

$$h(u_1, u_2, \dots, u_k) = \sum_{j=1}^{k} w_j \cdot u_j + b \tag{5.6}$$

for $\phi_j(u_1, \dots, u_k) = u_j$ for $j \in \{1, 2, \dots, k\}$, i.e. deactive feature map.

Such class can be extended rigorously, through a varied change in dynamics, specifically, considering the general feature representation on ϕ.

Definition 5.1.3 (General feature, kernel linear model). *Consider class $\mathcal{H}_{\mathsf{Lin}}$. For maximum flexibility, allow arbitrary feature maps $\phi : (\mathbb{R}^{d+1})^q \times \mathbb{R}^k$ for $k = \{1, 2, \dots\}$ as of hidden or feature mapping processor internally. The functional form is then expressed by*

$$h(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) = \sum_{j=1}^{d} w_j \cdot \phi_j(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k) + b \tag{5.7}$$

for $j \leq d$ as the iterative form.

Common choices for such kernel mapping includes:

- **Polynomial features**: $\phi_j(\mathbf{u}) = u_{i_1} u_{i_2} \cdots u_{i_m}$ for various index combinations
- **Radial basis functions**: $\phi_j(\mathbf{u}_1, \mathbf{u}_2) = \exp(-\gamma \|\mathbf{u}_1 - \mathbf{u}_2\|^2)$
- **Fourier features**: $\phi_j(\mathbf{u}) = \sin(\boldsymbol{\omega}_j^\top \mathbf{u})$ or $\cos(\boldsymbol{\omega}_j^\top \mathbf{u})$
- **Neural network activations**: $\phi_j(\mathbf{u}) = \sigma(\mathbf{v}_j^\top \mathbf{u} + c_j)$ for activation σ, though we note that this is a very simplistic removal of structural basis class. Rather, it is performing shallow network in linear approximation on kernel system. The reason most of the time we do not use this is because of complexity in multi-kernel distortion.

From such, it basically hints at the generality of, at least being a *linear approximation* on various hypothesis spaces, while not actually being similar. Then, our setting of sparsity being synthetically induced introduces the basin of subclasses. For any $p \in \{1, 2, \dots, d\}$, we can then simply define the sparse subclass $\mathcal{H}_{L,p} \subseteq \mathcal{H}_L$ as the set of all functions $h \in \mathcal{H}_L$ where the weight vector $\mathbf{w} \in \mathbb{R}^d$ has exactly p non-zero entries. Formally, per Lafon and Thomas [2024],

$$\mathcal{H}_{L,p} = \{h \in \mathcal{H}_L : |\{j : w_j \neq 0\}| = p\} \tag{5.8}$$

The subset $S \subseteq \{1, 2, \ldots, d\}$ of indices with $|S| = p$ corresponding to non-zero weights is assumed to be fixed throughout the learning process (i.e., the sparsity pattern is predetermined). This constraint yields:

$$h(\mathbf{u}_1, \ldots, \mathbf{u}_k) = \sum_{j \in S} w_j \cdot \phi_j(\mathbf{u}_1, \ldots, \mathbf{u}_k) + b \tag{5.9}$$

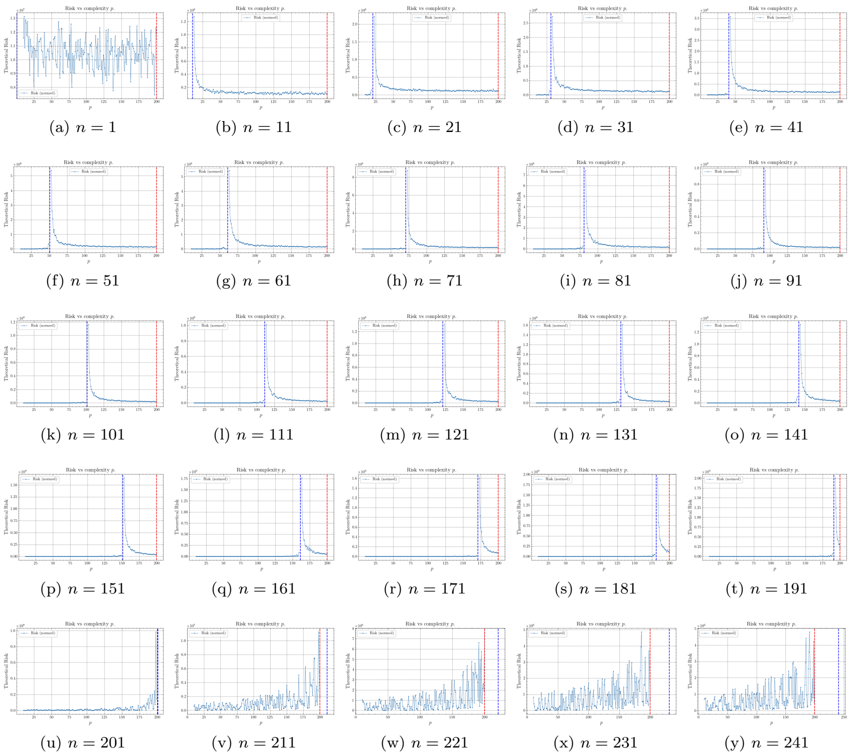
where only p features contribute to the final prediction. This effectively masks the data randomly, simulating the general observation-based information corruptions, without inducing specific bias like *dropout*, though potentially leads to strong perturbation in cases of meaningful information threshold for different function classes.

Baseline investigations

Preliminary model learning then will relies on theorem 5.1.1 for its initial prediction, using least square approximation results. The testing is adversarial and consistent, of which we have $c \in \mathcal{H}_L$ and $h \in \mathcal{H}_{l,p}$ for $p = d$, which can then further be modified. Because of their equal expressive capacity, respect to independent consideration of the parameter space $\mathbf{X}$, we define the complexity, which will be used later on, simply as

$$\mathbb{C}(h) = \frac{1}{||\overline{\mathbf{x}} - \underline{\mathbf{x}}||} \left[1 + p + q + \mathop{\mathbb{E}}_{i \in S}(w_i x_i) + b \right], \quad S = \{p_i\} \tag{5.10}$$

Functional prediction on the classical theorem is presented below, for $n = \{1, \ldots, 241\}$ samples, and with $p \in [1, 200]$. We set the target degree to the maximal 200, since the theorem is not defined on the overparameterized region.



(a) $n = 1$ (b) $n = 11$ (c) $n = 21$ (d) $n = 31$ (e) $n = 41$

(f) $n = 51$ (g) $n = 61$ (h) $n = 71$ (i) $n = 81$ (j) $n = 91$

(k) $n = 101$ (l) $n = 111$ (m) $n = 121$ (n) $n = 131$ (o) $n = 141$

(p) $n = 151$ (q) $n = 161$ (r) $n = 171$ (s) $n = 181$ (t) $n = 191$

(u) $n = 201$ (v) $n = 211$ (w) $n = 221$ (x) $n = 231$ (y) $n = 241$

Figure 5.5: Extension general experimentation, using the prediction made of theorem 5.1.1 for verification. Here, $q = 1$ as the general basis case of verification.

A design of the above class will require, in general, a much less straightforward implementation than the least square method, at least when it comes to circumstances and case fielding. The previous experimental result was done using a straightforward, single-time calculated theorem. To verify such, we need to adopt the non-zero class in a way that is static of the entire training procedure, and so on, for larger and larger q. First, we discuss of the mismatch calculation and thereof taken from the iterative error segmentation. In our cases, we have h and c with *model mismatch*, that is, there might exist certain parameter that h has, but the concept c of its originality does not, and vice versa for the concept model of which the hypothesis does not have, or attain certain parameter of interest. This prompts us to make use of a type of **penalty** for such case. With respect to the mean square error on parameter, from the perspective of the supervisor, we have two ways to implement this, either of the truncate or padding mode. The truncate mode simply calculates parameter expression of those that can be compared. By default, for the learning problem to be consistent of the space, there must be at least one parameter that is not inhibited. Then,

$$\text{MSEP}_{\text{truncate}}(h, c) = \frac{1}{k}\sum_{i=1}^{k}(p_i - q_i)^2 \tag{5.11}$$

For $K = \min(n, m)$ of the overlap, or can be changed to be the set $k = \{(h_i, c_i)\}$ of which matches each other. Such can be changed to absolute error, if one does not want to inflate the error measure beyond its distance metric with respect to certain axis. The padding mode simply takes into account of all missing or surplus information and of the optimization phase. By initialization, from the hypothesis side, all surplus parameters are padded with input 1. Hence, the measure is defined as

$$\text{MSEP}_{\text{pad}}(h, c) = \frac{1}{k}\sum_{i\in k}^{k}(p_i - q_i)^2 + \lambda\left(\frac{1}{n-k}\sum_{i\notin k}^{n} p_i^2 + \frac{1}{m-k}\sum_{i\notin k}^{m} q_i^2\right) \tag{5.12}$$

The total error then is simply the above error, is simply calculated toward all layers possible or any architectural design parameters. This design will be also used in subsequent testing, including polynomial, in which the effectiveness is enhanced significantly.

5.2 The polynomial transition problem

Polynomial space is one of those famous example that is often cited of the bias-variance tradeoff, for example, in Goodfellow et al. [2016], Sugiyama [2015]. Indeed, it is considered, usually, a very effective and high-volatility construct class. Let us reiterate what we know about polynomial class.

There exists two type of elementary polynomial classification, based on their (i) variable count or (ii) its domain range. We first define the univariate polynomial function.

Definition 5.2.1 (Univariate polynomial). *A function of a single variable t is a polynomial on its domain if we can put it in the form*

$$a_n t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0 t^0 \tag{5.13}$$

where $\{a_i\}$ are constants, and of a singular variable t.

From this, we can extend to polynomial of multiple variables. In general, a function $f(x, y)$ is a *multivariate polynomial* in x, y if and only if it can be represented as a finite sum of terms of the form $cx^k y^m$, where c is a coefficient and k an m are nonnegative integers. The number $k + m$ is called the degree of the term, and the degree of the polynomial $f(x, y)$ is equal to the highest degree of its own terms. We can then generalize multivariate polynomial and univariate polynomial in the same kind.

Definition 5.2.2 (Polynomial, multivariate generalization). *Let K be a field. We then denote $K[x_1, \ldots, x_n]$ the space of all polynomials in n variables, $x_1, \ldots, x_n$. A **multivariate monomial** is then an expression $x_1^{\alpha_1} \ldots x_n^{\alpha_n}$ where $\alpha_1, \ldots, \alpha_n$ are integers. The degree ò such multivariate polynomial is the largest degree of terms of any monomial term.*

Effectively, we can see that multivariate polynomial can take on many forms, and also unconventional forms. In order to capture any reasonable details, most of the time focus are on symmetric polynomials $f(x, y)$ in which $f(x, y) = f(y, x)$, or homogeneous polynomials in which all the terms are of the same degree.

Given such, operationally, polynomial is used for expressing other functions, due to their usefulness because of such simplicity and versatility. In fact, applications ranges, even to a polynomial basis on n-dimensional space, $(p_1, \dots, p_n) \in K[x, \dots, x_n]$; their growth rate being polynomial also added more to their strength of approximating fast increasing function, usually expressed and used in setting, for example, polynomial growth class $\mathcal{O}(p(x))$ in computer science. Two applications stand out for such properties: function approximation of arbitrary criteria ℓ, or polynomial interpolation on arbitrary points. Those two are interconnected, though their objective and object of concern is much different.

We can define the application of analytical polynomial on arbitrary function by using the criteria on approximation. Usually, this is expressed in the theory of approximation, and requires a guarantee that polynomial can indeed approximate to arbitrary accuracy, of arbitrary function on specific interval. This is the statement of Weierstrass theorem.

Theorem 5.2.1 (Weierstrass Approximation Theorem). *Let $f \in C[a, b]$ of all continuously differentiable functions on the closed interval $[a, b]$. Then, for every $\epsilon > 0$, there is a polynomial p such that $||f - p|| < \epsilon$, for $||f - p|| = \sum_{i=1}^{\infty} |(f(x_i) - p(x_i))|$* [3]

Proof. This is one of the many standard proofs for WAT (Schep [2007] to be specific). We begin by extending f to a bounded uniformly continuous function on $\mathbb{R}$, defining $f(x)$ as to be $f(a)(x - a + 1)$ on the domain $[a - 1, a)$, for $f(x) = -f(b)(x - b - 1)$ on the domain $(b, b + 1]$, and $f(x) = 0$ on $\mathbb{R} \setminus [a - 1, b + 1]$. There exists, in particular, $R > 0$ such that $f(x) = 0$ for $|x| > R$. Let $\epsilon > 0$ and M such that $|f(x)| \leq M$ for all $|x|$. let $\epsilon > 0$ and M such that $|f(x)| \leq M$ for all x. Then by the above theorem there exists $h_0 > 0$ such that for all $x \in \mathbb{R}$, we have

$$|S_{h_0} f(x) - f(x)| < \frac{\epsilon}{2} \tag{5.14}$$

Since we also have $f(u) = 0$ for $|u| > R$, we can write

$$S_{h_0} f(x) = \frac{1}{h_0\sqrt{\pi}} \int_{-R}^{R} f(u) \exp - \left(\frac{u - x}{h_0}\right)^2 du \tag{5.15}$$

On $\left[-\frac{2R}{h_0}, \frac{2R}{h_0}\right]$ the power series of e^{-v^2} converges uniformly, so there exists N such that

$$\left|\frac{1}{h_0\sqrt{\pi}} e^{-\left(\frac{u-x}{h_0}\right)^2} - \frac{1}{h_0\sqrt{\pi}} \sum_{k=0}^{N} \frac{(-1)^k}{k!} \left(\frac{u - x}{h_0}\right)^{2k}\right| < \frac{\varepsilon}{4RM}. \tag{5.16}$$

for all $|x| \leq R$ and all $|u| \leq R$, since in that case $|u - x| \leq 2R$. This implies that

$$\left|S_{h_0} f(x) - \frac{1}{h_0\sqrt{\pi}} \int_{-R}^{R} f(u) \sum_{k=0}^{N} \frac{(-1)^k}{k!} \left(\frac{u - x}{h_0}\right)^{2k} du\right| < \frac{\varepsilon}{2}. \tag{5.17}$$

for all $|x| \leq R$. If we put

$$P(x) = \frac{1}{h_0\sqrt{\pi}} \int_{-R}^{R} f(u) \sum_{k=0}^{N} \frac{(-1)^k}{k!} \left(\frac{u - x}{h_0}\right)^{2k} du, \tag{5.18}$$

then $P(x)$ is a polynomial in x of degree at most $2N$ such that $|S_{h_0} f(x) - P(x)| < \varepsilon/2$ for all $|x| \leq R$. This implies that $|f(x) - P(x)| < \varepsilon$ for all $x \in [a, b]$. ☐

[3] We again note the disparity between $h(x) - c(x)$ or $c(x) - h(x)$ for clarity. In essence, $c(x) - h(c)$ is looking from the perspective of the concept itself. While $h(x) - c(x)$ is comparison on difference from the perspective of the hypothesis.

Here, ϵ is defined to be the *global error* of the polynomial with respect to the arbitrary function, and with uniform approximation. Another variation works on the squared error, such that $||f - p|| = \sum_{i=1}^{\infty}(f(x_i) - p(x_i))^2$, or the mean absolute

$$||f - p|| = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} (f(x_i) - p(x_i))$$

Weierstrass Approximation Theorem provides tighter bound for the mean absolute error, but not the square error, or the mean square case. Nevertheless, they are natural measure on the uniform metric space $C(X)$ with measure $d(f, g) = \sup_{x\in X} |f(x) - g(x)|$. If said metric also use the L^p metric instead, which is defined by

$$d_p(f, g) = \left(\int_X |f(x) - f(g)|^p \, dx \right)^{1/p} \tag{5.19}$$

we can also see that it is weaker than the bound given by Weierstrass theorem. Generalization of Weierstrass theorem can be made by Stone-Weierstrass theorem instead, of which is state in real space as followed, though generally Weierstrass theorem is sufficed.

Theorem 5.2.2 (Stone-Weierstrass Theorem (real version)). *Let S be a locally compact Hausdorff space. Let A be the subset of bounded continuous real functions on X, or of subalgebras $\mathcal{C}(X, \mathbb{R})$ which contains a non-zero constant function, then A is uniformly dense in $\mathcal{C}(X, \mathbb{R})$ if and only if it separates points.*

In the rigorous proof, or so is the proof for Theorem 5.2.1, the only objective is to establish the existence in which such polynomial might exist, but talk nothing effectively about how to obtain such polynomial, or the feasibility of such model. Indeed, in Bernstein's construtive proof (Sury [2019], Muresan [2023]), the proof utilized the construction of the monomial-basis spanning sub-family of the Bernstein polynomial, and show that converges uniformly to f arbitrarily defined on the mapping space. Such is a sufficient proof, in which Bernstein's family is dense in $C[0, 1]$ and thus specify the polynomial structures itself; but overall, there exists no necessity proof on such matter. Such a necessity theorem does not exist, because WAT is usually expressed as *far too weak* to enforce such generality. A heavier version must be used, for example, Korovkin's on the sequence of positive linear operators, but otherwise this is inherently lacking. Expressively, this is the same for the dynamics and how different families of polynomial will emerge or moves under dynamical process and true approximation. However, the theorem is still useful in its sense of the universality, thus we will still use it, but **only** so.

The setting of polynomial interpolation is rather different from the setting of function approximation. The setting of interpolation gives an n samples set $\mathcal{S}$, of which in this case we restrict to $\mathbb{R} \to \mathbb{R}$. Then, we are required to fit exactly the entire dataset, such that the following setting is sufficed.

Definition 5.2.3 (Interpolation). *A function class $\mathcal{F}$ is said to* ***interpolate*** *a distinct dataset $\mathcal{S}_n = (x_n, y_n)$ of n samples, if there exists $f \in \mathcal{F}$ such that $\mathbb{E}||S_i - |f(x_i)|| = 0$. Such f is then called the* ***interpolating function*** *of $\mathcal{S}_n$.*

Under such definition, the question is then if the polynomial class $\mathcal{P}_n$ of any given polynomial with the highest degree n is sufficed to approximate such dataset. The answer can be found using unisolvence theorem.

Theorem 5.2.3 (Unisolvence theorem). *Given $n + 1$ bivariate data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n) \in \mathcal{S}_n \subset \mathbb{R}^n$, with all x_i distinct, there exists a unique polynomial $p_n(x)$ of degree at most n such that $p(x_i) = y_i$ for $i = 0, 1, \dots, n$.*

In determining such problem, we can also say that there exists a unique n-saddle point function represented here as polynomial, such that to fit n points perfectly by the definition of the saddle point. The process in obtaining such formula is difficult, however, and finding the perfect interpolation polynomial would be computationally significant, and erroneous. The following defintion specifies the concept of such saddle point.

Definition 5.2.4 (Saddle point,Nie et al. [2021]). *Let $X \subseteq \mathbb{R}^n$, $Y \subseteq \mathbb{R}^m$ be two sets (for dimension $n, m > 0$) and let $F(x, y)$ be a continuous function in $(x, y) \in X \times Y$. A pair $(x^*, y^*) \in X \times Y$ is said to be a saddle point of $F(x, y)$ over $X \times Y$ if*

$$F(x^*, y) \le F(x^*, y^*) \le F(x, y^*), \quad \forall x \in X, \forall y \in Y \tag{5.20}$$

Effectively speaking, each interpolation degree of the polynomial interpolation scheme admits a specific (x^*, y^*) of which that the convexity is absolved. Thus, the reason why you require n saddle points for n points, or d-degree polynomial for $d+1$ points. The landscape of such interpolation is "wiggling", as indicted, since for any two interpolation point, the macroscopic details would be similar as Figure 5.6, of such extend for any local monotonicity on n saddle points, or basically a secant line.

x	a		x_i^*		x_{i+1}^*		b
$\mathcal{D}_1[p]$		$-$	0	$+$	0	$-$	
$p_1(x)$	$p(a)$	↘	y_i^*	↗	y_{i+1}^*	↘	$p(b)$
$\mathcal{D}_2[p]$		$+$	0	$-$	0	$+$	
$f_2(x)$	$p(a)$	↗	y_i^*	↘	y_{i+1}^*	↗	$p(b)$

Figure 5.6: Table of monotonicity for the saddle point landscape between $[a, b]$, for both cases of the landscape.

Polynomial regression is then one such analytical solution to function approximation task, using Weierstrass theorem as its basis. From such, we can also observe that polynomial regression, and hence polynomial interpolation, are correlated and can be reduced to each other. Polynomial regression can be reduced to interpolation via stating the criteria such that, denoting it as ℓ, then $\ell(f, p) = 0$, and interpolation can be reduced to regression via the condition that $p(x_i) - y_i \leq \epsilon$ for arbitrary $\epsilon > 0$. Hence, the regression formulae lie between the two extreme: the Weierstrass setting and the interpolation setting.

We then have to identify such extreme and what kind of extreme they represent. Fortunately, we know this as the problem of **transparency**, in the setting of approximation, which certainly correlates to the learning theory. According to such, the Stone-Weierstrass represents the total transparency case, where the function f is known to the approximator with polynomial class. Interpolation represents total black box, where the goal is only to interpolate the sample point by itself. In between such interpretation, there exists a fundamental assumption, hence the structural assumption.

Assumption 5.2.4 (Transparency). *For any bivariate pair (x, y), we assume each pair depends on and originate from a based concept $y = f(x)$.*

Two kinds of knowledge is then available, which correlates to the degree of freedom the observation space can have. First is the knowledge of f. This knowledge of f is encoded directly by the number of sample point and their distribution in certain case. Accordingly, the worst case is as followed.

Conjecture 5.2.1. *If $\forall x$ there exists y corresponding to such, and $(x, y) \in \mathcal{S}$, then we say we have total transparency to the concept f. If there exists only finite x that we can have y corresponding to, then we say we have limited transparency to the concept f. If we assume no knowledge of f, then we say the setting is called the **black-box setting**.*

Secondly is the action on the dataset itself. One of the main assumption that is usually observed is that

Assumption 5.2.5 (Fallacy). *We assume there exists every specific bivariate pair $(x_i, y_i) \in f[\mathcal{S}_n]$, such that $f(x_i) = y_i$.*

However, usually, this is not the case, as we have **noise** in addition from the dataset. The origin of the noise cannot be tracked. However, we can interpret what happens when such noise is introduced to the dataset. The noise transitions the transparency of the learned setting, on the field $\mathbb{R}$, by utilizing the dense property of the real space. Assuming we operate on $\mathbb{Z}$. Then, each discrete noise will introduce abnormality point such that no function on $\mathbb{Z}$ will be able to match, up to certain distance on $\mathbb{Z}$. This is because on such space, there exists no point x_1, x_2, for example, that lies between $d(x_1, x_2) = 1$. If the noise is randomized on ± 1, then there will exists no function that can approximate such disparity better than a random fluctuating function. However, on $\mathbb{R}$, the real number is inherently dense, such that there will always be certain point x_3 such that $x_3 \in [x_1, x_2]$, by the density of real number. By the distance metric on $\mathbb{R}$, we define $d(X, Y) = |X - Y|$ for distance between two data points. Noise intrinsically increases such distance, such that $d'(X,Y) > d(X,Y)$. Hence, the space in which density can appear increase, thus there then exists more fluctuation, and thus more uncertainty between such two points. This notion can be increased from local uncertainty to global uncertainty, hence the *noisy dataset*.

We then can measure or aggregate some properties of the dataset itself, including the **uncertainty** of the dataset. We define the simple uncertainty of the dataset as followed.

Definition 5.2.5 (Dataset uncertainty). *Given a closed interval $[a,b]$, for the sample set $\mathcal{S}_n = (X,Y) \in [a,b]$, the uncertainty $\mathcal{U}(\mathcal{S}_n)$ is defined as*

$$\mathcal{U}(\mathcal{S}_n) = \left[\mathop{\mathbb{E}}_{x_i \in X}\Big(x_{i+1} - x_i\Big)^2 + \mathop{\mathbb{E}}_{y_i \in Y}\Big(y_{i+1} - y_i\Big)^2\right]^{1/2} \tag{5.21}$$

We assume X, Y are totally ordered sets.

Hence, the assumption of fallacy assumes $\mathcal{U}(\mathcal{S}_n) = 0$, though in reality it is often $\mathcal{U}(\mathcal{S}_n) > 0$. We would like to clarify, that such definition is currently reserved for only polynomial-like functional class; though, this can be extended pretty easily for any $\mathbb{R}^n \to \mathbb{R}^m$ functional concepts. Furthermore, it is *operational*, in the sense that it measures the numerical encoding inherent uncertainty, of which is also the space used of algorithms traversing in stochasticity. It affects cross-validation, gradient-based algorithm path finding, and roughness of the learning processes. Certain concerns must also be addressed, such as the definition, which assumes *sequential processing* and pair-wise measurement (which can also be extended to tri-fold measurement and so on). Indeed, modern deep learning use parallel batch processing, mini-batch sampling, or distributed learning across devices for optimization. Nevertheless, such includes sequential-based learning optimization, and so on, where mini-batch still inherently includes batch ordering; algorithms like RNNs and so on also are sequential. The computational cost per dataset, if not, is $\mathcal{O}(n)$ after initial order, or with mundane immediate measurement on the way that the dataset was configured at start, since the system and stochastic algorithms sequentially go through them.

Certain others aspects can be expected to be clarified. While indeed, this is order-dependent measure, and one might expect to go against it and say it is not real dataset property, but the key point is that dataset are measurements, captured and recorded into the encoded space that is used upon the learning regime (supervised or semi-supervised or so on), or the learning algorithms. In a classical algorithmic world, such must be done in discrete steps, and hence ordering strongly matters as well as the processing pipeline relevant to it. There exists natural ordering of data within noise ϵ, because if now, the projection on embedded space $\mathbb{R}^n \times \mathbb{R}^m$ of any given concept functional c holds no value, as the function itself, plus the distribution, with respect to the encoding space where the function mapping is defined already contains structural information encoded within 'value range' as the most basic form. A weaker posture we can adopt is that it is an operational uncertainty measure, given presentation order to a finite discrete processing machine, and is indeed *pre-training* diagnostic. There is also the issue in which X and Y might not be measurable or comparable. Indeed, since the range of value can jump intermittently between different concept class and form, but even then, it will only highlight the disparity between the contributing terms. For example, if we have three pairs, $(130,1)$, $(150,1)$ and $(225,0)$ as the data, then the volativity is not that bad, and stable on Y, but *very high on X*, as there are missing information in large gap. From the machine-algorithmic, this matters. if we take the $130, 225$ points as a case of shuffling, then the machine experiences a 95-to-1 volatility ratio. Nevertheless, one can attempt a normalized metric, i.e.

$$\mathcal{U}_{\text{norm}}(\mathcal{S}_n) = \sqrt{\mathbb{E}[(\Delta x_i)/r(X)]^2 + \mathbb{E}[(\Delta y_i)/r(Y)]^2} \tag{5.22}$$

for relative jumpiness accorss different problems with normalized morphism.

With this, we can quantify how the dataset is behaving on the scale of the bivariate form typically seen. Noise intrinsically increases this distance, usually by the y-axis. This dataset uncertainty increases with multivariate dataset, for example, $(\mathbf{x}, y)$ of $\mathbb{R}^n \to \mathbb{R}$. We call the new metric as the *empirical uncertainty metric*. The basis of such argument on the concept of a tradeoff comes from Nakkiran [2019a], in which linear regression is affected, at least in such setting and consideration, such that dataset will both hurt and benefit the learning process.

Overall, however, this allows us to state of the following theorem:

Theorem 5.2.6 (Separation encoding). *We fix the encoding space on $\mathbb{D} = \mathbb{R}^q \times \mathbb{R}^p$ for a (q,p) conjunction space. For $\mathcal{S}_n \in \mathbb{D}$ of n samples (x_i, y_i), we assume such set is output from an oracle $EX(\theta)$. Then, for a diffused set $\mathcal{S}'_n = \mathcal{S}_n + \eta$, for $\eta = \mathcal{D} \approx \mathcal{N}(0, \lambda\sigma^2)$, then*

$$\mathop{\mathbb{E}}_{\eta \in \{\mathcal{N}\}}[\mathcal{S}'_n] \geq \mathcal{U}(S) \tag{5.23}$$

for $\lambda\sigma^2$ the x-aware deviation. Fix $\lambda = 1$ for consistent enclosure 0-based error, or $\mathcal{N}(\cdot, \sigma^2)$ for consistent data deviation.

Proof 1. Simple proof. Using normed metric, $\mathcal{U}_{\text{norm}}$. For noised data, that is,

$$\mathcal{S}'_n = \{(x_i + \eta_i^{(x)}, y_i + \eta_i^{(y)})\}_{i=1}^n \tag{5.24}$$

where the noise vector on encoding space is orthogonal, such that

$$\eta_i = (\eta_i^{(x)}, \eta_i^{(y)}) \sim \mathcal{D} \approx \mathcal{N}\left(a, \lambda \text{Cov}(\eta)\right) \tag{5.25}$$

where we can fix $\mathcal{D} = \mathcal{N}(a, \lambda\sigma^2)$ for $a = 0$ as normalized around the zero value. If we assume independent noise, then we have the covariance as

$$\lambda \text{Cov}(\eta) = \lambda \begin{pmatrix} \sigma_x^2 I_q & 0 \\ 0 & \sigma_y^2 I_p \end{pmatrix} \tag{5.26}$$

We can simplify slightly, as to have the case in consideration that $\sigma_x^2 = \sigma_y^2 = \sigma^2$, then the notation would effectively be such that $\lambda\sigma^2 I_{d+p}$ is enough. The uncertainty metric then should be expressed as

$$\begin{aligned} &\mathbb{E}[\mathcal{U}(\mathcal{S}_n)^2] \\ &= \mathbb{E}\left[\mathop{\mathbb{E}}_{x_i \in X}\Big(x_{i+1} - x_i\Big)^2 + \mathop{\mathbb{E}}_{y_i \in Y}\Big(y_{i+1} - y_i\Big)^2\right] \\ &= \mathbb{E}_i \end{aligned} \tag{5.27}$$

□

Thirdly, is the treatment on such information. Even though information is known, such as from the assumption of transparency and the fallacy conjecture, it relies on the method in which the model uses that make use of such information. Polynomial interpolation effectively removes information about transparency, and thus the interpolation setting is a kind of **black-box absolute setting**. This setting is *extremely brittle*, as to see that if y_i is slightly wrong due to measurement noise, the interpolant oscillates wildly, which is explained by Runge phenomenon (Runge [1901], Corless and Sevyeri [2018], Boyd [2009]). Probability then arises from such setting naturally, though this particular degree of probabilistic setting must be quantified, which comes from the process that the model simulates upon. Regression is inherently not probabilistic, but treatment about the probabilistic assumption, for example, such that (X, Y) inherently as random variables. We can then separate this into three parts - the data-level probabilistic layer (data uncertainty), the model-level layer, and the algorithm-layer that determines the solution solver within such model. We can define that as followed.

Conjecture 5.2.2 (Probabilistic layer). *Given a learning setting with dataset $\mathcal{S}$ of bivariate pairs, $h \in \mathcal{H}$, $c \in \mathcal{C}$ as the hypothesis and concept, we have the following conceptual probabilistic setting:*[4]

1. *Data-level probability: Sampling assumptions (observation sample points) such that there exists the distribution $\mathcal{D} \sim (X, Y)$, intrinsic data uncertainty $\mathcal{U}(\mathcal{S}_n)$.*

2. *Model-level probability: Structure of the dataset is controlled by the concept structure $\{\theta\} \in c$, c is modelled as a target sampler $c \equiv \mathcal{D}$, knowledge about the structure of c, Bayesian prior-posterior assumptions, $\mathbb{P}(Y \mid X, \{\theta\})$. For Bayesian polynomial regression, the model-level probability is formulated as*

$$\mathbb{P}(Y \mid X) = \int \mathbb{P}(Y \mid X, \theta)\mathbb{P}(\theta)\, d\theta, \quad \mathbb{P}(\theta_j) = \mathcal{N}(0, \sigma^2) \tag{5.28}$$

of which assumes aleatoric uncertainty and epistemic uncertainty from the weight.

3. *Algorithm-level probability: Stochastic method, decision conditional (dropout, MCMC), ensembles criteria, Monte-Carlo initialization, variational inference.*

[4] Currently, we have no way to **quantify** the 'probabilistic degree' of the system setting itself

Dynamically, in this paper, we would like to investigate this particular junction between uncertainties, to see what happens when the model transparency, the treatment of data, and so on, that exhibits the observations that we would receive from the model itself. The experimental details will infer such result.

Generally, we would be entitled to see particular trajectory specifically designed around the point of interpolating match, or $n = m$ of n degree of freedom, technically, and m data points. After this point, there exist two possibility of dynamic - either the model dilutes, in which case result in *double descent*, or the model fluctuate strongly afterward, leading to *bias-variance emergence* again later on. This conclusion however, would have to be analysed within the current framework of the possible interjection to its fluctuation strength. For example, in the monomial basis $\{x^i\}^n$, any given fluctuation contributes to the entire polynomial. In such case, fluctuation of the other $[0, n-k]$ of arbitrary k and large enough n would be unaccounted for by the scale. Conversely, restrict them to $[1, 0]$ range will reveal something about the system, however will destroy particular properties of the prediction, which is not recommended. Nevertheless, we must conduct the test before giving in any reasonable conclusion.

Previously, we said that we did indeed focus on the interpolation problem. Now, assuming we have given the interpolating algorithm $\mathcal{A}$ to approximate the interpolation, what will this look like? Firstly, solving this is *very computationally expensive*, because it is a dense, $(n+1) \times (n+1)$ linear system, and even more so if we consider the overparameterized outcome. However, this can be reduced further down if we attempt to solve them using convergence bound, but will be still very expensive. That and incremental soft interpolation is somewhat out of the question since it will be kind of the same. Aside from the insight, we would still have to know what to do. Then there exists two ways - we have some version of this to optimize of simply evaluating by gradient descent optimization, or, we can use somewhat analytical fit for the case of soft interpolation modification.

A theory on such problem can be obtained of its mechanics, as will be a fairly intuitive way in consideration of the theory of interpolation, as well as the technique of interpolation by itself. For a fixed-range interpolation threshold problem, a hard-encoded interpolation solution requires n-degree polynomial, of which has $n+1$ degree of fitness, to fit $n+1$ distinct points. Suppose all points are distinct of the set S which is finite, then this is the case. If we switch to the problem of interpolation but induce a more forgiving bound, there exists a possibility that it induces a relative *density averaging path* for the points, especially if we are considering the case where $p > n$ for p saddle points or degree of fitness, and n the number of point to interpolate. In a more nuanced formulation, we have

$$\mathsf{Flatten}\left(\begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^p \\ 1 & x_1 & x_1^2 & \cdots & x_1^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_p & x_p^2 & \cdots & x_p^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^p \end{bmatrix}_{n\times p} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} - \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ f(x_3) \\ \vdots \\ f(x_n) \end{bmatrix}\right) \leq \epsilon \tag{5.29}$$

or equivalently, $\mathsf{Flatten}(\mathbf{X}_{n\times p}\mathbf{w}_n - \mathbf{Y}_n) \leq \epsilon$ where $\mathsf{Flatten}(\cdot)$ is the compression of vector to scalar, or so $\mathsf{Flatten} : \mathbb{R}^n \to \mathbb{R}$. This essentially is the problem of solving n points approximation, toward using only p saddle points available. If $\epsilon = 0$, it is well known that interpolation is impossible. However, if $\epsilon \neq 0$, there are possibilities, depends on the shape of the dataset, that there will exist a data-dependent property or more that indict a particular *density preference* toward those saddle points - or rather, the act of density averaging. This is reflected rather implicitly in the structure itself. In machine learning, the error measure is usually the mean average, such is to say that the criterion now would look more like $\mathbb{E}\Big[\mathsf{Flatten}(\mathbf{X}_{n\times p}\mathbf{w}_n - \mathbf{Y}_n)\Big] \leq \epsilon$. Formally, we have the following hypothesis

Hypothesis 5.2.1 (Polynomial double descent). *For a polynomial $\{a_i x^i\}_n$ on Vandermonde basis, approximating on the set $\{(x_k, y_k)\}_m$, then*

1. *For $n < m$, we expect the polynomial form factor to be inherently sparse, such that for an iterative approximation, for each p_i saddle point of the polynomial expansion form, there exists an ϵ-ball B_ϵ for $\epsilon > 0$ with at least one data point satisfying $(x_k, y_k) \in B_\epsilon$.*

2. *For $n = m$, we expect the polynomial to be able to fit all points, up to hard interpolation threshold such that $\hat{R}_S \approx 0$.*

3. *For $n > m$, we expect the polynomial form factor to be locally dense then turning to globally dense, over random ensembles. That is, for every point (x_k, y_k) there exists a β-ball B_β such that there is at least one saddle point of the polynomial belong to the ball.*

The shift that will be hypothesized to appear here is expected to be from sparse in the polynomial term sense, out to globally dense with enough model complexity, over an iterative process. One also main point of clarity is, at certain point right now, we haven't discovered the effect of error deviation, yet. For such, we theorize the following.

Conjecture 5.2.3 (Double descent). *Double descent happens, for a polynomial class model $h_n \in \mathcal{H}_p$, is upper bounded to certain error deviation $\epsilon > 0$ from the mean only of the generalization (test) dataset verification.*

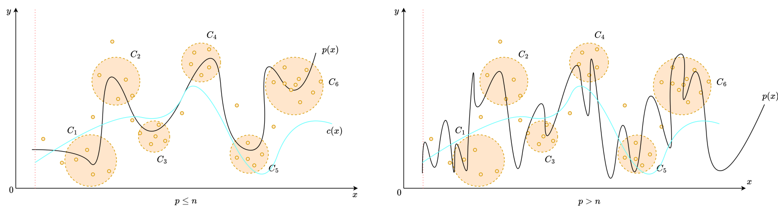


Figure 5.7: **Intuition of Hypothesis 5.2.1.** Effectively, in the overparameterized range ($p > n$), the problem turns into soft interpolation, where the polynomial is possible to be dense enough such that relative error remains internally low. This can also happen if the dataset itself is intrinsically large.

That is, the disparity between the test and training dataset helps in such regard. If the two dataset is partitioned in the sense that their deviation is practically the same (which can be done using synthetic generation for a control group), then this will amount to not much of it. After all, it will be the consequence of the algorithm and error landscape interpretation used.

For this, we propose another speculative observation beside such, as to consider adjourned transform of the Vandermonde basis to the Newtonian or similar basis, in which the effects and motion can be easily observed, as least easy. This is because aside from mapping down to $[-1, 1]$, normal polynomial on monomial basis gives suboptimal result because of intrinsic high-degree term domination. This will become more relevant later on in the analysis.

5.2.1 Structure

For mathematical analysis and experimental design, we would like to state the current main structure of a polynomial model. Specifically, in such example, we are concerned of the shallow polynomial neural network, of which is stated as such in Arjevani et al. [2025]. Notice that it is defined on the normal Vandermonde basis, where an expression of such polynomial model under Newtonian or Lagrangian basis is not so clear-cut of an analysis.

Definition 5.2.6 (Shallow polynomial neural network). *A shallow polynomial is a neural network of one layer that are functions $f_{\mathcal{W}}$, defined such as: $\mathbb{R}^d \to \mathbb{R}$ of the form*

$$
\begin{aligned}
f_{\mathcal{W}}(x) &= \sum_{i=1}^{r} \alpha_i (w_i \cdot x)^d, \\
\mathcal{W} &= (\alpha, w_1, \dots, w_k) \in \mathbb{R}^r \times (\mathbb{R}^n)^k,
\end{aligned}
\tag{5.30}
$$

where $d \in \mathbb{N}$.

A modification is then available, in which we simply add $b(\cdot)$ for random initialization process. We note that this introduces certain flexibility to the model initial interpolation curvature.

Definition 5.2.7 (Polynomial neural network, general). *The polynomial network based on Model 5.2.6 is modified as*

$$
f_\Theta(x) = \sum_{i=1}^{d} \theta_i \left(\mathbf{w}^\top x\right)^i + b,
\tag{5.31}
$$

Table 5.1: Controllable parameters and hyperparameters for polynomial class analysis

Information	Notation	Details
Hypothesis model	$h[\mathcal{W}_h, \mathbf{H}_h] \in \mathcal{H}$	Hypothesis model. For simplicity, it contains $\mathcal{W}$ as the configuration parameter set, and $\mathbf{H}$ the fixed hyperparameters
Concept model	$c[\mathcal{W}_c, \mathbf{H}_c] \in \mathcal{C}$	Concept or target model. Similar configuration as the hypothesis model
Complexity measure(s)	$\{\mathbb{C}(\cdot)\}$	Set of complexity measure of choice for any arbitrary model taken in.
Sample count	m	The number of sample available for hypothesis model of c, for $m = \lvert\mathcal{S}\rvert$ dataset
Hypothesis dimension (hyperparameter)	$[d_h]$	An explicit hyperparameter contains the list of all in-testing polynomial degree for the hypothesis model.
Concept dimension (hyperparameter)	d_c	An explicit hyperparameter of the concept's true dimension. Theorized such that $d_c = d_{h,i}$ for arbitrary i will reduce to possible low-error problem.
Data noise	$\epsilon_{\mathcal{S}}$	Dataset-wide noise with relative strength to the value of the sample point. This is to reduce vanishing noise with higher-degree polynomial sample points.
Learning parameters	epoch, lr, bs	Two main parameters for gradient-descent, iterative-based algorithm - the epoch, the learning rate, and the batch size.

Notes: We output to $\{\mathbb{E}[\ell]\}$ as the expected value of the error measure on h, c. There are a few things possible, but under approximation setting, we use MSE-on-parameter and MSE-on-prediction for two different insights: internal representation and external representation.

where $\Theta = (\{\theta_i\}, \{w_k\}, b)$.

From this, we can state the model within a testing scenario listed as such, to fully capture the structural foundation of the related testing model class. For the notion of complexity, naively, we employ two kinds of complexity measure. First is the usual complexity measure via parameter counts. Second, is the expressive complexity on the basis of the linear transformation space. That is, an expressive complexity that take into account the fact that $\mathcal{O}(wx^n)$ will always be stronger than $\mathcal{O}(wx^{n-1})$ for $x > 1$. All of which can be contained in a singular expression, as seen in definition 5.2.8. However, we would have to analyse what is happening inside the metric itself. The model itself is illustrated in Figure 5.8.

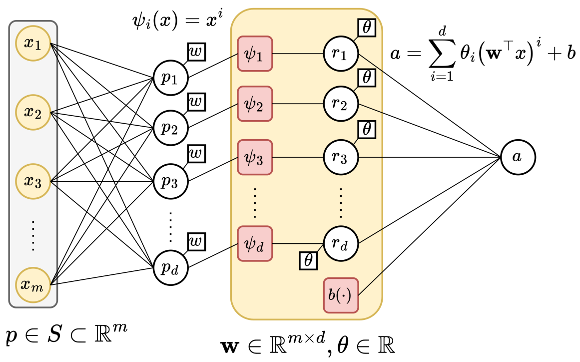


Figure 5.8: Illustration of polynomial model in definition 5.2.7.

In the equation for expressive complexity, the first term is the normalized linear parameters θ_i fixing and controlling the shift of the polynomial inning parameter, the second term is the sum of weight-aware complexity with degree factor, and the degree factor over operating range $||x_{max} - x_{min}||$ of the dataset, for the degree complexity. The final term is the last weight-aware complexity for the dominating degree n of the polynomial function. Let us begin with the gauge on coefficient norm, i.e. the term $||\alpha_d|| + 1$. Here, we set 1 as baseline, and the arbitrary norm on α there coefficient effect. The terms coefficient in this expression matters because, for a function $f : [a, b] \to \mathbb{R}$, the Lipschitz constant L for polynomial f goes by

$$\begin{aligned}
L_p &= \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|} && (5.32)\\
&= \sup_{x \neq y} \frac{|f_\Theta(x) - f_\Theta(y)|}{|x - y|} && (5.33)\\
&\leq \sup_{x \in [a,b]} \left\| f'_\Theta(x) \right\|_2 && (5.34)\\
&= \sup_{x \in [a,b]} \left\| \sum_{i=1}^{d} i\, \theta_i \left(\mathbf{w}^\top x\right)^{i-1} \mathbf{w} \right\|_2 && (5.35)\\
&\leq \|\mathbf{w}\|_2 \sup_{x \in [a,b]} \sum_{i=1}^{d} i\, |\theta_i|\, |\mathbf{w}^\top x|^{i-1} && (5.36)\\
&\leq \|\mathbf{w}\|_2 \sum_{i=1}^{d} i\, |\theta_i|\, \max\{|\mathbf{w}^\top a|, |\mathbf{w}^\top b|\}^{i-1}. && (5.37)
\end{aligned}$$

Where we note that $f_\Theta(x) = \sum_{i=1}^{d} \theta_i \left(\mathbf{w}^\top x\right)^i + b$, and $||\mathbf{w}||_2$ is the Euclidean ℓ^2 norm. The magnitude of $f'_\Theta(x)$ depends directly on the coefficient size α_i, and also on how big is x on the domain. All of this mean for large expression and value range on f_Θ, the Lipschitz constant of that model is accordingly large. Defining α-norm that way is a cheap proxy for Lipschitz constant, since it gauges stability, and rate of change of the functional expression itself.

Definition 5.2.8 (Polynomial expressive complexity). *For $x > 1$ or $\mathbf{x}_i > 1$ of $\mathbf{x} \in \mathbb{R}^n$ as input. Denote the polynomial class as $\mathcal{P}_n$ of their highest degree n, then we define the total expressive complexity of $p \in \mathcal{P}_n$ as followed.*

$$\begin{aligned}
&\mathbb{C}_E(p)\\
&\quad = \mathbb{E}\Bigg[||\alpha||_d + 1\\
&\quad + \frac{\sum_{i \in S \setminus \{p_n\}} w_i\, i}{||p|| - p_n} \\
&\quad + \frac{1}{||\overline{x} - \underline{x}||} \left(\frac{\sum_{i \in S} i}{||p||} + w_n n \right) \Bigg],
\end{aligned} \tag{5.38}$$

where $S = \{p_j\}$, $||p||$ denotes the amount of different degree separable terms and $||x_{max} - x_{min}||$ is the observable range of the dataset.

5.2.2 Scenario schema

The scenario schema will be as followed. The following set of parameters are required for controlling the testing case, aside from different system measures. For parameter error, we reuse equation from Definition 5.2.8 but with degree aware terms. This is covered in Table 6.1.

We note a few problems with both interpolation interpretation and polynomial regression of higher-degree terms. Firstly, is the intrinsic interpolation error and numerical, software issue error. Higher degree polynomial can be represented as solving for a wide Vandermonde system of n degrees for m sample points. Using monomial basis for interpolation, either for soft interpolation also exhibits instability and numerical garbage within the operations, as indicated by Shen and Serkh [2025]. They can also be a potential source of double descent, as the majority of models tested around this range are not coded effectively for handling large data operations as such, and artifacts from such choices might be taken as the indication of double descent in some particular cases.

Aside from it, and aside from the fully-connected case, we have the particular testing scenario that can be expected of the ease of implementation. First, is the mundane implementation that is of gradually increasing degree mismatch. That is, for polynomial of h_p and c_p, one requires only d_h and d_c to figure out the structure as there will exist exactly $d+1$ with the constant term, terms of which of finite many degrees. The second case is where for p chosen randomly, such that aside from such p entry, then the other degree or nodes of input are simply scattered dependent nodes, linear algebraically. That is, for a uniformly constructed d-degree network, choose at random p particular degree to be fixed, while the other of $d-p$ are assigned randomly of belonging to some $p_i \in \{p\}$ nodes given the random choice on p above. Linearly algebraically, scattered and dependent nodes then mean for example, node 4 might use node 35's degree, and vice versa (if they were both non-fixed). The dependency structure is deliberately randomized rather than organized. For overparameterized case, we simply design the hypothesis by the following line of justification. For a mapping $\mathbb{R}^d \to \mathbb{R}$, the mapping $\mathbb{R}^{d+n} \to \mathbb{R}$ for $n=1$ requires additional information, since the source provision does not contain information to handle additional n dimensional input. Hence, we design it the following. For the hypothesis h, such unit of which does not fit to the original input space, the *internal input space* that h 'hypothesize' contains, for the other n entries, $q \leq d$ per unit input. That is, each input of n overparameterized case receives at random, q inputs can be received, for example, $w_{n1}x_j + w_{n2}x_k$ for k, j specific input numbering. By the simplicity argument, then $q=1$ for the simplest case. However, this creates a skewed polynomial network, of which informally, the provided structural bias or additional information, is the fact that there exists overparameterization. To prevent this, instead, the final case considers the most general structure — for each unit, each one can take q inputs controlled as hyperparameter of the network, then the set $\{p\}$ of chosen degree influence their degree choice, and is uniform if the sampling process it is not included.

5.3 Graph Neural Network (GNN) - Shi et al. [2024]

To confirm and observe particular insights on the problem, experimentally, we focus more on the results given by the analysis on a particular type of neural network, the graph neural network (GNN). However, preliminary experiments are also needed, for example, with various complexity and function class, though most of them are algorithm based on ERM, which typically can be called derivation of gradient descent.

According to major literature, for example, Shi et al. [2024], double descent is absent usually in GNN, for example, GCN networks. However, in fact, it is relatively unstable, as for certain papers and researches point out its existence and variation of it being ubiquitous to the network, some reports none of such occurrence in their experiments. Thereby, our first strategy would be to encounter this absence, and the way to force it to exhibit itself, if the phenomenon is perceived non-existent. In experiments by Shi et al. [2024], Buschjäger et al. [2020], we know that graph network are inherently sensitive to data configuration. The graph MPNN in **?**, for example, depends heavily on the configuration of data to present its neighbourhood aggregation for each layer. However, we would like to first claim the dichotomy of bias-variance holds for it to be presented as the first interpolation point.

We would use, and utilizes a hybrid setting between MPNN, GCN and GAN (attention network), to configure out network in itself. However, heterogeneous network - either consists of only MPNN or GCN layers, is somewhat preferred for ease of analysis, and potency of experimental learning control. Because double descent lies between the concept of test error and training error, either supervised or semi-supervised would be used. According to Shi et al. [2024], semi-supervised setting gives better flexibility and overall 'range' of operation. Other than GNN, certain more abstract, 'vanilla' neural network formalism as specified in the above section treatment will also be conducted, in a supplementary manner.

5.3.1 Methodolody

Because of the irregularity in the statement that GNN does not exhibit any phenomena that is related to double descent Shi et al. [2024], we would be investigating this phenomenon the most. This particularly means that a lot of our focus will be to analyse the GNN network by itself.

Quick introduction to graph theory

A **graph** G is a 2-tuple (V, E) where V is the set of **vertices**(or nodes) and E is the set of **edges**. The set $ne[n]$ stands for the neighbour of vertex n, while $co[n]$ is the set of edges that have n as vertex. Edge is often denoted by (u, v) for vertices u and v. We said that (u, v) **joins** u and v, and it can be directed or undirected. A graph is called a *directed graph* if all edges are direct or *undirected graph* if all edges are *undirected*. The **degree** of vertices v, denoted by $d(v)$, is t number of edges connected with v. The graph data can be loosely (not specifically in cases) as followed. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{W})$ be a graph, where $\mathcal{V} = \{1, \dots, n\}$ is the set of nodes, $\mathcal{E} = \{1, \dots, M\} \subseteq \mathcal{N} \times \mathcal{N}$ is the set of edges, and $\mathcal{W} : \mathcal{E} \to \mathbb{R}$ is the edge's weight function. There is also the optional weight $\mathcal{W}' : \mathcal{V} \to \mathbb{R}$ for each vertex. In this case, it is for certain problem like TVP, where all cities have certain properties for the path. We say that a data sample $\mathbf{x}$ is a graph data, if its entries are related through the graph $\mathcal{G}$.

Our 2-tuple (V, E) and the collection of all such tuple forms a group of *simple graph* (undirected) and *directed graph*, where the only change is that $(u, v) \neq (v, u)$ for any given edge on a graph. From such, a typical setting, if we are to apply a machine learning setting on graph-theoretical problems, can be described as the following

Much information can be packed on a graph. Typically, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{M})$ will contains node information and edge information. Edge information can be the path-only, or weight-specific, that is, there exists for an edge (u, v) a property $\phi_{uv} \in [0, 1]$ acting like a weight to gauge a walk on the graph. This can also be used for the *distance measure* of a graph, for such graph being represented with values per position, that is, each node/vertice has position, and similarly there exists arbitrary distance $|(u, v)|$ between u and v. Any space that encode a graph's information in such way is called an **embedding space**. For node information, connections to other nodes are one type of information; however, it also retains its own properties, typically inferred through the *node embedding space* which contains features of a node and the graph itself. There are a few restrictions of such embedding space, such as to define the node embedding space to be of the same shape n, to avoid 'edge cases' in the process.

The reason for choosing an encoding can be justified as followed. Typically, graphs can be classified and categorized into a specific type of data representation, called *non-Euclidean data*. More specifically, there exists a topological space encapsulating the graph, but exists no fundamental measure on such topological space housing it. For example, there exists no notion of a *discrete distance* on a graph, but only the induced notion of *graph distance* by counting the shortest path from one node to another in a specific graph. Because machine learning works on the assumption that the space of the system gives *measurable space*, a natural response is to encode the system into an encoding space of arbitrary meaning. One, for example, simple encoding is the degree map, where nodes are mapped into an $n \times n$ matrix of nodes and valued by the number of neighbour they have.

This type of encoding then creates, for each graph, a versatiel number of arbitrary properties space they can have, stacked on each other. Typically, for a hypothesis h to work on a graph $\mathcal{G}$, then it will have two main targets that act directly on the graph, as illustrated in Illustration 5.9. There are many ways or aspect of a graph to be 'extracted'

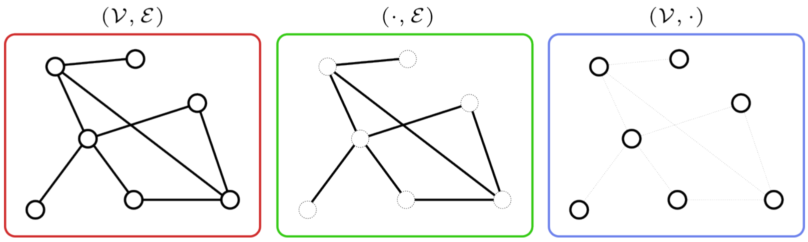


Figure 5.9: Illustrative decomposition of the graph on its simplexes edges and vertices: Each of the decomposed space $\mathcal{V}$ and $\mathcal{E}$ can be encoded separately to their own ordeal, for example, of the edge connection through incident matrix.

from the graph features itself. Classically, this is done using feature engineering (?) and hence the learner is also configured to utilize such features. Nowadays, it is mostly abstracted into an arbitrary representation space of the graph encoding, used in neural network architectures (Oono and Suzuki [2020], ?], Scarselli et al. [2009], ?). The only different between the classical encoding and the deep learning encoding is on the arbitrary properties of deep learning encoder, of which the transformed feature might not be well-defined feature engineering as the classical case.

This prompts two types of main learning targets: either to have the hypothesis h to learn the edge connections and properties, or to learn on the vertices/node. They are, however, not separated, as to learn about which connections $e \in \mathcal{E}$ to make between two nodes requires information about the general graph's node itself, to predict which node would be connected under obscured information (removing edges). So, there are:

- **Edge reconstruction**: predict if there exists an edge at an arbitrary segment of the graph, for example, between $u, v \in \mathcal{V}$.
- **Edge classification**: An edge $e \in \mathcal{E}$ is endowed a label, and we will then try to classify it based on the actions, or properties it has on the graph.
- **Node classification**: Classify a node $v \in \mathcal{V}$ of certain class of labels.
- **Node regression**: Estimate values of certain nodes. In turns, it might also be used for ranking certain nodes based of specific criteria.

It is helpful to see that a graph can also be classified into two more main archetypes, based off their actions: either **static**, where the node and edges remains the same, or *dynamic*, where deletion of nodes and edges and vice versa are taken into account. The problem revolving a dynamically changing graph (as more prominent in real-life cases) is resolved using temporal methods for graph learning (?), however, it is much more difficult to analyse, and hence we would restrict ourselves in the case of static graph analysis.

We then state our problems in regard.

Definition 5.3.1 (Graph learning problem). *Given a graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, for any type of graph-theoretical subtype (multigraph, digraph, etc.) by m contraints. There then exist two topologies: the* ***graph-theoretical topology*** *of connections and nodes appearances, and the* ***encoded topology*** *of various measured properties for all $v \in \mathcal{V}, e \in \mathcal{E}$, denoted by (ϕ_G, ϕ_E). A hypothesis $h \in \mathcal{H}$ is then tasked to learn its own encoding, or loosely speaking* ***interpretation*** *of the hypothesis about $\mathcal{G}$ based on the topological space given by $\mathcal{G}$, and use it to target or solve problems related to both ϕ_G and ϕ_E.*

From this, we might derive various problems setting as it is, and considering the set $\{\phi\}$ of encoding intrinsic to the graph.

Graph Neural Network

We target the graph neural network structure (GNN), specifically a neural network implementation for solving graph data problems. A more detailed description can follow from ?Scarselli et al. [2009]. Typically, graph neural network (Scarselli et al. [2009], ?], ?], ?) follows the instruction flow of the *encoder-decoder* architecture. A GNN is formulated and structured by analysing a graph data system by conceptually apply a *neighbour-dependent* neural input arrangement on top of the data. That is, in principle, it reserves on its own a particular extractor and modifiers on the graph interpretation itself, through its layers.

Structurally, the description of a GNN is defined by the overall *message-passing neural network* (MPNN), defined by (??):

$$\mathbf{x}_i^{(k)} = \gamma^{(k)} \left(\mathbf{x}_i^{(k-1)}, \bigoplus_{j \in \mathcal{N}(i)} \phi^{(k)} \left(\mathbf{x}_i^{(k-1)}, \mathbf{x}_j^{(k-1)}, \mathbf{e}_{j,i} \right) \right), \tag{5.39}$$

for $\bigoplus$ the differentiable, permutation invariant function, usually called the aggregator, $\mathbf{x}_i^{(k-1)} \in \mathbb{R}^F$ the node features of node i in passing layer $k-1$, $\mathbf{e}_{j,i} \in \mathbb{R}^D$ the optional edge features from node j to node i. Additionally, γ denotes the nonlinearity differentiable function (usually ReLU, or sigmoid) ϕ denote the MLP layer accompanied They are often called the **updater** and the **messenger**, respectively. A simplified example of this structure in Scarselli et al. [2009], ? as

$$\mathbf{x}_i^{(k)} = \sigma \left(\mathbf{W}_{\text{self}}^{(k)} \mathbf{x}_i^{(k-1)} + \mathbf{W}_{\text{neigh}}^{(k)} \sum_{v \in \mathcal{N}(i)} \mathbf{x}_v^{(k-1)} + \mathbf{b}^{(k)} \right) \tag{5.40}$$

where $\mathbf{W}_{\text{self}}^{(k)}, \mathbf{W}_{\text{neigh}}^{(k)}$ are trainable parameter matrices, and σ denotes an elementwise nonlinearity, and an optional bias term $\mathbf{b}^{(k)}$. Different flavours of the differentiable aggregator create the *graph convolutional networks* (GCNs), defined by:

$$\mathbf{x}_i^{(k)} = \sigma \left(\mathbf{W}^{(k)} \sum_{v \in \mathcal{N}(i) \cup \{i\}} \frac{\mathbf{x}_v}{\sqrt{|\mathcal{N}(i)||\mathcal{N}(v)|}} \right) \tag{5.41}$$

A somewhat popular approach is to apply attentional layer and weights to facilitate neighbourhood attention, which is called graph attention network.

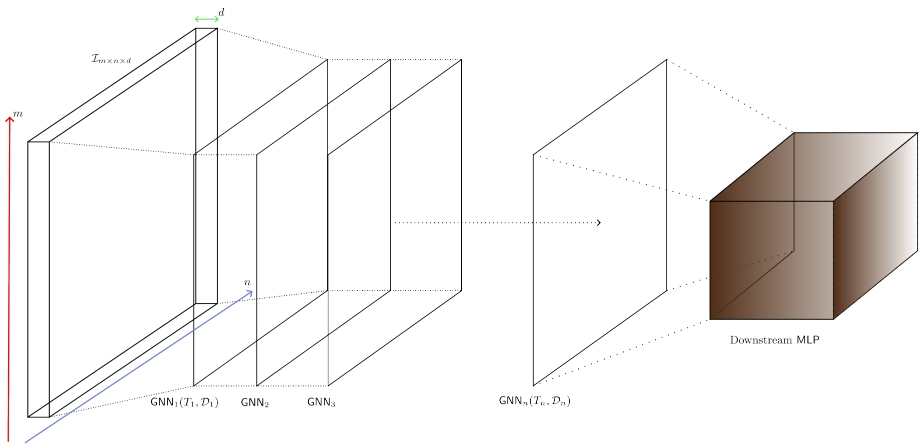


Figure 5.10: A conceptual illustration on the running flow of an n-layer GNN on particular structure of interest. Note that the data section itself has particular embedding structure on its own.

As noted by ?, node features are often required to be inputted into the GNN. Usually, this is of the form $\mathbf{x}_u$ for all $u \in \mathcal{V}$. The same can be said for edge in specific tasks required of them. However, if there are no sufficient node representation possible in the first glance, then we can use node/edge statistics as classical techniques.

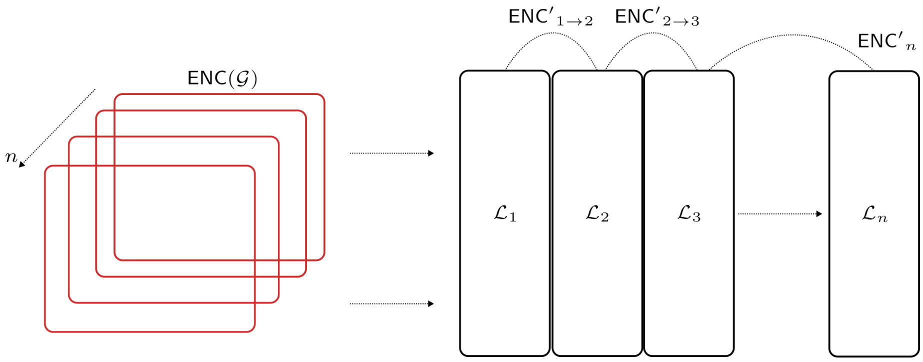


Figure 5.11: Encoding space and the change of encoding inbetween a static graph (or snapshot-wise) GNN. The encoding space is warped toward arbitrary notions inside a GNN, to the point that after certain layers of processing, the encoding outputs different from expected encoding space (native to the model, not to the designer), though there might be universal notions preserved, like the degree of node represented in a different way.

A GNN is, when fitting into the framework of learning system and modeling, is a feature mask generation and adaptation mechanism of the modelling pipeline. What this means is that GNN *assumes* every data point has its own feature-embedded space, and according relationships. By this, we further mean that it creates an embedding space of the aggregator, or the GNN-embedding space within such. While input embedding space captures and represents the best possible interpretation of the input space, GNN-embedding space captures what is required of the aggregation properties that the GNN layers specify. Hence, we can assume relative fallacy in such embedded space. Those n-embedded space of data aggregated at the nth final layer, would then be fed into another straightforward - task-based and structured network, such as a pass to a single-layer sigmoid network, Hamming network, or generally an FCN.

5.3.2 Double descent and GNN

As of date, there has been no reports on double descent on GNN. This can be taken accounted of the complex and often difficult analysis for graph and the neural network adopted for such task. Because of this, we will have to attempt to force double descent appears instead of simply observing it. To do this, we need to specify both the task for evaluation for the error measure, and model complexity for the other axis. For such, we would often reserve to node-level or edge-level task, and not graph-level tasks.

Aside from layer-wised complexity, GNN also has two types of complexity: the size of the graph itself, and the size of the encoding in specification. Thence, the complexity would then at least be the tuple (m, n, l) for m-size of the original graph, n-depth of the GNN layers, and l layers of GNN. If n varies between layers, then we will have the index set instead. However, for simplicity, we would likely want to have constant n throughout all layer.

Chapter 6. Intersectional insights

It is evident that we need another theory to explain what we are seeing, of plain sight. We still emphasize the need toward which the new theory can go, and the backward compatibility that is needed of such. Thus, this section will going through ground zero. We would not try to create the theory right away, but reformulate what has been seen first; this includes even the classical generalization and empirical optimization process, under different interpretations. Some of those results might not fit the formalism, or rather rigorously proven system, yet will ultimately form a logical framework on which developments can be made, and perhaps later on theorems to reaffirm the thesis. But the general idea is simple: *learning is not just approximation*, but an entirely paradoxically simple yet complex construct.

Specifically, in the first few sections, the fragmented insight would have to be considered, namely of the following dilemma:

1. We need a more general consideration between the environment and the agent, i.e. between the structures c required to extrapolate, and the modulus in which h explicitly exhibits such 'replica' trace.

2. Information destruction between encoding and modelling assumptions must be addressed, to accommodate the argument about cross-system information divergence or convergence, and so on. This includes also distribution shift and different kind of mechanistic fits.

Those are fragmented realization of the development of learning theory, especially with more experiments conducted ever since the developmental framework (Vapnik [1999], Valiant [1984] are now at least two decades old). The new theory must then, seed such understanding and factors into places, thus also hints at certain type of structural analysis that can conduct such approach. Prerequisition on such structural conditions, the firstly-analysed aspects of consideration includes:

1. To capture uncertainty naturally of mathematical modelling descriptions. That is, the mechanistic to probabilistic transition is inherently decomposable, and such transition from white-box and black-box can be captured by different configurations of space.

Before moving on, we emphasize the importance of realizing the gap between theory and practice. This is inherently serious, and if not for context, should have been included at the very top of the paper. Nevertheless, it should be emphasized here. Critically, this is not rare, and is inherently a complex matter, as seen in many tangent fields and analysis, of Semmelrock et al. [2023], Kapoor and Narayanan [2023] on reproducibility and experimental leakage, Kliemann and Sanders [2016], Tamassia and Vismara [1997] on the engineering reality of algorithms (or geometric algorithms), bgp [2024] goal on determining the gap between theory and practices, Revol and Theveny [2014], Collange et al. [2015], Diethelm [2012] on numerical reproducibility limits and overall empirical safety, Kaskenpalo and MacDonell [2012] on theoretical-practice bridge. On the specific side of the theory, Holk et al. [2025] analysed such statistical guarantee between the theoretical and empirical mistmatch, of which found certain problems regarding such gap, for example, the convergence rate in difussing controlled setting (statistically and system-addess wise, stable); so is Adcock and Dexter [2021] on the gap between theory and practice in function approximation with deep neural networks; and so is also Saxena et al. [2025], but on the examination of AI mismatches, which usually is concerned of the *implementation details*. Interestingly, Behrouz et al. [2025] paper on nested learning exploits the line of argument on the mismatch, but however, is troublesome in their formulation and the disparity between theory and practice is not actually addressed, but simply metrics in which the new model is compared to the previous models.

That said, it is clear, that any attempt to build theory will inherently attain certain amount of presumptions, implementation decisions, language, memory hierarchy, compiler behaviours, cache, practical limitation on hardware and platorm, and the environment tested in which the model is captured of its results. Even when the theory can be correct, such can also be a factor, coupled with the side of uncertain given when learning process or any stochastic, long-runtime system run for a long time. Such also suggests us to consider the agenda of the **ergodic theory** in touch.

6.1 Interpreting the learning theory

Most of the paper will argue about different features of statistical learning theory (Sterkenburg [2024], Vapnik [1999], Hajek and Raginsky [2021]). It is then imperative to discuss the background setting of such. On one hand, when talking of the details on statistical learning theory and the understanding of the encoded language to describe the learning action, typically we are given the observations, or dataset of the form $\mathcal{S} = (\mathcal{X}, \mathcal{Y}) \subset \mathbb{R}^n \times \mathbb{R}^m$, of all 2-tuple pairs, assumed to be sampled or observed and governed by a distribution $\mathcal{D}$. On the other hand, there is also the push toward a more general and sophisticated view, in which might as well reveal deeper or richer structures on the theoretical framework to realize. Note that this contains reformulation, or unification, reinterpretation or patchworks of existing structures too, and not simply claimed to be entirely new.

6.1.1 Domain and process

In dataset, between the *actual concept* and the observable spaces, lies the empirical and generalization setting. While it is not often mentioned in literatures, the switch to direct observable evaluation of statistical learning, induces certain limitations.

Let us assume the following preliminary in discussion. We have two polars, on the mechanistic absolute, or *white box* knowledge, with full expressions on the underlying mechanics of a certain system concept. Explicitly, this can be written as $y = F(x;\theta)$ under standard morphism, for any kind of description θ. Extension of such notion can also include what we call as the mechanism, or $F(x;\theta,\Gamma)$ of which complies both the parameters, and the formulations. The other polar is the *probabilistic domain*, of which then, we have the inference $\sim$ operator instead. Thus, we denote $y \sim \mathbb{P}(Y \mid X)$ in typical notation, in which the dependency is now Y on the basis of X. One can also do the reverse, as to generate the inverse conditions, but this is more probably general, if one switch to joint distribution (X, Y) under statistical schemes. Abstractly speaking, the dual structure contains what we call temporary, as the **system action** A_S. The notation then becomes

$$yA_S x \to f(x;\theta,\Gamma), \quad yA_S \to \mathcal{S}[\mathbb{P}(X,Y)] \tag{6.1}$$

For $\mathcal{S}$ here temporarily denotes the *sampling operations*, of observing randomly on the basis of said probability without any knowledge but the presumed parameters. If such process is with control parameter, from the optimization perspective, we have the notation as different to be $\mathcal{S}[\mathbb{P}(X,Y);\theta_P,\Gamma_P]$, where now Γ_P is the probabilistic process, i.e. distributions or other random structure, for example as stochastic processes as a finite-dimensional distributions plus regularity structure, for random variables θ_P. In general, we call it the **black-box** domain.

If we are to recover θ from θ_P, or at least on the similarity metric, typically or ML encoded by $\ell(\theta,\theta_P)$, where $\ell(\theta,\theta_P) \approx 0$, then we say that the system experiences a *white transition*. Note that we do not take into account of Γ and Γ_P, since they are inherently different in this white transition, but mechanistically recoverable; thus, *total white transition* means $\Gamma \approx \Gamma_P$ of the mechanistic patterns. If the reverse happens, where mechanistic system turn to probabilistic (X,Y) distribution on θ_P and structure $\Gamma \to \Gamma_P$.

Probability inherently, similar to mechanistic Γ, enforces causal or rather transformational relation (simply the dual $x \to y$ is transformational within f). There, the probability can be interpreted, for either Γ explicitly states that they are conditional, for X to be deterministic and explicitly stated, or joint distribution and so on. However, the interference in such recovering mode comes from observational space inconsistency. This is usually expressed of *noise*, where if we admit (x,y) of the mechanistic system, then for $x + \epsilon_x$ and $y + \epsilon_y$ we have instead a *noisy mechanistic system*. We would note that this is a larger class than pure mechanistic system, basically $x \to [\epsilon, f(x;\theta,\Gamma)]$ instead. Thus, for intrinsic noise of measurements or statistical information, probabilistic transition in recovery of mechanistic systems admits the recovery of the noisy system instead. Depends on the severity of ϵ, such system can be irrecoverable. If such approximation is impossible within certain margin, then probabilistic approximation remains in probable the only manageable system of approximation, rather than a quasi mechanistic recovery mode (we will prove this in a later theorem).

If we to accept such view, then it means that in certain failure modes, you *cannot move a mechanistic (deterministic) system into probabilistic then back*, given enough disturbance and noise. This quantifies to the *information loss* system-wide, where we can create a prototypical functional, denoted $\mathcal{R} : F \to P$, where

$$\mathcal{R}[F](A) = \int_{\Theta} \mathbf{1}_{\{\mathbf{F}(\mathbf{x};\theta,\mathbf{\Gamma})\in\mathbf{A}\}}\mu(\mathbf{d}\theta) \tag{6.2}$$

for such transition, of the probabilistic measure space $(\Omega, \mathcal{F}, \mathbb{P})$, for $\mu \in \mathcal{F}$ and $A \in (\mathcal{X}, \mathcal{Y})$ of the observational space. The implication here is that, under no information (information-agnostic), every noisy concept must meet certain limit on the basis and resultant structure itself. There exists, presently, no sufficient guarantee to return P to F, except the same operation in reversal, is forced on ideal setting, for infinite-range coverage, and approximately dense observational space. The limit is, inherently, enforced by arbitrary ϵ.

It is to note that we do not claim that mechanistic systems are ontologically privileged; we claim that the mapping from mechanisms to observations is not generally invertible under disturbance.

6.1.2 The learner's encoded setting

Previously, we say that we have the dataset, or rather called as **observations** of a given system c, $\mathcal{S}_m$ of size m, such that $(x, \cdot) \sim \mathcal{D}$ of a distribution, optionally of also y to be sampled from such distribution. The dataset itself contains pairs, as

$$\mathcal{S} = \{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\} \subset \mathcal{X} \times \mathcal{Y}$$

where the dataset is assumed to be i.i.d. sampled according to $\mathcal{D}$, which is unknown by the priori. Hence, we have the first set of assumption. This assumption, based on earlier insight, situated our finding or analysis into the range of black-box and probabilistic models.

Assumption 6.1.1. *The observational space is governed by a particular sampling distribution $\mathcal{D}$, for fixed y-response according to the concept $c \in \mathcal{C}$. The sampling process is further assumed to hold no further structure, hence is identically and independently distributed (i.i.d.).*

The learning problem is then formulated as followed. Given the machine learning model expressed a hypothesis h of the hypothesis class $\mathcal{H}$, the learning theory aims for creating a procedure to learn either elements of the concept class $\mathcal{C}$ of all concepts $c : \mathcal{X} \to \mathcal{X}$, or the function class $\mathcal{F}$ of all functions $f : \mathcal{X} \to \mathcal{Y}$, which is usually set to $\{0, 1\}$ or $[0, 1]$. This distinction is trivial, hence, if it is clear, we will talk about the concept class $\mathcal{C}$ only as representative form. The learner $\mathcal{L}(h)$ consider the set of possible hypothesis $\mathcal{H}$, in which might not coincide with $\mathcal{C}$. It receives a partial image of sample $S = (x_1, \dots, x_2, \dots, x_n)$ drawn i.i.d. according to $\mathcal{D}$ as well as the label $(c(x_1), \dots, c(x_n))$. This constitutes the dataset $\mathcal{S}$, which are based on specific concept $c \in \mathcal{C}$ for the hypothesis to learn. The task is then to use (or *extract*) meaningful information to select a hypothesis $h_S \in \mathcal{H}$ that accurately mimic c, with marginal error Θ. The notation h_S stands for all hypothesis that can be inferred from the range of the dataset (the first argument, $\mathcal{X}$).

The marginal error Θ is considered of two parameters, the empirical error $\hat{R}(h)$ and the generalization error $R(h)$. We give the following definition of them.

Definition 6.1.1 (Empirical risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and a sample $S = (x_1, \dots, x_m)$. For some particular $\epsilon > 0$, the* **empirical error** *or empirical risk of h is defined by*

$$\hat{R}_S(h) = \mathop{\mathbb{P}}_{x \in S \sim \mathcal{D}}[\ell\{h(x), c(x)\} \geq \epsilon] = \frac{1}{m}\sum_{i=1}^{m} \ell\{h(x_i), c(x_i)\} \tag{6.3}$$

The generalization risk is then defined as the extension of the empirical risk, of which instead of being evaluated on the set $\mathcal{S}$, it is evaluated on arbitrary infinite density sample space. The notation for the risk calculation can also be $R_{\mathcal{S}}$ instead, of which the need is to classify the set of which it is evaluated from.

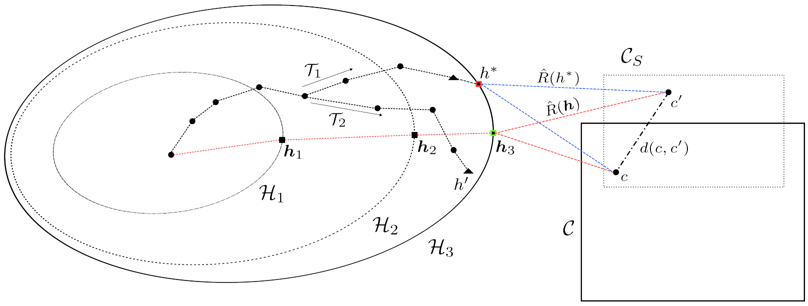


Figure 6.1: **Illustrative dynamic of the learning problem**. For an incremental hypothesis space sequence $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$, we aim to obtain the procedure $\mathcal{T}$ that would either reach the *generalization solution* $\boldsymbol{h}$, or the *empirical solution* h^* for a particular hypothesis class, with respect to the concept c and its observed concept c'. Model selection hence dictates how an algorithm or procedure than choose the best possible hypothesis to approximate the generalization solution from the empirical solution set. Do notice that here we explicitly state that the data would create a **proxy concept** that overlaps with the concept class and of some arbitrary 'distance' from the true concept by $d(c, c')$. In case the two classes overlap, there still exists the arbitrary distance. Also, we can also observe the intuitive notion of increasing hypothesis class - while it indeed can help in getting closer to the concept class, the somewhat intrinsic property of current learning theory being the relative random procedure class make it probabilistically unstable.

Definition 6.1.2 (Generalization risk). *Given a hypothesis $h \in \mathcal{H}$, a target concept $c \in \mathcal{C}$, and an underlying distribution $\mathcal{D}$ on $\mathcal{X}$. For some particular $\epsilon > 0$, the generalization error or risk of h is defined by*

$$\begin{aligned} R(h) &= \mathop{\mathbb{P}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\} \geq \epsilon] \\ &= \mathop{\mathbb{E}}_{x \sim \mathcal{D}}[\ell\{h(x), c(x)\}] \\ &= \int_{x \in \mathcal{D}} \ell\{h(x), c(x)\}\, dP(x) \end{aligned} \tag{6.4}$$

In essence, the empirical risk measure the hypothesis to observation error, and the generalization risk captures the difference of the hypothesis to the actual concept. Notice that by doing this, we have implicitly assumed that the observations and the concept is not the same thing, under the majority of scenarios. For the observations to be *approximately equal* or sufficiently captures the concept, there then would have to be various criteria to fulfil. This can be illustrated as Illustration 6.1. For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation (data) errors, the empirical risk is the generalization risk. These two measures between h and c constitute the learning problem, which can also be separated into both cases - either empirical learning or generalization learning, one to minimize $\hat{R}(h)$, and one to minimize $R(h)$. Here, we emphasize such hierarchy by defining the generalization goal as the **nested criteria** for empirical risks minimization.

Definition 6.1.3 (Empirical learning problem). *We present the formal form of the empirical learning. Suppose we have a **target**, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. The problem is to use certain algorithm $\mathcal{A}$ using $\mathcal{S}$, to obtain a hypothesis h^* for a fixed $\mathcal{H}$ such that:*

$$R(h^*) = \min_{h \in \mathcal{H}} \hat{R}(h) = \min_{h \in \mathcal{H}} \mathop{\mathbb{E}}_{x \sim \mathcal{D}, x \in \mathcal{S}} \ell\{h(x), c(x)\} \tag{6.5}$$

The generalization best $\boldsymbol{h}$ is also called in literature as the *Bayes model*, such that it reduces the Bayes risk that is the infimum of the generalization risk, $R^* = \inf_{h \in \mathcal{H}} R(h)$. The hypothesis h^* is often called the *empirical best*, for it being the minimal, finite hypothesis of the lowest loss evaluation on the entire observation space $\mathcal{S}$. There exists no certified assumption regarding whether h^* aligns with the minimal generalization error.

Definition 6.1.4 (Generalization learning problem). *We present the formal form of the generalization learning problem. Suppose we have a **target**, $c \in \mathcal{C}$, where $\mathcal{C}$ is an arbitrary concept class that captures targets of the same type. Suppose we are provided a set of observations $\mathcal{S}$. Supposed we have an algorithm $\mathcal{A}$ that for fixed hypothesis space $\mathcal{H}$, equation 6.5 holds true. The problem is to use certain algorithm $\mathcal{A}'$ such that, under limited availability, to obtain $\boldsymbol{h}$, satisfies:*

$$R(\boldsymbol{h}) = \min_{h \in \mathcal{H}} R(\boldsymbol{h}) \leq \{\epsilon\}, \quad \epsilon > 0 \tag{6.6}$$

For a set of risk bounds ϵ. If the setting is deterministic, then there exists $\epsilon = 0$ accordingly.

Then, we can partially say that generalization problem is an advancement from empirical solution. As such, empirical solution imposes the observed concept, and not the concept itself. This is reflected in modern machine learning landscape, by modelling the concept as distributions, and then using notions such as KL-divergence to measure such relative disparity.

The process of generalization goal can be, described in certain form as the global optimization objective in which empiricism can obtain from reasonable set of assumptions and conditions. The following algorithmic framing indeed aims to focus and extradite such sense of operational goal.

Optimization problem 2: Empirical-generalization learning dynamics (non-proxy)

Data: $(x_i, y_i) \in \mathcal{S}, \Gamma \supset \mathcal{S}$, objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot,\cdot)$, dependent on $\mathcal{S}$.
begin
  $\Gamma \supset \mathcal{S}$, the general set;
  $R(\cdot), \hat{R}(\cdot) \leftarrow \ell$ as objective functional;
  for *Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$* **do**
    Find $h \to \boldsymbol{h}'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A}\left(\min_{h \in \mathcal{H}} \hat{R}_{\mathcal{S}}(h) + R(h)\right) \leq \epsilon \tag{6.7}$$

    for procedure $\mathcal{A}(\cdot,\cdot)$ of structural constraints.;
  end
end
return *Optimized model h on arbitrary objective space (ℓ, ∇).*

These two stages of learning procedure comes of as the intrinsic design of machine learning setting. As opposed to interpolation, where empirical learning problem would have to be put into first priority to achieve the best possible point-through model, machine learning leans on approximation more and a separated criterion that is only visible in the learning setting - the fact that we are using c' to extract c. This disparity is exactly the **generalization problem**, of which is then also the goal of machine learning, to approximate the general solution using the observed solution. One then can ask what is the difference between the two notions, and when we can guarantee that both will 'converge' to the same point. The process of **model selection** is then the procedure that will optimize $h \in \mathcal{H}$ to a given target fixture, such as the empirical best.

One of the major assumption or observation from the point of the hypothesis class, is that the hypothesis cannot know the generalization error. Thereby, the more efficient and often used procedure/algorithm in training, is then the *empirical risk minimization* (ERM) method, minimizing only the empirical risk to the empirical best; then, certain measures to 'generalize' this heuristically is apprehended on the hypothesis. The process of model selection is also the place that we get the concept of **bias-variance tradeoff**, and its longstanding form to be a foundational metric on model selection.

6.1.3 Overfitting and underfitting

Following the above framework of statistical probabilistic learning theory, one understanding about the concept of underfitting and overfitting (similar to James et al. [2009]) can be reached, using the notion of *partitioning* of observation space. Such work is typically done because of **practical condition**, in which the following constraints is applied.

Assumption 6.1.2. *No knowledge of $c \in \mathcal{C}$ is provided.* [1]

Without such knowledge, $R(h)$ cannot be computed, accordingly. We specify such approach as *heuristic*, as it is only an approximation, and for better definition, we define it as followed.

Definition 6.1.5 (Heuristic empirical risk). *Given a dataset $\mathcal{S}$ of size m, supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$, then the **heuristic empirical risk** take a partitioning $k < m$ with $k/m \geq 1/2$, hence defining the heuristic empirical loss on k, denoted $\hat{R}_k(h)$.*

Definition 6.1.6 (Heuristical generalization risk). *Given a dataset $\mathcal{S}$ of size m, supposedly of the concept $c \in \mathcal{C}$, and the hypothesis $h \in \mathcal{H}$. Fixed $\hat{R}_k(h)$, for the final optimization of algorithm $\mathcal{A}$, then the **heuristic generalization risk** take the remaining 2-partition of portion $p = k - m$, hence defining the heuristic generalization loss on p, denoted $\hat{R}_p(h_{\mathcal{A}})$.*

The two heuristic definition however, can be noted to not be equal to the true generalization risk - empirical risk pair in reality. Nevertheless, one can then define overfitting and underfitting in such manner.

Definition 6.1.7 (Underfitting). *A model h is called **underfitting** if for a constant $\epsilon > 0$ chosen for the setting, then $\hat{R}_k(h_{\mathcal{A}}) > \epsilon$, $\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.*

Usually, underfitting is defined to be such that test error is smaller than training error partition, however, we do not think that is a good definition. Especially since, at least in this setting, the empirical error under algorithm $\mathcal{A}$ is reduced to the representative smallest error that $\mathcal{A}$ can achieve on that particular model, within particular setting of h, then come the heuristic generalization error. Considering also that $k/m \geq 1/2$, that is, at least the training partition is half of the observation set, then only under very specific case will $\hat{R}_p(h_\alpha)$ be smaller than $\hat{R}_k(h)$, statistically speaking. Thence this line of thought also reinforces the idea that both of them will be larger than the threshold ϵ. Usually, heuristically, this threshold of error is 0.3, normalized. Next, we define the notion of overfitting.

Definition 6.1.8 (Overfitting). *A model h is called **overfitting** if there exists a specific constant $\epsilon > 0$ chosen for the setting, such that $\hat{R}_k(h) < \epsilon$, $\hat{R}_p(h_\alpha) > \epsilon$ over the entire observation space.*

Namely, this mean that the static learning makes the generalization error worse. This draws the analogy of the student - studying explicitly for the past paper tests, yet unaware of the larger margin of questions in the large set of the test paper pool, leading to subpar performance. Similarly, overfitting is theorized to be fitting too fit to the reduced observation space - hence when it comes to the generalization observation space, it fails within such heuristic. The notion works on the mask of the risks, $\hat{R}_k$ and $\hat{R}_p(h_\alpha)$.

In practice, Assumption 6.1.2 means we cannot have $c \in \mathcal{C}$, or its probabilistic interpretation $\mathcal{D}$ such that $x \sim \mathcal{D}$, or in the extreme non-structured case, $(x, y) \sim \mathcal{D}$. In such case, the typical practical fix relies on the partitioning of the available partitioning space, that is, the empirical-generalization proxy. For large enough singleton count of $|\{(X_k, Y_k)\}|$, we assume that $\hat{R}_{\mathsf{Gen}}(h) \approx R(h)$. Thus, the algorithm can be stated in a form as Algorithm 3.

The assumption above of the generalization set to the generalization truth error is apparent in the usual system theorem, Theorem 6.1.3.

Theorem 6.1.3 (Mohri et al. [2012]). *For fixed $\mathcal{H}$, for fixed and sufficiently large $\mathcal{S}$, and no observation errors, the empirical risk is the generalization risk:*

$$\mathop{\mathbb{E}}_{\mathcal{S}\sim\mathcal{D}^m}[\hat{R}_{\mathcal{S}}(h)] = R(h) \tag{6.9}$$

[1] Few points need to be said about this assumption. This assumption represents the *worst-case possible* of a system dynamic. From the perspective of a designer, inherent leakage of information can be provided in the design of a model. In such, case we are operating in the mode of a *resource-constrained, closed* game, in which the model with further assumptions of fixed class, operates on. The problem space itself is inherently large and penalize any additional structural changes. If one is to occur, we usually mention it as *implicit bias*, but another name that would be more fitting would be *structural attractor*, after the notion of gradient attractor in chaotic systems.

Optimization problem 3: Empirical-generalization learning dynamics (proxy)

Data: $(x_i, y_i) \in \mathcal{S}, y_i = \{\pm 1\}$, hinge objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot,\cdot)$, dependent on $\mathcal{S}$.
begin
$((X_1, Y_1), \dots)_n \leftarrow \mathsf{Partition}_n(\mathcal{S})$;
$\{(X_j, Y_j)\} \leftarrow \mathsf{Train}$;
$\{(X_k, Y_k)\} \leftarrow \mathsf{Gen}$ for $k \neq j$;
for *Fixed $\mathcal{H}$, with $h \in \mathcal{H}$, for $\epsilon > 0$* **do**
(Optimistically) achieve $h \to \boldsymbol{h}'$, using algorithm $\mathcal{A}$ for minimization objective, such that

$$\mathcal{A}\left(\min_{h\in\mathcal{H}} \hat{R}_{\mathsf{Train}}(h) + \hat{R}_{\mathsf{Gen}}(h)\right) \leq \epsilon \tag{6.8}$$

for procedure $\mathcal{A}(\cdot,\cdot)$ of structural constraints.;
end
end
return *Optimized model $h_{\mathcal{A}}$ on arbitrary objective space (ℓ, ∇).*

Proof. By linearity of the expectation, and the fact that we assume the sample is given i.i.d., we can write:

$$\begin{aligned}\mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}[\hat{R}_S(h)] &= \frac{1}{m}\sum_{i=1}^{m} \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathbb{1}_{h(x_i)\neq c(x_i)}\right] \\ &= \frac{1}{m}\sum_{i=1}^{m} \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathbb{1}_{h(x)\neq c(x)}\right],\end{aligned} \tag{6.10}$$

for any x in sample S. Thus, we result in

$$\begin{aligned}\mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}[\hat{R}_S(h)] &= \mathop{\mathbb{E}}_{S\sim\mathcal{D}^m}\left[\mathbb{1}_{h(x)\neq c(x)}\right] \\ &= \mathop{\mathbb{E}}_{x\sim\mathcal{D}}\left[\mathbb{1}_{h(x)\neq c(x)}\right] \\ &= R(h).\end{aligned} \tag{6.11}$$

which yields the desired result. □

This theorem can be transferred to a more generalized conjecture, of which fit the generalized notion of the empirical proxy and so on.

Conjecture 6.1.1. *Supposed $\mathcal{H}, \mathcal{C}$ are fixed, for $\mathcal{S}$ is the empirical c^* for $c \in \mathcal{C}$ the true concept. Then, for algorithm $\mathcal{A}(\cdot,\cdot)$ enforcing minimization apparatus, and the specific objective on the second argument, we have:*

$$\begin{aligned}&\mathop{\mathcal{A}}_{j\to\infty, \mathcal{S}_i\neq\mathcal{S}_k}\left(\min_{h\in\mathcal{H}} \hat{R}_{\mathsf{Train}}(h) + \hat{R}_{\mathsf{Gen}}(h), [\ell]\right) \\ &\approx \mathcal{A}\left(\min_{h\in\mathcal{H}} \hat{R}_{\mathcal{S}}(h) + R(h), [\ell]\right) + \epsilon_{\mathcal{S}}\end{aligned} \tag{6.12}$$

For $\mathcal{S}_i \neq \mathcal{S}_j$ such that there are only distinct samples, j the observation set size, $[\ell]$ the set of structural or empirical objectives enforced on the problem landscape, and $\epsilon_{\mathcal{S}}$ is the empirical minimum that the algorithm result can achieve assuming sufficient 'perfect interpolation'.

By the end of this section, we would also want to note that we deliberately choose an expression basis of $\mathcal{A}$ via the couple tuplet $(\min, [\ell])$ as a choice for *algorithm-agnostic class* of algorithm satisfies the dynamic only, not just implementations. The algorithm, in certain sense, depends much more on structural assumptions, of which is not accessible via observations but via design, and intrinsic information specified in the observation class itself. This is a requirement for fine-tuning analysis in the long-run, as there exists a myriad of different approaches. Indeed, the classical theory of learning simply does not have much to offer in terms of a unified perspective, in which important factors are analysed of its realization. Specifically speaking of such, classical theory does do worse in many parts, for example, the aspect of transparency given the model, supervisor and the concept, of which their roles are not fully understood or formulated correctly, both in controlled setting and extension to replication of real life examples. Thereby, the problem requires a newer theory of learning to come forward. A new learning theory must extend beyond what constitute the basic learning model, such as the idealized PAC-learning, and statistical learning setting in general. More specifically, we want to design a learning theory that is more akin to **modelling theory** of its internalization, but also attain the learning of machine learning theory.

Within the constraints of such theoretical framing, we can then construct a different notion in which the generalization and the empirical learning problem can possibly relate to each other, at least from the statistical framework. Let us consider the problem of mathematical modelling in such sense. Beside the model class construction, which will be discussed in further sections, the entire idea of this structuralization is to refine the framework of which learning and so on can be defined itself.

6.2 System-wide structuralization

From which we can draw out of the above discussions, concurrent theories are inherently unstable and with many variations, making them ill-suited for instant application on analysis of different models without losing major information about the structural details and expressions. Hence, we propose a novel analysis on the basis of the axiomatization of the learning setting, under various structural assumptions, which lies in the implementation sections. For now, we defer to the insight in which justify such movements.

6.2.1 The idea of neural network formalism

Previously, we have been discovering, examining past results and testing on models of different types, generally classical models. While decisive, each modelling setting requires different analysis and is often not so streamline, typically fragmented as per different functional schemes, probabilistic schemes, optimization and algorithmic-based choice of designs, and so on. They have barely any unified view or general patterns for extension, thus making analysis inherently difficult. Within such, our interest, and discovery, posit that the formalism or the idea of neural network (accordance to designs and patterns of McCulloch and Pitts [1943], Hebb [1949], Rosenblatt [1958b], Widrow and Hoff [1960b], Minsky and Papert [1969], Fukushima [1980], Kohonen [1982], Hopfield [1982], Werbos [1974], Rumelhart et al. [1986], Cybenko [1989], Hornik [1991], Hochreiter and Schmidhuber [1997], LeCun et al. [1998], Bishop [1995], Haykin [1994]) and some of its modern understanding (as of Goodfellow et al. [2016], Zhang [2023], Hinton et al. [2006], LeCun et al. [2015], Goodfellow et al. [2014], Srivastava et al. [2014b], Szegedy et al. [2014], Simonyan and Zisserman [2014], Vaswani et al. [2017b], He et al. [2016], Devlin et al. [2018], Brown et al. [2020b]) practiced and exhibited, to be, in its general form, the correct but ill-defined and poorly developed framework. Thus is our goal in this section to layout such task, and the development that will be described in this section.

Basic understanding

In modern form, a neural network (Goodfellow et al. [2016], Zhang [2023]) N is typically specified by the dual $(L, [S])$ where L is the number of processing layer of a neuron, and $[S]$ is the list of neurons configuration per layer - for example, the number of neurons per layer, and so on. For such network, the definition can be simple as followed

Definition 6.2.1 (Standard multilayer network, Zhang [2023]). *We define a K-layer fully-connected deep neural network with real-valued output. Let $m^{(0)} = d$ and $m^{(K)} = 1$ for m the width of the network, and d the shape of the input vector. We then recursively define:*

$$x_j^{(0)} = x_j \quad (j = 1, \ldots, m^{(0)}), \tag{6.13}$$

$$x_j^{(k)} = h\left(\sum_{j'=1}^{m^{(k-1)}} \theta_{j,j'}^{(k)} x_{j'}^{(k-1)} + b_j^{(k)}\right) \quad (j = 1, \ldots, m^{(k)}), \tag{6.14}$$

$$f(x) = x_1^{(K)} = \sum_{j=1}^{m^{(K-1)}} u_j x_j^{(K-1)} \tag{6.15}$$

where $k = 1, 2, \ldots, K-1$, for the model parameters can be represented by $w = \{[u_j, \theta_{j,j'}^{(k)}, b_j^{(k)}] : j, j', k\}$ with $m^{(k)}$ being the number of hidden units at layer k; $\theta \in \mathbb{R}^m$ the weight of the neuron.

Here, each neuron is denoted by their expression as $f(\mathbf{W}^\top \mathbf{x} + b)$ for the activation function f. We can simplify a lot of those structures above into the neural network schemes. For example, the linear function class case can be simplified to a single $(2, [n, 1])$ neural network, such as:

$$x_j^{(0)} = x_j \quad (j = 1, \ldots, n) \tag{6.16}$$

$$x^{(1)} = \sum_{j'=1}^{n} w_{j'} x_{j'}, \tag{6.17}$$

without the activation function h. The polynomial neural network can be represented by simply a shallow, $[3, p]$ neural network. More concretely, if we take the *input channel* as a (pre-)layer, then

$$x_j^{(0)} = x_j \quad (j = 1, \ldots, m^{(0)}), \tag{6.18}$$

$$x_j^{(1)} = h\left(\sum_{j'=1}^{m^{(0)}} \theta_{j,j'}^{(1)} x_{j'}^{(0)}\right) + b_j^{(k)} \quad (j = 1, \ldots, m^{(1)}), \quad h(x_j) = x^j \tag{6.19}$$

$$f(x) = x_1^{(2)} = \sum_{j=1}^{m^{(1)}} u_j x_j^{(1)} \tag{6.20}$$

for a neural network of shape $(3, [n, n, 1])$, and of polynomial custom scaling function. For support vector machine, this is even harder to do, since its definition varies between different notion of the model itself. However, it is interesting because we can treat it as an arbitrary generation of a neural network structure. There are papers about formalizing SVM in the sense of neural network, including making it to be an infinitely-wide shallow neural network, for example, Chen et al. [2021], Wang et al. [2019], Zhang and Wang [2014], however, the structure that is formed from the SVM learner can be expressed as followed:

Setting 6.2.1 (Support vector machine). *For any given optimization problem on input space $x_i \in \mathbb{R}^d$, and label $y_i = \{\pm 1\}$, the shallow neural network resultant from the training procedure is always of the form:*

$$x_j^{(0)} = x_j \quad (j = 1, \ldots, m^{(0)}), \tag{6.21}$$

$$x_j^{(1)} = \sum_{j' \in m^{(1)}} \theta^{(1)} x_{j'}^{(0)} K(x_{j'}^{(0)}, x) \tag{6.22}$$

$$f(x) = \text{sign}\,(x_1^{(2)}) \tag{6.23}$$

for a specific kernel $K(x, x')$.

Those three fundamental conversions to neural network raise interesting insight into the neural network formalism. For linear and polynomial shallow neural network, we design the network and then provide it with a training scenario, evaluated of the proxy generalization. SVM however can be considered an adaptive data construction, where the neural network is constructed from the dataset or observation set itself, using minimization on RKHS and with constraints. This prompts exactly as such, on one side the pre-optimization construction, and one is the post-construction optimization of a neural network architecture. This interpretation is surprisingly useful when talking about neural network's unit-like structure. Hence, from such evidence prompts us to analyse a neural network, not in only the sense of managing its layers, but also by its unit components together.

This way of analysing unit-wise gives us much more interpretability, and also potential for a newer kind of network specification. As far as we are concerned of, a neural network can represent an active *mathematical modelling system*. This then indicates, for example in adversarial learning case between the target neural network tailored to specific concept, and the learning hypothesis network, then the process of learning itself can be loosely interpreted as a concept of mathematical modelling representation learning, even in the case of black-box model, and not just simply phenomenological modelling of statistical data. To do this, we will examine the structure from the perspective of a **generator**.

The SVM generator

Support vector machine, first formally originated from Vapnik [1999], is a model inherently, specifically defined for pattern recognition task, and binary classification via an *optimal separating hyperplane*. There are two variants for SVM, namely, for linear and nonlinear hyperplane, but the main general aspect of SVM is its reversal of the direction on model building - instead, it aims to minimize the structure of a model, using the data-dependent structure, or *support vectors*.

The SV machine implements the following two precursor ideas: It maps the input vectors x, supposed of the setting, into a high-dimensional feature space Z through some nonlinear mapping, chosen a priori. In this space, an optimal separating hyperplane is constructed. By statistical learning theory, Vapnik restricted the function class (as for infinite hypothesis it is impossible to learn) to the class of hyperplanes by

$$\langle \mathbf{w} \cdot \mathbf{x} \rangle + b = 0; \quad \mathbf{w} \in \mathbb{R}^n, b \in \mathbb{R} \tag{6.24}$$

which $\mathbf{w}$ is the controlling weight, $\mathbf{x}$ is the input space to the space of all binary $\{-1, +1\}$ category. Hence, this basically divide the input space into two: one part containing vectors of the class -1 and the others being $+1$. If there exists such a thing, then it is said to be *linearly separable*. To find the class of a particular vector $\mathbf{x}$, we use the following decision function

$$f(\mathbf{x}) = \operatorname{sgn}[\langle \mathbf{w} \cdot \mathbf{x} \rangle + b] \tag{6.25}$$

This is called the **hyperplane classifier** class[2]. The above form of finding final classification can then be presented in dual form, which then depends only on dot products between vectors, as

$$f(\mathbf{x}) = \operatorname{sgn}\left(\sum_{i=1}^{\ell} y_i \alpha_i \langle \mathbf{x} \cdot \mathbf{x}_i \rangle + b\right) \tag{6.26}$$

where $\alpha_i \in \mathbb{R}$ is a real-valued variable that can be viewed as a *measure* of how much informational value $\mathbf{x}_i$ has (for x_i the support vectors we get from training)[3] The second idea is of the *kernel method*. This particular method, useful and well-known of the time SVM was created, gain a more prominent position among other techniques, including neural network. Specifically, this method is characterized by the mapping of the input vectors into a richer (usually high-dimensional) feature space where they satisfy *linear separable* criteria. This prompted the possibility to solve nonlinear problem through such mapping, and indeed yields a nonlinear decision surface in the input space, which is linear in the feature space. A **kernel** (function) is then a function $k(\mathbf{x}, \mathbf{y})$ that given two vectors in input space, return the dot product of their images in feature space such that $k(\mathbf{x}, \mathbf{y}) = \langle \phi(x), \phi(y) \rangle$ for the nonlinear mapping ϕ. The general form of SVM is then usually cited as

$$f(\mathbf{x}) = \operatorname{sgn}\left(\sum_{i=1}^{\ell} y_i \alpha_i k \langle \mathbf{x} \cdot \mathbf{x}_i \rangle\right) \quad \ell \leq n \tag{6.27}$$

[2] As can be understood simply from such, there exists many hyperplanes that can correctly classifies the classes. It has been then shown that the hyperplane that guarantees the best generalization problem performances is the one with the maximal margin of separation between two classes (?).

[3] The *support vector* of a particular hyperplane $\mathbf{y}$ is the collections of all points that lies closest to the hyperplane, which is specified as condition for maximization on both side of maximal margin vector machine.

for any particular final categorization. One of the major problem for analysing SVM, is in the properties of it potentially having infinitely many parameters, depends on the size of the dataset. This makes the curve of both bias-variance and double descent not applicable in such regard – infinitely many parameters' complexity, yet still finite complexity class[4].

With respect to the observable space, SVM and neural network, we can call the construction scheme of SVM a type of *backward construction* from data $\mathcal{S}$. Or rather, the following algorithm represents SVM construction problem:

Factory 4: Support Vector Machine – backward construction.

Data: $(x_i, y_i) \in \mathcal{S}, y_i = \{\pm 1\}$, hinge objective $\ell(y, f(x))$, $\nabla(\cdot)$ differentiable gradient field on ℓ.
Result: Expected optimized model to the minimal error on the objective $\ell(\cdot,\cdot)$, dependent on $\mathcal{S}$.
begin
  $d \leftarrow \dim |\mathcal{S}|$;
  $n \leftarrow |\mathcal{S}|$;
  $\ell(y, f(x)) = \max(0, 1 - yf(x))$;
  Kernalization $\phi : \mathbf{x} \to \mathbf{x}'$;
  $f(\mathbf{x}) \leftarrow \mathbf{w}^\top \phi(\mathbf{x}) + b$;
  Objective
    Attain the minimization

$$\min_{w,b \in f} \frac{1}{2}||w||^2 + C \sum_{1 \leq i \leq n} \ell(y_i, f(x_i)) \tag{6.28}$$

    Ideally for $y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1,\ \forall i, y_i \in \mathcal{S}$, penalized by ℓ.;
end
return *Optimized model* $\mathsf{SVM}(f)$ *on objective* ℓ *of* ∇ *field.*

Such a factory is reversed inherently, and with its formulation constrained in the form $\mathbf{w}^\top \phi(\mathbf{x}) + b$, of which is equivalent to a non-linear shallow neural network with infinite width (Wang et al. [2017], Chen et al. [2021]), but is not a full-fledged neural network in technicality, hence it is only a functional equivalence. That is, such result only indicates that for $N \to \infty$ of the number of neurons of a single-hidden layer neural network, $\sum_{j=1}^{N} a_j(w_j^\top x + b_j)$, then

$$\mathsf{NN}(x) \approx \mathsf{SVM}(x) = \sum_{i}^{\ell} \alpha_i k\langle x, x_i \rangle, \ell \leq n \tag{6.29}$$

The support vector, ℓ list, is inherently implicit of the factory itself. On the other hand, typical deep learning or neural network constructions are rather *forward-constructed*, that is, the model is in a sense, prefabricated then tested of its efficacy, and thus the effective performance of the model is limited by iterations of its restricted frame. We can easily visualize the factory of which neural network are constructed, different of the SVM factory case:

6.3 Complexity notion

Assume the working space is the real field $\mathbb{R}^n$. Let us recall a model $h \in \mathcal{H}$ of certain hypothesis class $\mathcal{H}$. Then, the **complexity** refers to the complexity of the hypothesis h itself; there is also the notion of class maximal complexity for $\mathcal{H}$. Complexity in this case refers to the mass of the model, or rather, to answer the following questions:

1. What is the effective expression range of a hypothesis?
2. What is the hypothesis structured in?
3. If the model is constructed by parameters, how many parameters are there?

---

[4]From such observation, it would seem imperative that the notion of complexity based solely on the parameter counts of certain model construction is ill-equipped for certain flavours of machine learning model, in this case support vector machine (in the same type is the Gaussian Mixture Model). No double descent has been observed by empirical reports record per the author's knowledge.

4. How many components are there in specific model h, and how would another model $h' \in \mathcal{H}$ differs from h?

Those questions are not so easily answered. Specifically, analysis often puts the hypothesis class $\mathcal{H}$ as a class of functions. So, there exists the class $\mathcal{H}_n$ of n-th degree polynomials, or else. Effective complexity in such case means the range of expression a function can give - for example, for n-th polynomial class. For the class $\mathcal{H}_n$ of polynomials of degree n on a closed interval $[a, b] \subset \mathbb{R}$, the Weierstrass approximation theorem guarantees that, as $n \to \infty$, any continuous function $f : [a, b] \to \mathbb{R}$ can be approximated arbitrarily closely in the uniform norm. Hence, the expressive complexity of $\mathcal{H}_n$ grows with n, but for fixed n only a finite-dimensional subset of $C([a, b])$ is exactly representable. This distinction highlights the difference between *approximation capacity* of a hypothesis class and the *exact representational capacity* captured by its parameterization. These expressions depend on the field or space the hypothesis class lives in, for example, if the hypothesis is configured to work in the set of all functions in $\mathbb{F} = \mathbb{R}$ of algebraic field, then we call it the algebraic family. There are various hypothesis classes in such case, and usually, not a lot of them can be analysed and represented in the same way, and hence also the analysis of the 'size' of the hypothesis, also between different hypothesis classes, prompted the usage of a more general notion, called **representation complexity**[5].

Table 6.1: Descriptions of complexity measures applied to models and learning processes, from Hajek and Raginsky [2021], Sugiyama [2015], Mohri et al. [2012], Kearns and Vazirani [1994] and Vapnik [1999] with some custom notions.

Type	Notation	Purposes	Notes
Representation	$\mathcal{M}_r$	Complexity of encoding a model (size of representation, alphabet-dependent).	Model-level measure.
Expressive	$\mathcal{M}_e$	Complexity of the model's observable behaviour (operational expressivity of the mapping it implements).	Model-level measure.
Structural	$\mathcal{M}_s$	Complexity of the model's internal architecture (how components compose and interact).	Model-level measure.
Structural mass	$\mathcal{M}_\Sigma$	Aggregate representation size derived from representation complexity across the model.	Sum of $\mathcal{M}_r, \mathcal{M}_e, \mathcal{M}_s$
Sample	$\mathcal{M}_{p\in\mathcal{S}}, \mathcal{M}_p$	Number of samples required to reach a prescribed performance level (statistical sample-count).	Process-level measure.
Sample space	$\mathcal{M}_\mathcal{S}$	Intrinsic complexity of the data distribution / sample domain (richness, diversity, structured-ness).	Global process measure.
Optimization	$\mathcal{M}_\epsilon$	Complexity of the learning algorithm (convergence rate, computational/iteration cost, sensitivity) or worst-medium-best learning cost.	Process-level measure.

Notes: The notion of structural mass is convoluted. In certain setting, it is simply the parameter space $\theta \in \Theta$, but in some setting, there exists fundamental difference than identically weighted parameter space complexity, as seen in multi-basis polynomial models, not taking into account homogeneous deep learning networks. Optimization complexity coincides with PAC-learning bounds for polynomial runtime, and *sample complexity* sometimes can only be the dimension of the input/observation space.

The above complexity (and so is in Table 6.1), aside from sample space-related measure, can also be applied onto the concept class's members itself. We also take the class complexity in such case, as the argument that maximize those measure: that is, the largest value of the measure, taken on the class as the class's complexity.[6,7,8,9,10]

The next complexity measures that can be taken of, comes from the learning process structure itself. Usually, in statistical learning theory (Hajek and Raginsky [2021], Mohri et al. [2012], Shalev-Shwartz and Ben-David [2014]), we often use the setting that is related to **supervised process**. This usually means the process can be configured in a

[5]Representation complexity refers to the number of components needed to express h and generally, any hypothesis in $\mathcal{H}$. This is often the parameters count, since the parameters themselves are used in constructing the function in algebraic system. Thus, again, an n-th degree polynomial might have $2(n + 1)$ parameters, for $n + 1$ variational input (or variable), and $n + 1$ coefficients. Notice that we specifically at the beginning, use the notion of algebraic structure $\mathbb{R}$ because of it. If we restrict ourselves to the expression, for example, for $\mathbb{B} = \{0, 1\}$, then it can be expressed in various ways in logic theory, such as conjunctive normal form (CNF), disjunctive normal form (DNF), and else. This forbid the analysis based on the effective process itself, but rather the number of components required, hence taking them as complexity of construction.

[6]Expressive complexity measures the variety of input-output behaviours the model can implement.

[7]Structural complexity is decomposable into component-wise measures (e.g., per-layer, per-module) and useful for architectural comparisons.

[8]Sample complexity depends on hypothesis class, loss, and desired error/confidence. Further clarification must be made of a grounded definition for utilization.

[9]Sample space complexity infers the statistical measure over the sample space; affects learnability and effective sample size.

[10]We also note that the notion of optimization is usually hard to pin down analytically, often characterized empirically or by worst-case bounds.

feedback loop, or induced response toward the model - often come from the observational space, where the concept that is set to be the target of the process - then called as learning process - is conjured. For now, we reserve this to the classical learning theory and process, and will not inquire further of such direction.

The learning process assumes an operational space containing the structure, for example, $\mathbb{R}$-encoding space of real values, and a set of **samples** living in such space, denoted $\mathcal{S}$. Those samples are often referred to as the **observation set** of a particular objective of learning. As mentioned above, $\mathcal{S}$ belongs to the concept c, assumed to be of certain concept class $\mathcal{C}$. $\mathcal{S}$ is obtained by using an observing process to create or generate the result. This process can be determined or considered in multiple way, however, generally can be considered by having a deterministic interpretation (using an explicit function $c(x)$) or by probabilistic approach (modelling a distribution $\mathcal{D}$). Hence, we gain another type of complexity, with regard to the process called **sample complexity** - usually defined of cardinality - the amount of sample needed for particular purpose, for particular objective. For the complexity category up to the structure of the learning setting by itself, we call it the **sample space complexity**.

One final complexity that is worth mentioning is then the **optimization complexity**, measure on the action of the learning algorithm. By such, it investigates the construction of an optimization algorithm solely dependent on its size of the algorithm, and other non-process related measures. Optimization complexity can also be related or expressed, in situation of time complexity, which is typical in computing system.

One might ask why we need to determine and categorize complexity measures and the need to generalize and differentiate of each one. Indeed, while in usual use cases, such complexity notion can be considered sufficient in controlling and gauging the model itself. However, just as James et al. [2009] indicated in the section on deep learning double descent, such complexity might eventually, of some regions, inadequate to express and evaluate the complexity of the model in its operations any more. Such transition is unanswered, and while the founding of complexity notion is still fragmented, it is indeed difficult to not paying attention specifically to those notions themselves. Considering such of the learning framework and the more sophisticated goal that we push forth of reforming the learning theory, it is then natural to define such.

6.4 Attractor-based insights

The above reformulation in this chapter, giving insights of the learning formulation typically observed, arguably still attains a lot of assumptions, of which we settle for the next section. Optimization procedure 2 and procedure 3 define the precursor tradeoff in the problem space, or rather, the heuristic-generalization tradeoff. We can state it as the following to clarify such problem in plain sight.

Theorem 6.4.1 (Heuristic-generalization tradeoff, model selection). *An algorithm $\mathcal{A}$ on the system $(\mathcal{H}, \mathcal{C}, \mathcal{A})$, for the expected empirical heuristic $\hat{R}_{\mathsf{Train}}(h)$ and generalization heuristic $\hat{R}_{\mathsf{Gen}}(h)$ on $\mathcal{S}$, on the measure landscape of $\mathcal{L}[\ell]$ for $\hat{R}(\cdot)$, of which $\mathcal{A}$ runs separately on $\mathcal{A}'(\mathsf{Train}, h)$ and $\mathcal{A}'(\mathsf{Gen}, h)$ for the above heuristic, there exists*

$$\left\{ \arg\min_{h \in \mathcal{H}} \hat{R}_{\mathsf{Train}}(h), \arg\min_{h \in \mathcal{H}} \hat{R}_{\mathsf{Gen}}(h) \right\} \in \mathcal{L}[\ell] \tag{6.30}$$

*as **convex attractor** on the measure landscape. Then, algorithm $\mathcal{A}$ **solves** the generalization if for the transition $\mathcal{T} : \mathcal{L}_{\mathsf{Train}}[\ell] \to \mathcal{L}_{\mathsf{Gen}}[\ell]$ for separated loss-landscape, it satisfies for $\mathcal{A}(\mathcal{S}, h) \to h'$, that*

$$h' = \arg\min_{h' \in \mathcal{H}} \left(\hat{R}_{\mathsf{Train}}(h) + \hat{R}_{\mathsf{Gen}}(h) \right) \tag{6.31}$$

This tradeoff relies on the precursor concept of the **attractor** landscape, of which is observable within the objective-space expansion from $\mathcal{L}_{\mathsf{Train}}[\ell]$ to $\mathcal{L}_{\mathsf{Gen}}[\ell]$ of the heuristic space. The partition $\{(X_j, Y_j)\}$ and $\{(X_k, Y_k)\}$ covers the practical notion of cross-validation, though the cross-validating procedure itself works much different, as it covers multiple re-shuffling, i.e. unordered re-partition of the default permutating groups. This is the theorem of purpose in which we are to prove. Such requires further clarification, and thus form the basis for this particular len.

Chapter 7. Theoretical reformulated frameworks

Since the previous analysis of the learning model, we can then notice a few disadvantages and a few very subtle points that we omitted. It is apparent that the analysis is somewhat entangled, with various situations, interpretations, setting and assumptions together, and with a non-unified structure of models in question. The system in which the model is considered and analysed is also missing in essence. There are a lot of ambiguities around what is considered and what is not, even with probabilistic nature and the deterministic aspect. Terms like **latent spaces** and many do not capture the entire dynamic or meaning of the system. It is then we introduce our novel analytical framework, namely the **Unified Unitary Construct** framework. It is however, can be said otherwise that the learning theory is rather not so novel, but approaching the problem is a more distinct way of the already existing framework.

7.1 Motivation

The predecessors of neural network, for operations and purposes, take in the form of a specific application of function constructs, finite singleton algorithms. For such purposes, a specific classical model m takes the form of a single structure, with complex construction, often offering fixed architecture of its representation. Neural network is an attempt, for its philosophy, to change the aforementioned philosophy of constructing machine model by considering compute unit ?Rosenblatt [1958a].

Motivated by earlier analysis and observations, it is apparent that the analysis is entangled, with various situations, interpretations, setting and assumptions together, and with a non-unified structure of models in question. The system in which the model is considered and analyzed is also ineffective when considering larger structures or more 'realistic' consideration aside from very simplified, strong system itself. There are a lot of ambiguities around what is considered and what is not, even with probabilistic nature and the deterministic aspect. Terms like **latent spaces** and many do not capture the entire dynamic or meaning of the system. It is then we introduce our novel analytical framework, namely the **Unified Unitary Construct** framework. It is however, can be said otherwise that the learning theory is rather not so novel, but approaching the problem is a more distinct way of the already existing framework.

Recalled the classical learning framework, for h. We know and construct the principle of the system by settle on $h : \mathcal{X} \to \mathcal{Y}$, for any arbitrary defined in-out 2-tuple. Hence, the overall construction of a given model can be abstracted into layers of responsive results. This in particular draws some advantages, for example, can be easily approximated or expressed in the black box model style, or empiricism approach. However, it also comes with the drawback that we have seen, in the sense that the interpretation of such black box model is very narrow, even after we add different kind of flavours, for example, the assumptions of the Gaussian noise. This is majorly because of applying and addition of different components without a general framework, leading to a lot of messiness when looking directly into the theory itself. Specifically, for example, is the fact that there exists two kinds of learning analysis, one considers the information-theoretic approach of information exchange, while the other is simply the classical approach given in such the intrinsic noise error given by the observation set itself, keeping the classical theory intact. How to formalise this is a big hurdle to incorporate each other in, without considering the wider framework - one of the main problem with the speciality of learning theory. Nevertheless, at least accordance to general theory of the current date, formalization in to unit-like processing unit as *neural network components* is one thing that should be considered for the learning framework. This is also one of the main point for the word **Unified** in our theory.

7.2 The concept of learning

Within the authoritative sources on the modern deep learning theory, like Neyshabur et al. [2015, 2017], ?], Arora et al. [2019], Lee et al. [2019], Chizat and Bach [2018], Zou et al. [2019], Bartlett et al. [2021], Roberts et al. [2022], Wei et al. [2022a], Yang [2019] in comparison, the new theory of modern networks, or deep learning, remains a very troublesome regime with a lot of unspecified behaviours or outcomes leading on its structures. Unsolved problems emerged from the structural usage, for example, double descent, grokking, neural scaling limits, etc, all of which hinders and expose the immaturity of the concurrent theory. As such, a direction to take on to resolve such issue must be contained, effectively nonetheless.

7.3 Neurons perspective

The formalization to be *atoms* comes from the biological inspiration of exactly the biological neurons in human and species' brain. However, the mechanism and how they operate is a bit more interesting.

For the biological more advanced equivalent, the brain consists of a very large number of highly connected elements, called *neurons*. The simplified structure of such neuron consists of three components: the *dendrite*, which is tree-like receptive network that carry signals (electrical) to the cell body; the *cell body* effectively processes these signals, usually as sums and threshold response; and the *axon*, a single long fibre that carries signal from the cell body to other neurons. The point of contact between an axon and a dendrite of another cell is called a synapse. The neural network, hence, is established by the synapses and various arrangements of the neuron.

This configuration of biological neuron is particularly powerful. Not only that they can process information allowing someone to write this text, but also the fact that of billions or neuron, they parallelly work at the same time. Furthermore, the intricacy lies further ahead than the above formation - the entire structure is of too much complexity, for the brain itself. However, a principle can be seen - they are contained of *building blocks* together, in which case is neuron. And **artificial networks** also take this approach.

Note 7.3.1. *From such, we can conclude that, we have to have some classification that requires the smallest component neuron to be the atoms that everything else is formed upon.*

One of the most fundamental thing of the structure of the *neural network formalism*, hence, is the notion of a *unit neuron*. Specifically, a neuron x is defined, and designed to be a discrete operating unit on its own. The word *operating* here means that it has all the facility requires for an input-output process - the most simple one includes the input receiver, the processing formulae, and the output transmitter.

The neural network formalism is best expressed by the following principle. This theorem is by design, and thus our design is to fix the foundational minimization to a specific neuronal basis module. Such is to say, we state the principle for *inductive recursion* of neuronal system collapse.

Theorem 7.3.1 (Fundamental theorem of neural formalism). *Let $\mathcal{S}$ be an automated operational system with internal composition system, admitting*

1. *nontrivial input-output interaction,*
2. *internal closure under composition, and*
3. *irreducibility with respect to its operational primitives.*

Then any minimal operational component of $\mathcal{S}$ must admit a representation whose cardinality and arbitrary (analytic, topological, etc.) structures are equivalent to that of the minimally defined neural structure $\mathfrak{N}(\mathcal{N}, \mathcal{C})$.

We begin by examining the model of a typical neuron. Note that, this is the simplest one, but also is the fundamental block. We have the following.

Definition 7.3.1 (Neuron). *Given the neural network formalism. Define a construct x. Then x is called a* **neuron** *if it belongs component-wise to the class $\mathcal{N}$ of all neurons, which is minimally expressed by $\mathcal{N}(n_i, n_o, M(\dots))$, where n_i is the input handler, n_o is the output handler, and $M(\dots)$ is the internal system.*

From this, we then come up with the definition of the minimally defined neuron - the single most iteration of the above definition.

Definition 7.3.2 (Minimally defined neuron). *Given the neural network formalism. Then, for $x \in \mathcal{N}$, we call a neuron **minimally defined** if it belongs to the class $\mathcal{N}_0 \subset \mathcal{N}$ of $\mathcal{N}_0(n_i[\mathbf{w}], n_o[\mathbf{w}'], M(f, b))$ for $f : n_i[\mathbf{w}] \times b \to \mathcal{O}$ is the internal function of M. $\mathcal{O}$ is the domain of n_o in which it receives the value, and in cases, there exists no $\mathbf{w}'$ configured for the output channel.*

The notation $n_i[\mathbf{w}], n_o[\mathbf{w}']$ inherently indicate the notion of *control* of the neuron on the two gate of input and output. More specifically, a given neuron, aside from the upper-level expression of $\mathcal{N}(\dots)$, can also be realized by its *parameters*, or rather, its configuration of the neuron itself. By that, for $x \in \mathcal{N}$, the *configuration parameters* for the minimally defined neuron can be taken in the form $\mathcal{C}(\mathbf{w}_i, \mathbf{w}_o, \mathbf{w}_M)$, here we use instead the pairing notation. The letter $\mathbf{w}$ used here is historical by certain accounts: the original idea comes from the term **weight**, in which the neuron can conceptually influence the input, for example, the signal received or information received, by certain amount, either *downplay* it or *signify* it.

We clearly clarify the need for both $\mathcal{N}$ and $\mathcal{C}$ here. We know, that we want the neuron to be an *operating unit*. This means that for it to be fully realized, it needs to also operate, and hence, to be 'observed'. Hence, you need both the description of the parameters which defines it - taken the interpretation where each parameter and configuration is one specific building block on its own, then the block of all those parameters form the shape of the neuron. Similarly, the *operations* on such neuron assure the interaction and working mechanism of all those components together, inside the neuron, by itself. Hence, to fully, minimally express a minimalized neuron, you need to have both $\mathcal{N}$ and $\mathcal{C}$ as its minimality requirement. We then revise it to the following definition.

Definition 7.3.3 (Classical neuron class). *Given the neural network formalism. Then, we define a **neuron** A to be minimally defined if it is equivalent to any unit x of the class $\mathfrak{N}_m(\mathcal{N}, \mathcal{C})$, called a neuron class, where the 2-tuple expanded to $\mathcal{N}(n_i, n_o, M)$ and $\mathcal{C}(\mathbf{w}_i, \mathbf{w}_o, \mathbf{w}_M)$.*

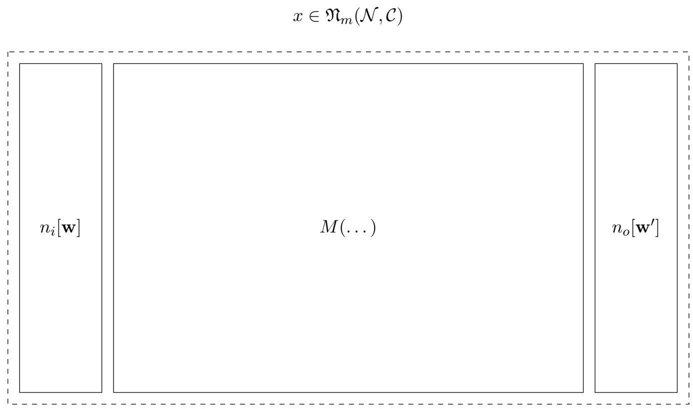


Figure 7.1: Minimal neuron structure

With this, we formalize the abstract neuron into two descriptions. Note that, however, the *minimally defined* neuron has a parameterized exposure protocol to the construction. In practice, for any neuron $a \in \mathfrak{N}$, the tuple $(\mathcal{N}, \mathcal{C})$ can be different, not to say complex. One of the strong points of this construction of the neural class is that it is *recursive*. The following proposition might help aid in understanding why it is recursive.

Proposition 7.3.2 (Compound construction). *An abstract neuron A can be dissected into three most important compartments $(\mathrm{I}, \mathrm{O}, M)$, where I, O is respectively the in-out interface. Hence, a cluster of neurons $\{A_i\}_{i \leq n}$ for such $|\{A_i\}| \geq 3$ can be compartmentalized into a new neuron unit, that is, $\{A_i\} \in \mathfrak{N}$, if and only if $|\{A_i\}| \geq 3$.*

Proof. This mostly comes of as definition (2.1). Notice that for any structure of a neuron to be minimally defined, the tuple $\mathcal{N}$ must minimally contain (n_i, n_o, M). We want them to be discretely defined, then a cluster of neuron will have to have at least one neuron such that to satisfy the requirement of the 3-tuple component. If $n = |\{A_i\}| = 1$, it reduces to a singular neuron; for $n = 2$, at least one compartment do not have any component, hence minimally we need 3 neurons to effectively be arranged as a neuron unit. $\square$

Hence, we can recursively define neuron, either going up or down, if they satisfy such condition. However, usually, we will only go up, as we will have to define in detail the set $\{A_{M,i}\}$ of all possible *minimally defined* neuron. [1]

There are several assumptions and properties accompanied by this type of neuron structure that we would like to note.

1. This neuron structure works *sequentially*. A minimally defined neuron, such as in a practical setting, will have $\mathcal{C}$ to support sequential operation. What this means is present in the definition of the minimal neuron, such that $\mathcal{C} = (\mathbf{w}_i, \mathbf{w}_o, \mathbf{w}_M)$, and the operation follows $i \to M \to o$ for an input-process-output session.
2. By extension of the point above, the neuron order in a cluster will also be *fixed*. Usually, this follows a single dimension, or, for example, if there exists a singular span directional vector $\vec{dr}$ that indicates the direction of operation in the cluster.
3. We suppose the *independence* of any components inside a neuron. This is meant for both *operational independence* for non-connective and resultant actions case. If subcomponents are connected in a causal form and so on, then such independence is violated, but not otherwise.
4. The neuron is expressed in a *parameterized manner*. That is, the behaviour of the neuron can be expressed, controlled, and designed by modifying and constructing a set of non-realized parameters. If this set of parameters is finite, we say that it is *closed parameterization*. If not—i.e., there can be infinitely many parameters—we call it *open parameterization*.
5. This neuron structure is called the *forgetful neuron*, simply because it works, in principle and in actuality, as a processing unit, in the minimally defined case. We will soon touch upon this definition for the detailed structure of the minimally defined neuron.

When we structure our neuron in this way, it is natural to ask whether the neural unit can be expressed similarly to a *finite automaton* or not. But aside from such an analogy, our minimal neuron is already very powerful. Certain clusters of such neurons, embedded with structure $\mathcal{L}$ of *layer-order clustering* (which can be thought of as a more formal notion for a layered neural network), can approximate any continuous representation scheme $c : \mathcal{X} \to [\cdot, \Sigma]$, given an arbitrary closed interval and to an arbitrary degree of accuracy. We often refer to this as the *universal approximation theorem* for the minimally defined neuron system, and it will serve as one of the focuses of the future analysis.

7.3.1 Analysis

The above construction of the fundamental neuron fundamental create the component model of the minimum processing unit which will be used in various collections of system of related mean. In mathematical modelling context, the neuron model is *descriptive*, meaning that it aims specifically not only for the input-output protocol, but also the unit neuron itself. To do this, we recall any given concrete, functional structure under mathematical formalism is given of the basis, on which abstraction and quantification (without physical realization). Hence, our model here aligns well with such mathematical formalism, in the sense that they end up with abstraction without the impromptu need to signify the inner physical realization - or existence itself. And doing this leaves us with formalizing the *minimally defined neuron* of the class $\mathfrak{N}(\mathcal{N}, \mathcal{C})$ into specific requirement.

The inspiration for the concept of a neuron is a biological one: it stems from the study of the brain or any given processing organs of species with specialized region to be called brain. Considering such, we found out that the brain mostly consists of the fundamental component called biological neuron and its supporting structures - other resource matters of the inner brain that helps to make it function. Without physical realization depends on many things, but most of the time, it is expressed by the proposition that is to remove the physical attachment from any given object of interest, giving it a more quantified look. For a neuron, it can be treated in such way by formalize the neuron component without its supporting structure, replacing electrical signal and other innate signalling mechanism with abstract 'data flow' and 'operations', without much worry about the actual physical realization of chemical reaction and channel - as we do not need such.

Back there, we all said of the neuron class. What does this mean? For starter, this section on the neuron is the relatively minimal construction we can get it onto. Recall that we say mathematical formalism is to put objects in quantifiable notions, and abstractions of the given subject; also, to express them in a language (or framework, depends

[1] At this point, I am considering using *group theory* to aid such set compartment.

on your choice of lexical sense) that support such. Hence, one of the first thing we would like to do, is to formalize the neuron into specific *description scheme*; that is, we would like to treat our objects - the neurons - specifically in certain descriptions that captures the essence of the neural unit, without much ambiguity, and an overhead view of the neuron. The tuple $(\mathcal{N}, \mathcal{C})$ did exactly that - it tells us that the neuron is expressed by that separation of descriptions - in the language of mathematical quantification; for then, $\mathcal{N}$ stands for the quantitative *parameters* that characterize the resources, component scale, or the **mass** of a given neuron, which $\mathcal{C}$ stands for the operations that governs the neuron's internal dynamics. One feasible assumption here is that the neuron does not exhibit any of its operation, that is, interfering with the macroscopic world outside itself, aside from the input-output formalism. Those two descriptions, one governs the mass, one govern the internal mechanics that confounds the subject's behaviours, is what the mathematical description do - and it works very well as it is.

In such formalism, $\mathcal{N}$ is typically easier to specify than $\mathcal{C}$ - which is trivial since the internal mechanics is harder to define and construct, for there exists many relationship and connection between components in the same system. And, considering that for any working system, especially for an interactive one, which inherently dependent on the scale t, or given any reference point of operation to be referred upon, there can be either infinite configuration, or infinite type of ordering of the system itself. We may come to the analysis on *invariant structure* of the space of configuration, but otherwise, it is uncommon to have invariant structure presented in the majority inside a typical system, except fitting of certain criteria.

Overall, the neuron formalism of capturing them into a neural class also helps to figure out the *macroscopic configuration* and the *microscopic configuration* of any given neuron. Under a certain instance of the neuron class $\mathfrak{N}$, iterations of individual neurons can be vastly different, depends on their representation scheme $\mathcal{R}$ for individual sub-description of subcomponent - for example, if there exist the component called the *loss function*, then the macroscopic behaviour will only be configured up to certain accuracy, the rest is left for the detail microscopic setting. Hence, we can partition $\mathcal{N}$ into

$$\mathcal{N} \supset \mathcal{N}_{\mathcal{H}}(\dots) \times \mathcal{N}_{\mathcal{P}}(\dots), \quad \mathcal{N}_M \supset \mathcal{N}_{\mathcal{H},M}(n_i, n_o, M) \times \mathcal{N}_{\mathcal{P},M}(n_i, n_o, M)$$

of which here we use the word *hyperparameter* for $\mathcal{H}$ a bit different from its usage of literature, and $\mathcal{P}$ for microscopic *parameters*, under the proposition that the properties can be configured using parameters as abstraction. Similar notion goes for $\mathcal{C}$, though it is less apparent.

Therefore, the minimally defined neuron, or what we can take as the **fundamental model of neural processing unit** works well, and provides us with the first iteration of the constructing brick for our larger, wider implementation. Though our aim is to advance this structure, first, we must take a look to the formalization and propositional-nation of the rudimentary concept of the neuron, even of the fundamental, minimal neuron. And this will happen at the end of this chapter. Until then, examining the maximal usage of the current structure will be more than enough as it aligns with historical literature's neuron and neural network formalism. Following sections will see how far we can go with this.

7.3.2 Why we call it minimal

It seems a bit weird why we call our construction on its own a *minimal construction*. To see this, though, we have to go back to the definition of a neuron, and the historical viewpoint on the constructing principle. The need for structuralization and complexity measure is impending on the theory itself, for example, the fact that there exists no current singular class of measure for any specific structures of feedforward networks, as seen of Hu et al. [2021b]. Often, the minimal construction in typical layered network is the neuron unit only, of which is simply understood as a function with changing activation function. However, this is a limited approach, as it only satisfy mathematical elegance or simplicity of form, which it does well on standard systems, without targeting specific automaton-reliant and actual ergodic properties. For example, take the polynomial neuron. By certain structural minimization as we defined of the fundamental theorem

7.4 Formalization

The structure of our theory on the neural formalism is influenced by the object-abstracted treatment of mathematically embedded structures, and the unit-wise principle of particular neuron. Before we meet ourselves into the notion of epistemic circularity[2] problem, we might as well clarify a few prerequisites for such structure to exhibit.

First, we indict on the fundamental encoding environment that any object can take. The main point of any structure here is that there exists fundamentally the encoding space of two types. First is the object's cardinality space, denoted $\Gamma = (\mathbb{N}, F)$ for any given categorization F. Second is the encoding primitive of the field $\mathbb{R}$ for generality - in general any field is alright, and they define the analytic structure of the system itself. Any extension, for example, the $\mathbb{R}$-algebra of complex number $\mathbb{C}$ is then the primitive field's extension. [3]. Any type of data or system can then be decomposed to such, with additional structure on top of such primitive. Such is then called the *primitive framework*.

Definition 7.4.1 (Primitive framework). *Let us define the primitive framework $\mathcal{P}_0$ of the dual $(\Gamma, \mathbb{R})$ where $\Gamma = (\mathbb{N}, F)$ is the cardinality encoding space, and $\mathbb{R}$ is the base field primitive of the analytical encoding. An object $X \in \mathcal{P}_0$ admits a dual representation,*

$$X \mapsto (\gamma(X), \rho(X)) \tag{7.1}$$

for $\gamma : \mathrm{Obj} \to \Gamma$ of cardinality encoding, and $\rho : \mathrm{Obj} \to \mathcal{E}(\mathbb{R})$ for the field extension of $\mathbb{R}$ category, or the category of all $\mathbb{R}$-algebras. For $\mathcal{E}(\mathbb{R})$ without extension, then $\mathcal{E}(\mathbb{R}) \cong \{\mathbb{R}\}$ of all $\mathbb{R}$-algebra.

For a structure that is supposed to be unit-wise constructed like neural network and the like, one of the main principle is the principle of abstraction. With this, come the idea of layer. Specifically, a *layer* separates abstraction in terms of subspace. Let us take an example of such kind for clarification. Let us define the primitive framework $\mathcal{P}_0$ as now the layer L_0 of this layering scheme. Then, we define the layer $\mathcal{L}_1$ of all unit-wised neuron-like units taking over the representation scheme on $\mathcal{L}_0$. We then define the structure of the neuron class $\mathcal{N}_1$ upon such as followed.

Definition 7.4.2 (Base neuron class). *We define the base neuron class $\mathcal{N}_1 \equiv L_1$ as followed. For any $\mathcal{U} \in L_1$ for $\mathcal{U}$ as unit-wise construction over L_0, then every $\mathcal{U}$ satisfies the input signature $\mathcal{I}_{sig} : \mathcal{E}'(\mathbb{R}) \to L_0$, output signature $\mathcal{O}_{sig} : L_0 \to \mathcal{E}'(\mathbb{R})$, and $\mathcal{C}_{ext}$ as extensible construction over L_0, for $\mathcal{E}'(\mathbb{R})$ particular extension on analytical encoding of L_0. The* **type** *template of a neural unit $\mathcal{U} \in L_0 \equiv \mathcal{N}_1$ is then defined as $\mathcal{U} = \langle \mathcal{I}_{sig}, \mathcal{C}_{ext}, \mathcal{O}_{sig} \rangle$ where $\mathcal{I}_{sig}$ and $\mathcal{O}_{sig}$ are invariant as type.*

Those definitions directly link it to type theory, while such development is perhaps more complex. In general, this specifies the *type* of a particular unit using the invariant analytical typing in the general environment linking to it. This is the *outer typing* of a given model construct , of which defers the type of which interaction between the environment, or the *global space state* happens and what is received of such. While the ambiance space can be forgiving, such cannot be said for the typing of given structure, since it must have the correct typing for identification. The extension is then is fairly simple, but the more important notion is the equivalence of the variant C_{ext} extensible unit of the inner structure itself. Hence, the equivalence will be of the following type:

outer typing

Let's take an example on it. The first example that we can think of refers to the multiple input neurons previously introduced. How can it be expressed under this framework?

[2]Also called bootstrapping, where to understand or define certain notion, requires the knowledge of the object wishes to be defined itself - for example, defining the size of a finite natural number set, using the elements of the number set itself.

[3]This corresponds to a variety of idea. For example, see ? for the idea of discrete and continuous representations and processing, and ? for the same idea, but used on computer graphic process where it is constructed as discrete indexing structure (hash tables) with continuous neural fields. Cardinality space takes inspiration from combinatorial species, sheaf theory of global-local set, and Grothendieck's approach to algebraic topology. Extension is the idea from field extension itself, for structures that can be extended. The general idea of this section relies on interpretation of category theory.

Chapter 8. Statistical physics

It is not surprising that the modern landscape, insofar of analysis and deep learning architectures, inspired a particular cohort of researchers with ideas on using **statistical physics** to infer on a learning theoretic system as for it to be a *statistical physics-induced complex system analysis*, i.e. a complex multi-degree of freedom system that statistical physics can take a hold on. The other direction is to apply artificial neural network to simulate a complex interacting systems, and thus verifying a version of mathematical descriptions on it, such as the **duality** of statistical physical descriptions. The list of principle illustrative works on this is not lacklustre in the least, such as

1. Ferrari et al. [2024]: Machine learning and optimization-based approaches to duality in statistical physics, i.e. where two mathematical descriptions can describe the same system. The paper used artificial neural network, which benefits from arbitrary component encoding, thus allow for representation on arbitrariness. Similar analysis is used of ML-assisted system in Hou and You [2023], for the renormalization group.

2. Barbier et al. [2025]: Works on the statistical physics of deep learning, and sought to analyse deep neural network similar to statistical system, and thus derive an optimal learning scheme on such. Works similar to this can be of comparison, toward **statistical physics interpretation**, such as Pei [2025], Faulkner and Livingstone [2024], or specialized case, most often as **Graph Neural Network** (GNN) like in the case of Duranthon and Zdeborová [2025]. An interesting phenomenon or observation on deep learning system, and marginally classical learning systems, is **phase transition**, which is explored in Ziyin and Ueda [2022] for example, and Ziyin and Ueda [2023] for a recent concept of first-order and second-order understanding of deep neural network interpretation for phase transitions.

Utilization of such statistical physics structures onto deep learning includes Wainwright and Jordan [2008] of the Ising or Markov Random Field models, Sherrington-Kirkpatrick (SK) spin glass in Sherrington and Kirkpatrick [1975] for nonconvex loss landscape in ML (as per direct interpretation language); similar purpose is presented by using **p-spin or spherical p-spin models** (Crisanti and Sommers [1992]). Specifically on neural network and deep learning, Hopfield [1982], Ackley et al. [1985b], Hinton [2002] and Gardner [1988] utilized many statistical physics analogues onto neural networks, though in Gardner case it is a **perceptron optimal-capacity model**. In modern time, it is the **cavity model** on high-dimensional understanding, as in Donoho et al. [2009], Jacot et al. [2018] Neural Tangent Kernel (NTK) with statistical inspirations, Langevin mechanics of stochastic dynamics analogue as in Mandt et al. [2017]. Energy-based models, as in Ho et al. [2020], Song and Ermon [2020] and entropy-SGD of Chaudhari et al. [2016], mean-field analysis (Mei et al. [2018]), and a rather modern **free-energy interpretation** of modern ML, in Kingma and Welling [2013]. It is fair to say that statistical physics abstraction itself gives plenty applications and inspirations on the current modern ML landscape.

However, it is our subjective opinion that is to validate, that statistical physics application, and most of its concurrent analysis are inherently messy, disorganized, and of intrusive mapping from statistical physics onto deep learning or classical systems. We notice a central pattern that typically, statistical analogues are applied onto the machine learning or dynamic model system, without cares and consideration for a direct *formal and conceptual mapping* between the two theories. For example, it is not sure how well all different axiomatic presumption of statistical physics, statistical field theory works on machine learning system, or simple as old classical models with human-aided designs in the 1970s-1980s, or symbolic branches of machine learnable methods. Concepts like temperatures and entropy are typically presented as fact, or rather a list of intrinsic properties gave rise to a 'natural measure', yet is not so applicable to a wider degree of the deep learning landscape. Interpretation exploded during the rise of deep learning models, as Goodfellow et al. [2016] also introduced some in his book, since neural network produces a corridor for *isolated component structures*, treated similar to statistical physics without structural difference between the two. As such, there exists the decoupling, that leads to Hinton's both 'failure' and 'success' of the Boltzmann neural network (Ackley et al. [1985b], Hinton [2012]), being a failure since it requires to express neural network in a niche, toy-liked, explicitly redesigned, narrow structure, and closed end-to-end loop for equilibrium evolution in-system. Such is often considered to be utilizing a representation scheme, this time, neural network, with fixed descriptions on encoding choice, rather than neural network in general.

Thus, we lay our goal to identify this gap, clarify concepts and deliver a closest possible working the problem, of applying statistical physics scheme onto the ontologically different space of objects. Within reasons, we also utilize such onto the set of **double descent problem**, in which way interpretations make sense and historical patterns or theories, and some experiment to test the hypothesis.

8.1 Note for team writing member

Please skip this page, right of the `clearpage` command right behind this. It is for organization during paper writing, and will be removed afterward.

8.1.1 Main structure

The following outlines the organizational structure in proposal specifically for the statistical physics paper. There are two parts, few claims that we posit in such discussion. Those are the main central point, but can also be mutated when further evolutions are possible.

1. We posit that statistical physics is a quite formal and general method, yet unrefined, and often is applied incorrectly in the framework of perspective of machine learning. There are certain situations (or rather frequently) that statistical physics do describe a deeper interpretation of a dynamical system, but on the other hand, forces on machine learning the **physical perspective** and description, and thus enables false claims, restricted claims, setup that is impossible, of assumptions that never satisfy the theory itself, to force everything into a physical system instead of machine learning, then claiming the reverse is the truth.
2. We claim that to understand statistical physics interpretation to machine learning or system, we need more than apply superficially, that is, to simply do the one-one mapping of this framework's concept to another.
3. We will highlight such mismatch in the analysis, both in details of interpretation, and both formal way of mathematical expression, proofs, or so on, concretely.
4. We will propose a way to look at such, particularly through the problem of double descent, where other theories or statistical physics interpretation fails, and the analogous similarity to other previous theories and attempts historically.
5. We then bring up forth a new theory, or rather, a new abstract framework to explain it. Couple with that are some toy experiments, to explain of the system we want to highlight.

Thus, the following sectioning is adopted as the formal form for the document. This includes sections on the introduction of statistical theory, the argument and concurrent (present theory) analysis, then reformulation, then transition to a new theory with experiments.

1. Introduction to Statistical Physics: Simply the introduction of the discussion about the main position, as to what is statistical physics, and the morphism to machine learning, or the learning theory.
   a) Reasons of statistical physics. This is to dissect *what*, and why statistical physics is even applied, and what does it mean or how to specifically connect to machine learning. This includes the analysis of different *abstraction* level that must be made to apply certain parameters or so on of ML, to fit statistical physics (for example, what is required to *simplify* or *add details* into it such that the notion of temperature is borderline acceptable in ML algorithms?).
   b) Axioms of statistical physics. What are the axioms and formulation that statistical physics use, from assuming properties of multi-particle interpretation, the dynamic allowed, what kind of dynamics internally and externally, etc., such that statistical physics is expressed most generally? And what can be of the application and mismatch of between?
   c) Assumptions of statistical physics. Explicit assumptions list, and incompatibility explanation, derivation, analysis. For example, it can be like that independent of components system i.e. the molecular behaviours, which is often not true in an integrated ML system.

d) Relevant concepts that it uses in formation, assumptions, already established notions.

2. Problem and settings. Here, it is simple as to *identify the errors*, speaking out loud of the problems, the problems of people solving the problems, the fallacy that make everything fails, what can be retained, what actually works, its limitation and so on.
3. Identification: Strictly identify the problem solutions and identification on "what exactly the problem can be categorized", proposing hypothesis, solutions, there of.
4. Reformulation: Reformulate it using a new theory. Chooses two mainline approach. One on the **predictive patterns**, and one on the **structural formation**, i.e. not just predicting *the peak or rise* in double descent, but to identify the structural reason why that happens, and so on.

All of this works in conjunction with double descent as a medium of illustration. Specifically, the *problem and setting* part would have to utilize in-context illustration, which then would be the double descent experimental cases. So is *identification* and reformulation. The **reformulation** part is however much more general, but use double descent as a toy problem for test bed.

8.1.2 Notices

There are a few points that must be addressed. This will help and will be included in a much more sophisticated writing style prefix than before, ensuring that details will be added now and not later where necessary.

1. **Results are written as if they are already settled theory**. The entire thing contains intermediate reasoning, failed paths, test cases, and scope limits are erased (actually provided later but still, it is *not finalized but pushed as if is*). You need **expendable structure** instead of **finished fragmented endpoints**.
2. The current working of the statistical physics treatment below is *suboptimal*. You jump to the conclusion too fast, and whenever submissions are in, it is in the state of already finalized semi-manuscript with conclusion. Elaborate on details, testing out cases, and so on, and make incremental gains as cumulative on a main document, not simply 'separated documents' that cannot be used in tangent with other team members.
3. Hiding too many stuffs. For example, even though it is said that there is isomorphism, you **must** clarify what is isomorphism here, why it is here, what is it, what isomorphic to what. All concepts are the same. Phase transition? If you list it out or seriously consider it part of the theory, include it or explain in great details.
4. Even though you stated, for example, the **ergodic theorem** or axioms or principle, you must *explicitly explain them*, with **full setting** of what object to consider, what is the central lining of the axioms, what it asserts, why is it there, what situation makes it to not *work* any more, and vice versa, add many more of the full details as to then justify the content itself.
5. Identify **actual notational system** and how they mean, as ontology and notations must be clarified of for one symbol contains many contexts and so on, especially with a long manuscript. Even Langevin dynamics will have to be clarified as to **why it is defined like that**, and thus why machine learning people use it in the theory. Exploit many topics, for example, Hamiltonian of a system but now ask 'since we are working in machine learning, why don't we test how and who the heck did the Hamiltonian of a ML system, or are there any *analogous* notion in here?' so on an so forth. Equilibrium assumption is very *structural*, and requires many knowledges beforehand, so **list them out carefully and elaborate**.

Overall, this section misses details. It misses the basic foundational sections, explanation, completeness of details in a sensitive setting that we **interpret** and *unify* a fragmented connection between two fields, of which one assumes higher authority (statistical physics) than the other (learning theory). If we keep doing this, everything will fail and misrepresentation will happen, overclaiming will happen, something that seems to be important and big will turn out to be a slight cheeky representation and nothing more. And all the good ideas might just be not developed in blindness of the other mediocre but 'nice-looking' and hand-wainable.

This is a setback and must be addressed. The concept itself is highly valuable in the followed idea templates, and frankly to say genuinely good, but the elaboration and formation of such is lacking seriously in several fronts. Also, the **styling** is also too 'short' and is too dense in the wrong way of dense - in the sense of compacting many technical jargons but confusing in many parts. For example, 'energy landscape as rugged, multi-valleyed, dominated by metastability' contains the concept of a **rugged affine** n**-hyperspace in a** $n+1$ **space**, convex peak or valley and **why is it even there** on the metric of the system, metastability but does not clarify whether it is a **global metastability** from chaotic distribution or something else. Spin-glass theory is also non-trivial, but is sighted without prior blanket. This standard **must be met** without question. Another thing to finally add as to why that kind of density is dangerous is that, from that simple sentence, you have several shots onto your leg:

1. What metric defines distance in the n-space? What is that space and the structures provided? Does it have inner product? Why the heck inner product is even in there? Suddenly now a configuration space is given pseudometric - why?
2. What notion of convexity is being used? What if it is not convex? What is the reference point (usually convexity is defined by 'probing two points' intuitively speaking)? What is alternative than convex? Why is convex there? Is the theory restricted to this convexity or that convexity and if it is non-convex in general, is it useless? Is convexity even a principle subject that defines the theory?
3. Is ruggedness quantified (e.g. Hessian spectra, barrier scaling)? What is rugged? Under what context? Is it emergent or inherent of a system up to certain threshold?
4. Is metastability dynamical (algorithm-dependent) or thermodynamic? What is being thermodynamic here means? Is it about a thermodynamic interpretation of a componential deep neural network structure? Is the algorithm being agnostic? Is the algorithm of different nature (physical operational algorithm or simply numerical or automata-kind one)?
5. What is the embedding space, what is encoded, how it is encoded, what is encoded? Even if information theoretic ideas are imported in interpretation from us researchers, **what does it mean for a system to have information?** Is it intrinsic, or observers' interpretation only?

Too many vector of analysis and thus too many fall-line that makes any further attempt to collapse under its own weight. We are not in the stage of building ideas any more in a fragmented way. There are guidelines which I provided, and there are structural in which we agreed upon. Ideas go in ideas, but structural must stay loyal to the structure and not to avoidance of details. The manuscript is already 150 pages long. There is nothing to shy from *keeping it short either* as the nature of this document. This is for anyone, so do **myself** on this section that I keep in my mind:

> You can't expect people to understand what you write when you yourself cannot even invoke it whenever asked. Don't delegate meaning to those you subconsciously know, because you also know consciously you don't. In foundational, unifying, or interpretive work, the opposite is true. Every unstated assumption **becomes a liability**, every borrowed term **must be re-earned**, and every failure mode **must be named**.

Keep on with this and it would be alright. Otherwise, foundational work will find its foundation in another corner of a dusty library and nothing more. If they cannot meet this standard, they are free to leave and fragments of what they have done will still be recognized in the manuscript as proof of their involvement, no question asked. However, this is imperative. They must adhere to this and so am I, also free to be criticized when wrong. Science does not await for anyone to be soft-talked into obscurity. When I asked of you this project being foundational, I said to beware of the contract you must sign before you commit. Whether you now feel it too harsh of your own or not, the judgement is yours. But if you still work, this will be the judgement I give before you improve on this, for the betterment of both our work and ourselves.

8.1.3 In detail systematic projection

Now then, for the main portions, we should go with the following list, though it can be modified any given time. If you find something lacking, please insert it in this frame on details more than the above conceptual schematic, providing it is a good addition to the fold. In general, there is

1. Introduction to statistical physics. Specifically, this part will grapple about the *foundational criteria of statistical physics*, i.e. what is the system of laws, how objects are presented. This includes the following topic to include in, not per specific order but relative conceptual order:
   - What is statistical physics? Why is it necessary?
   - What is the foundational framework that precedes it? I.e., for example, statistical physics of course goes before quantum but still applicable in quantum system with modification, so it effectively **switch register**

in between. So, let's say we go to the foundation on how Newtonian foundation, theoretical mechanics and the needs of statistical method influences such foundation and thus the assumptions that follow.

- The tools and perspective that statistical physics uses for such problem and so on. Especially on, for example, the **simulated scale problem** that statistical physics seems to work around, because of the requirement to *estimate large and multi-body* systems.
- What is the tradeoff in forming that system, what is assumed during the process of 'statisticalized' the system and a new second-layer ordering of axiomatic set, for example? We need to systematically identify this, by proposing your own questions, interpretations, aspects.
- Now, after we are done, move on to the typical and standard analysis, layout and rigorous analysis of statistical mechanics in actions, the principles, the theorems, the system analysis and observations in the form of physical theory that statistical theory has made. Going for this into the heart of the theory itself, especially those that would often be used in ML, like replica methods, and why it is there, is a crucial problem.
- Following the previous step, ask – why you use *replica method* and so on, but not the others' analysis? This is because theory in itself compounds from analysis and aspects, or descriptions and encoding in-field, but this is where you deliberately **map the theory to another place**, which means that this particular relationship is rather 'toxic' – the other might not have what you previously identify as *natural evolution in solution of methods*. For example, even the action of formalizing Newtonian mechanics to Lagrangian requires the presumption and properties by uniformity of system description – meta-theoretical duality, so to speak.

2. The connection to learning theory, and the modelling schema in general. This includes the following problem sections:

   - Current methods and categorization of them (with references as always) from statistical physics and was used in ML. Similarly, analogous concepts from statistical physics to ML, and the reverse relationship if any in the way back.
   - Present results and historical researches, caveats in great details, analysis of the results themselves, author's own observations, the scope limitation, the setting self-constraints, and so on.
   - Analysis of each fundamental case – what is used, what is the **object of concept** in statistical physics, and **object under transformation** of understanding in learning theory or computational model used, i.e. what is the core objects and so on that are mapped to *what object over there* in ML,TML, learning theory and so on.
   - Amplification on detail research of drawbacks, i.e. this is the section to limit the scope and range of applicability of those above concepts and researches, nitpick on them, and so on. The goal is to reframe their reach here, and to provide windows of opportunity resulted from such act.
   - Replication of old experiments and applications, and explore their specific edge cases.

3. Double descent on learning models and the statistical compounding explanation. This section is rather straightforward – find out what is going on with double descent and **the statistical physical explanation already presented.** This is crucial as to connect back to earlier sections on *why it works* and when it does not, and to analyse the specific double-descent specific categories here. That said, focus on the conclusion or methods used on the notion of **model complexity**, **model structure**, **error landscape and model evaluation**, then **algorithmic performance and operational awareness** (i.e. the difference between deterministic and probabilistic plus operational systems are that they are volatile enough for us to justify actual operational features), **algorithm complexity** both in the time-complexity and structural complexity, agnosticity of algorithm and operational processes of models, hidden assumptions and reduction of scalability, for example, in certain texts, they specifically *constraints* the appearance of double descent to a very niche portion of the domain – if they generalize it further than it should be – that is our cue. Additionally, if able, the structural changes and pathological pathways explored by those models.

4. The novel development. I will leave this to you and the collaborators else in your own judgement. However, as said, there are two types of theoretical development in question – the *predictive framework* and the *structural framework*. The first one succeed in *the prediction is inherent of this domain and system*, while failing at to point out specifically why and which components break, while the other one *can explain well the structural reasons*

that lead to this particular phenomenon, but phenomenologically performs worse do to the nature of structural conformation.

The general inquiry of this is to develop them in rigorous detail, for the repeated usage of this particular word as-is. We do not require being super-formal and mathematical, but the right way of being rigorous - as to hide nothing, unpack everything, precise language or can be often generating as the topic itself is largely uncharted domain, and mathematically *effective* - we don't want performative mathematical overformalism. The writing style fixture has been included in earlier notes, so we should not bring it here. However, if to add, then the mainline argument is to "don't overclaim where shouldn't, overcommit where trivial, under-communicate and describe where obscure, attack persons where should be targeted toward their works".

8.2 Approach A - Isomorphism

Isomorphism here, for the sake of clarify means the **structure preservation mapping** between two systems of abstract objects. The categorical error here is to assume isomorphism between two fields, and mapping them onto the general line of basis, and is a very famous *failure point* in the entire thesis of this section.

8.2.1 From Analogy to Isomorphism

IN here, we critically examine the intersection of statistical physics and machine learning. We posit that while statistical physics provides a rigorous framework for analyzing complex systems with many degrees of freedom, its application in machine learning has often been hindered by superficial analogies rather than mathematical isomorphisms. To truly leverage this framework, one must move beyond metaphorical comparisons and establish a dictionary of strict correspondence between physical quantities and learning dynamics.

Reasons for Statistical Physics in Machine Learning

The primary motivation for applying statistical physics to machine learning lies in the "curse of dimensionality" and the emergent phenomena observed in deep neural networks. Just as thermodynamics describes the macroscopic behaviour of 10^{23} particles without tracking individual trajectories, statistical learning theory seeks to describe the generalization performance of models with billions of parameters. However, a critical mismatch often occurs in the abstraction level.

Common approaches often equate the "learning rate" in Stochastic Gradient Descent (SGD) to "temperature" in Langevin dynamics. We argue that this abstraction is insufficient and often misleading. In equilibrium physics, temperature is an isotropic scalar field. In contrast, the noise introduced by SGD is inherently anisotropic and state-dependent, scaled by the gradient covariance matrix. Therefore, a direct application of equilibrium concepts (like canonical distributions) without accounting for this structural mismatch leads to incorrect predictions about the system's stationary state.

To be more precise, let us consider the standard Langevin dynamics:

$$dW = -\nabla L(W)\, dt + \sqrt{2T}\, dB_t \tag{8.1}$$

where T is the temperature and dB_t is a Wiener process. The stationary distribution is the Boltzmann distribution $\rho(W) \propto \exp(-L(W)/T)$. However, SGD dynamics are fundamentally different:

$$W_{t+1} = W_t - \eta \nabla L_{\mathcal{B}}(W_t) \tag{8.2}$$

where $\mathcal{B}$ is a mini-batch. The noise covariance is:

$$\Sigma(W) = \mathbb{E}_{\mathcal{B}}\left[(\nabla L_{\mathcal{B}} - \nabla L)(\nabla L_{\mathcal{B}} - \nabla L)^\top\right] \tag{8.3}$$

This covariance is **state-dependent** and **anisotropic**, fundamentally different from isotropic thermal noise.

Axioms and Formulation

Statistical physics is built upon fundamental axioms, most notably the *Ergodic Hypothesis*, which assumes that a system evolves over time to explore all accessible microstates with a probability proportional to their Boltzmann weight.

The Mismatch. Deep learning dynamics, particularly under SGD, are fundamentally non-ergodic on practical timescales. The system does not explore the entire weight space but is confined to specific trajectories determined by the initialization and the landscape geometry.

The Isomorphism. To resolve this, we propose a strict isomorphism:

- **Microstate**: A specific configuration of weights $W \in \mathbb{R}^P$.
- **Hamiltonian (Energy)**: The Loss function $L(W, \mathcal{D})$.
- **Quenched Disorder**: The dataset $\mathcal{D}$, which remains fixed during the training dynamics, analogous to the random interactions J_{ij} in spin-glass models.

This mapping situates machine learning not in the realm of simple ideal gases, but squarely within the physics of *disordered systems*, where the energy landscape is rugged, multi-valleyed, and dominated by metastability. The disorder average over datasets corresponds to the quenched average in spin-glass theory:

$$\overline{\langle O \rangle} = \int d\mathcal{D}\, P(\mathcal{D}) \langle O \rangle_{\mathcal{D}} \tag{8.4}$$

where $\langle \cdot \rangle_{\mathcal{D}}$ denotes the thermal average for fixed dataset $\mathcal{D}$.

Assumptions and Incompatibilities

Standard statistical mechanical treatments (e.g., the Replica Method) assume the system reaches thermodynamic equilibrium. This implies that the final solution is independent of the training path. However, the phenomenon of double descent—and specifically epoch-wise double descent—demonstrates that the "solution" is a transient state of a non-equilibrium process.

Assumption 8.2.1 (Equilibrium Assumption). *The system reaches a stationary distribution $\rho^*(W)$ independent of initial conditions and training trajectory.*

This assumption is violated in practical deep learning for several reasons:

1. **Finite training time**: Training is stopped before equilibration.
2. **Non-convex landscape**: Multiple local minima prevent ergodic exploration.
3. **Time-dependent dynamics**: Learning rate schedules break time-translation invariance.

Forcing an equilibrium assumption onto a dynamic process like SGD is the fundamental error we aim to correct in subsequent sections.

8.2.2 Limitations of Equilibrium Theories

Here, we identify the specific failures of current theoretical frameworks when addressing the Double Descent (DD) phenomenon. While significant progress has been made, a comprehensive theory remains elusive due to a reliance on static analysis.

The Failure of Classical Bias-Variance

As established in Theorem 3.3.2, the classical U-shaped curve predicts that as model complexity increases beyond the optimal point, the variance term dominates and test error monotonically increases. Formally, for the decomposition:

$$\mathbb{E}[\mathcal{M}(f, y)] = \mathcal{B}(f, y) + \mathcal{V}(f, y) + \sigma^2_{\text{noise}} \tag{8.5}$$

classical theory predicts $\mathcal{V}(f, y) \to \infty$ as model complexity $\to \infty$. However, empirical evidence from Belkin et al. [2019a], Nakkiran et al. [2019] shows a second descent in test error for over-parameterized models ($P > N$).

The Limitation of RMT and Replica Theory

Recent applications of Random Matrix Theory (RMT) and Replica Theory have successfully derived the "peak" of the double descent curve at the interpolation threshold ($P \approx N$). For linear regression with random features, the expected test error takes the form:

$$R_{\text{test}} = \sigma^2 \frac{P}{N-P-1} + \text{bias terms}, \quad P < N \tag{8.6}$$

which diverges as $P \to N$. However, these successes are limited to "static" or "shallow" models. They treat the solution as a global minimum (min-norm solution) found in a convex landscape:

$$W^* = \arg\min_W \|W\|^2 \quad \text{s.t.} \quad XW = Y \tag{8.7}$$

The Identification of Errors. The critical error in these approaches is the neglect of *feature learning dynamics*. Deep neural networks do not merely fit fixed features; they evolve their representation space. Furthermore, the existence of *epoch-wise double descent*—where the test error exhibits non-monotonic behavior over time even with fixed model size—proves that DD is a dynamical phenomenon. Equilibrium theories, by definition, cannot explain transient dynamical effects.

8.3 Identification: The Dynamic-Landscape Interaction

We categorize the Double Descent problem not merely as a statistical anomaly, but as a complex interaction between *multi-scale dynamics* and a *critical landscape*.

The Gap Between Statics and Dynamics

There exists a "gap" between the static description of the loss landscape and the dynamic trajectory of the optimizer. We identify three distinct regimes that must be unified:

Under-parameterized Regime ($P \ll N$): Dominated by simple signal learning. The landscape is relatively smooth with a unique global minimum.

Critical Regime ($P \approx N$) **— Jamming**: At $P \approx N$, the system undergoes a "Jamming Transition". The landscape becomes rugged with sharp minima, corresponding to satisfying a precise number of constraints equal to the degrees of freedom. The condition number of the Hessian diverges:

$$\kappa(H) = \frac{\lambda_{\max}}{\lambda_{\min}} \to \infty \quad \text{as} \quad P \to N \tag{8.8}$$

Over-parameterized Regime ($P \gg N$): The system becomes "floppy" or unjammed. There exists a continuous manifold of zero-loss solutions, and the geometry of this manifold determines generalization.

Hypothesis: Double Descent as Phase Transition Navigation

Hypothesis 8.3.1 (Dynamic Origin of Double Descent). *Double Descent is the result of the SGD algorithm navigating through phase transitions in the loss landscape. Specifically:*

1. *The "peak" at $P \approx N$ is a trapping event in the jammed phase (sharp minima).*
2. *The second descent is a relaxation process driven by the specific noise structure of SGD, which allows the system to escape sharp minima and settle into flat, generalizing minima.*

The key insight is that SGD noise is not merely a nuisance but an *implicit regularizer* that performs geometric selection among the manifold of solutions.

Reformulation: Non-Equilibrium Statistical Physics Framework

To address the identified gaps, we reformulate the learning problem using a Non-Equilibrium Statistical Physics (NESP) framework. We move from describing static probability distributions to modeling the evolution of the probability density $\rho(W, t)$.

The Fokker-Planck Description

The evolution of the probability density over weight space is governed by the Fokker-Planck equation:

$$\frac{\partial \rho(W,t)}{\partial t} = \nabla \cdot (\rho \nabla L) + \nabla \cdot (D(W)\nabla \rho) \tag{8.9}$$

Crucially, the diffusion tensor $D(W)$ is not constant (as in thermal equilibrium) but is **state-dependent**, defined by the covariance of the stochastic gradients:

$$D(W) = \frac{\eta}{2B}\Sigma(W) \tag{8.10}$$

where η is the learning rate, B is the batch size, and $\Sigma(W)$ is the gradient noise covariance.

Noise-Driven Entropic Selection

This reformulation leads to a new mechanism for generalization. The SGD noise is anisotropic and approximately proportional to the Hessian (curvature) of the loss landscape:

$$\Sigma(W) \approx \frac{1}{N}\sum_{i=1}^{N} \nabla \ell_i \nabla \ell_i^\top \sim H(W) \quad \text{near minima} \tag{8.11}$$

This leads to curvature-dependent diffusion:

- **In Sharp Minima** (associated with the interpolation peak/jamming): The curvature is high, so $D(W)$ is large. This creates a strong effective "repulsive force" that drives the system out of these non-generalizing states.
- **In Flat Minima** (associated with the second descent): The curvature is low, so $D(W)$ is small. These states are dynamically stable under SGD.

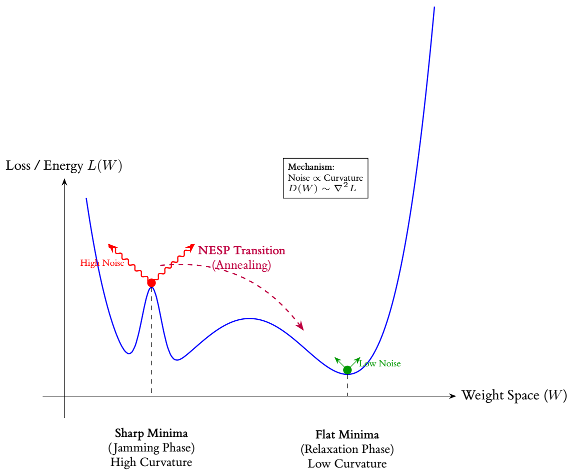


Figure 8.1: **NESP Mechanism of Double Descent**. At the interpolation threshold (Sharp Minima), high curvature induces strong anisotropic SGD noise (red wavy arrows), driving the system to escape. The system then relaxes to flat minima where low noise ensures stability. This process explains the second descent in test error.

The Mechanism of Double Descent

This framework reinterprets the Double Descent curve as a dynamical timeline:

1. **Descent 1 (Signal Learning)**: Fast learning of dominant signal components (low-frequency modes). The model captures the primary structure in the data.

2. **Ascent (Peak/Jamming)**: The system attempts to fit high-frequency noise but encounters the "Jamming transition" at the interpolation threshold. The landscape is rugged, and the trajectory is temporarily trapped in sharp, overfitting minima.
3. **Descent 2 (Relaxation)**: In the over-parameterized regime, the "entropic selection" mechanism of SGD dominates. The landscape-dependent noise "anneals" the system, driving it out of sharp minima and into flat minima. This *implicit regularization* via non-equilibrium dynamics is the physical origin of the second descent.

Mathematically, the escape rate from a sharp minimum with curvature λ scales as:

$$\Gamma_{\text{escape}} \sim \exp\left(-\frac{\Delta L}{D_{\text{eff}}}\right) \sim \exp\left(-\frac{\Delta L \cdot B}{\eta\lambda}\right) \tag{8.12}$$

where ΔL is the barrier height. For sharp minima with large λ, the escape rate is high, explaining the instability of overfitting solutions.

Predictions and Implications

The NESP framework makes several testable predictions:

1. **Batch size dependence**: Larger batch sizes reduce noise ($D \propto 1/B$), making sharp minima more stable and potentially suppressing the second descent.
2. **Learning rate dependence**: Higher learning rates increase noise ($D \propto \eta$), accelerating escape from sharp minima and enhancing the second descent.
3. **Epoch-wise double descent**: The non-monotonic behavior over training time is a natural consequence of the system first falling into sharp minima (fast convergence) then slowly escaping (relaxation).

These predictions align with empirical observations in Nakkiran et al. [2019] and provide a unified explanation for both model-wise and epoch-wise double descent phenomena.

8.3.1 Experimental Verification: A Toy Model Proposal

The theoretical framework developed in preceding sections—positing that Double Descent arises from a Non-Equilibrium phase transition between Jammed and Relaxed states, mediated by anisotropic SGD noise—demands empirical validation. However, the complexity of modern deep neural networks obscures the fundamental mechanisms we seek to isolate. Therefore, we propose a series of *Gedankenexperimente* utilizing minimal, analytically tractable toy models that preserve the essential physics while eliminating confounding factors.

The Teacher-Student Protocol

We propose the **Teacher-Student framework** as our primary experimental testbed. This paradigm, extensively employed in the statistical physics of learning Engel and Van den Broeck [2001], Advani et al. [2013], provides the crucial advantage of a known ground truth, enabling precise measurement of generalization error without resort to heuristic train-test splits.

Setup. Consider a *Teacher network* $f^*(x) = \sigma(W^*x)$ with fixed, randomly drawn weights $W^* \in \mathbb{R}^{P^* \times d}$, where $\sigma(\cdot)$ denotes a nonlinearity (e.g., ReLU or $\tanh$). The Teacher generates labels:

$$y = f^*(x) + \xi, \quad \xi \sim \mathcal{N}(0, \sigma_\xi^2) \tag{8.13}$$

where ξ represents irreducible label noise—a critical ingredient for inducing the interpolation peak.

The *Student network* $\hat{f}(x; W) = \sigma(Wx)$ with $W \in \mathbb{R}^{P \times d}$ is trained on N samples $\{(x_i, y_i)\}_{i=1}^N$ drawn i.i.d. from the Teacher's distribution. By varying the Student's capacity P while holding N fixed, we sweep across the interpolation threshold $P/N \approx 1$.

The Critical Control Parameter. The ratio $\alpha = P/N$ serves as our control parameter, analogous to the "packing fraction" in jamming physics. Our NESP framework predicts:

- $\alpha \ll 1$: Under-parameterized regime. Classical bias dominated behavior.
- $\alpha \approx 1$: Critical/Jammed regime. Interpolation threshold.
- $\alpha \gg 1$: Over-parameterized regime. Entropic selection dominates.

8.3.2 Physical Observables: Beyond Test Error

To validate the NESP mechanism—rather than merely confirming the existence of Double Descent—we must measure *physical observables* that directly probe the landscape geometry and noise structure. We propose three such observables:

Observable I: The Hessian Trace (Sharpness)

The trace of the Hessian, $\mathrm{Tr}(H) = \mathrm{Tr}(\nabla^2 L)$, provides a scalar summary of the local curvature at the solution W^*_{SGD}. This quantity is computationally accessible via Hutchinson's trace estimator:

$$\mathrm{Tr}(H) \approx \frac{1}{K} \sum_{k=1}^{K} v_k^\top H v_k, \quad v_k \sim \mathcal{N}(0, I) \tag{8.14}$$

where the Hessian-vector products Hv can be computed efficiently via automatic differentiation.

Hypothesis 8.3.2 (Sharpness Peak at Jamming). *The Hessian trace, measured at the end of training, exhibits a pronounced maximum at the interpolation threshold:*

$$\mathrm{Tr}(H)|_{\alpha\approx 1} \gg \mathrm{Tr}(H)|_{\alpha \ll 1} \quad \textit{and} \quad \mathrm{Tr}(H)|_{\alpha\approx 1} \gg \mathrm{Tr}(H)|_{\alpha \gg 1} \tag{8.15}$$

Furthermore, in the over-parameterized regime, we predict a ***monotonic decay****:*

$$\frac{d}{d\alpha}\mathrm{Tr}(H) < 0 \quad \textit{for} \quad \alpha > 1 \tag{8.16}$$

This decay directly evidences the system's migration from sharp (Jammed) to flat (Relaxed) minima as capacity increases.

Observable II: Noise-Hessian Alignment

The central claim of our NESP framework is that SGD noise is *not* thermal (isotropic) but rather *geometric* (aligned with the Hessian). To test this, we propose measuring the alignment between the noise covariance matrix $\Sigma(W)$ and the Hessian $H(W)$:

$$\mathcal{A}(\Sigma, H) = \frac{\mathrm{Tr}(\Sigma H)}{\|\Sigma\|_F \|H\|_F} \tag{8.17}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. For thermal noise, $\Sigma = \sigma^2 I$ and thus $\mathcal{A} \propto \mathrm{Tr}(H)/\|H\|_F$, which is independent of the noise structure. For geometry-dependent noise, we expect strong positive correlation.

Hypothesis 8.3.3 (Anisotropic Noise Alignment). *The alignment $\mathcal{A}(\Sigma, H)$ is significantly elevated compared to the null hypothesis of isotropic noise:*

$$\mathcal{A}(\Sigma, H) \gg \mathcal{A}(\sigma^2 I, H) \quad \textit{across all } \alpha \tag{8.18}$$

Moreover, this alignment should ***strengthen*** *near the interpolation threshold, where the curvature structure is most pronounced. This would provide direct evidence that SGD performs geometric, not thermal, exploration of the loss landscape.*

Observable III: Eigenspectrum Evolution

The full eigenspectrum of the Hessian, $\{\lambda_i\}_{i=1}^P$, encodes richer information than the trace alone. We propose tracking the spectral density $\rho(\lambda) = (1/P)\sum_i \delta(\lambda - \lambda_i)$ across the α-sweep.

Hypothesis 8.3.4 (Spectral Signature of Phase Transition). *At the Jamming transition ($\alpha \approx 1$), the spectral density exhibits:*

1. *A* ***hard edge*** *at $\lambda = 0$, corresponding to the vanishing of the null space as constraints saturate degrees of freedom.*
2. ***Outlier eigenvalues*** *at large λ, corresponding to sharp directions associated with fitting noise.*

In the Relaxed phase ($\alpha \gg 1$), the spectrum should exhibit:

1. *An extensive* ***null space*** *(bulk of eigenvalues at $\lambda \approx 0$), corresponding to flat directions in the solution manifold.*
2. *Suppression of outliers, indicating the system has escaped sharp directions.*

8.3.3 Experimental Protocol

To empirically validate the Entropic Selection mechanism, we propose the following experimental protocol:

Experimental Protocol 5: NESP Validation via Teacher-Student Model

Data: Teacher weights W^*, noise level σ_ξ, sample sizes N, capacity range $P \in [P_{\min}, P_{\max}]$.
Result: Measurements of $R_{\text{test}}(\alpha)$, $\mathrm{Tr}(H)(\alpha)$, $\mathcal{A}(\Sigma, H)(\alpha)$, $\rho(\lambda; \alpha)$.
begin
  for $P \in \{P_{\min}, \ldots, P_{\max}\}$ **do**
    Initialize Student network with random weights W_0;
    Train via SGD: $W_{t+1} = W_t - \eta \nabla L_{\mathcal{B}}(W_t)$;
    `;// At convergence:`
    Compute $R_{\text{test}}(W_T) = \mathbb{E}_x[(f^*(x) - \hat{f}(x; W_T))^2]$;
    Compute $\mathrm{Tr}(H)$ via Hutchinson estimator;
    Estimate $\Sigma(W_T)$ from mini-batch gradient variance;
    Compute alignment $\mathcal{A}(\Sigma, H)$;
    (Optional) Compute top-k Hessian eigenvalues via Lanczos iteration;
  end
end
return *Observables as functions of* $\alpha = P/N$.

Predicted Signatures. If our NESP framework is correct, the experimental results should exhibit the following correlated signatures:

1. $R_{\text{test}}(\alpha)$ displays the characteristic Double Descent curve.
2. $\mathrm{Tr}(H)(\alpha)$ peaks at $\alpha \approx 1$ and decays for $\alpha > 1$, *anti-correlated* with the second descent.
3. $\mathcal{A}(\Sigma, H)$ remains elevated throughout, peaking near $\alpha \approx 1$.
4. The spectral density transitions from hard-edge (Jammed) to bulk-zero (Relaxed) as α crosses unity.

The simultaneous observation of these four signatures would constitute strong evidence for the NESP interpretation of Double Descent as a dynamical phase transition driven by geometry-dependent noise.

Alternative Minimal Model: Two-Layer Linear Networks

For maximal analytical tractability, we also propose the **two-layer linear network**:

$$\hat{f}(x; U, V) = UV^\top x, \quad U \in \mathbb{R}^{d_{\text{out}} \times k}, \ V \in \mathbb{R}^{d_{\text{in}} \times k} \tag{8.19}$$

Despite being linear in the input-output map (equivalent to $W = UV^\top$), this model exhibits *nonlinear dynamics* under gradient descent due to the factorization. Recent work Advani and Saxe [2017b], Saxe et al. [2014] has shown that this model captures essential features of deep learning dynamics, including staged learning and critical slowdowns.

The advantage of this model is that the Hessian and noise covariance can be computed *analytically* in certain regimes, enabling precise theoretical predictions against which experimental measurements can be compared. The NESP predictions for this model are:

- The "effective temperature" $T_{\text{eff}} = \eta \cdot \mathrm{Tr}(\Sigma)/P$ should exhibit non-monotonic behavior, peaking at the interpolation threshold.
- The condition number $\kappa(H) = \lambda_{\max}/\lambda_{\min}$ should diverge as $\alpha \to 1$, signaling the Jamming transition.

8.3.4 Extension

The idea itself offers many extensions for deployments.

Synthesis: From Equilibrium to Non-Equilibrium

In this work, we have undertaken a critical re-examination of the theoretical foundations underlying the Double Descent phenomenon. Our central thesis is that the apparent "anomaly" of Double Descent—its violation of the classical bias-variance tradeoff—is not a failure of statistical physics *per se*, but rather a failure of **equilibrium** statistical physics inappropriately applied to an inherently **non-equilibrium** dynamical process.

We have argued that the dominant theoretical approaches—Random Matrix Theory, Replica Methods, and classical bias-variance decomposition—share a common, often implicit, assumption: that the learning algorithm converges to a stationary state characterized by a Boltzmann-like distribution over weight space. This equilibrium assumption, while mathematically convenient, fundamentally mischaracterizes the nature of Stochastic Gradient Descent.

Our reformulation, grounded in Non-Equilibrium Statistical Physics (NESP), makes three key departures from the equilibrium paradigm:

1. **From static distributions to dynamical evolution**: We model the probability density $\rho(W, t)$ via the Fokker-Planck equation, treating the "solution" as a transient state rather than an equilibrium configuration.
2. **From isotropic to anisotropic noise**: We recognize that SGD noise is not thermal but geometric—its covariance $\Sigma(W)$ is state-dependent and aligned with the loss landscape curvature $H(W)$.
3. **From energy minimization to entropic selection**: We posit that generalization arises not from finding the global minimum, but from the *dynamical stability* of flat minima under geometry-dependent noise. Sharp minima, despite potentially lower loss, are rendered unstable by their own curvature.

Bridging the Gap: The Play and The Stage

A persistent tension in theoretical machine learning has been the disconnect between two perspectives:

- **The Stage (Landscape)**: Static analysis of the loss surface—its minima, saddle points, and global structure.
- **The Play (Dynamics)**: The trajectory of the optimizer through weight space—its convergence, oscillations, and implicit biases.

Equilibrium theories privilege the Stage: they characterize the set of possible solutions and assume the optimizer will find them. But they cannot explain *which* solution is selected, nor *how* the selection depends on hyperparameters like learning rate and batch size.

The NESP framework bridges this gap by coupling the Stage to the Play. The landscape geometry (Hessian) determines the noise structure (diffusion tensor), which in turn shapes the dynamical trajectory. Double Descent emerges naturally from this coupling: the interpolation peak corresponds to a *Jamming transition* where landscape constraints saturate available degrees of freedom, and the second descent corresponds to *entropic relaxation* as the system escapes sharp minima driven by its own noise.

This perspective offers a resolution to the longstanding puzzle: *Why do over-parameterized models generalize?* Our answer: not because they find better minima (they often do not), but because SGD dynamics are *geometrically biased* toward flat regions of the solution manifold. Generalization is an emergent property of non-equilibrium dynamics, not a static property of the loss landscape.

Moreover, if the NESP framework, is formally validated, it opens several exciting avenues for future research, as followed.

Grokking as Extreme Relaxation

The phenomenon of "Grokking"—wherein models exhibit delayed generalization long after achieving zero training error Power et al. [2022b]—presents a striking challenge to equilibrium theories. From the NESP perspective, we speculate that Grokking represents an *extreme* instance of the Jamming-Relaxation transition.

Specifically, certain data structures (e.g., modular arithmetic) may induce loss landscapes where the Jammed phase is *metastable* over extended timescales. The system achieves interpolation (zero training loss) but remains trapped in sharp, non-generalizing minima. Generalization occurs only after sufficient time for the noise-driven annealing process to escape these metastable states.

This interpretation predicts that Grokking should be:

- **Accelerated** by increasing learning rate or decreasing batch size (higher effective noise).
- **Suppressed** by explicit regularization that pre-selects flat minima.
- **Accompanied** by a monotonic decrease in Hessian trace during the "grokking" phase.

Transformers and Large Language Models

The training dynamics of Large Language Models (LLMs) represent perhaps the most consequential—and least understood—application of deep learning. The NESP framework suggests several speculative connections:

Temperature in Sampling vs. Training. LLMs employ a "temperature" parameter T during inference to control the sharpness of the output distribution. Intriguingly, our framework identifies an "effective temperature" $T_{\text{eff}} = \eta \cdot \text{Tr}(\Sigma)/(2B)$ that governs training dynamics. We pose the question: *Is there a principled relationship between these two temperatures?*

One speculative hypothesis: models trained with higher T_{eff} (larger learning rate, smaller batch) may naturally produce sharper internal representations, requiring lower inference temperature for coherent outputs. This would establish a non-trivial link between training dynamics and inference behavior.

Scaling Laws and Phase Transitions. Empirical "scaling laws" Kaplan et al. [2020] describe how LLM performance improves with model size, data, and compute. From the NESP perspective, these smooth power-law scalings may obscure underlying *phase transitions* in the training dynamics. As model capacity crosses critical thresholds relative to data complexity, the system may undergo Jamming-Relaxation transitions analogous to Double Descent.

Understanding these transitions could inform more efficient training schedules: rather than smooth scaling of learning rate, one might employ *punctuated* schedules that exploit phase boundaries—high noise to escape Jammed phases, low noise to stabilize in Relaxed phases.

Toward a Unified Theory of Implicit Regularization

The NESP framework provides a candidate mechanism for "implicit regularization"—the empirical observation that SGD finds generalizing solutions without explicit regularization terms. Our entropic selection mechanism offers a *dynamical* explanation: SGD implicitly regularizes by destabilizing sharp minima.

A unified theory would connect this mechanism to other forms of implicit bias:

- **Minimum-norm solutions** in linear regression.
- **Max-margin classifiers** in linearly separable problems.
- **Low-rank bias** in matrix factorization.

We conjecture that all these biases arise from the same underlying principle: the geometry-dependent noise of SGD preferentially stabilizes solutions with specific structural properties (low norm, large margin, low rank) because these properties correlate with flat loss landscape geometry.

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Reevaluation of theorem 5.1

To verify the theorem itself, we test it with a standard setting without the theorem results. This is reflected in Figure 5.1. The result is somewhat consistent with the theorem and its latter parameter error and prediction error results, however, this time, it correlates exactly as the prediction error instead. Additionally, peculiarity occurs in the non-double descent region, with much less volatility than in the prediction of the theorem.

Algorithm 6: Linear regression with Gaussian white noise, full control.

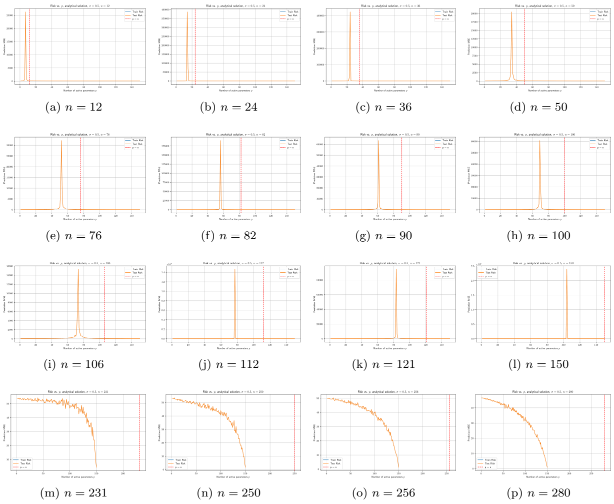
Data: n, p, d, σ, r, fixed range $[a, b]$, $u \in [0, 1]$
Result: Threefold error measure - MSE on parameters p, d, MSE on prediction Y', Y, and theoretical risks measure by prescription.

begin
- $\text{Randomizer}(\cdot) \leftarrow \text{PCG64}(n \leq 100)$; `Randomizing pseudo-random engine`
- $\sigma \leftarrow 0.5, d \leftarrow 150$; `Common presetting for parameters`
- $nl \leftarrow \{12, \ldots, 280\}$; `Common full-list` n `count of datapoints`
- $p \leftarrow 150$; `Underparameterization to equal`
- $\mathbf{p} \leftarrow \text{range}(1, p)$;
- $[a, b] \leftarrow [-100, 100]$;
- $w_c \sim \text{PCG64.Uniform}([1, d], 150)$;
- $b_c \sim \text{PCG64.Uniform}([1, 10])$;
- $w_p \sim \text{PCG64.Uniform}([1, p], p)$;
- $b_p \sim \text{PCG64.Uniform}([1, 10])$;
- $X \sim \text{PCG64.Uniform}(\text{range} \leftarrow [a, b], \text{size} \leftarrow [n, d])$; n `samples of` d `dimensions.`
- $\text{concept}(\cdot, \cdot, \cdot) \leftarrow (d, w_c, b_c)$;
- $Y \leftarrow \text{concept}(X, w_c, b_c)$;
- $\sigma_{\text{scaled}} \leftarrow \sigma \cdot \mathbb{E}[|Y|]$;
- $Y \leftarrow Y + \text{PCG64.}\ldots\text{Normal}(\mu \leftarrow 0,\ \sigma \leftarrow \sigma_{\text{scaled}},\ \text{size} \leftarrow |Y|)$; `Noisy featuring`
- **for** $n \in nl$ **do**
  - $w_p^* \leftarrow \text{PCG64.Permutation}(d, p)$;
  - **for** $i \leftarrow 1$ **to** $|w_p|$ **do**
    - **if** $i \neq w_p^*$ **then**
      - $w_p[i] \leftarrow 0$, fixed zero point.
    - **end**
  - **end**
  - $\text{model}(\cdot, \cdot, \cdot) \leftarrow (d, w_p, b_p)$;
  - $Y \leftarrow \text{model}(X, w_p, b_p)$;
  - $\text{approx} \leftarrow \text{LeastSquare}(\text{model}, Y', X, Y)$, fixed w_p.
- **end**
- $\text{MSE}_{\text{param}}(\cdot, \cdot, \cdot) \leftarrow \text{MSE}(Y, \{\text{approx}\}, \text{model})$;
- $\text{MSE}_{\text{pred}}(\cdot, \cdot) = \text{MSE}(Y, Y')\ \hat{R}_{\text{Thm}} = \text{TR}(w_c, \sigma, p, d, n)$;

end
return $\text{MSE}_{\text{param}}$, MSE_{pred},$\hat{R}_{\text{Thm}}$

We use the Algorithm 6 for verification, including three modes of error measure, which will then be calculated accordingly. For gradient descent, we refer to another experiment. For such algorithm, we notice that the operational space of the model is $\mathbb{R}^n \rightarrow \mathbb{R}$. The masking of parameters then is not well-received, since the output space is far more skewed - single dimension resultant with no scaling difference because the model in question is supposed to be linear. Regarding such, it makes sense to realize that the model itself can still attain relatively stable error, because its feature compounds on the scaling parameters

To test the stability of the model, we run for a stable section of the permutations done on p. The result is outlined in 5.1. Interestingly, while the interpolation threshold theorized being $p = n$ does not hold, the pattern of double descent without head descent still fundamentally exists, albeit in a much more constrained region, as can see from $n = 12$ to $n = 150$. The pattern from then onward follows a reduced trend, without the chaotic behaviours seen. The result is shown in Figure 1.



(a) $n = 12$ (b) $n = 24$ (c) $n = 36$ (d) $n = 50$

(e) $n = 76$ (f) $n = 82$ (g) $n = 90$ (h) $n = 100$

(i) $n = 106$ (j) $n = 112$ (k) $n = 121$ (l) $n = 150$

(m) $n = 231$ (n) $n = 250$ (o) $n = 256$ (p) $n = 280$

Figure 1: Risk curves for varying sample size n (dimension $d = 100$, noise $\sigma = 0.5$). Double descent is determined in similar sequence, even though the interpolation theoretical $p = n$ is actually shifted over to the right from the actual interpolation observables, in parameterization. Afterward, correlation fails, and the theoretical p is ineffective, similar to previous theorem-based test run. Sample set size range is $n \in \{12, 24, 36, 50, 76, 82, 90, 100, 106, 112, 121, 150, 231, 250, 256, 280\}$. One additional remark is that the error landscape is relatively thin in either side, making it abnormal in comparison to actual bias-variance curvature.

Remark

At least from this experiment, overall, results gathered using Theorem 5.1.1 and variations of it suggests inconsistent results. The supposed interpolation threshold has different interpretation and indication in different scenario, including both parameter-wise error and prediction-only error measure. Furthermore, verification also shows inconsistency with the interpolation threshold, albeit given the fact that the margin of error from the highest peak to the interpolation point is consistent up to certain n. Breakdown of the result from theorem, as well as double descent behaviour starts with high n counts, as well as interesting outlook of bias-variance reappearance for $p \sim 5000$, far more than what is tested in normal testing cases. Such result suggests further experiments to validate the observations, as well as statistical analysis of the entire system.

Specifically, the interpretation of the result is not perfect. Under such setting, then for the concept of the same class, so $d = 150$, any concept of $p > d$ would result in the metric taking zero value of the 'truncated region' where d is literally flat on the $d + 1$ onward space. Hence, the error being added on top of such error measure is the component distance of p to the flat sub $p - d$ space itself - which explains the curve going upward, as more and more risks are configured. Nevertheless, the amount of data provided, at least in terms of this model, does indeed have non-trivial effect on the model itself, and the least square solution of best fit which gives the potential minimal, hence the empirical best. Considering the results, we can see regions where the theorem 'breaks' of its bias-variance pattern. Further experiment to verify such claim and reconfiguration is needed. Worryingly, in this particular case, there exists no saddle point that was theorized by bias-variance tradeoff, which must then be explained accordingly. Furthermore, we also notice the troublesome fact that this particular pattern works on *same operational space* models, that is, linear models approximating linear models. This setting is fairly limited, and would not generalize well. However, we also do not know what kind of measure to take in such case. Conflicting notions and reports aid into the confusion of such topic more than can be normally observed, and the anomalous behaviours in some researches (for example, in Shi et al. [2024], we received that in GNN, there exists no trace of double descent) makes it even harder. As such, both the status of bias-variance being false, albeit in a given range only, and double descent in masking certain behaviours of the modelling process, have taken a mysterious stance in the modern machine learning community.

While discovering the double descent phenomena, it also exposes theoretical concepts that are not organized or formalized, portions of the theory and empirical observations that are within conflict of each others, issues with

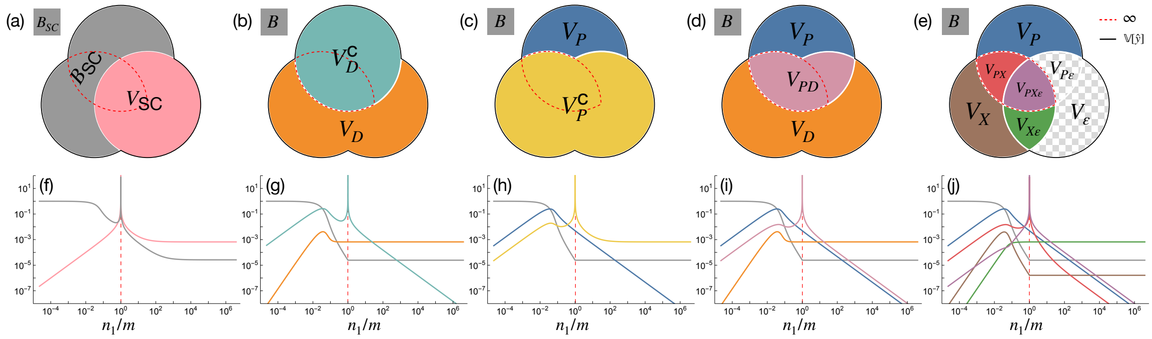


Figure 2: **(a–e) The different bias-variance decompositions**. (f-j) Corresponding theoretical predictions for $\gamma = 0$, $\phi = 1/16$ and $\sigma = \tanh$ with SNR $= 100$ as the model capacity varies across the interpolation threshold (dashed red). (a,f) The semi-classical decomposition of Hastie et al. [2019], Mei and Montanari [2019a] has a nonmonotonic and divergent bias term, conflicting with standard definitions of the bias. (b,g) The decomposition of Neal et al. [2018] utilizing the law of total variance interprets the diverging term V_D^c as "variance due to optimization". (c,h) An alternative application of the law of total variance suggests the opposite, *i.e.* the diverging term V_P^c comes from "variance due to sampling". (d,i) A bivariate symmetric decomposition of the variance resolves this ambiguity and shows that the diverging term is actually V_{PD}, *i.e.* "the variance explained by the parameters and data together beyond what they explain individually." (e,j) A trivariate symmetric decomposition reveals that the divergence comes from two terms, V_{PX} and $V_{PX\boldsymbol{\varepsilon}}$ (outlined in dashed red), and shows that label noise exacerbates but does not cause double descent. Since $V_\varepsilon = V_{P\varepsilon} = 0$, they are not shown in (j). Taken from Adlam and Pennington [2020]

definitions, theorems, and interpretation of them. Those problems would then ultimately hinder further research in such direction, as well as broader theoretical requirement of the theory of machine learning, and specifically to a larger and more complex system of deep learning.

Observing gradient descent decomposition

While the interpretation of what the bias-variance tradeoff, the usual training-versus-test curve, and statistical learning consideration, another interesting aspect to consider (as well for double descent), is how bias-variance decomposition acts on gradient descent and its components. This claim is perhaps reinforced by the work of Adlam and Pennington [2020] which indicates that different bias-variance decompositions effects differently on the total system mechanism.

The main term of a gradient descent algorithm is the gradient of the loss function $\mathcal{L}(h(x), y)$. For the hypothesis h, the number of instances supplied denoted by k has three main cases: $k = 1$ for stochastic descent, $k \in (1, m)$ for batch gradient descent, and $k = m$ for standard gradient descent, for a given space of m samples. The gradient in such case is calculated using the empirical risk on m size, that is:

$$\mathcal{L}_k(h(x), y) = \hat{R}_{S[k]}(h) = \frac{1}{k}\sum_{i=1}^{k} \ell(h(x_i), y_i) \tag{20}$$

We use the classical bias-variance tradeoff at first. Since the gradient is a linear operator, we have:

$$\begin{aligned}\nabla_k(\mathcal{L}) &= \nabla_k\left((\mathbb{E}\hat{y}(\mathbf{x}) - \mathbb{E}y(\mathbf{x}))^2 + \mathbb{V}\left[\hat{y}(\mathbf{x})\right] + \mathbb{V}[y(\mathbf{x})]\right) \\ &= \nabla_k\left(\mathbb{E}\left[\hat{y}(\mathbf{x})\right] - \mathbb{E}\left[y(\mathbf{x})\right]\right)^2 + \nabla_k\mathbb{V}\left[\hat{y}(\mathbf{x})\right] + \nabla_k\mathbb{V}\left[y(\mathbf{x})\right] \\ &= \nabla_k\left(\mathbb{E}\left[\hat{y}(\mathbf{x})\right] - \mathbb{E}\left[y(\mathbf{x})\right]\right)^2 + \nabla_k\mathbb{V}\left[y(\mathbf{x})\right]\end{aligned} \tag{21}$$

This effectively decompose the gradient into three parts that influence the overall gradient calculation. Because $\nabla_k\mathbb{V}\left[y(\mathbf{x})\right]$ calculates the gradient of irreducible error, which is supposed to be constant of the sample space, it is then equal 0. Suppose a sample $\mathbf{x}_k$ of size km, of partition k. Hence, we have gradient descent based on kth run over the entire dataset, permuted. Then, the loss function calculates in the form

$$
\begin{aligned}
\nabla_{k,n}(\mathcal{L}) = \nabla_{k,n} &\left[\frac{1}{n}\sum_{i\le n}\hat{y}(x_{kn+i}) - \frac{1}{n}\sum_{i\le n} y(x_{kn+i})\right]^2 \\
+ \nabla_{k,n} &\left[\frac{1}{n}\sum_{i=1}^{n} y(x_i)^2 - \left(\frac{1}{n}\sum_{i=1}^{n} y(x_i)\right)^2\right]
\end{aligned}
\tag{22}
$$

The indices is now changed to (k, n) for $m/k = n$ as the ith partitioned, ordered run, thereby $kn + i$ runs for the index. If we assume a simplified structure θ_t of parameters $y(x_i)$ depends on, we can reduce it further.

$$
\begin{aligned}
&\nabla_{k,n}(\mathcal{L}) \\
&= 2\left(\frac{1}{n}\sum_{i=1}^{n}\hat{y}(x_{kn+i}) - \frac{1}{n}\sum_{i=1}^{n} y(x_{kn+i})\right) \\
&\times \left(\frac{1}{n}\sum_{i=1}^{n}\nabla_k\hat{y}(x_{kn+i}) - \frac{1}{n}\sum_{i=1}^{n}\nabla_k y(x_{kn+i})\right) \\
&+ \frac{2}{n}\sum_{i=1}^{n} y(x_i)\,\nabla_k y(x_i) - \frac{2}{n^2}\left(\sum_{i=1}^{n} y(x_i)\right)\left(\sum_{i=1}^{n}\nabla_k y(x_i)\right) \\
&= 2\Big[\big(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\big)\big(\mathbb{E}_n[\hat{G}_k] - \mathbb{E}_n[G_k]\big) \\
&\qquad + \mathbb{E}_n[YG_k] - \mathbb{E}_n[Y]\,\mathbb{E}_n[G_k]\Big] \\
&= 2\Big[\big(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\big)\big(\mathbb{E}_n[\nabla_k\hat{Y}] - \mathbb{E}_n[\nabla_k Y]\big) \\
&\qquad + \mathrm{Cov}_n(Y, \nabla_k Y)\Big] \\
&= 2\big(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\big)\big(\mathbb{E}_n[\nabla_k\hat{Y}] - \mathbb{E}_n[\nabla_k Y]\big) \\
&\quad + 2\,\mathrm{Cov}_n(Y, \nabla_k Y)
\end{aligned}
$$

for the expectation over Z_i, and covariance,

$$
\mathbb{E}_n[Z] = \frac{1}{n}\sum_{i=1}^{n} Z_i, \quad \mathrm{Cov}_n(U, V) = \mathbb{E}_n[UV] - \mathbb{E}_n[U]\,\mathbb{E}_n[V]. \tag{23}
$$

Here, we can see two terms multiplication, for the first one being the correlation between the expected error $\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]$ and the gradient $\mathbb{E}_n[\nabla_k\hat{Y}] - \mathbb{E}_n[\nabla_k Y]$. This measures the correlation between the expected error, and the gradient measure. The second term is exactly the empirical covariance between the target Y and the true gradient component. Since we are taking a fixed concept case, we can simply disregard the truth gradient hence $\mathbb{E}_n[\nabla_k Y]$, for which the term is reduced to $2\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)\left(\mathbb{E}_n[\nabla_k\hat{Y}]\right)$. Under this same assumption applied to the covariance term, we obtain

$$
\begin{aligned}
\nabla_{k,n}(\mathcal{L}) &= \frac{2}{n}\sum_{i=1}^{n}\big(\hat{Y}_i - Y_i\big)\,\partial_k\hat{Y}_i \\
&= 2\left(\left(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\right)\mathbb{E}_n[\nabla_k\hat{Y}]\right)
\end{aligned}
\tag{24}
$$

Looking into some particular algebraic representation, we can write this simply as

$$
\nabla_{k,n}\mathcal{L} = 2\,\mathbb{E}_n\left[(\hat{Y} - Y)\,\nabla_k\hat{Y}\right] \tag{25}
$$

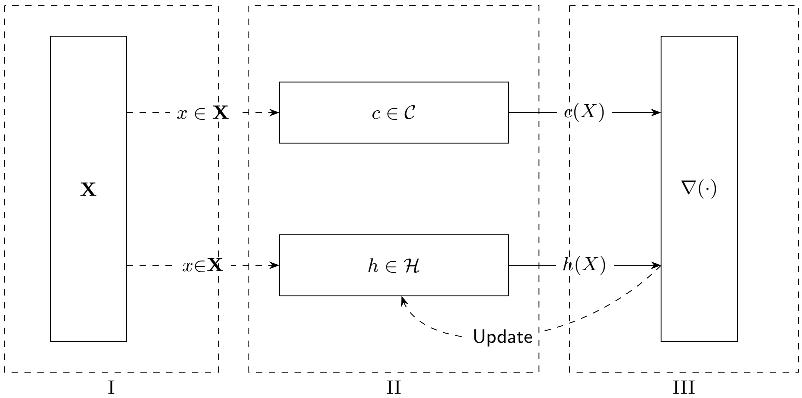


Figure 3: An illustration of the (supervised) statistical process. Phase III contains two parts: First is the evaluation $\nabla(h, c)$ according to the data $\mathcal{D}$, and second is the Update process to re-align c to the actual target.

$$
\begin{aligned}
&= 2\Bigg(\underbrace{\big(\mathbb{E}_n[\hat{Y}] - \mathbb{E}_n[Y]\big)}_{\left[\mathbb{B}(\hat{Y},Y)\right]^{1/2}} \mathbb{E}_n[\nabla_k \hat{Y}] && (26)\\
&+ \underbrace{\mathbb{E}_n[\hat{Y}\,\nabla_k \hat{Y}] - \mathbb{E}_n[\hat{Y}]\,\mathbb{E}_n[\nabla_k \hat{Y}]}_{\mathrm{Cov}_n(\hat{Y},\nabla_k \hat{Y})} && (27)\\
&\quad - \underbrace{\mathbb{E}_n[Y\,\nabla_k \hat{Y}] - \mathbb{E}_n[Y]\,\mathbb{E}_n[\nabla_k \hat{Y}]}_{\mathrm{Cov}_n(Y,\nabla_k \hat{Y})} \Bigg) && (28)
\end{aligned}
$$

The first term can be attributed to self-correct of bias-error gradient correlation, while the two covariance term suggests correlation between model randomness and its gradient, and the last one deal with the truth-gradient correlation. What those gradients mean will have to be tested in some preliminary models. Generally, we should not expect the result to be quite effective. Similar decomposition can be conducted for the other decomposition, as suggested, however, the gradient effect is particularly interesting, and will have to be resolved.

Details of supervised learning structure

In a typical machine learning setting, we obtain assume the following figure 3. This includes the ground, input space (the setting of which the observations are observed), the concept $c \in \mathcal{C}$, the hypothesis $c \in \mathcal{H}$, and a supervisor ∇ that 'correct' the behaviours of the hypothesis to match with c. This is called a supervised learning setting, in which the learning part is governed by the supervisor and the observations from the concept.

Phase 1.

Begin with the construction and initialization of the feature space $\mathcal{X}^{\infty}$. By the assumption of a probabilistic process, $\mathcal{X}_m$ or simply $\mathcal{X}$ representing the dataset is assumed to be sampled, or appeared from the distribution $\mathcal{D}(p, \cdot)$ for p an arbitrary probabilistic system.

The *ground space*, or in literature generally called **feature space**, is created. This results in the first component of the tuple D, being $\mathcal{X} \subseteq \mathbb{R}^{m \times k}$, where $|\mathcal{X}| = m$, but $size(x) = k$, for $x \in \mathcal{X}$. We assume there exists the random variable x of a given distribution $\mathcal{D}$, such that the set $\mathcal{X} \sim \mathcal{D}$ of all observation is given by an underlying distribution. A well-known assumption in this case is that $x \in \mathcal{X}$ has no dependent components. That is, for example, there does not exist any function $\psi_x : \{x_i\} \to \{x_j\}$, for $i \neq j$. (Thought, this only helps in Bayesian priori analysis). Furthermore, by specifying that $\mathcal{X}$ belongs to a specific distribution, we argue of the assumption that $x \in \mathcal{X}$ is *independently and identically distributed* (i.i.d.), that is, for a given random sampling interpretation $S \subset \mathcal{X} \sim \mathcal{D}$, all data is sampled with the events space governed by the distribution $\mathcal{D}$ for every instance, and the existence of x_i to x_j for

$i \neq j$ is none.

The **role** of the distribution configuration is important. If $\mathcal{D}$ is uniform, that is, $D \sim \mathcal{U}(a, b)$, then we can disregard the aspect of $\mathcal{X}$ with random variables. For $[a, b]$ to be $(-\infty, +\infty)$, the problem changed to a *regression problem*. If $\mathcal{D}$ is some other distribution function, then the problem of *representative data* and *population size* becomes apparent, which requires statistical analysis to be taken into consideration.

Phase 2.

For the constructed feature space $\mathcal{X}$, let $EX(c, \mathcal{D})$ be a procedure (or *oracle*) acting on $\mathcal{C}$ and $\mathcal{X}$ to output $\langle x, c(x)\rangle$. The hypothesis then contains a similar procedure $EX_{\mathcal{H}}(h, \mathcal{D})$, such that to output $\langle x, h(x)\rangle$. We call this the *inference phase*.

The feature space $\mathcal{X}$ is provided to h. We assume c is processed of the same process, resulted in $\mathcal{Y}_c$. Then, $h(\mathcal{X})$ outputs the set of hypothesis' target $\mathcal{Y}_h$. Thereby, we are approximating the *process* c itself. We can approach this in two main ways[1]: either *deterministic* or *probabilistic*[2]. We define such as followed. For the constructed feature space $\mathcal{X}$, let $EX(c, \mathcal{D})$ be a procedure (or *orcale*) acting on $\mathcal{C}$ and $\mathcal{X}$ to output $\langle x, c(x)\rangle$. The hypothesis then contains a similar procedure $EX_{\mathcal{H}}(h, \mathcal{D})$, such that to output $\langle x, h(x)\rangle$. We call this the *inference phase*.

Definition .0.1 (Deterministic - *discriminative modelling*). *A **deterministic model** h_d assumes its internal space as followed. For any $h \in \mathcal{H}_d$ of deterministic model, h is characterized by the mapping $h_d : \mathcal{X} \to \mathcal{Y}_h$, such that $h \in C^n$ of n-differential space.*

Similarly, if the interpretation of the model is *probabilistic*, we have the following definition of probabilistic-based model.

Definition .0.2 (Probabilistic - *generative modelling*). *A **probabilistic model** h_p assumes its internal space as followed. For any $h \in \mathcal{H}_p$ of all probabilistic hypothesis, h is characterized by the probability distribution $\Gamma(p_X(X))$, where Γ is the discrete decision process outside the probabilistic interpretation.* [3]

Notice that the process and the learning setting is probabilistic, however, the interpretation of the hypothesis h can be either deterministic or probabilistic, or anything else. This puts the flexibility into the setting of the model, and permits us to consider various representation of the model structure. Furthermore, the hypothesis for the learning process is evaluated relative to the same probabilistic setting, in which the training takes place, and we allow hypotheses that are only approximation of the target concept Sugiyama [2015], Kearns and Vazirani [1994].

The similarity between both of the two general types is that the model is characterized by their **parameters**, both **model-specific parameters** and **hyperparameters**.

Phase 3.

There exists an evaluator (or *supervisor*) $\nabla(h, c)$ that evaluates specific *loss framework* $\ell\{h(x), c(x)\}$ based on available information, and a *update modifier* $U(\ell, \mathcal{A})$ of certain objective to algorithm $\mathcal{A}$.

In this step of the procedural setting, the *supervisor* includes a loss function, ℓ, which takes discrete arguments, and 'binarily' evaluate the discrete result between h and c. This loss function is supposed to have a global minimum, or at least a certain region of minimal approximation. We have the following definition.

Definition .0.3 (Loss function). *A loss function is a non-negative function $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, +\infty)$.*

In general, the following is supposed of the loss function:

---

[1] Note that, in some aspect, this is similar to the usage of mathematical simulation of certain dynamical system, or *any* system that has certain patterns or relations observable.

[2] The model itself can be entirely deterministic, while the learning process is not. in fact, one of the very first assumption and axiomatic view of the learning problem is that learning occurs in a *probabilistic setting*.

[3] The discrete decision process Γ is very simple to argue. For example, given a Naive Bayes model with $p(C_k \mid D[i])$ of the dataset, the probability distribution is regarded as the argument

$$Y^* = \arg\max_{c_k \in C} P_\theta(c_k \mid x) = \arg\max_{c_k \in C} P_\theta(x \mid c_k) \cdot P(c_k)$$

Conjecture .0.1 (Loss function convexity). *For any supervisor ∇ of a learner framework $\mathcal{L}(\mathcal{H}, \mathcal{A})$, the loss function ℓ is assumed to be a p-converging function, or as a convex function. That is, for $\ell : \mathbb{R}^n \to \mathbb{R}$, then ℓ is convex on its domain, or is locally convex on $[a, b] \subset \mathbb{R}^n$, if, for $x, y \in \mathbb{R}^n$ of such interval, and for $\lambda \in [0, 1]$, it satisfies*

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \tag{29}$$

Under a single structure, that is, for example, a class of mapping $\mathbb{R}^n \to \mathbb{R}$, there can exist many loss functions to be considered. Take for example the setting of regression analysis. Then, ℓ can be either the absolute error, $|h(x) - c(x)|$, or the L^2 norm, $||h(x), c(x)||_2$. Different loss function provides different path and landscape, though they still share somewhat similar interpolation patterns, or either some might be subjected to, under the consideration of a loss landscape, local minima than others. Furthermore, the form and representation of loss functions is not discrete - but rather based of its interpretation and the supposition of the underlying notion required for the loss function to operate in. For example, if the setting is *generative*, the loss function would take the form of a *divergence measure* between two probability distribution. Notice however, that the action potential provided by the model in their operation dissapears, or rather, is not considered, when using the loss function.

Using such loss function, for the case of no noises or interference uncertainty, if the goal is to minimize the empirical risk on such dataset, then we call this the method of *empirical risk minimization*, or ERM. This is born out of the consideration that in practice, the oracle $EX(c, \mathcal{D})$, and the actual generalization error $R(h)$ is not obtainable.

Appendix D - Bias-variance decomposition derivation in Geman et al. [1992]

Proof. We present the original derivation of bias-variance tradeoff and its analysis from Geman et al. [1992]. Geman's paper is, by his own word, concerned of *parametric models*, i.e. hypothesis class without strong assumption of parameters defined, though the definition of bias-variance tradeoff afterward and its justification is made in the sense of parametric model. We partially clarify[4] this with the following definition as customary.

Definition .0.4 (Parameterization). *A **parametric model** is one that can be parameterized by a finite number of parameters. In general, for a hypothesis class $\mathcal{H}$of all parametric hypothesis is then expressed as:*

$$\mathcal{H} = \{f(x; \theta) : \theta \in \Theta \subset \mathbb{R}^d\} \tag{30}$$

*where Θ is called the parameter space. A **nonparametric model** is one which cannot be parameterized by a fix number of parameters.*

Suppose of a training dataset $\mathcal{D}$ of N 2-tuples $(\mathbf{x}_i, y_i)$, that is:

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\} \quad |\mathcal{D}| = N, \mathbf{x} \in \mathbb{R}^d, y \in \mathbb{R} \tag{31}$$

The regression problem is to construct a function $f(\mathbf{x}) \to y$ based on $\mathcal{D}$, which is denoted $f(\mathbf{x}; \mathcal{D})$ to show this dependency. Then,

$$\begin{aligned}
&\mathbb{E}\left[(y - f(\mathbf{x}; \mathcal{D}))^2 \mid \mathbf{x}, \mathcal{D}\right] \\
&= \mathbb{E}\Big[\big((y - \mathbb{E}[y \mid \mathbf{x}] + (\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x}; \mathcal{D}))\big)\big)^2 \mid \mathbf{x}, \mathcal{D}\Big] \\
&= \begin{aligned}[t] &\mathbb{E}\Big[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D}\Big] + \Big(\mathbb{E}[y \mid \mathbf{x} - f(\mathbf{x}; \mathcal{D})]\Big)^2 \\ &+ 2\mathbb{E}\big[(y - \mathbb{E}[y \mid \mathbf{x}]) \mid \mathbf{x}, \mathcal{D}\big] \cdot \big(\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x}; \mathcal{D})\big)^2 \end{aligned} \\
&= \mathbb{E}\Big[(y - \mathbb{E}[y \mid \mathbf{x}]) \mid \mathbf{x}, \mathcal{D}\Big] + \big(\mathbb{E}[y \mid \mathbf{x}] - f(\mathbf{x}; \mathcal{D})\big)^2 \\
&\geq \mathbb{E}\big[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x}, \mathcal{D}\big]
\end{aligned} \tag{32}$$

[4] Generally, there is no definite definition on what would be considered non-parametric. Though, an example might be drawn from the (vanilla) neural network and Gaussian Process Regression (GPR)

Hence, we decompose it to:

$$\begin{aligned}&\mathbb{E}\left[((y - f(\mathbf{x};\mathcal{D})))^2 \mid \mathbf{x},\mathcal{D}\right] \\ &\quad= \mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right] + (f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2\end{aligned} \tag{33}$$

Where $\mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right]$ is regarded to be a relative constant of variance on y given $\mathbf{x}$. Taking the expectation on $\mathcal{D}$, we gain:

$$\begin{aligned}&\mathbb{E}_{\mathcal{D}}\Big\{\mathbb{E}\left[((y - f(\mathbf{x};\mathcal{D})))^2 \mid \mathbf{x},\mathcal{D}\right]\Big\} \\ &= \mathbb{E}_{\mathcal{D}}\Big\{\mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right] + (f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2\Big\} \\ &= \begin{array}{l}\mathbb{E}_{\mathcal{D}}\Big\{\mathbb{E}\left[(y - \mathbb{E}[y \mid \mathbf{x}])^2 \mid \mathbf{x},\mathcal{D}\right]\Big\} + \\ \qquad \mathbb{E}_{\mathcal{D}}\Big\{(f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2\Big\}\end{array}\end{aligned} \tag{34}$$

The second term is of importance, and is further decomposed to:

$$\begin{aligned}&\mathbb{E}_{\mathcal{D}}\Big\{(f(\mathbf{x};\mathcal{D}) - \mathbb{E}[y \mid \mathbf{x}])^2\Big\} \\ &= \mathbb{E}_{\mathcal{D}}\Big\{(f(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] + \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}])^2\Big\} \\ &= \mathbb{E}_{\mathcal{D}}\Big[(f(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})])^2\Big] \\ &\quad + \mathbb{E}_{\mathcal{D}}\Big[(\mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}])^2\Big] \\ &\qquad + 2\mathbb{E}_{\mathcal{D}}\Big[f(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})]\Big] \\ &\qquad \cdot (\mathbb{E}_{\mathcal{D}}[f(\mathbf{x};\mathcal{D})] - \mathbb{E}[y \mid \mathbf{x}]) \\ &= \underbrace{\{\mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})] - \mathbb{E}[y \mid x]\}}_{\text{bias term}} \\ &\quad + \underbrace{\mathbb{E}_{\mathcal{D}}\left\{(f(x;\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[f(x;\mathcal{D})])^2\right\}}_{\text{variance term}}\end{aligned} \tag{35}$$

which yields the desired bias-variance tradeoff. □

Practical consideration

While we have been talking in the learning setting pretty loosely on the side of the data itself, it is the data that gives us the problem setting. For example, if the data is a pair (X, Y) of input output representation, then we know that our working space would be $\mathbb{R}^n \times \mathbb{R}^m$ or any of its subset, given the fact that data must be encoded into numerical representations, even in discrete or logical case (in such case, typically, $0 - 1$ pair is enough). In more complex and complicated setting, for example, on the structure of a graph and its embedded representation, it is much more difficult to see the *form of dataset* to be, hence, also the working space.

Another thing with data is the *availability of the data* and the *distribution of the data* also matters. In some cases, our data consists of only skewed or partitioned sections of the actual concept range, at least up to the observable criterion of the concept (some concept does not exist in certain range, at least in a numerical example, and some other reasons). In the other case, the availability of data, meaning the density of data, as well as added *observational effect* also alters the learning setting, simply because learning theory relies on, at least from our glance, simple statistical, observational learning from observation space, without the added structures. An actual, practical concern however, lies in the range of **practical empirical-generalization** setting.

Observational space partitioning

Previously, we established that at least in said formal setting, generalization errors cannot be known beforehand. Thereby, all of our operations, procedures, processes has to be conducted using the observed data, or the sample space $\mathcal{S}$. A solution to utilize that is to consider the generalization region as the region of *new, unseen observations* that while still is of the same concept c, but it is indeed, unseen if the model is never provided of such instances. Hence, it prompts the utilization of limited resources by then to partition the sample space into various subspaces, that would be then fed into the learning process as observational space. This is simply called **sample set partitioning**. Denoted the number of partition (discrete) as k, then for $k = 2$, then this is called the **train-test partition** (or dataset). This contains a dataset $\mathbf{x}_1$ and $\mathbf{x}_2$ of the same original sample set, such that $\mathbf{x}_1/\mathbf{x}_2 \geq 1$ (one is bigger than another, usually). The ratio does indeed affect the overall running procedure, accordingly [5].

Based on the above conclusion, the empirical error and generalized error can be re-calculated to fit the general scheme. For $|\mathcal{S}| = m$ and partition $k + p = m$ for train and test partition respectively,

$$\hat{R}_k(h) = \frac{1}{k}\sum_{i=1}^{k} \ell\{h(x_i), c(x_i)\}, \tag{36}$$

$$\hat{R}_p(h_{\mathcal{A}}) = \frac{1}{k}\sum_{i=1}^{p} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\} \tag{37}$$

where $h_{\mathcal{A}}$ is the result from the learning algorithm (procedure), for $h^* \neq h_{\mathcal{A}}$. Then, the empirical learning seeks to optimize the first term, that is, for the objective

$$\arg\min_{h\in\mathcal{H}} \hat{R}_k(h) = \arg\min_{h\in\mathcal{H}} \frac{1}{k}\sum_{i=1}^{k} \ell\{h(x_i), c(x_i)\} \tag{38}$$

and the generalization learning as:

$$\begin{aligned}
&\arg\min_{h\in\mathcal{H}} \hat{R}_p(h_{\mathcal{A}}) \\
&= \hat{R}_k(h_{\mathcal{A}}) + \arg\min_{h\in\mathcal{H}} \hat{R}_p(h_{\mathcal{A}}) \\
&= \arg\min_{h\in\mathcal{H}} \left[\frac{1}{k}\sum_{i=1}^{k} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\} + \frac{1}{k}\sum_{i=1}^{p} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\}\right] \\
&= \arg\min_{h\in\mathcal{H}} \left[\epsilon + \frac{1}{k}\sum_{i=1}^{p} \ell\{h_{\mathcal{A}}(x_i), c(x_i)\}\right]
\end{aligned} \tag{39}$$

By the definition order, the generalization error can only be calculated post-$\mathcal{A}$. Hence, we often called the empirical, or ERM-generalizer case to be dynamic, and the solution for ERM is tested statically on generalization set. This partition of order of computation though, brings up the problem of updating factors and others contributing potential that will damage or diffuse the measure.

For cross-validation and other randomize partition, the formulation requires more care in the notion by itself. The structural risk minimization scheme dilemma can be understood from the above consideration naturally. We have effectively defined the two proxies, **heuristic empirical risk** and **heuristic generalization risk**.

[5] If $k > 3$ and above, we call it the nth partitioning. If we further consider the choice of observational spaces being provided, then if $k > 3$, and subspaces are chosen randomly, it is called **cross-validating partition**. The effect of them are fairly well-understood, and we might as well refer to authoritative sources on such issue.

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